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CHAPTER

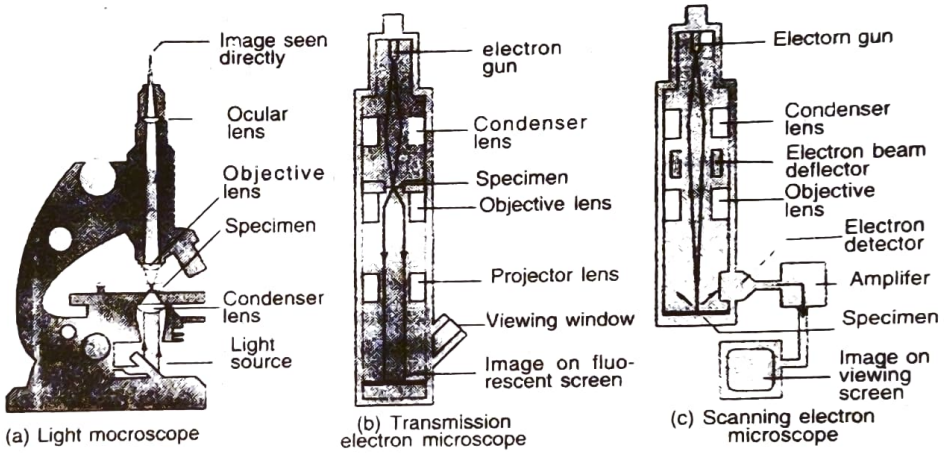
Biophysical and Biochemical Techniques

Microscopy

A microscope may be defined as optical instrument consisting of a lens, or combination of lenses for making enlarged or magnified images of minute objects.

Light Microscope or Compound Microscope

The compound microscope has two set lenses. One is known as objective and the other eye piece. These are mounted in a holder commonly known as body tube. The lens system nearest to the specimen is called objective which magnifies the specimen to a definite number of times. The second lens system is called eye piece which is the nearest to eye. It further magnifies the image formed by the objective. Accurate focusing is attained by a special screw appliance called as fine adjustment.



Types of Microscopes.

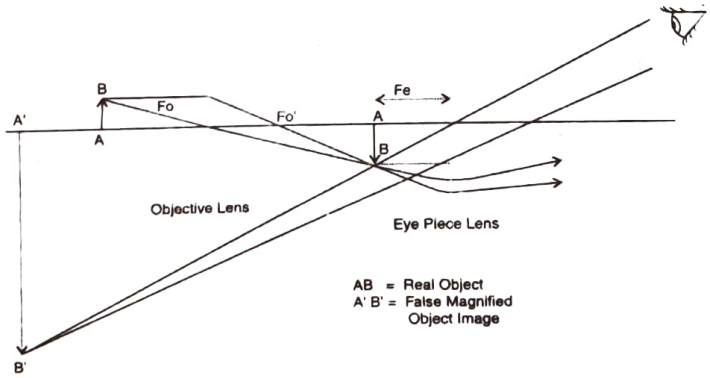


Fig. Image formation in compound microscope.

(*i*) Parts of Compound Microscope :

A compound microscope consists of several parts as described below:

(*a*) Objective Lens:

The chief characteristics of the objective lens are: (a) to gather the light rays coming any point to the object, (b) to unite the light in a point of image, (c) to magnify the image.

There are three types of objectives, namely (a) achromatic which are simplest in construction and adequate for most purpose, (b) flourite are those in which the aberrations are largely eliminated by use of these objectives, (c) apochromatic which are most corrected with respect to aberrations.

(b) Eye piece: The eye piece is also known as oculars. The main functions of the eye piece are: (a) to magnify the real image of the object formed by the objectives, (b) to correct the defects of the objective.

There are three types of eye pieces namely: (a) huggenian eye piece in which two simple planoconvex lenses are employed, one of which is below the image plane. The convex surfaces of both layers face downwards. The huggenian eye piece works well with the low power achromats, (b) hyper-place eye piece are those which may be employed with the high power acromatic flourite and apochromatic objectives without introducing chromatic aberrations in the image; they give much flatter fields, (c) compensating eye piece consists of a chromatic triplet combination of lenses. These eye piece are more perfectly corrected then the other two described above.

(c) Condensers: Several methods are employed for illustrating the object under examination two methods are commonly used: (a) illumination by transmitted light, and (b) dark field illumination. Condensers may be defined as a series of lenses for illumination to the object on the stage of microscope by transmitted light.

(ii) Resolving Power:

It is defined as the power of an objective able to separate distinctly two adjacent points. The resolving power of a microscope is a function of the wavelength of light used and the numerical aperture (NA) of the lens system. The larger the NA, the greater the resolving power of the objective and the finer the detail it can reveal.

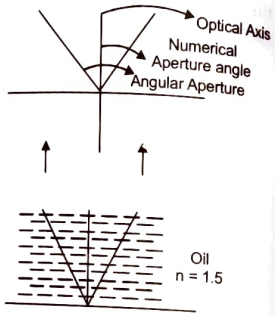


Fig. : Function of oil immersion

(iii) Numerical Aperture (NA):

The angle subtended by the optical axis and the outermost rays still covered by the objective is the measure of aperture of the objective. It is half-aperture angle. The magnitude of this angle is expressed as a sine value.

The sine value of the half aperture angle multiplied by the refractive index (n) of the medium (the space between the front lens and cover slip) gives the numerical aperture (Fig.)

Hence NA = n sine θ

With dry objectives the value of n is 1 because 1 is the refractive index of air. When immersion oil is used as the medium, n = 1.56 and if $\theta = 58°$, then NA = n sin θ = 1.56 × sine 58 = 1.56 × 0.85 = 1.33.

If objects are smaller, the smallest wavelength of visible light cannot be seen. In order to see such minute objects it would be necessary to use rays of shorter wavelength.

(iv) Limit of Resolution:

It is the smallest distance by which two objects can be separated and distinguished as two separate objects. The greater resolution in light microscopy is obtained with the shortest wavelength of visible light and an objective with maximum NA. The relationship between NA and resolution can be expressed as follows:

$$d = \frac{\lambda}{2NA}$$

where, d = resolution, λ = wavelength of light, NA = numerical aperture

(v) Magnification:

Magnification is the increase in size of the object. Magnification beyond the resolving power is of no value. It is because the larger image will be less distinct in detail and fuzzy in appearance.

Bright Field Microscopy

In bright-field microscopy, the microscope field (the area observed) is bright and the microorganisms appear dark because they absorb some of the light. Normally, microorganisms do not absorb much light, but staining them with dye greatly increases their absorbing ability. Generally microscope of this type produce useful magnification of about X 1000 to X 2000. At magnification greater than X 2000, the image appeears fuzzy. It is also called microscopy by transmitted light.

Dark Field Microscopy

In this type of microscopy, a dark back ground is produced against which objects are brilliantly illuminated. For this purpose the light microscope is equipped with a special kind of condenser that transmits a hollow core of light from the source of illumination. Thus, if the aperture of condenser is allowed to open completely, and a dark field stop inserted below the condenser, the light rays reach the objects form a hollow core. Any object within this beam of light will reflect some light into the objective and will be visible. This method of illuminating an object where the object appears self-illuminous against a dark field, called dark-field illustration. The condensers used are Abbe condenser, paraboloid condenser and cardoid condenser. Dark field microscopy is particularly valuable for the examination of unstained microorganisms suspended in fluid wet mount; and hanging drop operations.

Phase Contrast Microscopy

The phase contrast principle was discovered by Fritz Zernike who was awarded Nobel prize in physics in 1953. According to this principle, light waves have variable character for frequency and amplitude. Human eyes cannot perceive a phenomenon when two light rays have similar amplitude and frequency but different phases (Fig.).

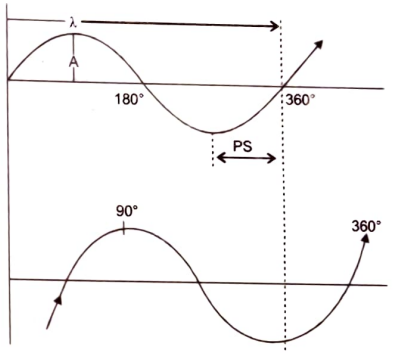


Fig. : Light waves exhibiting similar properties of amplitude (A), and wavelength (λ) but deffering from one another by being out of phase (Ps)

The phase contrast microscope is an ordinary bright field microscope with two additional plates, namely annular diaphragm and phase shifting plate, which enables the usage forming rings to be phase shifted with respect to others. Annular diaphragm allows only a ray of light to pass through the condenser and then to object.

The phase shifting plate is placed at the rear focal plane of objective lens. This disc has a ring of optical dielectric material on which the ring of light from annular diaphragm is focused. The ring has the property of retarding or advancing the phase of light of a quarter of a wavelength traversing it.

From each translucent or transparent particle in the object, consider a single ray of incident light. From this two rays result, one, the direct (undiffracted) ray comes through the annular diaphragm passes through the objective and focussed on phase-shifting ring which either retards or advances the ray one quarter wavelength with respect to second ray. The second ray is also derived from the incident ray, but modified by being scattered or diffracted in passing around margin of the object. This ray does not pass through the phase-shifting ring but traverse the other area of transparent disc. Its wavelength is neither retarded nor advanced. Thus, there is one quarter wavelength out of phase. The contrast between the two is called phase-contrast. It is valuable device in wet mounts and hanging drops. It increases visualization of cellular structure and traverse the other area of transparent disc.

Fluorescent Microscopy

Many chemical substances absorb light. After absorbing light of a particular wavelength and energy, some substances emit light of larger wavelength and lesser energy content. Such substances are called fluorescent and the phenomenon is termed asflourescence. This is the phenomenon which is applied in fluorescent microscopy. In practice, microorganisms are stained with a fluorescent dye and then illuminated with blue light. The blue light is absorbed and green is emitted.

Blue and green fluorscence can be excited by ultraviolet radiation as well. A high intensity mercury lamp is used as light source which emits white light. Two filters are used, one is the exiter filter which transmits only blue light to pass to specimen and blocks out all other colours; the second is barrier filter which blocks out blue light (excitation radiation) and allows green light (or other light emitted by fluorescing specimen) to pass through and reach the eye.

A high intensity mercury lamp is used as the light source and it emits white light. The excited filter transmits only blue light to the specimen and blocks out all other colours. The blue light is reflected downward to the specimen by a dichromatic mirror (which reflects light of certain colours but transmits light of other colours). The specimen is stained with a fluorescent dye: certain portions of the specimen retain the dye, others do not. The stained portion absorbs blue light and emit green light which pass upward, penetrate the dichromatic mirror and reaches the barrier. This filter allows the green light to pass to the eye and blocks out the resident blue light from the specimen which may not have completely deflected by the dichromatic mirror. Thus, the eye perceives the stained portions of the specimen of glowing green against a black background.

Some suitable fluorescence are : acridine orange, acridine yellow; acriflavine auramine 0; fluorescence titan yellow G; rhencine A. etc.

Electron Microscopy

In 1931 Knoll and Ruska. German scientists discovered electron microscopy. Von Borries and Ruska (1938) in Berlin constructed first practical electron microscope.

In electron microscope the source of illumination is electron beam. The construction and principle of electron microscope are easily related to those of light microscope. The range of wave length of visible light used in light microscope is 4000 Å - 7800 Å, while with an electron microscope employing 60-80 KV electron, the wave length is only 0.05 Å. In the instrument, the electron gun generates electron beam. These electrons are concentrated by other components of electron gun producing a fast moving narrow beam of electron. Electrons are focused by electromagnetic lenses. Electromagnetic lens consists of wire encased in soft iron casing. When electric current is passed through the coil, it generates an electromagnetic field through which electrons are focused.

There are three general types of electromagnetic lenses. The one is placed between the source of illumi-nation and the specimen. This focuses the beam of electron on specimen functions in a similar manner as that of light microscope. The other two lenses are on the opposite side of specimen which magnify the image in similar fashion as objective and ocular in light microscope.

Transmission Electron Microscopy

The electron source is commonly a tungsten filament of 30-150 KV potential. The electron beam passes through the centre of ring like magnetic condenser and becomes converged on the specimen. After being transmitted through the specimen (hence transmission electron microscope (TEM), the magnetic objective focuses the electron into a first (real) image of the object which is enlarged (2000 times). The magnetic projector lens then magnifies a portion of the first image producing mangnification upto 240,000 or more.

The final enlarged image can be reviewed by striking a fluorescent screen which makes it visible. The image can also be thrown upon a photographic plate for permanent record. Portions of the photographs may be enlarged four to six times giving the picture in the range of two million times as large as the object.

Molecules in the microscope interfers with the movement of electrons. To prevent this, the interior of the microscope is kept in the state of high vacuum, around 10-4—10-6 mm Hg. It is also necessary to have specimen ultra thin. The electron beams has very poor penetrating power, therefore, only small objects or very thin sections of the specimen can be examined.

Scanning Electron Microscopy (SEM)

Scanning electron microscopes combine the mechanism of electron microscopy and television. SEM became commercially available in early 1960's and the researchers were Knoll, Von Ardenne, Zworytein etc.

In SEM, electrons are not transmitted through the very thin specimen from below but impinge on its surface from above. The specimen may be opaque and of any manageable thickness and size. If the specimen is an electron conductor, it needs only to be held on an appropriate support. If it is non-conductor, it is allowed to dry but if moist, freeze dried in liquid nitrogen is necessary. The specimen is then coated with metal vapour (gold) in vacuum.

The electrons originate at high energy (20,000 V) from a hot tungsten or lanthanum hexoboride cathode "gun". These electrons are sharply focused, adjusted and narrowed by an arrangement of magnetic fields. Instead of forming a broad inverted cone of rays, in SEM a needle sharp probe (about 5 - 10 mm in diameter) is made. This primary beam (probe) acts only as an exciter of image forming secondary electrons emerging from the surface of the specimen.

The probe scans the specimen like that on a blank TV screen. The probe can impinge on depth and heights with equal speed and accuracy giving great depth of field and producing images with three dimensions. Images are elicited from wherever the probe strikes the metal coated areas of the specimen. Magnification is the ratio of final image to the diameter of area scanned.

Any of the secondary electrons with sufficient energy can emerge from the surface. Those that emerge not too far from the point of impact of the probe, can be used to form an image. The useful secondary electrons are magnetically deflected to a collector or detector. Here, they produce a signal that represents at any single moment, only 5-10 mm area or spot of impingement of the probe on the specimen. The successive signals from the collector are amplified and transmitted to a cathode ray (TV) tube. The scanning beam and TV tube beam are synchronized. The image scan by the eye on TV screen is thus the sequence of signals representing in *araster pattern*, the successive areas traversed by the primary probe beam. Exposure may range from a few second to one-half hour or more. The TV image may be photographed, video taped or processed in motion on a computer.

Centrifugation

Centrifuge is a device that spins liquid samples at high speeds and thus creates a strong centripetal force causing the denser materials to travel towards the bottom of the centrifuge tube more rapidly than they would under the force of normal gravity.

Basic Principles of Sedimentation

The rate of sedimentation is dependent upon the applied centrifugal force, directed radially outwards.

Relative Centrifugal Force (RCF)

When an object moves in a circle at a steady angular velocity, it experiences a force, F, directed outwards. F is determined by the square of the angular velocity of the rotor ω, in radians per second) and the radial distance of the particle from the axis of rotation (r, in cm).

$$\text{Centrifugal force} = (\text{angular velocity})^2 \times \text{radius}$$

$$F = \omega^2 r$$

The common way of expressing rotor speed is in terms of revolutions per minute (rev min^{-1} or rpm). Since one revolution of the rotor is equal to 2n radians, its angular velocity can be expressed as,

$$\omega = \frac{2\pi \text{ rev min}^{-1}}{60} \qquad \text{...(2)}$$

Substituting the value of ω in equation 1,

$$F = \frac{4\pi^2 (\text{rev min}^{-1})^2 r}{3600}$$

However, F is expressed as a multiple of the earth's gravitational field (g = 981 cm s^{-2}).

The ratio of the weight of the particle in the centrifugal field to the weight of the same particle when acted by gravity alone is known as the relative centrifugal field (RCF) which is commonly referred to as the 'number times g'. Hence,

$$RCF = \frac{4\pi^2 (\text{rev min}^{-1})^2 r}{3600 \times 981}$$

$$RCF = (1.118 \times 10^{-5}) (\text{rev min}^{-1})^2 r$$

From the above relationship, it is clear that RCF depends upon the rpm and the radius of rotation, r. If r is a constant for a given rotor, then variations in rpm alone determine the variations in RCF.

Sedimentation Rate

The sedimentation rate of a given particle depends upon a number of factors such as density of the particle (ρ_p), radius of the particle (r_p), the density of the medium (ρ_m), and the viscosity (η) of the suspending medium.

Rate of sedimentations of a rigid spherical particle is given below:

$$v = \frac{2}{9} \times \frac{r^2(\rho_r - \rho_m)}{\eta} \times g$$

where, v = rate of sedimentation,

g = gravitational field and

$\frac{2}{9}$ = shape factor constant for a spherical particle.

Svedberg Unit or Sedimentation Coefficient

The sedimentation rate or velocity (v) of a particle can be expressed in terms of its sedimentation rate per unit of centrifugal field. It is commonly referred to as its sedimentation coefficient, S. Since sedimentation rate studies are performed using a wide variety of solvent-solute systems or at different temperatures, the value is affected by temperature, solution viscosity and density. Therefore, these values are corrected to the sedimentation constant theoretically obtainable in water at 20°C. The equation for standard coefficient is

$$S_{20w} = \frac{S_{obs}(1 - \bar{v}\rho_{20w})}{(1 - \bar{v}\rho_T)} \times \frac{\eta_T}{\eta_{20}} \times \frac{\eta}{\eta_{20}}$$

where, S_{20w} = the standard sedimentation coefficient,

S_{obs} = the experimentally measured sedimentation coefficient,

η_T/η_{20} = the relative viscosity of water at temperature T compared with that at 20°C,

η/η_{20} = the relative viscosity of the solvent to that of water,

ρ_{20w} = the density of water at 20°C,

ρ_T = the density of the solvent at temperature T(°C) and

$\bar{v}$ = the partial specific volume of the solute.

The basic unit of sedimentation coefficient is 1×10^{-13} sec. This is also termed as one Svedberg unit (S), in recognition of T. Svedberg's pioneering work. Generally, the larger the molecule or particle, the larger is its Svedberg unit and hence the faster is its sedimentation rate. Svedberg units for viruses are 40 to 1000S, for lysosomes 40000S and for mitochondria 20×10^3S to 70×10^3S.

Analytical Centrifugation

The analytical ultra centrifuge has found many applications in biology, especially in the field of protein chemistry and nucleic acid chemistry. This technique is used to determine the sedimentation coefficient, relative molecular mass and also to test the purity of macromolecules.

Sedimentation velocity method : In sedimentation velocity method the sedimentation coefficient of the molecule is initially determined either by boundary sedimentation or zonal sedimentation. The equation used for calculating molecular weight is

$$M = \frac{RTS}{D(1 - \bar{v}\rho)}$$

where, M = relative molecular weight of the molecule,

D = diffusion coefficient of the molecule,

$\bar{v}$ = partial specific volume of the molecule,

ρ = density of the solvent at 20°C,

R = molar gas constant,

T = absolute temperature and

S = sedimentation coefficient.

Sedimentation equilibrium method : In this method Molecular weight can be calculated using the equation

$$M = \frac{2RT \ln(C_2/C_1)}{\omega^2(1 - \bar{v}\rho)(x_2^2 - x_1^2)}$$

where,

C_1 and C_2 are the concentrations of solute at distances X_1 and X_2 respectively from the centre of rotation,

R = molar gas constant,

T = absolute temperature,

ω^2 = square of the angular velocity of the rotor,

$\bar{v}$ = partial specific volume of the molecule and

ρ = density of the solvent.

This technique does not require diffusion coefficient and hence the method is more convenient and widely used for the determination of molecular weight of proteins.

Preparative Centrifugation

There are many types of preparative centrifugation such as differential and density gradient centrifugation.

Differential Centrifugation (Cell Fractionation)

The process of separation of cell organelles is known as subcellular fractionation. To isolate a specific organelle, initially, the organs (liver, brain, or kidney) are homogenged in a suitable homogenizing medium at 4°C. The resulting suspension, containing many intact organelles, is known as a homogenate.

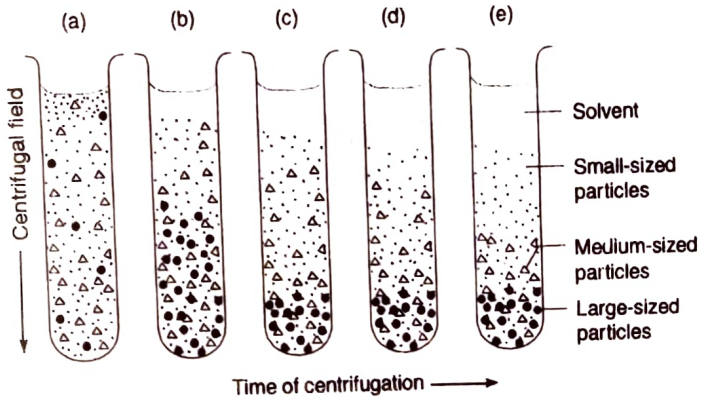


Fig. : Differential sedimentation of a particulate suspension in a centrifugal field. (a) Particles are uniformly distributed throughout the centrifuge tube. (b) to (e) Sendimentation of particles during centrifugation is dependent upon their size, shape and density.

Fractionation of the contents of a homogenate is done by a classical biochemical technique called differential centrifugation. This method is based upon the differences in the sedimentation rate of particles of different sizes and density (Figure). This method uses a series of four different centrifugation steps at successively greater speeds. Each step yields a pellet and a supernatant. The supernatant from each step is subjected to centrifugation in the next step. This procedure provides four pellets, namely, nuclear, mitochondrial, lysosomal and microsomal fractions (Figure).

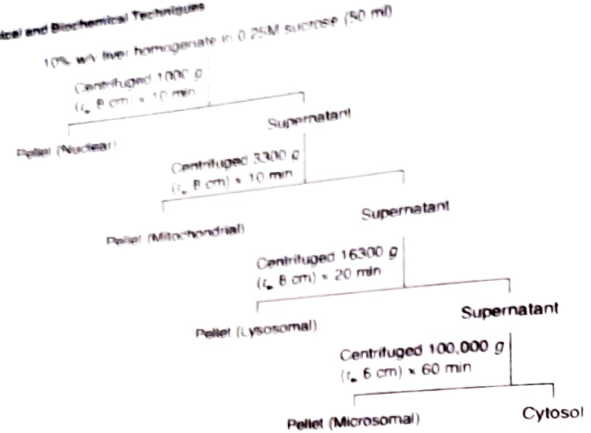


Fig. Cellular fractionation

Density Gradient Centrifugation

Differential centrifugation uses a homogeneous medium for separation whereas density gradient centrifugation uses a medium that has gradients. The commonly used gradient materials and their applications are given in Table

Table: Commonly used gradient materials and their applications

Gradient materials	Applications
Caesium chloride	Banding of DNA, nucleoproteins, viruses, plasmid isolation
Caesium sulphate	Banding of DNA and RNA, purification of proteoglycans
Sodium bromide	Fractionation of lipoproteins
Sodium iodide	Banding of DNA and RNA
Glycerol	Banding of membrane fragments, protein separation
Sucrose	Separation of subcellular particles, proteins, viruses and membranes
Ficoll	Separation of whole cells, subcellular particles and viruses
Dextran	Separation of whole cells, banding of microsomes
Bovine serum albumin	Separation of whole cells

In density gradient centrifugation, separation depends upon the buoyant densities of the particle. This method gives a much better separation than differential centrifugation. There are two types of density gradient centrifugation, namely, the rate-zonal technique and the isopycnic technique.

(*i*) Rate-zonal centrifugation technique

Rate-zonal centrifugation is a type of density gradient centrifugation in which particle separation is based upon

1. Differences in the size, shape and density of the particle,
2. The density and viscosity of the medium and
3. The applied centrifugal field

The gradient used in rate-zonal centrifugation technique has the maximum density at the bottom but its density is lesser than the most dense sedimenting particle to be separated. The density gradient is shallow. It is produced by layering different samples of very narrow densities in decreasing order. Thus, a gradient is produced whose density continuously increases from the top towards the bottom of the sample tube. Centrifugation is then performed at a comparatively low speed for a short time. The sample particles travel through the steep gradient and form discrete zones depending upon their sedimenting rate (Figure). The centrifugation must be terminated before any of the zones reaches the bottom of the centrifuge tube. This method is extremely useful for the separation of proteins possessing nearly identical densities but differing slightly in their size. It is used for the separation of RNA-DNA hybrids, ribosomal subunits and subcellular organelles

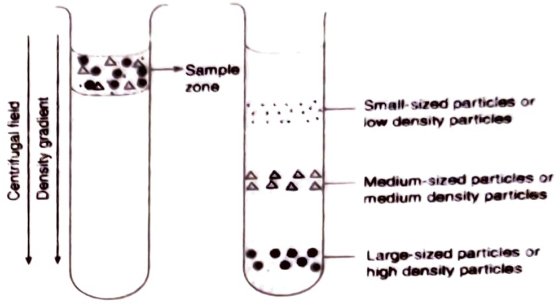


Fig. Rate zonal centrifugation

(*ii*) Isopycnic centrifugation

Isopycnic centrifugation depends only upon the buoyant density of the particle. It does not depend upon the shape or size of the particle and is independent of time. In isopycnic centrifugation, the maximum density of the gradient always exceeds the density of the densest particle. A continuous density gradient is always used. During centrifugation, sedimentation of the particle occurs until the buoyant density of the particle and the density of the gradient are equal. At this point of isodensity, no further sedimentation occurs, irrespective of how long centrifugation continues. This technique is used to separate particles of similar size but of differing density (Figure). Subcellular organelles such as golgi apparatus, mitochondria and peroxisomes can be effectively separated by this method.

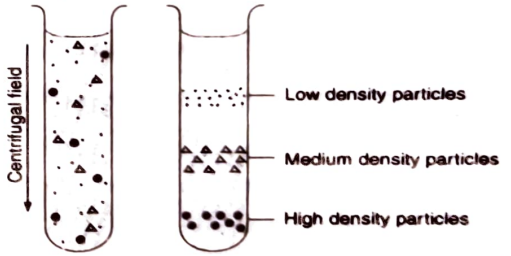


Fig. Isopycnic centrifugation using the equilibrium isodensity method.
(a) Particles distributed homogeneously throughout the tube prior to centrifugation.
(b) During centrifugation the gradient is allowed to establish itself, sample particles redistribute and band in a series of zones at their respective isopycnic positions.

Dialysis and Molecular Filtration

In common filtration, the solid particles are separated from suspension by passing it through a porous material. By using a semi-permeable membrane, the dissolved materials can be separated on the basis of their molecular dimension. **Dialysis** is a very common method, which uses semi-permeable membrane for separation. In 1861, T. Graham first introduced it. An aqueous solution containing both macro- and micromolecules is placed in a tubular cellophane or collodion (cellulose nitrate) bag which, in turn, is placed in a large reservoir of a given buffer or simple water. Small solute molecules and water freely pass through the bag, but molecules larger than the pore size of the membrane are confined to the tubing. It causes either dilution or concentration of non-dialysable materials until equilibrium is reached. The external fluid must be repeatedly changed to maintain the required concentration of macromolecules inside the bag. The semi-permeable membranes can be rendered chemically or physically to alter the size of the pores so that solutes of varying molecular weights are rendered permeable.

To concentrate a material inside dialysis bag, aother approach is follwed, called reverse dialysis, bag is put inside a water-soluble polymer, for example, polyethylene glycol (PEG). The water molecules from the dialysis bag come out to establish equilibrium with dry external fluid. Sucrose also can be used in place of PEG, but since it is dialyzable substance, it will enter the bag as water is removed. Excess sucrose can be removed by ordinary dialysis. It should be noticed that in reverse dialysis, equilibrium is never reached and the salts are continually removed until the sample is totally dry. If this stage is reached, large molecules irreversibly will bind with the dialysis bag and it cannot be utilized further.

Chromatography

It is a perfect analytical technique for the identification, isolation and separation of compounds based on differences in affinity for a stationary phase and a mobile phase. In chromatography, the stationary phase may be a solid, liquid, gel or a solid/liquid mixture and the mobile phase may be a liquid or a gas.

Principles of Chromatography

The basic principle of chromatography is adsorption and partition. When a mixture of compound is allowed to pass through stationary and mobile phases, the compounds become separated according to their Rf (Resolution front) values.

$$R_f = \frac{\text{Distance travelled by the sample solute}}{\text{Distance travelled by the solvent front}}$$

The interactions during chromatographic separation of substances include the following:

Partition When a solute is dissolved in two immiscible liquids of equal volume, the ratio of its concentration in two phases at a given temperature and equilibrium is called partition coefficient. The partition chromatography and gas-liquid chromatography are mainly based on this interaction.

Adsorption Separation of substances can be achieved by differential adsorption of substances by the mobile and stationary phases.

Paper Chromatography

In paper chromatography, the substance is analysed by allowing the flow of solvents on a specially designed filter paper. The solvent rises up on the paper by capillary action and the separation is effected by differential migration of the substances of a mixture. In paper partition chromatography, paper is used as immobile phase and the solvent as mobile phase. In paper adsorption chromatography, the paper is impregnated with an adsorbent like alumina and the solvent is allowed to flow over the paper.

In paper chromatographic technique, Whatman paper I which is available in different grades is extensively used. The sample is prepared by using proper solvent and is applied on the paper as spots or bands.

The sample is applied in drops on the spots with the help of a capillary tube or a microsyringe or a platinum loop. The spotting is done slowly and carefully in such a way that the spots occupy only small areas.

Selection of Solvents

The number of solvents to be used depends on the nature of the substances to be separated. The use of minimum number of solvents is preferable. When more than one component of the solvents is mixed together, two distinct layers would result, so the solvents should be shaken well before the use. Moreover, the solvents should be used in specific ratios to separate different substances as shown in Table

Table: Solvent systems to detect substances.

Substance	Components of solvents	Ratio
Amino acids	n-butanol-ethanol-water	4:1:1
	n-butanol-acetic acid-water	4:1:5
	phenol-water	80/20 v/v
	alcohol-water-ammonia solution	20:2.5:2.5
Sugars	n-butanol-acetic acid-water	4:1:5
Lipids	chloroform-benzene-ethanol	48.5:48.5:3.0
	chloroform-methanol-water	70/22/3 v/v/v
Organic acids	n-butanol-formic acid-water	10:2:15
Group I metallic ions	distilled water or ethanol	—
Group II metallic ions	ethanol-water–lN HCl	22.5:1.25:1.25
	Saturated n-butanol with 1NHCl	—
Group IV metallic ions	acetone-water-conc. HCl	21.75:1.25:2.0

Development of Chromatogram

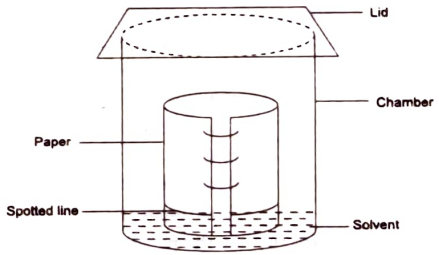


Fig. Diagram showing chromatographic setup.

Location of Spots

The developed paper is kept in an oven or on a hot plate for few minutes for drying

In physical methods, ultraviolet light and radioactive elements are used. In chemical methods, the colourless spots are made coloured by reagents called locating reagents which may be gases, liquids or solids. The locating reagents for various substances are given in Table Now the spots are marked on the paper. Rf values are calculated and compared with standard Rf values.

Table: Locating reagents to detect substances.

Substances	Locating reagents
Amino acids	0.1% ninhydrin in acetone or n-butanol
Sugars	0.5% benzidine mixture
Lipids	0.1 sudan black
Organic acids	0.05% bromocresol green
Group I metallic ions	0.25M potassium chromate solution
Group II and IV metallic ions	10 ml rubeanic acid + 10 ml ethanol

The method described above is ascending paper chromatography in which the solvent flows upwards on the paper by capillary action. In descending paper chromatography, the solvent is allowed to flow down by capillary action and gravitational pull on the paper by keeping the solvent at the top of the chromatographic chamber. The above method forms unidimensional or one-dimensional chromatography. Here the R_f values of substances depend on the size, nature of the substances, solvent systems, diffusion coefficient and so on.

Thin Layer Chromatography (TLC)

The principle and technique of TLC is very similar to paper chromatography. Here the sample is spotted at one end of a glass plate, which is coated, with the slurry of the adsorbents (Sodium acetate or silica gel). Using solvent systems develops the spotted plate and the spots are deducted as in paper chromatography. The adsorbents and solvent systems used to separate substances in TLC technique are shown in Table.

Table: Adsorbents and solvent systems in TLC.

Substances	Adsorbents	Solvents	Ratio (v/v)
Amino acids	Silica gel G	Ethanol-water	70/30
		n-Butanol-Acetk acid-water	60/20/20
Sugars	Sodium acetate	Ethyl acetate-n-propanol	65/35
		n-butanol-acetone-phosphate buffer (pH 5.0)	40/50/10
Lipids	Silica gel G	Petroleum ether-diethyl ether-acetone	89/10/1
		Chloroform-Methanol-Water	65/25/10
		Carbon tetrachloride-Chloroform	95/5

The separation of substances by TLC is faster than by the paper chromatographic method. TLC is commonly used to separate components in complex mixtures and the results obtained are more reproducible.

Gel Permeation Chromatography

When a gel having pores on its matrix is packed in the column and a heterogeneous mixture is passed through it, smaller ions enter the gel and come out of the column phase. But the bigger molecules cannot enter the gel, so they move only with the mobile phase through the inner phases of the gel. In this process, a type of sieving of molecules occurs and so this method is called molecular sieve chromatography or molecular exclusion chromatography. It is also called as gel nitration cnromatograpny.

Sephadex, agarose, polyacrylamide, porous glass and silica granules, polystyrene, etc. possess all the above characteristics and so have been widely used. Using the technique of column chromatography, gel permeation chromatography can be carried out.

Determination of Molecular Weight of Macromolecules by Gel Permeation Chromatography

By running the chromatogram, a fraction of macromolecules is collected and its volume is known. From this value, its distribution coefficient is calculated. In a given gel, the distribution of solute particles between inner and outer solvent is referred to as its distribution coefficient (K_d). The volume of individual fraction (V_e) depends on the volume of outer solvent (V_o) and volume of inner solvent (V_i)

$$V_e = V_o + K_d V_i$$

i.e.,

The volume of the inner solvent can be calculated by the formula

$$V_i = a\, W_r$$

where, a = Dry weight of the gel and

W_r = water regain value

Therefore,

$$V_e = V_o + (K_d \times a W_r)$$

Then,

$$K_d = \frac{V_e - V_o}{a W_r}$$

A standard graph with distribution coefficient of standards plotted against their molecular weight is referred, to know the molecular weight of the fractioned compounds.

Ion-Exchange Chromatography

The separation is usually carried out in tall glass columns packed with grains ot a synthetic ion-exchange resin (polymers to which numerous ionizable groups have been chemically added). Resins bearing negative charges are called cation exchangers and positively charged resins are anion exchangers. As a solution of ions is passed through the column, the ions compete with each other for the charged sites on the resin. Consequently, the rate of movement of any ion through the column depends on its affinity for the resin sites, its degree of ionization, and the nature and concentration of competing ions in the solution. The differential rates of movement of ions through the column arc the basis for protein and nucleic acid separations, because these molecules possess a variety of positively and negatively charged groups.

Table : Commonly used ion exchangers

	Polymer	Functional (Ionic) Group
Anion exchangers		
Diethylam noethylcellulose	DEAE-Cellulose	$-O-(CH_2)_2-N^+H-(CH_2H_5)_2$
QAESephadex	Cellulose	$-CH_2-N^+-(C_2H_5)_2\,CH_2\,CHOHCH_3$
Dowex-2; Bio-Rad AG2	Polystyrene-divinyl benzene	$-N^+-(CH_3)_2C_2H_5OH$
Polyiysine—kieselguhr (PLK)	Polylysine	$-(CH_2)_4-NH_3^+$
Cation exchangers		
Carboxymethykellulose	Cellulose	$-O-CH_2-COO^-$
Zeocarb 225, Bio-Rad AG50	Polystyrene	$-SO_3^-$
Hydroxyl apatite	–	$-PO_4^{3-}$
Amberlite IRC 50	Polymethacrylate	$-COO^-$
Sephade Se	Dextran	$-(CH_2)_2-SO_3^-$

Affinity Chromatography

It exploits the unique property of extremely specific biological interactions to achieve separation and purification. As a consequence, affinity chromatography is theoretically capable of giving absolute purification, even from complex mixtures, in a single process. The technique was originally developed for the purification of enzymes, but it has since been extended to nucleotides, nucleic acids, immunoglobulins, membrane receptors and even to whole cells and cell fragments.

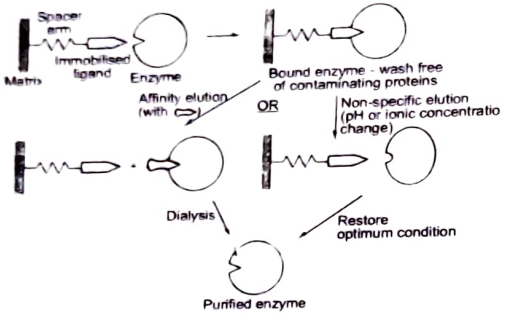


Fig. Purification of an enzyme by affinity chromatography

(*a*) affinity precipation, in which the ligand is attached to a soluble carrier that can be subsequently precipitated by, for example, a pH change; and

(*b*) affinity partitioning in which the ligand is attached to a water-soluble polymer such as polyethylene glycol and which, with the ligand bound, preferentially partitions into an aqueous polymer phase that is in equilibrium with a pure aqueous phase. In all cases, for effective chromatography, the association constant K_a, for the complex should be in the region 10^4 to 10^8 M^1

Matrix

The most common matrix are : cross-linked dextrans (Sephacryl S), agarose (Sepharose, BioGel A), polyacrylamide gels (Bio-Gel P), polystyrene (Bio-Beads S), cellulose and porous glass antisilica.

Examples of group-specific ligands commonly used in affinity chromatography

Ligand	Affinity
Nucleotides 5'-AMP	NAD^+ dependent dehydrogenases, some kinases
2'5'-ADP	NADP+ dependent dehydrogenases
Calmodulin	Calmodulin-binding enzymes
Avidin	Biotin-containing enzymes
Fatty acids	Fatty-aced-binding proteins
Heparin	Lipoproteins, Upases, coagulation factors, DNA polymerases, steroid receptor proteins, growth factors, serine protease inhibitors
Proteins A and G	Immunoglcbulins
Concanavalin A	Glycoproteins containing a-D-mannopyranosyl and α-D-glucopyranosyl residues
Soybean lectin	Glycoproteins containing N-acetyl-α (or β)-D-galactopyranosyl residues
Phenylboronate	Glycoproteins
Poly(A)	RNA containing poly(U) sequences, some RNA-specific proteins
Lysine	rRNA
Cibacron Blue F3G-A	Nucleotide-requiring enzymes, coagulation factors

Applications

A wide range of enzymes and other proteins, including receptor proteins and immunoglobulins, has been purified by affinity chromatography. The application of the technique is limited only by the availability of immobilised ligands. The principles have been extended to nucleic acids and have made a considerable contribution to developments in molecular biology. Messenger RNA, for example is routinely isolated by selective hybridisation on poly(U)-Sepharose 4B by exploiting its poly(A) tail. Immobilised single-stranded DNA can be used to isolate complementary RNA and DNA, Whilst this separation can be achieved on columns, it is usually perfomed using single-stranded DNA immobilised on nitrocellulose filters. Immobilised nucleotides are useful for the isolation of proteins involved in nucleic acid metabolism.

Gas-Liquid Chromatography(GLC)

This technique is another form of column chromatography in which the sample is carried out as vapour in an inert gas such as argon, nitrogen or helium over liquids like silicone oils or lubricating greases held by solids like diatomaceous earth or ground firebrick. That is, a non-volatile liquid is the stationary phase, which is dispersed on the surface of solid support (holder of the liquid phase), as shown in the Figure. The carrier gas stored in a tank is passed into the sample injection chamber. From the chamber, the sample is carried by the gas into the column.

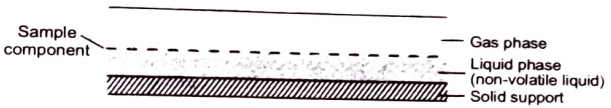


Fig. Distribution of sample in the column of GLC.

GLC is useful in the separation of the components of smoke, atmospheric pollutants, plant extracts, organic acids, etc.

High Performance Liquid Chromatography (HPLC)

HPLC is also a column chromatography and is designed to increase the flow rate of the solutes in order to reduce time consumption.

The pumping system generates high pressure and directs a constant flow of the solvent to the column. The solvents used as mobile phase include ethanol, methanol, propanol and acetonitrate. Pure water or buffers can also be used. The sample is dissolved in the mobile phase and filtered. Using a microsyringe or a metal loop which would withstand high pressure, the sample is injected. The column is straight with 20-50 cm length and 1-4 mm diameter. The supporting media in HPLC are rigid, solid column materials, which include three types namely micro porous supports, pellicular supports and bonded bases. In microporous supports, the particle has micropores having a diameter of 5-10 mm. The pellicular supports have a solid inert core surrounded by porous particles having a diameter of 40 mm. The bonded bases are the supports in which the particles are bonded to an inert support. Highly sensitive detectors such as UV photometers, spectrophotometers, refractive index monitors and fluorescence detectors are used to record the results.

HPLC is used for the separation of proteins, amino acids, polysaccharides, nucleic acids, plant pigments, pesticides, lipids and so on.

Electrophoresis

Electrophoresis is the study of movement of charged particles in an electric field. The earliest technique, pioneered by A. Tiselius in 1937, was a moving boundary electrophoresis The technique is widely used for separation and analysis of a large number of biomolecules such as amino acids, peptides, protens. nucleotides and nucleic acids.

Principle of Electrophoresis

Any charged ion or molecule migrates when placed in an electric field. The rate of migration of a compound depends on its net charge, size, shape and the applied current. This can be represented by the following equation:

$$v = \frac{E.q}{f}$$

where v = velocity of migration of the molecule

E = electric field in volts/cm

q = the net electrical charge on the molecule

f = the frictional coefficient which is a function of the mass and shape of the molecule

The movement of a charged molecule in an electric field is often expressed in terms electrophoretic mobility (μ) which is defined as the velocity per unit of electric field.

$$\mu = \frac{v}{E} = \frac{E.q}{f.E} \left(\text{since } v = \frac{E.q}{f}\right)$$

$$\mu = \frac{q}{f}$$

For molecules with similar conformation, f varies with size but not with the shape. Thus electrophoretic mobility (μ) of a molecule is directly proportional to the charge density (charge/mass ratio). Molecules with different charge/mass ratio, migrate under the electric field at different rates and hence get separated.

Depending upon the nature of support medium, zone electrophoresis is of different types such as paper, starch, polyacrylamide and agarose gel electrophoresis.

Paper electrophoresis

In this form of electrophoresis, the sample is applied as a circular point or spot on a strip of Whatman filter paper or cellulose acetate paper moistened with the buffer solution.

This method is suitable for separation of low molecular weight compound such as amino acids, small peptides and nucleotides.

Polyacrylamide gel electrophoresis

Electrophoresis in polyacrylamide gel is the most widely used technique for analysis and characterization of proteins and nucleic acids. Use of polyacrylamide gels has several advantages such as: it is chemically inert, gives superior resolution, is amenable to preparation of gels with wide range of pore sizes and that the gels are stable over a wide range of pH, temperature and ionic strength.

Polyacrylamide gel is prepared by polymerization of acrylamide monomers into long chains and cross-linking these by bifunctional reagents such as N, N′-tetramethylethylene bisacrylamide.

Polymerization is initiated by ammonium persulfate or riboilavir:. N, N, N′, N′-tetra:nethylethylene diamine (TEMED) catalyses formation of free radicals from persulfate which in turn initiate polymerization. In the case of riboflavin-TEMED system, light causes photodecomposition f riboflavin and production of necessary free radicals. Oxygen inhibits polymerization. Gel solutions have, therefore, to be degassed prior to the polymerization step.

The pore size of polyacrylamide gels can be varied by changing total concentration of acrylamide and bis-acrylamide monomers in a fixed volume of gelation solution. Gels ranging from 3 to 30% acrylamide concentration can be made and these can be used for separation of molecules of size upto 1×10^6 daltons. A gel with low percentage has larger pore size and is suitable for separation of high molecular weight compourds and a high percentage gel has smaller pore size and is used for separation of relatively low molecular weight compounds. Alternatively, gels of different pore sizes may be prepared by varying the relative concentrations of acrylamide and bis-acrylamide. Gradient gels with linear gradients of increasing acrylamide concentration give better resolution and are quite commonly used.

Buffer systems

Gel electrophoresis can be carried out in buffers ranging from pH 3-10. Ionic strength of the buffers is generally kept low (0.01-0.1 M) to reduce heat production during electrophoresis to a minimum. Depending upon the types of buffer systems used, polyacrylamide gel electrophoresis may be of the following kinds:

(*a*) Native continuous polyacrylamide gel electrophoresis

(*b*) Native discontinuous polyacrylamide gel electrophoresis or Native disc gel electrophoresis

(*c*) SDS—polyacrylamide gel electrophoresis (reducing)

(*d*) SDS—polyacrylamide gel electrophoresis (non-reducing)

(a) Native continuous polyactylamide gel electrophoresis (native PAGE)

In a native continuous polyacrylamide gel electrophoresis same buffer ions are present throughout the sample, gel and electrode vessel reservoirs at constant pH such that proteins retain their native conformation and biological activity and hence it is known as native continuous PAGE or simply native PAGE. In this system, protein sample is loaded directly onto the gel, referred to as resolving gel, in which separation of the sample components occurs.

(b) Native discontinuous polyacrylamide gel electrophoresis (Disc-gel electrophoresis)

Discontinuous or disc gel electrophoresis embodies the following features:

(i) contains different buffer ions in the gel than those in the electrode reservoirs

(*ii*) consists of two types of gels—a large pore stacking gel which is polymerized on top of small pore 'resolving gel'. The buffers used to prepare these two gels are of different ionic strength and pH

(*iii*) most discontinuous buffer systems have discontinuities of both buffer composition and pH (Table)

Discontinuous buffer systems have an advantage over continuous systems as relatively large volumes of dilute solutions of protein samples can be applied onto the gels and yet good resolution of sample components can he obtained. This is because the proteins in applied sample get concentrated into extremely narrow zone or get stacked during migration through large pored stacking gel prior to their entry and separation in small pored resolving gel.

Table: Buffers for native discontinuous polyacrylamide gel electrophoretic systems

Type of buffer	Preparation of buffer
(*a*) *High pH buffer system*	
(i) Stacking gel buffer (Tris-HCl, pH 6.8)	Tris 6.0 g 1 M HC1 48.0 ml Adjust pH to 6.8 and make final volume to 100 ml with water
(ii) Resolving gel buffer (Tris-HCl, pH 8.8)	Tris 36.3 g 1 M HC1 48.0 ml Adjust pH to 8.8 and make volume to 100 ml with water
(iii) Reservoir buffer (Tris-glycine, pH 8.3)	Tris 3.0 g Glycine 14.4 g Adjust pH to 8.3 and make volume to 1 L with water
(*b*) *Neutral pH buffer system*	
(i) Stacking gel buffer (Tris-phosphate, pH 5.5)	Tris 4.95 g Water 40.0 ml Adjust pH to 5.5 with 1 M orthophosphoric acid and make final volume to 100 ml with water

Type of buffer	Preparation of buffer
(ii) Resolving gel buffer (Tris-HCl, pH 7.5)	Tris 6.85 g Water 40.0 ml Adjust pH to 7.5 with 1 M HC1 and make volume to 100 ml
(iii) Reservoir buffer (Tris ethyl barbiturate, pH 7.0)	Diethylbarbituric acid 55.2 g Tris 10.0 g Add water to make 1 L
(c) *Low pH buffer*	
(i) Stacking gel buffer (Acetic acid— KOH, pH 6.8)	1 M KOH 48.0 ml Glacial acetic acid 2.9 ml Adjust pH to 6.8 and make volume to 100 ml with water
(ii) Resolving gel buffer (Acetic acid— KOH, pH 4.3)	1 M KOH 48.0 ml Glacial acetic acid 17.2ml Adjust pH to 4.3 and make final volume to 100ml
(iii) Reservoir buffer (Acetic-acid-B-alanine, pH 4.5)	β-alanine 31.2g acetic acid 8.0 ml Add water to make 1 L

(c) SDS-polyacrylamide gel electrophoresis (non-reducing) (SDS-PAGE)

In this system, proteins are dissociated into their constituent subunits using anionic detergent such as sodium dodecyl sulfate (SDS) or sodium lauryl sulfate. The protein mixture is denatured by heating the sample at 100°C in the presence of excess SDS. Under these conditions most polypeptides bind SDS in a constant weight ratio (1.4 g SDS per g of polypeptide). The intrinsic charges of the polypeptides are insignificant compared to negative charges provided by the bound detergent. Consequently SDS polypeptide complexes have essentially identical charge densities (charge/mass ratio) and these migrate in polyacrylamide gels of an appropriate porosity strictly according to polypeptide size or their molecular mass. Thus, in addition to the analysis of polypeptide composition of the sample, the molecular weight of the polypeptides of the sample protein can also be determined by using standard molecular weight markers.

SDS-PAGE can be carried out using either continuous or discontinuous buffer systems (Table). Resolution is, of course, superior in a discontinuous system.

Table: Buffers for SDS-polyacrylamide gel electrophoresis

(a) Continuous system

Resolving gel buffer

0.1 M sodium phosphate, pH 7.2

Reservoir buffer

0.1 M sodium phosphate, pH 7.2
0.1% SDS

(b) Discontinuous system

Stacking gel buffer

0.125 M Tris-HCl, pH 6.8

Resolving gel buffer

0.375 M Tris-HCl, pH 8.8

Reservoir buffer

0.025 M Tris
0.192 M glycine
0.1% SDS
pH 8.3

(d) SDS—polyacrylamide gel electrophoresis (reducing)

It is similar to SDS-PAGE except that in addition to SDS, a thiol reagent such as β-mercaptoethanol is included which cleaves the disulfide bonds. Hence, polypeptides in a native protein which still remain associated even in the presence of SDS get separated. So in this method a protein sample is boiled in presence of excess SDS and β-mercaptoethanol to denature proteins to their individual polypeptides. SDS polypeptide complexes (1.4 SDS: 1 protein) have net negative charge and migrate towards anode at rates based solely on size or molecular weight of the polypeptides.

Staining of proteins in polyacrylamide gels

After electrophoresis, protein bands in gels can be visualized by staining with dyes such as amido black 10B, coomassie brilliant blue R-25000 silver staining. Coomassie brilliant blue R-250 is the most commonly used stain though silver stain is about 100 times more sensitive than it and can detect upto 1 ng protein/band.

Protein (Western) blotting

The transfer of proteins from an acrylamide gel onto a more stable and immobilizing support such as nitrocellulose filter is called as protein blotting or Western blotting.

Blotting of protein bands allows them to become more accessible for detection and identification using a variety of methods such as immunological detection, autoradiography, analysis of DNA binding proteins, glycoproteins etc. and also the filters can be stored for much longer periods. A number of support matrices such as nitrocellulose, diazobenzyloxymethyl cellulose, nylon membrane, Hybond membrane etc. with high protein binding capacity are available, but nitrocellulose which is available in a range of pore sizes (0.2-0.45 ; m), is sufficiently effective for most purposes and is commonly used. The transfer of proteins from gels can be achieved by any of the methods such as simple capillary action, application of vacuum or electrophoretically. Thus based on the technique employed Western blotting is of different types namely, capillary blotting, vacuum blotting and electroblotting, the latter being the most efficient and has become a standard method.

Agarose Gel Electrophoresis

Agarose gels are more porous and have a larger pore size as compared to polyacrylamide gels and are, therefore, used to fractionate large macromolecules such as nucleic acids that cannot readily penetrate into and move through other types of supporting materials.

Agarose is a linear polymer of D-galactose and 3,6-anhydro-L-galactose. Agarose gels are cast by boiling agarose in presence of a buffer, then poured into a mold and allowed to harden to form a matrix. Porosity of the gel is determined by concentration of agarose. Higher the agarose concentration, smaller the pore size and lower the agarose concentration, larger is the pore size. When an electric field is applied across the gel, DNA molecules that are negatively charged at neutral pH, migrate towards oppositely charged electrode at rates determined by their molecular size and conformation. Since charge/mass ratio in nucleic acids is one, rate of migration of DNA molecules is inversely proportional to $\log_{10}$ of their molecular weights, i.e. smaller DNA molecules will travel faster as compared to the larger ones. Further, DNA molecules of the same size but with different conformations travel at different rates. The order of migration velocity in the increasing order of various forms of DNA is: supercoiled DNA > linear double stranded DNA > open circular DNA.

Buffers

Commonly used buffers include Tris-acetate (TAB), Tris-borate (TBE) or Tris-phosphate (TPE) at a concentration of approximately 50 mM, over a pH range of 7.5-8.0. EDTA is almost invariably incorporated in these buffers to suppress the activity of DNase.

Isoelectric Focussing (IEF)

It is an electrophoretic method for separation of proteins which differ in their isoelectric points. The proteins are made to migrate in a pH gradient maintained by a mixture of low molecular weight ampholytes. Proteins migrate depending upon their charge till they reach a region where the pH corresponds to respective isoelectric point at which pH proteins possess no net charge and hence get focussed. The method has high resolution and separation of proteins that differ in isoelectric boints by as little as 0.01 pH units can be achieved.

Two Dimensional Gel electrophoresis

The technique was developed by O'farrell (1975) and combines the resolving power of isoelectric focussing (IEF) and SDS-polyacrylamide gel electrophoresis (SDS-PAGE). Proteins in the first dimension are separated according to their charge by IEF and in second dimension SDS-PAGE facilitates their separation on basis of their size. It is a standard method for judging purity of a protein preparation.

Immunoelectrophoresis

It is a type of agar/agarose gel electrophoresis which exploits the separation of proteins on the basis of their charge to mass ratio and their antigenicity. Protein mixtures are first separated by electrophoresis on agarose gel and are then allowed in interact with a specific antibody preparation placed in a trough cut in the center of the gel. The antibody diffuses through the gel and forms a visible precipitate with the electrophoresed proteins for which the antibody has affinity. The technique allows the identification of the number of antigens in a sample and also indicates the relative amount of imunoreactive material.

A modification of the above technique is rocket immunoelectrophoresis so called because the precipitin patterns that form on the gel after electrophoresis look like rockets. This technique is used in the determination of antigen concentration. The area of the peak of the rocket serves as a measure of antigen concentration.

Capillary Electrophoresis (CE)

It is a more recent technique that has a very high resolving power and speed, is versatile and automated. CE involves electrophoresis of samples in small narrow capillary tubes and a high voltage of 10-50 kV is applied. The technique can be used to separate a wide spectrum of biological materials including amino acids, peptides, proteins, oligonucleotides and nucleic acids as well as small organic molecules such as drugs or even metal ions. Very small amounts of samples (5-10 nl) are required for analysis.

Pulse-field Gel Electrophoresis (PFGE)

The technique was developed by Schwartz and Cantor (1984). In this technique electric field is changed repeatedly (pulsed) in direction and strength during th seperations such that the DNA molecules are forced to,continuously orient and re-orient themselves. Larger is the DNA molecule, longer is the time, required for its reorientation. Very large DNA molecules of size upto 5000 kb can be seperated by this technique.

Spectroscopy

The interaction between electromagnetic radiation and matter forms the basis of spectroscopy when the electromagnetic radiation having a particular frequency is passed through a sample, the frequency which is absorbed or emitted by the molecules of the sample, is determined at molecular energy level by analysing the intensity of frequency emerging out of the sample. The instrument called spectrometer measures the change in intensity of electromagnetic radiation. The results are interpreted in terms of stereochemistry of the molecule (three-dimensional spatial arrangements of atoms in a molecule).

The spectrum obtained by the interaction in which a sample itself emits radiation is called emission spectrum. The spectrum obtained by the interaction in which the sample absorbs radiation is called absorption spectrum. When atoms absorb electromagnetic radiation, their electronic energy is increased resulting in a spectrum in the form of spectral lines of varying wavelength. This is atomic absorption. On absorption of electromagnetic radiation, the molecules become excited. When the molecules return to their original ground state, they emit molecular spectrum in the form of bands, each line of which is composed of a large number of very fine lines. This process is called molecular absorption. The energy associated with the uniform motion of a molecule is called translational energy. A molecule also possesses some forms of internal energy including rotational energy, which is due to the overall rotation of molecules, vibrational energy which is associated with the oscillation of atoms of the molecule, and electronic energy which is due to the movement of electrons. In a molecule, the levels of rotational energy are closely spaced so that very little energy is required for rotational transitions. This occurs in infrared and microwave regions of the electromagnetic spectrum and is useful in the study of molecular structure. The vibration energy levels are spaced apart and occur in infrared region of the spectrum. When molecules absorb photons of electromagnetic radiation in the visible and ultraviolet regions, the electronic energy is increased causing rotational and vibrational changes. As a result, the molecular spectrum exhibits a band of wavelengths.

Electromagnetic Spectrum

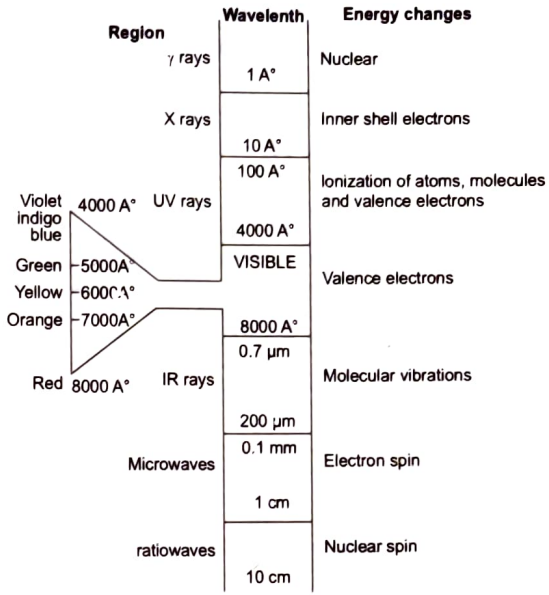


Fig. Schematic diagram of electromagnetic radiation and energy changes.

Beer-Lambert Law

The Beer-Lambert law is a combination of two laws, each dealing separately with the absorption of light, related to the concentration of the absorber (the substance responsible for absorbing the light) and the pathlength or thickness of the layer (related to the absolute amount of the absorber). Provided an absorbing substance is partially transparent it will transmit a portion of the incident radiation. The ratio of the intensities of transmitted and incident light gives the transmittance, T, expressed as :

$$T = \frac{I}{I_0}$$

where I_0 is the intensity of incident radiation, and I = intensity of transmitted radiation. (Note : Intensity = number of photons interacting in unit time (seconds).

A 100% value of T represents a totally transparent substance, with no radiation being absorbed, whereas a zero value of T represents a totally opaque substance, which, in effect, represents complete absorption. For intermediate values we can define the absorbance (A) or extinction (E), which is given by the logarithm (base 10) of the reciprocal of the transmittance.

$$A = E = \log(1/T) = \log(I_0/I)$$

Absorbance used to be called optical density (OD). Also, as absorbance is a logarithm it is by definition unit-less and has a range of values from 0 (=100% of T) to ∞ (= 0% of T).

It is now possible to define the Beer-Lambert law, which, as described above, states that the absorbance is proportional to both the concentration of absorber and thickness of the layer, as

$$\mathbf{A = \varepsilon_\lambda C l}$$

where ε_λ = molar absorbance coefficient (or molar extinction coefficient) for the absorber at wavelength λ

c = concentration of absorbing solution, and

l = pathlength through the solution (or thickness).

If l = 1 cm then $\mathbf{A \propto C}$

Colorimeter

Colorimeter is useful to determine the concentration of coloured samples by measuring the amount of light intensity actually absorbed by the sample. The components of a colorimeter are shown in Figure.

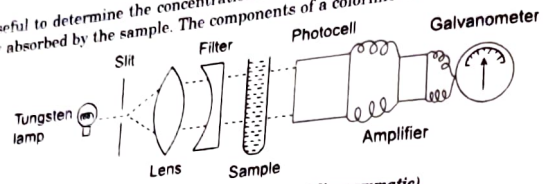


Fig. Components of a colorimeter (diagrammatic).

A tungsten lamp provides white light as a steady source of radiation through a slit. A condenser lens makes the light beam parallel and to fall on the solution. A coloured glass filter is used as monochromator, the colour of which is complementary to the colour of the sample. A photocell produces electrical current in proportion to the intensity of light falling on it. The current produced by the photocell is amplified by an amplifier. A galvanometer calibrated with optical density and percent transmittance in log scales gives absorbance readings directly.

A standard graph is plotted by using known concentrations of the substance relating the concentration and optical density. From this standard graph, the concentration of unknown solution or sample can be arrived at.

Ultraviolet and Visible Spectroscopy

When electrons in an atom are excited, radiations are absorbed resulting in the absorption spectrum of the atom. Similarly, when the electrons of a molecule are excited, the electronic spectrum originates due to a change in the arrangement of molecular electrons. The electronic spectra of molecules occur in the visible and ultraviolet regions of electromagnetic radiation. A small change in electronic energy results in a large change in vibrational energy, which in turn results in a change in rotational energy of the molecule. In other words, the molecular electronic spectra involve change in electronic, vibrational and rotational energy of the molecule. The UV absorption spectra are due to the transition state of an electron or electrons within a molecule or due to an ion, which occurs from a lower to higher electronic energy level. The UV emission spectra are due to the reversion of transition.

The ultraviolet spectrometer consists of the following components:

Radiation source

The source of radiation may be hydrogen discharge lamps, deuterium lamps, xenon discharge lamps or mercury arcs.

Monochromators

These are dispersion elements, which may be prisms or gratings made up of glass, quartz or fused silica.

Detectors

Three major types of detectors are used namely photovoltaic cell, photocell and photomultiplier tube.

Uses: Visible and UV spectroscopy are useful in determining the concentration of biological substances. As the bases of the nucleic acids absorb UV rays, UV spectroscopy is used to study denaturation of DNA double helix. It is also employed to characterize aromatic compounds, to detect impurities in organic substances, to determine molecular weight of many compounds, to know the dissociation constant of acids and bases and to study kinetics of reactions,

Infrared Spectroscopy

The basic principle of Infrared spectroscopy is selective absorption of chemical substances in the infrared region producing infrared absorption spectrum in the from of bands. The spectrum includes a wide range of wavelengths, the intensities of which can be measured as transmittance or absorbance. The IR spectrometer consists of the following components.

Radiation source

The source of radiant energy may be an incandescent lamp, Nernst Glower, or Globar source or mercury arc.

Sample and sample cells

As this technique is used to characterize solids, liquids and gases, these samples must be transparent to IR radiation. Therefore, the solid sample is prepared either by dissolving it in a suitable solvent or by grinding it with mineral oil into a paste. The liquid sample is directly taken in cells made up of NaCl or KBr. The gas sample is taken in a special cell, the end walls of which are made up of NaCl.

Detectors

Thermal detectors such as bolometers, thermocouples, thermisters, golay cell, photoconductivity cell, etc. are widely used along with an amplifier in order to identify even the low signals.

The IR spectrometer includes two types namely, single beam IR spectrometer and double beam IR spectrometer.

In single beam IR spectrometer, the radiation from the source is allowed to pass through the sample and then into a mirror through a prism. The mirror and prism select the desired wavelength, which is amplified and detected. By knowing the intensity of original radiation, the absorbed radiation can be measured. However, the spectrum obtained by this instrument is deformed due to variations in intensity of the emission. The spectrum includes bands of solvents also when the sample is analysed in solution. These disadvantages are overcome in double-beam IR spectrometer.

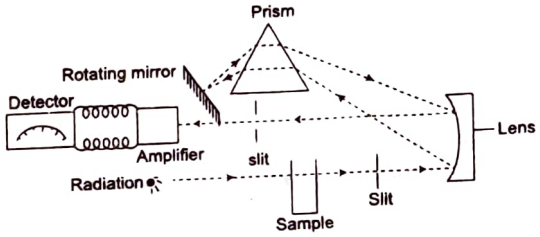


Fig. Optical system of a single-beam IR spectrometer.

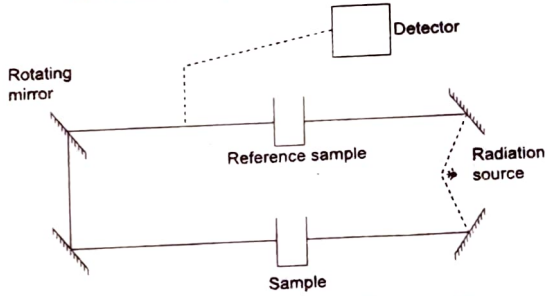


Fig. Optical system of a double-beam IR spectrometer

In double beam IR spectrometer, the radiation from the source is split into identical beams, one of which is passed through the sample and another one through reference sample. Then the two beams are recombined and passed to the detector.

Uses: IR spectroscopy is useful in identifying functional groups of organic compounds. It is used to determine steroids and hormones, to measure the rate of reactions, to study molecular conformations of biological molecules and to know the interactions between molecules.

Fluorescent Spectroscopy

Fluorescent spectroscopy is based on the absorption phenomenon of electromagnetic radiation. When a molecule absorbs electromagnetic radiation, a part of energy is dissipated as heat and the remaining part of the energy is radiated with lesser frequency than the incident radiation. This is called fluorescence. In other words, the absorption of light by a molecule with a specific wavelength and its emission at a different wavelength is called fluorescence. Fluorescent spectra are always the band spectra, which are independent of the wavelength of the absorbed radiation.

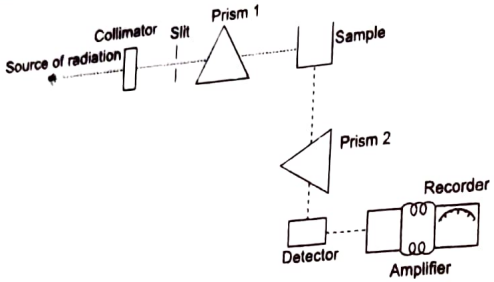


Fig. Optical system of a fluorescent spectrometer.

Fluorescent spectroscopy is mainly used for studying protein structure as well as for qualitative and quantitative analysis of compounds.

Atomic Spectroscopy

Atomic Absorption Spectroscopy

When a metallic solution is introduced into a flame, the metallic vapour is formed in which some metal atoms emit radiations characteristics of the metal. But a large percentage of atoms remain as non-emitting ground state. These ground state atoms are capable of receiving light of specific wavelength. When such a light is passed through the flame having the metallic atoms, a part of light is adsorbed and is proportional to the density of the atoms.

The important components in an atomic absorption spectroscope are given in the Figure

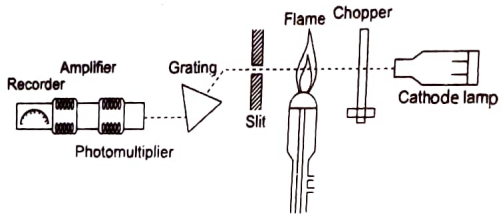


Fig. Optical system of an atomic absorption spectrometer.

Double Beam Atomic Absorption Spectroscopy

In this method, one part of the chopped light beam from the lamp is passed through the flame and the other part is made to bypass the flame. Then the two beams are recombined and passed through the monochromator. Therefore, the drifting effect of the lamp and the change in the sensitivity of the detector are eliminated.

Uses: This technique is mainly useful in determining metallic elements such as, Ca, Mg, Na, K, etc. in biological substances. As this method requires no sample preparation, both aqueous as well as non-aqueous solutions can be studied.

Atomic Emission Spectroscopy

The important components of an emission spectrometer are shown in the Figure.

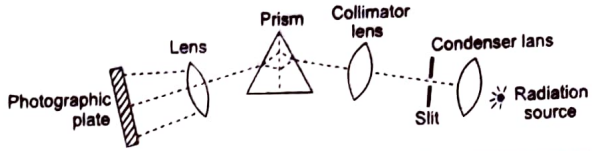


Fig. Optical system of an emission spectrometer (with prism as monochromator).

Uses This technique is used to analyse various elements in the tissues of animals to know the changes in the concentration of trace elements during ageing process. It is also used to determine the deficiency of elements in plants and soils.

Mass Spectroscopy

Mass spectroscopy helps to separate atoms or molecules in a compound on the basis of differences in their masses. When a compound is bombarded with a beam of electrons its ionic molecules are produced. These molecules separate according to their masses producing a spectrum called mass spectrum.

Diagrammatic structure of a typical mass spectrometer is shown in the Figure. It consists of the following components:

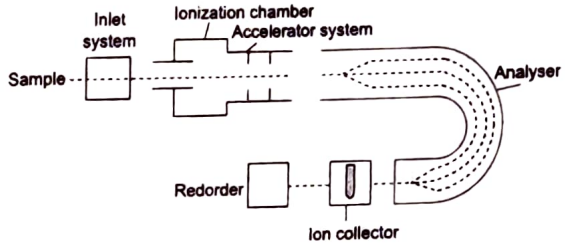


Fig. Diagrammatic structure of a mass spectrometer.

(i) ***Source of Ignition :*** It involves the gaseous inisation of analyte (the molecule to be analysed) to be examined. When the molecles gain or lose a charge (through elecron ejection, protonation or deprotonation), molecular ions generated.

(ii) ***Analyser :*** The ions generated as above are separated according to their mass-to-charge (m/z) ratio.

(iii) **Detector :** The molecular masses of separated ions are determined from the mass spectra obtained after mass spectrometry of analyte.

For identification, protein is separated from the crude extract on a two-dimentional gel. The protein spots are excised and then used a search for mass spectrometry or fragmented into small peptides. Protein fragmentation into peptides is done by proteases (e.g. trypsin). The peptides are separated by using liquid chromatography (such anion exchange or reverse phase column chromatography). Proteins are easily identified if these have been fragmented into peptides.

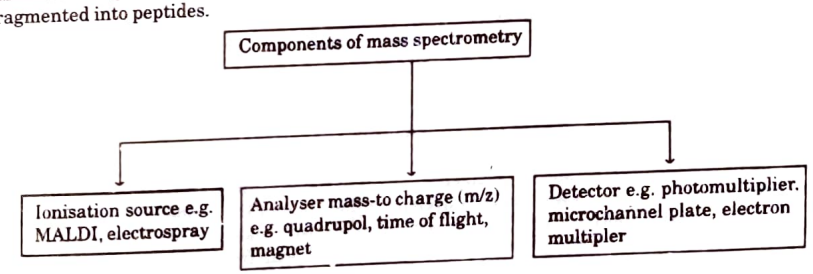


Fig. A simplified schema'ic diagram of mass spectrometry.

The MS requires that the molecules to be examined should be in the form of gaseous ions. There are several methods available for ionisation of biomolecules but MALDI and electrospray (ES) each coupled with TOF-MS are given below :

Matrix-Assisted Layer Desorption-Ionisation (MALDI):

MALDI was first developed by Karas and Hillenkamp in 1988. It co-precipitates the large excess of a matrix material with the analyte molecule. In this method a sub-microlitre of the mixture (matrix + analyte) is pipetted onto a metal substrate and allowed to dry. The dried mixture is irradiated by laser pulse (337 nm) which specifically absorbs the selected solid of matrix. The irradiation causes energy transfer and desorption resulting in gas phase matrix ions.

With the help of matrix ions the non-absorbing analyte molecules also get desorbed into the gas phase and ionised. The charged molecular ions of analyte are detected by MALDI-TOF mass spectrometer. Generally, MALDI is used for the study of molecules having mass above 500 Daltons (i.e. oligonucleotides and peptides). Proteome analysis will be cruicial for understanding different biological processes in the post-genomic era. To achieve this objective mass spectrometry is extensively used for :

(*i*) peptide sequencing

(*ii*) identification of protein

(*iii*) protein expression in different tissues and conditions

(*iv*) identification and post-transcriptional modification (e.g. phosphorylation, glycosylation, etc.) of proteins in response to different stimuli, and

(*v*) characterisation of protein interactions that include

(*a*) protein-ligand, (*b*) protein-protein and (*c*) protein-DNA interaction.

Recent improvement in MS has significantly improved its application in the study of protein structure and function.

Surface Enhanced Laser Desoption-Ionisation (SELDI):

It is a unique combination of affinity chromatography and time of flight (TOF) mass spectrometry on a single platform, a combination designated as SELDI-TOF MS (surface enhanced laser desoption-ionisation time of flight mass spectrometry). This system enables protein capture, purification, ionisation and analysis of complex biological mixtures directly on protein chips array surface and detection of the purified proteins by laser desorption-ionisation time of flight (LDI-TOF) MS analysis.

SELDI system consists of three parts :

(*i*) protein chip arrays (eight spots of 2 mm diameter),

(*ii*) protein chip reader (LDI-TOF ms)

(*iii*) specialised software.

Protons added during passage through the needle give additional charge to the macromolecule. The m/z of the nucleotide can be analysed in the vacuum chamber. ES produces many more ions than MALDI. These ions are transferred into mass spectrometer for analysis with high efficiency. ES MS can analyse a wide range of molecules including proteins, oligonucleotides, sugar and lipids. Since a MS measures m/z, it allows high molecular weight proteins to be observed on a low mass range equipment. Thus ESI generates multiple charged molecules. Peptides or proteins are identified on the basis of m/x. ratio. The m/z ratio of ions is analysed after their generation.

Uses: Mass spectroscopy is used to determine the molecular weight of the substances, the quantity of isotopes in vaporizable elements, ionization potentials of molecules and bonding nature of macromolecules.

Raman Spectroscopy

When a beam of monochromatic light ray is passed through a transparent substance (solid, liquid or gas), the incident light rays are mostly transmitted without alterations. However, some of the rays are scattered at different angles than their incident angles. This scattered light contains additional frequencies than that of the incident rays. This phenomenon is called Raman effect. The lines with modified wavelengths in Raman effect are called Raman lines. The lines with greater wavelength than that of the incident wavelength are called Stoke's lines and those with shorter wavelength are called anti-Stoke's lines. In Stoke's lines, the intensity is greater than the intensity of the corresponding anti-Stoke's lines. Raman frequencies are more or less similar to the frequencies of infrared regions of the spectrum.

The important components of Raman spectrometer are shown in the Figure.

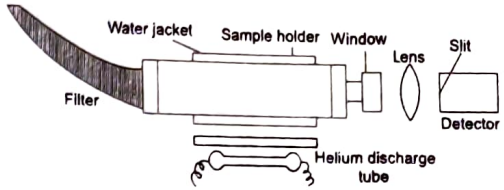


Fig. Diagrammatic structure of Raman spectrometer.

Raman spectroscopy is used to study the structure of gass, nature of bonding in molecules and the molecular structure of organic compounds. The technique is also used to study electrolytic dissociation, hydrolysis and change from crystalline state to amorphous state of molecules.

X-ray Spectroscopy

When accelerated electrons having high kinetic energy bombard on a metal, the energy is lost inside the metal atom, which emits the energy as X-ray radiation. X-rays are electromagnetic waves produced by electrons of an atom and are scattered in all directions. X-rays interact with matter either by absorption or scattering or diffraction. When an X-ray beam is passed through matter, part of its energy is lost by scattering and a part by absorption. Scattering of X-ray radiation forms the basis for diffraction.

X-ray spectroscopy includes three major types namely X-ray absorption spectroscopy, X-ray diffraction spectroscopy and X-ray fluorescent spectroscopy. All these instruments have more or less similar components but they differ only in their optical systems.

X-ray Absorption Spectroscopy

X-ray absorption spectrometer is designed specifically with a provision of ionization chamber. It consists of a tungsten X-ray tube, a motor-driven chopper to interrupt one half of the X-ray beam, an aluminium attenuator of variable thickness for making adjustment to balance absorption in the sample, a reference X-ray beam, a photomultiplier tube to receive X-ray beams, an amplifier and a detector. The results are interpreted through a calibration curve.

Uses: This instrument is used to analyse substances qualitatively and quantitatively, to locate trace elements and to measure the volume of liquids in closed vessels.

X-ray Diffraction Spectroscopy (X-ray Diffraction Crystallography)

When an X-ray beam is passed through a substance, the electrons of its atoms emit electromagnetic radiation in all directions like that of the incident X-radiation. These scattered waves from the electrons are arranged in the form of a crystal lattice. The interference of these waves cause diffraction by the crystal plane. Thus each crystalline substance scatters the X-ray in the form of its own diffraction pattern according to its atomic and molecular structure. X-rays are electromagnetic waves having definite wavelengths and the atoms are found in regular three-dimensional structure in crystals. Therefore, X-radiations resolve the atoms found in a crystal and the atoms of the crystals now scatter the X-rays, thus producing a diffraction pattern in the form of spots. The diffraction pattern thus obtained is recorded on an X-ray film or an electronic detector to determine the structure of the molecules.

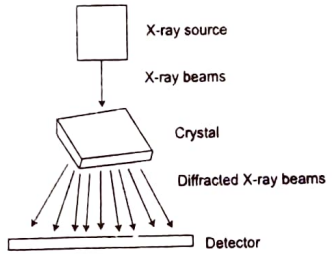


Fig. Schematic representation of X-ray crystallography.

The study of crystals constitutes crystallography. The crystals are three-dimensional space gratings and are used for diffraction of X-rays. When an X-ray beam having continuous range of wavelengths are allowed to pass through a crystal and the transmitted beam is made to fall on a photographic plate as in Figure, the photograph will show a central spot surrounded by a number of other weak spots in a definite pattern.

Thus the X-ray beam has been diffracted according to the geometrical pattern of the crystal in which the atoms are arranged in a regular three-dimensional lattice. The regular repetition of units in three dimensions is called lattice which may contain atoms arranged in many planes. Some planes will have more atoms than the others. When a beam of X-ray is passed vertically downwards on a molecule, there is a partial reflection of X-rays at each plane of atoms of the molecule. These reflected beams exhibit diffraction pattern.

The X-ray diffraction spectroscopy includes the following instrumentations:

(*i*) Transmission Photographic Method

(*ii*) X-ray Ionization Spectrometer

(*iii*) Rotating Crystal Diffraction Spectrometer

(*iv*) Powder Crystal Diffraction Spectrometer

Uses: X-ray diffraction spectroscopy is generally used to differentiate 'cis' and 'trans' isomers and linkage isomers of a complex. The transmission photographic method is useful to study the orientation of fibres and the symmetry of a single crystal. Ionization spectroscopy is used to study crystalline structures of molecules, while rotating spectroscopy is used to determine the size of unit cells. Powder crystal spectroscopy is useful to study cubic crystals.

Nuclear Magnetic Resonance (NMR) Spectroscopy

Radio waves are electromagnetic radiation with lowest energy and induced transition in magnetic level of the nuclei of atoms. Though the energy in radio frequency is very small, it affects the nuclear spin of atoms in a molecule so that the nucleus of an atom changes the direction of its spin on absorbing radiofrequency radiation. When a nucleus absorbs energy, it becomes excited and loses energy on return to unexcited state. The nucleus with alternate excited and unexcited conditions is said to be in a state of resonance. The absorption of radiation in the radiowave region of the electromagnetic spectrum by the nucleus in a magnetic field and its change from low energy state to high energy is called nuclear resonance. The energy absorbed by the nucleus is measured to determine the resonance frequency. When a nucleus absorbs energy, it produces a signal which is detected, amplified and recorded as a band of spectrum called NMR spectrum.

The important components of NMR spectrometer are diagrammatically represented in the Figure.

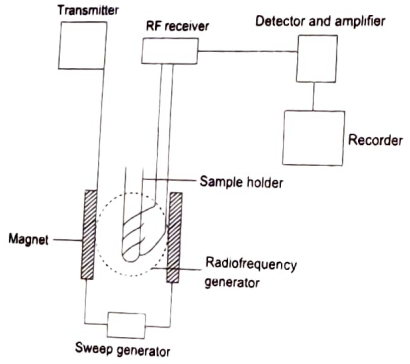


Fig. Diagrammatic structure of NMR spectrometer.

Uses: NMR spectrum of the sample is obtained by plotting the intensity of the signal against the strength of the magnetic field. By using this technique, the three-dimensional structure of small protein molecule can be determined. It is also useful to study the dynamics of protein molecules and small conformation changes, intracellular pH, membrane transport and concentration of metabolites in tissues and organs.

Electron Spin Resonance (ESR) Spectroscopy

The absorption of radiations of a specific frequency by the electrons under a magnetic field and their change from low-energy state to high-energy state is called electron spin resonance. The two states of energy levels are produced by the interaction between the magnetic moment of an unpaired electron and applied magnetic field. Substances with unpaired electrons absorb radiation having the frequency in the microwave region. These electrons produce transitions between magnetic energy level resulting in ESR spectrum, which is measured by plotting the intensity of the signal obtained by the spectroscopy against the strength of magnetic field.

The electron spin resonance spectroscopy is more or less similar to nuclear magnetic resonance spectroscope but the sensitivity of ESR spectroscope is comparatively higher than NMR spectroscope and it requires only very little quantity of the sample. The important components of ESR spectrometer are shown in the Figure.

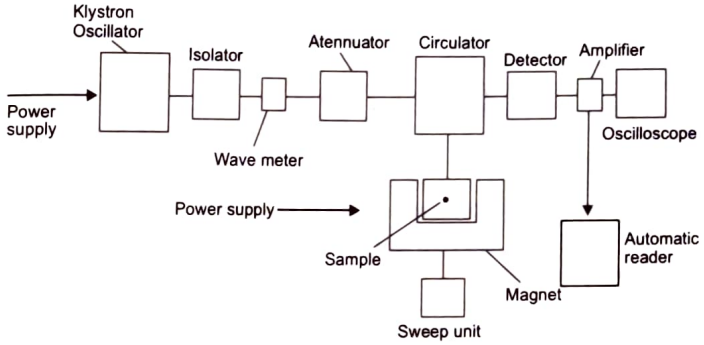


Fig. Diagrammatic structure of ESR spectrometer.

Uses: The biological macromolecules do not resonate due to the absence of unpaired electrons. They are 'spin-labelled' and are used in ESR spectroscopy. In this way, the technique is used to determine the rate of enzyme reactions, to find substrate-binding region of enzymes, to study denaturation and folding of protein molecules, enzyme-ligand interactions and symmetry and orientation of protein molecules. The apparatus is also used to study free radicals and inorganic compounds.

Optical Rotatory Dispersion (ORD) and Circular Dichroism (CD)

Most light as it occurs in nature is unpolarized and it consists of different wavelengths and vibrates in many different planes. If the magnetic field of light vibrates in a specific direction, then the light is said to be linearly polarized. If the linearly polarized light contains additional component of unpolarized light, it is partially polarized. If the electric field, instead of oscillating in a plane, proceeds in the form of a helix around the axis of propagation with a constant magnitude, then the light is referred to as circularly polarized. Here, the electric field completes one revolution within one wavelength. If the tip of the field rotates clockwise, then the light is called right circularly polarized light (RCPL). If it rotates anticlockwise, then it is left circularly polarized light (LCPL). Elliptical polarization is the intermediate condition between circular and linear polarizations. In other words, linear and circular polarizations are two extremes of elliptical polarization. Light can be polarized by scattering, reflection, transmission, selective absorption or double refraction.

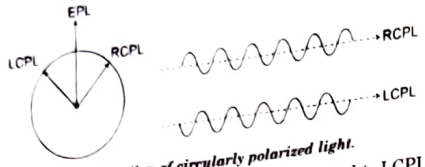


Fig. Rotation of circularly polarized light.

(CPL - Circularly polarized light; RCPL - Right circularly polarized light; LCPL- Left circularly polarized light)

Optical Activity

When a beam of linearly polarized light passes through a substance, the light remains linearly polarized but its plane of oscillation rotates, i.e., its plane of vibration becomes changed in orientation. Such substances are called optically active and the effect is called optical activity or optical rotation. When looked towards the source, the rotation may be clockwise and is called as right-handed rotation, and the substance is called dextrorotatory or it may be counter clockwise called as left-handed rotation and the substance is called as levorotatory (Figure). Most of the biological molecules can rotate the plane polarized light so that they are optically active. For example, most of the sugars are dextrorotatory while the proteins and amino acids as well as phospholipids are levorotatory. The secondary and tertiary structures of proteins and DNA are asymmetric and so they are optically active.

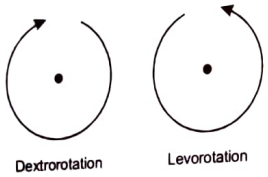


Fig. Rotation of light by optically active molecule.

Optical Rotatory Dispersion (ORD)

The optical activity of a substance is due to circular double refraction of light. In other words, the substance has a refractive index for right-circularly polarized light, which is different from the refractive index for left-polarized light. This means that the linear polarization of light is due to two circular polarizations of the same amplitude but with opposite rotation (circularly birefringence). The degree of rotation is proportional to the differences between two refractive indices and length of light velocities so that the emerging light beam will be out of phase with each other. Therefore, the light is rotated by an angle α to the original plane of polarization (Figure) and is dependent on wavelength. Thus at different wavelengths, the rotation of light is different and this effect is called optical rotatory dispersion (ORD).

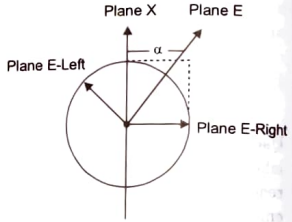


Fig. Rotation of polarized light from one plane to another.

Circular Dichroism (CD)

The absorption of light by a molecule at a given frequency is different in different directions. Therefore the molecule appears to have different colours when viewed with two kinds of plane polarized light. This phenomenon is called dichroism. If the wavelength of the plane polarized light falls within the range of the absorption maxima of the medium, the rays absorbed by the medium will prefer either left or right circularly polarized rays. Therefore the transmitted rays have lesser intensity than the incident rays. This differential absorption of circularly polarized light is called circular dichroism (CD).

Autoradiography

Autoradiography is an important technique by which the routes and conversion of molecules in biochemical reactions occurring in cells are traced.

In this technique, the sample containing radioactive substance (a chromatogram or a tissue slice or a section of a plant or animal) is kept in close contact with a photographic plate or an X-ray film. That is, the cells or tissues already exposed to radioactive isotopes are made to contact with a photographic emulsion for specific period as shown in the figure. The emanating radiations affect the emulsion in the same way like that of light. On development, the areas, which have been in contact with radioactivity, could appear as spots on the film, thus producing the image of the specimen.

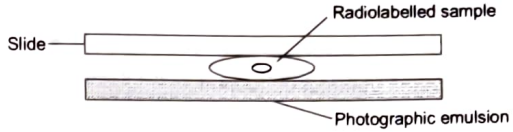


Fig. Contact between radiolabelled sample and photographic emulsion.

Specimen Preparation for Autoradiography

Typically, small precursor molecules, which are utilized by the cells as building blocks of macromolecules, are incorporated with radioactive isotopes. This is usually carried out by injecting suitable amount of a dilute radioactive solution into the animal (if required, repeatedly). Then the animal is anaesthetiazed and a specific tissue is dissected out after a period of time for fixation. The choice of fixation and fixatives depends on the chemical composition of the sample material to be studied. For example, nucleoproteins are fixed in Carroys acetic alcohol. The tissues are quickly frozen and dried in vacuum desiccators to obtain all the incorporated radioactive molecules. The incorporation of a radioactive isotope into RNA of cells requires the extraction of all DNA molecules from the tissue. Similarly, the study of DNA necessitates the removal of all RNA molecules from the tissues.

For tissues, the usual paraffin sections of 2 thickness are made by routine microtechnique method. Smears and frozen sections can also be radiolabelled and studied.

Uses: Autoradiography is useful in the study of the movement of solutes in tissues, to determine the rate of production and life span of blood cells and to identify thyroid

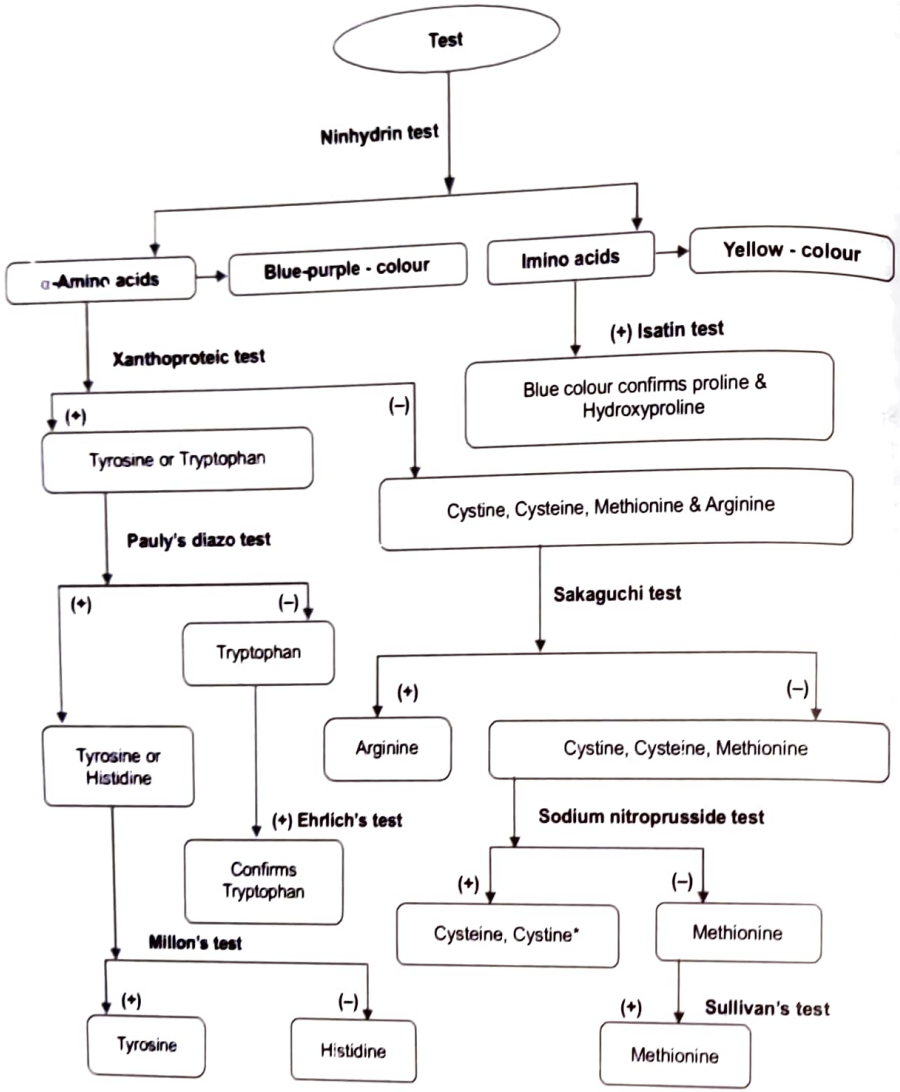
Table: Contributions of autoradiography to basic cell metabolism.

Radiolabelled substances	Events in cell metabolism
C^{14} labelled glycine	Synthesis of RNA from nucleolus, protein synthesis
H^3 labelled thymidine	Metabolism of chromosomal DNA, cell turnover times, DNA metabolism in salivary gland chromosomes
H^3 labelled cytidine	Nucleic acid metabolism during regeneration
H^3 labelled cytidine and C^{14} labelled glycine	Changes in nucleic acids and protein metabolism during meiosis in plants
C^{14} labelled adenine	nucleic acid metabolism during animal spermatogenesis

function and cancer cells. This technique has made many contributions in the study of cell metabolism. In medicine, this technique is useful to diagnose circulatory disorders, bone defects, tumours and so on. Some like I^{131} is used to treat thyroid defects, P^{32} to treat skin diseases and Co^{60} to treat tumours.

Biochemical Tests

Flow Sheet: Qualitative Analysis of Amino acids

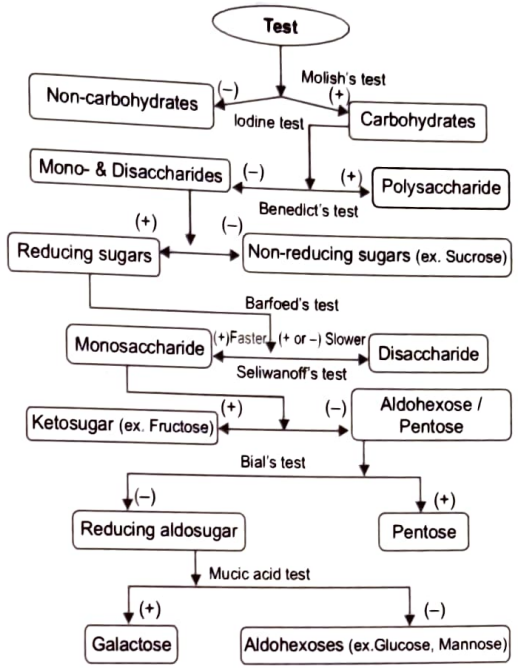


* Cystine shows a positive nitroprusside test after reduction of the disulphide bonds with reducing agents like KCN, or NaCN.

Note:

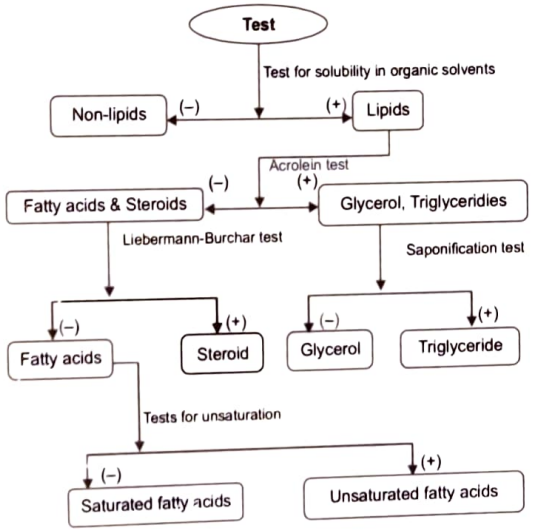
This flow sheet is applicable for qualitative analysis of eight amino acids. However, the rest of the amino acids which show positive ninhydrin reaction can be identified by paper, TLC and ion-exchange chromatography.

Flow Sheet: Qualitative Analysis of Carbohydrates



Biochemical test : (+) = Positive; (–) = Negative

Flow Sheet: Qualitative Analysis of Lipids



Biochemical test : (+) = Positive; (–) = Negative

EXERCISE - I

MCQ TYPE QUESTIONS

1. The optical density of a solution measured on a spectrophotometer using 1 cm path length is 1. If the solution is diluted 10 times and a cuvette 5mm length is used, the optical density will be:
   (a) 0.5 (b) 1.0
   (c) 10.0 (d) 0.05

2. Fluorescence microscopy is based on the ability of certain molecules to
   (a) Continuously emit light of a constant wavelength
   (b) Absorb light of many different wavelength
   (c) Absorb light of a given wavelength and then emit light of a longer wavelength
   (d) Absorb light of a given wavelength and then emit light of a shorter wavelength

3. Substitution of which amino acid in place of alanine would increase the absorbance of protein at 280 nm
   (a) Tyrosine (b) Leucine
   (c) Arginine (d) Proline

4. Proteins give fluorescence because of the presence of
   (a) Tyrosine (b) Phenylalanine
   (c) Tryptophan (d) All

5. Proteins give signal at far-UV. Which is used to determine secondary structure?
   (a) Peptide bond of the protein
   (b) Aromatic group of the protein
   (c) Peptide bond and aromatic side chain of the protein
   (d) None of the above

6. Fourier transform is used in NMR. What is its importance?
   (a) Convert electronic transition into signal
   (b) Convert time domain free induction decay pattern to frequency
   (c) Used to calculate chemical shift
   (d) Used to calculate spin-spin splitting

7. DOT525, QDOT565, QDOT585, QDOT605, QDOT625 and QDOT655 are semiconductor particles that also fluoresce at the wavelengths indicated in their names with bandwidths of emission being +/- 20nm. Emission filters are available for fluorescence microscope for detection of these fluorophores but unfortunately the bandwidths of the filters are also +/-20nm. If you want to detect 4 different proteins in a cell simultaneously, you will use the following sets of QDOTs for labelling the antibodies, so that there is minimum overlap in their detection using the filters.
   (a) QDOT525, QDOT565, QDOT605 and QDOT655
   (b) QDOT525, QDOT565, QDOT585 and QDOT605
   (c) QDOT535, QDOT605, QDOT625 and QDOT655
   (d) QDOT525, QDOT585, QDOT605 and QDOT655

8. Electrophoretic procedure which does not depend on the charge of a protein is:
   (a) Moving boundary electrophoresis
   (b) Isoelectric focussing
   (c) SDS-PAGE
   (d) Native PAGE

9. The pI (isoelectric point)of a protein is 6. At pH 7, when electric field is applied, the protein
   (a) Moves towards anode
   (b) Moves towards cathode
   (c) Does not move
   (d) Moves randomly

10. Antibody are separated by the
    (a) Isopycnic centrifugation
    (b) Differential centrifugation
    (c) Zonal centrifugation
    (d) Electrophoresis

11. Outer and inner membrane of bacteria are separated by
    (a) Sucrose gradient centrifugation
    (b) Electrophoresis
    (c) Sonication
    (d) Ultracentrifugation

12. The following statements are true about nucleotide analysis except
    (a) Duplex DNA can be detected by staining with intercalating dye like Ethidium Bromide
    (b) Very large DNAs are separated by pulse-field gel electrophoresis (PFGE)
    (c) Higher percentage of agarose gel is needed to resolve large fragments of DNA
    (d) mRNA can be isolated by affinity chromatography

13. After treatment of Cyanogen bromide and electrophoresis we find that there are 4 bands. What conclusion you draw from this experiment?
    (a) There is a his-tag of 4 residues at c-terminal
    (b) There are 3 Selanocysteine at c-terminal
    (c) There are 3 methionine residues
    (d) None of the above

14. DNA binding protein is detected by using which technique
    (a) Western blotting
    (b) Eastern blotting
    (c) Northern blotting
    (d) South-western blotting

15. Lectin is used as affinity column for purification of
    (a) Lipoprotein (b) Glycoprotein
    (c) Nucleoprotein (d) All of the above

16. DS and SS DNA are separated by which technique
    (a) CsCl density gradient centrifugation
    (b) Hydroxyapatite chromatography
    (c) Both (a) and (b)
    (d) Edman method

17. Number of subunits of a protein is determined by
    (a) SDS PAGE
    (b) Native PAGE
    (c) Reducing SDS-PAGE
    (d) Reducing SDS-PAGE and Gel filtration

18. Which of the following technique is used to study protein-protein interaction?
    (a) ChIP assay
    (b) Yeast one hybrid system
    (c) Yeast two hybrid system
    (d) All

19. Ethidium bromide is used to visualize nucleic acid because
    (a) It makes covalent bond with nucleic acid and fluoresces at 302 nm
    (b) It stacks between bases
    (c) Ethidium bromide gives fluorescence which is detected by UV-illuminator
    (d) Ethidium bromide-nucleic acid complex increase the fluorescence of the Et. Br.

20. Using site directed mutagenesis four mutants are generated. Which mutant has the largest difference in the number of atoms compared to the wild type?
    (a) Ser—Cys (b) Tyr—Phe
    (c) Lys—Ala (d) Arg—Lys

21. The molecular weight of an unknown protein can be best determined by
    (a) Ion exchange chromatography
    (b) Centrifugation
    (c) Mass spectrometry
    (d) Affinity chromatography

22. From the data given below, identify the protein pair that would (a)give the least mobility band on SDS-PAGE experiment and (b) elute last on an anion exchange e.g., DEAE column respectively

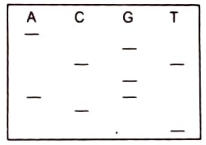
Protein	pI	Subunit M.W.	Native M.W.
A	9	10,000	20,000
B	8	35,000	35,000
C	6	15,000	90,000
D	5	20,000	80,000
E	3	30,000	30,000

    (a) Protein C, Protein A
    (b) Protein A, Protein E
    (c) Protein B, Protein E
    (d) Protein B, Protein A

23. On the completion of isoelectric focussing the net charge on a protein is
    (a) positive
    (b) negative
    (c) zero
    (d) indeterminate

24. Which technique is used to separate T_H cells from T_C cells?
    (a) Flow cytometry
    (b) RIA
    (c) ELISA
    (d) None of these

25. Shown below is the autoradiogram of an electrophoresis gel obtained during sequencing a single stranded DNA by Sanger's method



The base sequence of the DNA is
    (a) 3'-AGTCGAGCT-5'
    (b) 5'-TCGACTCGA-3'
    (c) 3'-TCAGCTCGA-5'
    (d) 5'-AGTCGAGCT-3'

26. You need to purify a specific antibody from serum for further experiments with Western blots. One of the following methods would give the purest antibody
(a) Affinity chromatography
(b) SDS PAGE
(c) Gel exclusion chromatography
(d) Reverse phase chromatography

27. General methods for the quantitative PCR
(a) Hydrolysis probes like TaqMan
(b) Hybridization probes like Light Cyber
(c) DNA-binding agents like SYBR Green
(d) All of the above

28. TEM is used for the study of
(a) topology of the bacteria
(b) internal structure of the bacteria
(c) Both (a) and (b)
(d) can be used for topology as well as internal structure

29. Hydrophobic interaction chromatography has a characteristic step
(a) Uses hydrophobic solvent for binding
(b) Uses hydrophilic solvent for binding
(c) Uses hydrophilic solvent for elution
(d) All of the above

30. Which of the following is used for the secondary structure determination?
(a) CD (b) UV-Vis
(c) Fluorescence (d) All

31. Ni-column is used for the purification of
(a) His tag protein
(b) GST tag protein
(c) Negatively charged protein
(d) Carbohydrate

32. Concentration of cytokine secreting cell is measured by
(a) RIA
(b) ELISA
(c) Eli-spot
(d) All

33. Where polymerase binds inside the cell, is determined by
(a) EMSA
(b) Super shift assay
(c) ChIP assay
(d) All of the above

34. Gel retardation assay is commonly used to monitor interactions between
(a) Proteins
(b) Drug and nucleic acid
(c) Nucleic acid
(d) Protein and nucleic acid

35. Which of the following fluorescent probes is used to monitor the progress of amplification in Real time PCR?
(a) Rhodamine (b) FITC
(c) Cyan Blue (d) SYBR green

36. A virus can be made radioactive by
(a) Culturing the virus in a medium of p32
(b) Culturing the virus in a medium of potato, dextrose and p32
(c) Providing p32 to virus when they are about to attack the bacteria
(d) Providing p32 to a bacterium which have been infected by a virus

37. An oligonulceotide DNA sequence tagged with fluorescent tag is used to identify unknown gene by hybridization is known as
(a) Probe (b) Reporter gene
(c) Ligand (d) cDNA

38. Restriction endonuclease generated DNA fragments separated by gel electrophoresis and blot transferred onto a membrane filter are probed with a radioactive DNA fragment. This procedure is called:
(a) Gene cloning (b) Southern technique
(c) PCR (d) Recombinant DNA

39. mRNA can be isolated in best form from total cellular RNA by
(a) Northern blotting
(b) Gel electrophoresis
(c) CsCl density gradient centrifugation gamma radioactive phosphate
(d) Affinity chromatography using oligo dT-tagged bead

40. FACS is used to
(i) Sort cells based on their cell surface marker-conjugated with fluorescently tagged antibodies
(ii) Sort chromosome based on hybridization with chromosome specific DNA probes tagged with fluorescent dye
(iii) Count cells with specific cell surface marker
(a) Only (i) (b) Only (ii)
(c) (i) and (ii) (d) (i), (ii) and (iii)

41. A particular cell type for a heterogeneous mix of cell types can be isolated by
(a) Antibody affinity chromatography
(b) Density gradient centrifugation
(c) Differential centrifugation
(d) High performance liquid chromatography

42. Using a PCR to amplify a given segment of DNA, after 30 cycles how many folds of the given segment has been amplified?
(a) 10^3
(b) 10^5
(c) 10^6
(d) 10^9

43. The unique feature of the enzyme Taq polymerase used in PCR is
(a) High speed
(b) High fidelity
(c) High thermal stability
(d) Low thermal stability

44. Which of the following DNA polymerase synthesizes DNA molecules containing a single 3′-Adenine at their termini?
(a) Taq polymerase
(b) Pfu DNA polymerase
(c) Both (a) and (b)
(d) None of the above

45. Match the following correctly

Group I	Group II
P. DNA footprinting	1. Protein-protein interaction
Q. Yeast two hybrid system	2. VNTR
R. DNA fingerprinting	3. DNA binding protein
S. SAGE	4. Transcriptome

(a) P-1, Q-2, R-4, S-3
(b) P-3, Q-1, R-2, S-4
(c) P-3, Q-4, R-1, S-2
(d) P-4, Q-2, R-1, S-3

46. MALDI is generally associated with which Analyser?
(a) Time of flight
(b) Triple Quadruple
(c) Ion trap
(d) All of the above

47. Which of the following is used for diagnosis of Kidney infection?
(a) I-131 (b) I-133
(c) Xe-133 (d) S-35

48. Desalting of the salt can be performed by
(a) Dialysis
(b) Gel filtration
(c) Both of the above
(d) None

49. Antigen interaction with antibody is measured by
(a) ELISA (b) RIA
(c) Immunodiffusion (d) All of the above

50. Reverse phase chromatography is known for its
(a) Separation of non-polar molecules
(b) Use of non-polar mobile phase
(c) Use of polar mobile phase
(d) Both (a) and (c)

51. Distance between active site and allosteric site is determined by
(a) CD (b) COSY
(c) FRET (d) NMR

52. Anchored PCR is used when
(a) Only single strand amplification is required
(b) Only information of one end of gene known
(c) When two gene is co-amplified
(d) All of the above

53. GOLD is a
(a) Alloy of metal Au
(b) Software used for metal analysis
(c) Docking program
(d) All

54. Which of the following is used for the liquid scintillation counting?
(a) PPO
(b) POPOP
(c) PPO and POPOP
(d) PPO, POPOP and aromatic

55. CM cellulose is used in
(a) Cation exchange chromatography
(b) Anion exchange chromatography
(c) Both (a) and (b)
(d) Affinity chromatography

56. Identical sized RNA transcript is detected by Northern blot analysis of UDP glucoronosyl transferase obtained from human liver and kidney. Microarray analysis of the same samples show equal intensity, whereas Western blot detects a 55 kDa strong band in liver, but a very faint band in kidney of same size. The regulation of UDP glucoronosyl transferase is

(a) Transcriptionally controlled
(b) Post-transcriptionally controlled
(c) Translationally controlled
(d) Post-translationally controlled

57. Taq DNA polymerase enzyme is associated with PCR, its role is

1. Initiation of DNA synthesis
2. Termination of DNA synthesis
3. Proof-reading of DNA synthesized by Taq polymerase 4. Inhibition of Taq polymerase

Choose the correct answer

(a) 1 and 2 only (b) 1 only
(c) 3 only (d) 1, 2, 3 and 4

58. A column packed with sieved particles are used in which of the following technique to separate smaller and larger protein molecules?

(a) Affinity chromatography
(b) Gel electrophoresis
(c) Molecular exclusion chromatography
(d) All of the above

59. In an expression where V= velocity, E_z = electric force and F = frictional force: the velocity with which the proteins move under electric field is given by

(a) V = Ez.F (b) V = F/Ez
(c) V = Ez + F (d) V = Ez – F

60. Affinity chromatography works on the basic principle that

(a) A specific protein with high affinity for the supporting system is retained and rest sample flows down
(b) Proteins with acidic charge separate first and basic protein later on
(c) Tertiary structural protein are retained and others flow smoothly through column
(d) All the above are partially true

61. CsCl gradient centrifugation helps in separation of DNA fragments, the basic principle involved is, that of

(a) DNA fragments can move and accumulate at a position where the density of CsCl and DNA is same
(b) DNA binds firmly to CsCl
(c) Only smaller garments of DNA can bind to CsCl
(d) CsCl degrades DNA into desired length fragments

62. Isoelectric focussing technique used to separate proteins works on the principle of

(a) Electrophoretic separation based on relative content of acidic and basic residues
(b) Mass of the protein molecules
(c) Number of amino acids containing sulphur
(d) Coagulation capacity of protein molecules

63. Amino acids that give blue coloration on treatment with ninhydrin are

1. α type 2. β type
3. γ type 4. δ type

Choose the correct answer

(a) 1 only (b) 1 and 2 only
(c) 1, 2 and 3 only (d) 1, 2, 3 and 4

64. Purines and pyrimidines are cleaved by

(a) Dimethyl sulphate
(b) Hydrazine
(c) Piperidine
(d) Dimethyl sulphate and hydrazine

65. Polyacrylamide gel electrophoresis is used for DNA sequencing because, it

(a) can separate different DNAs
(b) can resolve DNA molecules differing in length
(c) can break DNA into small fragments
(d) can separate purines and pyrimidines

66. Maxam Gilbert and Sanger dideoxy methods are concerned with

(a) DNA amplification (b) DNA damage
(c) DNA repair (d) DNA sequencing

67. Purity of particular antigen in sera, culture filtrates or tissues can be best obtained by

(a) Double immunodiffusion
(b) Simple immunodiffusion
(c) Both (a) and (b)
(d) None

68. Which of the radioisotope of iodine is generally used in autoradiography technique?

(a) ^{127}I (b) ^{125}I
(c) ^{126}I (d) ^{114}I

69. In autoradiography

(a) β-particles are used to expose silver halide grains of film
(b) α-particles used to expose silver halide grains of film
(c) Both (a) and (b)
(d) None of the above

70. Sucrose gradient centrifugation can be used to estimate the size of

(a) Proteins (b) RNA molecules
(c) Ribosomes (d) Ribosomal subunits

71. Fuelgen staining is specific for DNA only because

(a) Hydrolysis produces free aldehyde group in deoxyribose which can react with Schiff's reagent
(b) DNA is the true genetic material
(c) It stains a double stranded structure only
(d) DNA is found in nucleus where this stain is most active

72. Q-banding, the technique of staining chromosomes uses which of the following dyes?

(a) Acridine (b) Ninhydrin
(c) Quinacrine (d) Acetocarmine

73. R-bands of chromosomes are visualized as green with the help of _____ staining

(a) Acridine
(b) Acridine orange
(c) Giemsa
(d) Quinacrine

74. In the separation of different amino acids, _____ paper chromatography is clearly visualized by spraying to get a purple dot

(a) Acetocarmine (b) Giemsa
(c) Ninhydrin (d) Safranin

75. 60 S, the letter S represents the sedimentation coefficient and is expressed as Svedberg units. Greater the size of ribosome unit,

(a) Smaller will be the value of S
(b) Greater will be the value of S
(c) Same will be the value of S
(d) It has no direct correlation with the size

76. You have cut the genome of a double stranded virus with a restriction endonuclease and electrophoresed the products on an agarose gel. You observe only one band on the gel, equivalent to the size of the genome. This is because

(a) There are no introns in the genome
(b) The introns contain the recognition sites and have already been spliced out
(c) All of restriction fragments are too small to detect
(d) Restriction endonucleases do not cut RNA, and this virus has an RNA genome

77. In order to identify the person who committed a crime, forensic experts will need to extract DNA from the tissue sample collected at the crime scene, and conduct one of the following procedures for DNA fingerprinting analysis

(a) Cut the DNA and hybridize with specific micro-satellite probes
(b) Cut the DNA and subclone the fragments
(c) Determine the sequence of the subclones
(d) (b) followed by (c)

78. All of the following are true about DNA microarray technology except

(a) An electron microscope is use to gather data from the arrays
(b) The technology is used to assess transcription from multiple genes simultaneously
(c) The technology works best for organisms whose genome is completely sequenced
(d) The technology is derived from computer chip manufacture

79. Cibacron Blue dye affinity chromatography can be used for affinity purification of

(a) NADPH dehydrogenase
(b) Glucoamylase
(c) Subtilisin
(d) Caspase

NUMERICAL TYPE QUESTIONS

1. The molar absorption coefficient of NADH at 340 nm 6220 L/M/CM while that of NAD is negligible. If 10 M NADH and 10 M NAD are present in a cuvette of a path length 5mm, the absorbance would be-

2. Haemoglobin-A on SDS-PAGE gives n number of bands where n is

3. A beam of light passes through 1 cm of a coloured solution. Eighty percent of the incident light is transmitted. If the incident light passes through 2 cm of the same solution, the percentage of transmitted light is

4. A solution of tryptophan has an absorbance at 280 nm of 0.54 in a 0.5 cm path length cuvette. Given the absorbance coefficient (ε) for tryptophan is $5.4 \times 103\ M^{-1}cm^{-1}$, the concentration of the solution is

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2016

1. Cesium chloride density gradient centrifugation is commonly used for the separation of DNA molecules. The buoyant density, ρ, of a double stranded Cs^+DNA is given by the equation $\rho = 1.66 + 0.098X_{G+C}$ where X_{G+C} denotes

(a) total number of G and C

(b) mole fraction of G + C

(c) number of GC repeats

(d) ratio of G+C to A+T content

2. Which one of the following techniques relies on the spin angular momentum of a photon?

(a) CD spectroscopy

(b) Fluorescence spectroscopy

(c) IR spectroscopy

(d) Raman spectroscopy

3. In dead-end filtration, rate of filtration is

(a) directly proportional to the square root of pressure drop across the filter medium

(b) inversely proportional to the pressure drop across the filter medium

(c) inversely proportional to the viscosity of the solution

(d) inversely proportional to the square of viscosity of the solution

2014

4. If the dissociation constant for solute-adsorbent binding is K_D, the retention time of the solute in a chromatography column

(a) increases with increasing K_D

(b) decreases with increasing K_D

(c) passes through minimum with increasing K_D

(d) is independent of K_D

2013

5. Match the entries in the **Group I** with the elution conditions in **Group II.**

Group I

P. Ion-exchange

Q. Hydrophobic chromatography

R. Gel filtration column chromatography

S. Chromatofocusing

Group II

1. Isocratic solvent chromatography
2. Ampholytes
3. Increasing gradient of salt
4. Decreasing gradient of polarity

(a) P-4,Q-1,R-2,S-3

(b) P-4,Q-3,R-1,S-2

(c) P-3,Q-4,R-1,S-2

(d) P-3,Q-4,R-2,S-1

2012

6. In transmission electron microscopy, electron opacity is greatly enhanced by treating the specimen with

(a) ferrous ammonium sulfate

(b) uranium acetate

(c) sodium chloride

(d) basic fuchsin

7. The helix content of a protein can be determined using

(a) an infrared spectrometer

(b) a fluorescence spectrometer

(c) a circular dichroism spectrometer

(d) a UV-Visible spectrophotometer

8. Match the entries in Group I with the methods of sterilization in Group II.

Group I

P. Serum

Q. Luria broth filtration

R. Polypropylene tubes

S. Biological safety cabinets

Group II

1. Autoclave
2. Membrane
3. UV irradiation
4. Gamma irradiation
5. Dry heat

Codes :

	P	Q	R	S
(a)	5	3	1	4
(b)	1	4	5	3
(c)	2	1	4	3
(d)	4	1	3	5

2011

9. Match the items in **Group I** with **Group II.**

Group I

P. Circular dichroism

Q. X-ray crystallography

R. Freeze-drying

S. Ultracentrifugation

Group II

1. Concentration
2. Sedimentation coefficient
3. Secondary structure determination
4. Tertiary structure determination

	P	Q	R	S
(a)	4	1	2	3
(b)	1	4	3	2
(c)	2	3	4	1
(d)	3	4	1	2

2009

10. Match the properties in Group 1 with the downstream operation in Group 2:

Group 1	Group 2
P. Size	1. Extraction
Q. Density	2. Distillation
R. Volatility	3. Filtration
S. Solubility	4. Sedimentation

(a) P-3, Q-4, R-2, S-1

(b) P-4, Q-1, R-2, S-3

(c) P-4, Q-3, R-1, S-2

(d) P-3, Q-2, R-4, S-1

2008

11. Match the coefficients in group 1 with their corresponding downstream processing steps given in group 2

Group 1	Group 2
P. Sedimentation coefficient	1. Aqueous two phase extraction
Q. Partition coefficient	2. Ultrafiltration
R. Rejection coefficient	3. Dialysis
S. Activity coefficient	4. Centrifugation

	P	Q	R	S
(a)	3	1	4	2
(b)	2	1	4	3
(c)	4	3	1	2
(d)	4	1	2	3

2007

12. Match each parameter in group 1 with the appropriate measuring device in group 2

Group 1	Group 2
P. Pressure	1. Photometer
Q. Foam	2. Rotameter
R. Turbidity	3. Diaphragm gauge
S. Flow rate	4. Rubber sheathed electrode

(a) P-3, Q-4, R-1, S-2 (b) P-1, Q-3, R-2, S-4

(c) P-4, Q-1, R-2, S-3 (d) P-1, Q-2, R-3, S-4

2006

13. When electroporation is used for introducing DNA into mammalian cells

(P) a carrier for DNA is not required

(Q) the lipid bilayer (membrane) interacts with an electric pulse to generate permeation sites

(R) the viability of the cells becomes approximately zero percent

(S) the first step involves absorption of DNA on the cell membrane

(a) P, Q (b) Q, R

(c) P, S (d) Q, S

NUMERICAL TYPE QUESTIONS

2016

1. The molar extinction coefficients of Trp and Tyr at 280 nm are 5690 and 1280 $M^{-1}.cm^{-1}$, respectively. The polypeptide chain of yeast alcohol dehydrogenase (37 kDa) contains 5 Trp and 14 Tyr residues. The absorbance at 280 nm of a 0.32 $mg.mL^{-1}$ solution of yeast alcohol dehydrogenase measured in a cuvette of 1 cm pathlength will be _____.

(Assume that the molar extinction coefficient values for Trp and Tyr apply to these amino acids in the yeast alcohol dehydrogenase).

2. A protein is to be purified using ion-exchange column chromatography. The relationship between HETP (Height Equivalent to Theoretical Plate) and the linear liquid velocity of mobile phase is given by:

$$H = \frac{A}{u} + Bu + C$$

where H is HETP (m) and u is linear liquid velocity of mobile phase ($m.s^{-1}$). The values of A, B and C are 3×10^{-8} $m^2.s^{-1}$, 3 s and 6×10^{-5} m, respectively. The number of theoretical plates based on minimum HETP for a column of 66 cm length will be _____.

ANSWERS

EXERCISE - I

MCQ Type Questions

1. (d)	2. (c)	3. (a)	4. (d)	5. (b)	6. (b)	7. (a)	8. (c)	9. (a)	10. (c)
11. (c)	12. (c)	13. (c)	14. (a)	15. (b)	16. (c)	17. (b)	18. (c)	19. (b)	20. (b)
21. (c)	22. (d)	23. (c)	24. (c)	25. (a)	26. (a)	27. (c)	28. (d)	29. (a)	30. (a)
31. (a)	32. (d)	33. (d)	34. (d)	35. (d)	36. (d)	37. (a)	38. (b)	39. (d)	40. (d)
41. (a)	42. (d)	43. (c)	44. (a)	45. (b)	46. (a)	47. (a)	48. (a)	49. (d)	50. (b)
51. (c)	52. (b)	53. (c)	54. (d)	55. (a)	56. (d)	57. (b)	58. (c)	59. (b)	60. (a)
61. (a)	62. (a)	63. (a)	64. (c)	65. (b)	66. (d)	67. (c)	68. (b)	69. (a)	70. (c)
71. (a)	72. (c)	73. (c)	74. (c)	75. (b)	76. (d)	77. (a)	78. (a)	79. (a)	

Numerical Type Questions

1. 0.062 2. 4 3. 60 4. 0.2 mM

EXERCISE - II

MCQ Type questions

1. (b)	2. (a)	3. (c)	4. (b)	5. (c)	6. (b)	7. (c)	8. (c)	9. (d)	10. (a)
11. (d)	12. (a)	13. (a)							

Numerical Type Questions

1. (0.37 : 0.43) 2. (1000.0 : 1000.0)

EXPLANATIONS

EXERCISE - I

MCQ TYPE QUESTIONS

1. Given, optical density of a solution of 1 cm path length = 1

   Concentration of the solution (say) = C

   Molar absorption coefficient, ε (say)

   Therefore, $A = \varepsilon Cl$

   => $1 = \varepsilon C1$

   => $\varepsilon = 1/C$

   Now, when the solution is diluted 10 times, we have new concentration = 1/10 C = 0.1 C

   Given new path length = 5 mm = 0.5 cm

   Therefore the new optical density of the solution $= \varepsilon \times 0.1C \times 0.5 = 1/C \times 0.05\ C = 0.05$

2. Molecules in the excited state do not remain there long, but spontaneously relax to the more stable ground state. With most molecules, the relaxation process is brought about by collisional energy transfer to solvent or other molecules in the solution. Some excited molecules, however, return to the ground state by emitting the excess energy in the form of photon or light. This process is called fluorescence. Two important characteristics of the emitted light should be noted: (i) It is usually of longer wavelength (lower energy) than the excitation light. This is because part of energy associated with the stable state is lost as heat energy and (ii) the emitted light is composed of many wavelengths which results in a fluorescence spectrum.

3. Aromatic amino acids like tyrosine, tryptophan and phenylalanine can replace alanine to increase the absorption of the protein. Proteins in solution absorb ultraviolet light with absorbance maxima at 280 and 200 nm. Amino acids with aromatic rings are the primary reason for the absorbance peak at 280 nm. Peptide bonds are primarily responsible for the peak at 200 nm. Secondary, tertiary, and quaternary structure all affect absorbance, therefore factors such as pH, ionic strength, etc. can alter the absorbance spectrum.

4. Commonly, the optical absorbance of protein is measured at 280 nm. At this wavelength, the absorbance of protein is mainly due to the amino acids tryptophan, tyrosine and cysteine with their molar absorption coefficients decreasing in that order. Of course, the molar absorption coefficient of the protein itself at 280 nm will depend upon the relative concentrations of these three amino acids. Therefore, different proteins can have different absorption coefficients and even the wavelength of the maximum absorbance may differ. This fact can be used to help identify different types of proteins by relatively fast and simple optical tests.

5. Protein absorbs strongly at 280 nm due to a number of its constituent amino acids. The peptide bonds found in the amino acids also absorb at 205 nm. The UV absorbance of protein can be used both to quickly image and acquire spectra of microscopic samples non-destructively. The spectra can also be used to determine protein concentrations and the relative amounts of protein to DNA or RNA. Protein absorbs strongly at 280 nm due to a number of its constituent amino acids. The peptide bonds found in the amino acids also absorb at 205 nm. The UV absorbance of protein can be used both to quickly image and acquire spectra of microscopic samples non-destructively. The spectra can also be used to determine protein concentrations and the relative amounts of protein to DNA or RNA. Protein, including that in tissues and protein crystals, absorbs ultraviolet light quite strongly. Rather, it is the amino acids that make up the proteins that absorb the UV light. The strong absorbance of UV light by protein allows for rapid analysis of protein samples, including protein crystals, by microscopy and microspectroscopy. At far UV, mainly the protein structure is determined by the aromatic amino acids present in the protein.

6. The phenomenon of magnetic resonance results from the interaction of the magnetic moment of an atomic nucleus ( )with an external magnetic field. The cause of this magnetic moment is the quantum mechanical angular momentum (spin angular momentum)of all nuclei that are no (g,g) nuclei (even number of protons and neutrons). Due to this current the atomic nucleus behaves as a small electromagnet. A far better resolution and sensitivity in NMR was achieved by the introduction of pulsed Fourier transform techniques (FT-NMR). In FT-NMR the resonances are not measured one after another but all nuclei are excited at the same time by a radio frequency pulse: Normally a radio emitter works at fixed frequency nu0. However, if the radiation is emitted as a very short pulse (some s in NMR) the pulse frequency becomes 'uncertain'. A short radio frequency pulse contains many frequencies in a broad band around nu0 and thus excites the resonances of all spins in a sample at the same time. The pulses cause a signal to be emitted by the nuclei. This signal is measured as a function of time after the pulse. It cannot be interpreted directly. FT NMR leads to a tenfold, and sometimes 100-fold increase in sensitivity since the pulse response contains information on all resonance frequencies at the same time. During the same time needed to record a single conventional spectrum, the FT experiment can be repeated many times and the results summed by a computer.

7. Quantum dots (QDs) offer remarkable photostability and brightness. They do not exhibit photobleaching common to organic fluorophores. Moreover, the high emission amplitude for QDs results in a marked improvement in the signal to noise ratio of the final image. Protein microarrays allow highly parallel quantitation of specific proteins in a rapid, low-cost and low sample volume format. Furthermore the multiplexed assay enables detection of many proteins at once in one sample, making it a powerful tool for biomarker analysis and early cancer diagnostics. To encompass a a solution containing a number of proteins but with a limited bandwidth filter of +/- 20 nm, the QDs to be selected for antibody labelling must be such that they do have only minimal overlap in terms of their emission spectra. So, amongst the given QDs, the ones with the minimum overlap would be QDOT525, QDOT565, QDOT605 and QDOT655.

8. In SDS-polyacrylamide gel electrophoresis (SDS-PAGE), proteins are separated largely on the basis of polypeptide length, and so their molecular weight can also be estimated. SDS does however denature the protein, so activity stains cannot be used to identify particular enzymes. As a protein, concentrated into sharp bands by the stacking gel, travels through the separating gel, the narrower pores have a sieving effect, allowing smaller proteins to travel more easily and hence rapidly, than larger proteins. Because the charge-to-mass ratio is nearly the same among SDS-denatured polypeptides, the final separation of proteins is dependent almost entirely on the differences in relative molecular mass of polypeptides. In a gel of uniform density the relative migration distance of a protein (R_f) is negatively proportional to the log of its mass. If proteins of known mass are run simultaneously with the unknowns, the relationship between R_f and mass can be plotted, and the masses of unknown proteins estimated.

9. At pH of 7, the protein molecule which has an isoelectric point of 6 would become negatively charged due to the release of H^+ ions into the solution from the –COOH group of the amino acid. As such, the protein solution bears a slightly negative charge. Now, when electricity is applied to the solution, the negatively charged proteins would be attracted to the positive terminal of the electrode which would be the anode.

10. Separation of cells requires the removal of one cell type from another by physical means. Antibodies suit this application well owing to their great diversity and specificity. Antibodies can be attached to chromatography columns and used to bind a cell that possesses an antigen recognized by the specific antibody. For commercial scale

operations. IgM antibodies are usually purified by a combination of techniques, including ammonium sulphate precipitation followed by gel filtration, ion exchange chromatography or zone electrophoresis.

11. Membrane vesicles were prepared either by passing the bacteria through a French press or by conversion of the cells to spheroplasts by the lysozyme-EDTA treatment and disruption of the spheroplasts by sonication. Sonication using ultrasonic waves at a frequency of 20 kHz is normally employed to disrupt the cell membranes and separate out the outer cell membrane from the inner cell membrane.

12. The percentage of agarose used depends on the size of fragments to be resolved. The concentration of agarose is referred to as a percentage of agarose to volume of buffer (w/v), and agarose gels are normally in the range of 0.2% to 3%. The lower the concentration of agarose, the faster the DNA fragments migrate. In general, if the aim is to separate large DNA fragments, a low concentration of agarose should be used, and if the aim is to separate small DNA fragments, a high concentration of agarose is recommended.

13. Cyanogen bromide cleavage gives two peptide fragments, the longer of which has all the units on the C-terminal side of methionine. The C-terminal side of the methionine is obtained as a smaller peptide, which can be examined by any of techniques. The N-terminal side is characterized by a homoserine lactone at its C-terminus. Cyanogen Bromide is a chemical whose specificity is at the Carboxyl Side of Methionine. Thus, the four bands obtained are due to the presence of the 3 methionine residues in the protein.

14. Western blotting identifies with specific antibodies proteins that have been separated from one another according to their size by gel electrophoresis. The blot is a membrane, almost always of nitrocellulose or PVDF (polyvinylidene fluoride). The gel is placed next to the membrane and application of an electrical current induces the proteins in the gel to move to the membrane where they adhere. The membrane is then a replica of the gel's protein pattern, and is subsequently stained with an antibody.

15. Lectin affinity chromatography is a form of affinity chromatography where lectins are used to separate components within the sample. The most common application is to separate glycoproteins from non-glycosylated proteins, or one glycoform from another glycoform. Lectins are proteins that interact specifically and reversibly with certain sugar residues. Their specificity enables binding to polysaccharides and glycoproteins and agglutination of erythrocytes and tumour cells. Immobilized lectins are invaluable tools for isolating and separating glycoproteins, glycolipids, polysaccharides, sub-cellular particles and cells, and for purifying detergent-solubilised cell membrane components.

16. Fractionation of purified nucleic acid subtypes provides an effective mechanism by which to study viral assemblages without sacrificing a subset of the community's genetic signature. Hydroxyapatite, a crystalline form of calcium phosphate, has been employed in the separation of nucleic acids, as well as proteins and microbes. By exploiting the charge interaction between the positively-charged Ca^{2+} ions of the hydroxyapatite and the negatively charged phosphate backbone of the nucleic acid subtypes, it is possible to preferentially elute each nucleic acid subtype independent of the others. There is a method for the fractionation and recovery of ssDNA, dsDNA and RNA viral nucleic acids from mixed viral assemblages using hydroxyapatite chromatography. Macromolecules, such as DNA, will concentrate in the area of CsCl that has the same density as them. Hence, denser DNA will migrate downward and less dense DNA upwards forming bands.

17. Native PAGE uses the same discontinuous chloride and glycine ion fronts as SDS-PAGE to form moving boundaries that stack and then separate polypeptides by charge to mass ratio. Proteins are prepared in a non-reducing non-denaturing sample buffer, which maintains the proteins' secondary structure and native charge density. As such subunits of the proteins can be easily determined from the native gel.

18. The chromatin immunoprecipitation (ChIP) assay is a powerful and versatile technique used for probing protein-DNA interactions within the natural chromatin context of the cell. One-hybrid screening in yeast is a powerful method to rapidly identify heterologous transcription factors that can interact with a specific regulatory DNA sequence of interest (the bait sequence). Very stable, but also rather transient and dynamic protein-protein interactions generate new system properties at the level of multiprotein complexes, cellular compartments or the entire cell. Thus, interactomics is expected to largely contribute to emerging fields like systems biology or systems bioenergetics. The more recent technological development of high-throughput methods for interactomics research will dramatically increase our knowledge of protein interaction networks. The two most frequently used methods are yeast two-hybrid (Y2H)screening, a well established genetic in vivo approach, and affinity purification of complexes followed by mass spectrometry analysis, an emerging biochemical in vitro technique.

19. Ethidium bromide is a fluorescent dye used for staining nucleic acids. Ethidium bromide is a known mutagen and should be handled as a hazardous chemical and so gloves should be worn while handling. Ethidium bromide is a fluorescent dye that intercalates between bases of nucleic acids and allows very convenient detection of DNA fragments in gels. It can be incorporated into agarose gels, or added to samples of DNA before loading to enable visualization of the fragments within the gel. As might be expected, binding of ethidium bromide to DNA alters its mass and rigidity, and therefore its mobility.

20. In protein engineering, site-directed mutagenesis methods are used to generate DNA sequences with mutated codons, insertions or deletions. In a widely used method, mutations are generated by PCR using a pair of oligonucleotide primers designed with mismatching nucleotides at the centre of the primers. Site-directed mutagenesis (SDM)methods are used to generate cloned DNAs with modified sequences for examining the importance of specific residues in protein structure and function. SDM represents the primary rational method in protein engineering and for altering enzyme substrate selectivity.

21. In order to measure the characteristics of individual molecules, a mass spectrometer converts them to ions so that they can be moved about and manipulated by external electric and magnetic fields. The three essential functions of a mass spectrometer, and the associated components, are:

    1. A small sample is ionized, usually to cations by loss of an electron called the Ion Source
    2. The ions are sorted and separated according to their mass and charge called the Mass Analyzer
    3. The separated ions are then measured, and the results displayed on a chart called the Detector

    By varying the strength of the magnetic field, ions of different mass can be focused progressively on a detector fixed at the end of a curved tube (also under a high vacuum).

22. In reducing SDS-PAGE, the protein with the highest molecular weight will have the least mobility. So in this, the highest subunit molecular weight after being reduced by SDS and β-mercaptoethanol is Protein B (MW = 35,000). In case of anion exchange chromatography, the DEAE column will strongly bind to proteins which have the major amount of charge dispersed on its surface. In this case the protein with a native molecular weight of minimum value will only be able to have such way of charge dispersal and so the protein will be Protein A (MW = 20,000).

23. Isoelectric focusing (IEF) is an electrophoretic method that separates proteins according to their isoelectric points (pI). Proteins are amphoteric molecules; they carry either positive, negative, or zero net charge, depending on the pH of their surroundings. The pI is the specific pH at which the net charge of the protein is zero. Proteins are positively charged at pH values below their pI and negatively charged at pH values above their pI. The presence of a pH gradient is critical to the IEF technique. In a pH gradient and under the influence of an electric field, a protein will move to the position in the gradient where its net charge is zero.

24. Assay methods such as the Enzyme-Linked Immunosorbent Assay (ELISA) are used for protein identification with blood groups, cell surface markers, drugs, and toxins. Cell separation, based on intrinsic differences between cells, is becoming increasingly commonplace for researchers who study single cells or homogeneous cell populations to understand how the cells function in different situations. Separation of cells requires the removal of one cell type from another by physical means. Antibodies suit this application well owing to their great diversity and specificity. Antibodies can be attached to chromatography columns and used to bind a cell that possesses an antigen recognized by the specific antibody. T-cells are often selected for and enriched using these columns.

25. In Sanger Sequencing using dideoxy nucleotides, the autoradiogram generated after the electrophoresis is read from the bottom to the top which would give the sequence of the nucleotide. In this case, the sequence is: 3'-AGTCGAGCT-5'.

26. Antibodies that were developed as monoclonal antibody hybridoma cell lines and produced as ascites fluid or cell culture supernatant can be fully purified without using an antigen-specific affinity method (third type) because the target antibody is (for most practical purposes)the only immunoglobulin in the production sample. By contrast, for polyclonal antibodies (serum samples), antigen-specific affinity purification is required to prevent co-purification of nonspecific immunoglobulins. Immobilized metal chelate chromatography (IMAC) uses chelate-immobilized divalent metal ions (usually nickel, Ni^{2+}) to bind proteins or peptides that contain clusters of three or more consecutive histidine residues. The strategy is most often used to purify recombinant proteins that have been engineered to contain a terminal 6xHis fusion tag.

27. In quantitative PCR, DNA amplification is monitored at each cycle of PCR. When the DNA is in the log linear phase of amplification, the amount of fluorescence increases above the background. The point at which the fluorescence becomes measurable is called the Threshold cycle (C_T or crossing point). By using multiple dilutions of a known amount of standard DNA, a standard curve can be generated of log concentration against C_T. The amount of DNA or cDNA in an unknown sample can then be calculated from its C_T value. Two types of detection chemistries are used for quantitative PCR. The first uses an intercalating dye that incorporates into double-stranded DNA. Of these fluorescent dyes, SYBR Green I dye is the most common one used. This detection method is suitable when a single amplicon is being studied, as the dye will intercalate into any double-stranded DNA generated.

28. The transmission electron microscope (TEM) forms an image by accelerating a beam of electrons that pass through the specimen. In TEM, electrons are accelerated to 100 KeV or higher (up to 1MeV), projected onto a thin specimen (less than 200 nm) by means of the condenser lens system, and penetrate the sample thickness either undeflected or deflected. The greatest advantages that TEM offers are the high magnification ranging from 50 to 106 and its ability to provide both image and diffraction information from a single sample. In addition to the capability of structural characterization and chemical analyses, TEM also has been explored for other applications in nanotechnology. So it can be used to study the topology as well as the internal structure of the bacteria.

29. Hydrophobic interaction chromatography (HIC) separates molecules based on their hydrophobicity. HIC is a useful separation technique for purifying proteins while maintaining biological activity due to the use of conditions and matrices that operate under less denaturing conditions. Sample molecules containing hydrophobic and hydrophilic regions are applied to an HIC column in a high-salt buffer. The salt in the buffer reduces the solvation of sample solutes. As solvation decreases, hydrophobic regions that become exposed are adsorbed by the media. The more hydrophobic the molecule, the less salt is needed to promote binding. Usually a decreasing salt gradient is used to elute samples from the column in order of increasing hydrophobicity. Sample elution may also be assisted by the addition of mild organic modifiers or detergents to the elution buffer.

30. Circular dichroism (CD)spectroscopy measures differences in the absorption of left-handed polarized light versus right-handed polarized light which arise due to structural asymmetry. The absence of regular structure results in zero CD intensity, while an ordered structure results in a spectrum which can contain both positive and negative signals. Secondary structure can be determined by CD spectroscopy in the "far-UV" spectral region (190-250 nm). At these wavelengths the chromophore is the peptide bond, and the signal arises when it is located in a regular, folded environment. Alpha-helix, beta-sheet, and random coil structures each give rise to a characteristic shape and magnitude of CD spectrum.

31. For affinity purification, beaded agarose supports can be derivatized with chelating groups to immobilize the desired metal ions. This is know as immobilized metal affinity chromatography (IMAC). The chelators most commonly used as ligands for IMAC are nitrilotriacetic acid (NTA) and iminodiacetic acid (IDA). Once IDA-agarose or NTA-agarose resin is prepared, it can be "loaded" with the desired divalent metal. The resulting resin is typically called (using nickel as the example metal)Ni-chelate agarose or Ni-IDA agarose. Expressed His-tagged proteins can be purified and detected easily because the string of histidine residues binds to several types of immobilized metal ions, including nickel, cobalt and copper, under specific buffer conditions. In addition, anti-His-tag antibodies are commercially available for use in assay methods involving His-tagged proteins.

32. Quantitation of cytokine production is a valuable adjunct to standard immunologic assays in defining several pathologic processes. Cytokines often circulate as proteins bound to soluble receptors, carrier proteins, or inhibitors, which may mask their easy detection by enzyme-linked immunosorbent assay (ELISA)or radioimmunoassay. Secreted protein is typically measured by ELISA or radioimmunoassay due to the simplicity and sensitivity of these methods. Bioassays are available for some cytokines, and they have the advantage of being able to distinguish active cytokines from inactive cytokines, such as is seen with transforming growth factor β or when some cytokines are complexed with soluble receptors.

33. The gel electrophoresis mobility shift assay (EMSA)is used to detect protein complexes with nucleic acids. It is the core technology underlying a wide range of qualitative and quantitative analyses for the characterization of interacting systems. The chromatin immunoprecipitation (ChIP)assay is a powerful and versatile technique used for probing protein-DNA interactions within the natural chromatin context of the cell. This assay can be used to identify multiple proteins associated with a specific region of the genome, or the opposite, to identify the many regions of the genome associated with a particular protein. If the protein of interest binds to the target DNA, the antibody will bind to that protein:DNA complex, further decreasing its mobility relative to unbound DNA in what is called a "supershift". Supershift reactions need not be limited to antibodies, but could include other secondary or indirectly bound proteins as well.

34. The gel retardation assay, also known as the bandshift assay or electrophoretic mobility shift assay (EMSA), is used to detect DNA-binding proteins in crude cell extracts and to study the binding activity of purified DNA-binding proteins. It is also applicable to the study of RNA-protein interactions. The basis of the assay is that protein-DNA complexes remain intact when gently fractionated by gel electrophoresis and migrate as distinct bands more slowly than the free DNA fragment. Gel retardation assays can be used quantitatively to estimate dissociation constants for protein-DNA complexes.

35. Real-Time PCR (qPCR)technology accurately detects, in real time, the PCR amplification of nucleic acid targets. Through the use of various fluorescent detection chemistries, real-time PCR enables the quantification of DNA sequences as the PCR reaction proceeds. The machine contains a sensitive camera that monitors the fluorescence in each well of the 96-well plate at frequent intervals during the PCR Reaction. As DNA is synthesized, more SYBR Green will bind and the fluorescence will increase.

36. Virus is a pseudo living organism. It cannot be culture in natural and artificial media because it does not depend on any kind of support or nutrient. It mainly feeds on host and uses the host machinery to survive. So, in order to radiolabel a virus, the best way is to label the genetic material of the bacterium with p32 radiolabel which the virus infects.

37. DNA probes are small segments of DNA which help to detect the presence of a gene of a long DNA sequence, in a biological system. These DNA probes are prepared for commercial purposes and are believed to be the most sophisticated and sensitive means to identify genes or specific DNA sequences. The production of DNA probes can be done by any of the following methods. Such as (a) using a template DNA with the help of purified biological enzymes (b)DNA probe of specific sequence can also be obtained by using automated DNA synthesizers (c)DNA probe can also be included in viral DNA and may even multiply in bacteria, thus by this way many copies of DNA probe can be obtained. However, the DNA probe assay consists of the following steps. Sample to be tested is treated with detergents and enzymes to remove non DNA components. Then DNA is denatured by low PH. Single stranded DNA binds on filters and is exposed to excess of DNA probes but only one of which will hybridize. At the same time unbound DNA is detected by a variety of available methods using florescence and dye etc.

38. In Southern blotting, denatured DNA is transferred from an electrophoresis gel to a sheet of nitrocellulose or derived nylon. The exploded view at left shows that transfer is accomplished by the flow of buffer through wick, gel, and sheet and into adsorbent paper layers. DNA is retained by the derived nylon or nitrocellulose. After transfer, the DNA is permanently attached by heating or by cross-linking using UV radiation. Spatial arrangement of DNA in the gel is preserved in the sheet. In hybridization, the sheet with attached DNA can be incubated with a tagged denatured DNA population (probe). If probe molecules are complementary to molecules bound on the sheet, double-stranded DNAs will form, thus binding the probe to the sheet. After excess probe and poorly bound probe molecules are removed by washing, the only tagged molecules left are those that have formed hybrids with the target DNA.

39. Obtaining high-quality RNA is the first, and often the most critical, step in performing many molecular techniques such as reverse transcription real-time PCR (RT-qPCR), transcriptome analysis using next-generation sequencing, array analysis, digital PCR, northern analysis, and cDNA library construction. To generate the most sensitive and biologically relevant results, the RNA isolation procedure must include some important steps before, during, and after the actual RNA purification. The first option starts with isolating total RNA by conventional means, followed by mRNA enrichment from the total RNA pool. The second option is to lyse cells and purify mRNA directly from the starting sample. Three distinct technologies used to accomplish this include: oligo(dT)cellulose/spin columns, oligo(dT)/ magnetic beads, and coated plates.

40. In multicellular organisms, all the cells are identical in their DNA but the proteins vary tremendously. Therefore, it would be very useful if we could separate cells that are phenotypically

different from each other. In addition, it would be great to know how many cells expressed proteins of interest, and how much of this protein they expressed. Fluorescence Activated Cell Sorting (FACS)is a method that can accomplish all these goals. The cell suspension is entrained in the centre of a narrow, rapidly flowing stream of liquid. The flow is arranged so that there is a large separation between cells relative to their diameter. Just before the stream breaks into droplets, the flow passes through a fluorescence measuring station where the fluorescent character of interest of each cell is measured. An electrical charging ring is placed just at the point where the stream breaks into droplets. A charge is placed on the ring based on the immediately-prior fluorescence intensity measurement, and the opposite charge is trapped on the droplet as it breaks from the stream. The charged droplets then fall through an electrostatic deflection system that diverts droplets into containers based upon their charge.

41. Most affinity purification procedures involving protein: ligand interactions use binding buffers at physiologic pH and ionic strength, such as phosphate buffered saline (PBS). This is especially true when antibody: antigen or native protein: protein interactions are the basis for the affinity purification. Once the binding interaction occurs, the support is washed with additional buffer to remove unbound components of the sample. Nonspecific (e.g., simple ionic) binding interactions can be minimized by adding low levels of detergent or by moderate adjustments to salt concentration in the binding and/or wash buffer. Finally, elution buffer is added to break the binding interaction and release the target molecule, which is then collected in its purified form. Elution buffer can dissociate binding partners by extremes of pH (low or high), high salt (ionic strength), the use of detergents or chaotropic agents that denature one or both of the molecules, removal of a binding factor or competition with a counter ligand.

42. PCR begins with a mixture containing a dsDNA template, a pair of short ssDNA oligonucleotide primers, a pool of the four dNTPs, and a heat-resistant DNA polymerase, Taq Enzyme. Because a heat-resistant polymerase is used, the reaction can be repeated continuously without addition of more enzymes. Each cycle doubles the copy number of the amplified gene: ten cycles ideally produces 2 → 4 → 8 → 16 → 32→ 64 → 128 → 256 → 512 → 1,024 (2^{10}) copies. Thus, 30 cycles yields a $(2^{10\times3}) = 10^9$-fold amplification.

43. Taq DNA Polymerase is a thermostable DNA Polymerase isolated from an *E. coli* strain that carries the Taq DNA polymerase gene. Taq DNA Polymerase is the most common polymerase used for PCR reactions.

44. Taq DNA Polymerase has terminal transferase activity which results in the addition of a single nucleotide (adenosine)at 3'- end of the extension product. The exonuclease proofreading ability is absent in Taq polymerase. This means that Taq has low replication fidelity, as it cannot correct any errors that may occur during amplification. These errors limit the use of Taq polymerase where high accuracy is required.

45. Footprinting is a method for determining the exact DNA sequence to which a particular DNA-binding protein binds. Serial analysis of gene expression (SAGE)is a powerful tool that allows the analysis of overall gene expression patterns with digital analysis. Because SAGE does not require a pre-existing clone, it can be used to identify and quantitate new genes as well as known genes. The use of a sample of DNA to determine the identity of a person within a certain probability is DNA fingerprinting. DNA fingerprinting is done by analyzing repeating patterns of base pairs in DNA sequences that are known to vary greatly among individuals. Yeast two-hybrid is based on the reconstitution of a functional transcription factor (TF) when two proteins or polypeptides of interest interact. This takes place in genetically modified yeast strains, in which the transcription of a reporter gene leads to a specific phenotype, usually growth on a selective medium or change in the colour of the yeast colonies.

46. Matrix-assisted laser desorption/ionisation-time of flight mass spectrometry (MALDI-TOF MS) is a relatively novel technique in which a co-precipitate of an UV-light absorbing matrix and a biomolecule is irradiated by a nanosecond laser pulse. Most of the laser energy is absorbed by the matrix, which prevents unwanted fragmentation of the biomolecule. The ionized biomolecules are accelerated in an electric field and enter the flight tube. During the flight in this tube, different molecules are separated according to their mass to charge ratio and reach the detector at different times. In this way each molecule yields a distinct signal.

47. Lymphoid tissue tumour/hyperthyroidism treatment, antibody labelling, brain biochemistry in mental illness, kidney agent, thyroid problems (C), alternative to Tl-201 for radio-immunotherapy are some of the uses of I-131 radiolabel.

48. Desalting and buffer exchange involve the chromatographic separation of soluble macromolecules from smaller molecules based on gel filtration or molecular sieve chromatography. Desalting and buffer exchange are two of the most widely used gel filtration applications, and both can be performed using the same materials. Applications for desalting include the removal of salts from protein solutions, removing phenol or unincorporated nucleotides from nucleic acid preparations and separating excess cross-linking, labelling or derivatization reagents from conjugated proteins. Most common method of desalting is through dialysis.

49. In radial immunodiffusion antibody is incorporated into the agar gel as it is poured and different dilutions of the antigen are placed in holes punched into the agar. As the antigen diffuses into the gel, it reacts with the antibody and when the equivalence point is reached a ring of precipitation is formed. Radioimmunoassays (RIA)are assays that are based on the measurement of radioactivity associated with immune complexes. In any particular test, the label may be on either the antigen or the antibody. Enzyme Linked Immunosorbent Assays (ELISA)are those that are based on the measurement of an enzymatic reaction associated with immune complexes. In any particular assay, the enzyme may be linked to either the antigen or the antibody.

50. The opposite of normal phase, or Reversed Phase Chromatography, results from the adsorption of hydrophobic molecules onto a hydrophobic solid support in a polar mobile phase. Decreasing the mobile phase polarity by adding more organic solvent reduces the hydrophobic interaction between the solute and the solid support resulting in de-sorption. The more hydrophobic the molecule the more time it will spend on the solid support and the higher the concentration of organic solvent that is required to promote de-sorption. A non-polar solvent (usually referred to as Buffer B)is then usually mixed with buffer A in slowly increasing proportion.

51. Fluorescence resonance energy transfer (FRET) is a distance-dependent interaction between the electronic excited states of two dye molecules in which excitation is transferred from a donor molecule to an acceptor molecule without emission of a photon. The efficiency of FRET is dependent on the inverse sixth power of the intermolecular separation, making it useful over distances comparable to the dimensions of biological macromolecules. Thus, FRET is an important technique for investigating a variety of biological phenomena that produce changes in molecular proximity.

52. Anchored PCR is a variation of the PCR technique, similar to ligation-mediated PCR that is applied to double-stranded DNA fragments for which the sequence at only one end of the gene is known. The technique allows amplification of a complete sequence of a gene when only the N-terminal sequence of a protein is known. A short polynucleotide of known sequence is ligated to the 3'-ends of the double-stranded DNA so that a primer complementary to it can be added, along with the primer determined from the partial protein sequence.

53. GOLD is highly configurable allowing to take full advantage of knowledge of a protein-ligand system in order to maximise docking performance. GOLD enables complete user control over speed versus accuracy settings, from efficient virtual screening of large compound libraries, to highly accurate exhaustive sampling for lead optimisation. With a wide range of available scoring functions and customisable docking protocols, GOLD provides consistently high performance across a diverse range of receptor types.

54. Liquid scintillation counting (LSC) is a standard laboratory method in the life-sciences for measuring radiation from beta-emitting radioactive isotopes. Scintillating materials are also used in differently constructed scintillation counters in many other fields. In the majority of LSC facilities, the scintillation solvent is benzene (C_6H_6) or a mixture of benzene and toluene ($C_6H_5CH_3$). Benzene has been chosen because of its excellent light transmission properties and the high chemical conversion yield of sample C to benzene. The sample is first converted to CO_2, and then reacted with molten lithium to form lithium carbide (Li_2C_2), before being catalytically trimerised to benzene. The most commonly used scintillator in 14C dating is PPO + POPOP, either dissolved in toluene or directly in sample benzene.

55. Ion-exchange chromatography separates proteins based on molecular charge. Protein mixtures are applied to in oppositely charged, chromatographic matrix and the various proteins are bound by reversible, electrostatic interactions. The adsorbed proteins are eluted in order of least to most strongly bound molecules (by increasing the ionic strength or varying the pH of the elution buffer), collected as individual chromatographic fractions, and analyzed separately. Carboxy methyl (CM)cellulose matrix is commonly used as resin in cation-exchange chromatography of protein separation.

56. The identical sized RNA transcripts that are generated from Northern Blot give the analysis of UDP glucoronosyl transferase enzyme of human liver and kidney. After microarray analysis, the cDNA developed giving the gene expression shows the same level of fluorescent intensities of the cy3 and cy5 dyes. This indicates that the control or regulation of the enzyme's expression is not transcriptionally and translationally controlled. However, after Western blot analysis of the enzymatic protein from liver and kidney, the liver protein gives a 55kDa band while the kidney protein gives a faint band. This indicates that the enzyme undergoes post translational modification which regulates the function of the enzyme.

57. Taq DNA Polymerase is a thermostable DNA polymerase that possesses a 5 → 3 polymerase activity and a 5 flap endonuclease activity. Taq DNA Polymerase is a thermostable DNA Polymerase isolated from an *E. coli* strain that carries the Taq DNA polymerase gene.

58. Size-exclusion chromatography (SEC) is a chromatographic method in which molecules in solution are separated by their size, not by molecular weight. It is usually applied to large molecules or macromolecular complexes such as proteins and industrial polymers. Typically, when an aqueous solution is used to transport the sample through the column, the technique is known as gel-filtration chromatography, versus the name Gel permeation chromatography, which is used when an organic solvent is used as a mobile phase. The main application of gel-filtration chromatography is the fractionation of proteins and other water-soluble polymers, while gel permeation chromatography is used to analyze the molecular weight distribution of organic-soluble polymers.

59. Electrophoresis is a technique used to separate and sometimes purify macromolecules - especially proteins and nucleic acids - that differ in size, charge or conformation. When charged molecules are placed in an electric field, they migrate toward either the positive or negative pole according to their charge. In contrast to proteins, which can have either a net positive or net negative charge, nucleic acids have a consistent negative charge imparted by their phosphate backbone, and migrate toward the anode.

60. Affinity Chromatography (AC) separates molecules based on the reversible interaction between target protein and the specific ligand attached to a chromatography matrix. The interaction can be specific (such as antibodies binding to protein A or protein G), or nonspecific (e.g., a histidine-tagged protein binding to metal ions).

61. Under high centrifugal force, a solution of cesium chloride (CsCl) molecules will dissociate, and the heavy Cs+ atoms will be forced towards the outer end of the tube, thus forming a shallow density gradient. DNA molecules placed in this gradient will migrate to the point where they have the same density as the gradient (the neutral buoyancy or isopycnic point). The gradient is sufficient to separate types of DNA with slight differences in density due to differing [G+C] content, or physical form (e.g., linear versus circular molecules).

62. Isoelectric focusing (IEF) is an electrophoretic method that separates proteins according to their isoelectric points (pI). Proteins are amphoteric molecules; they carry either positive, negative, or zero net charge, depending on the pH of their surroundings. The pI is the specific pH at which the net charge of the protein is zero. Proteins are positively charged at pH values below their pI and negatively charged at pH values above their pI. The presence of a pH gradient is critical to the IEF technique. In a pH gradient and under the influence of an electric field, a protein will move to the position in the gradient where its net charge is zero.

63. Ninhydrin is also used in amino acid analysis of proteins. Most of the amino acids, except proline, are hydrolyzed and react with ninhydrin. Also, certain amino acid chains are degraded. Therefore, separate analysis is required for identifying such amino acids that either react differently or do not react at all with ninhydrin. The rest of the amino acids are then quantified colorimetrically after separation by chromatography. It is often used to detect α-amino acids and also free amino and carboxylic acid groups on proteins and peptides.

64. Allan Maxam and Walter Gilbert developed a method for sequencing single-stranded DNA by taking advantage of a two-step catalytic process involving piperidine and two chemicals that selectively attack purines and pyrimidines. Purines will react with dimethyl sulfate and pyrimidines will react with hydrazine in such a way as to break the glycoside bond between the ribose sugar and the base displacing the base. Piperidine will then catalyze phosphodiester bond cleavage where the base has been displaced.

65. The most common technique for determining DNA sequence is the Sanger method, which generates fragments that differ in length by a single nucleotide. High-resolution polyacrylamide gel electrophoresis is then used to separate the fragments and to allow the sequence to be determined. This is because the different lengths of DNA fragments generated after Sanger's dideoxy termination would be differentially resolved in the gel.

66. The Sanger method uses a DNA polymerase to synthesize a new DNA strand using a template and a specific primer, in the presence of dideoxynucleotides that terminate the synthesis specifically following the addition of an A, C, G, or T. The Maxam-Gilbert method takes a DNA fragment that has been labelled at the 5'-end, usually with radioactive phosphorus, and subjects to chemical cleavage with reagents that cut at specific bases. Both are concerned with DNA sequencing.

67. The Ouchterlony double diffusion (ODD) technique is one of the simplest techniques extensively used to check antisera for the presence of antibodies for a particular Ag and to determine its titre. This method has been widely used for detection and qualitative diagnostic procedures. The method is called "double" referring to the fact that in this procedure, antigen and antibody are allowed to migrate towards each other in a gel and a line of precipitation is formed where the two reactants meet. This precipitation reaction is highly specific. Single radial immunodiffusion is used extensively for the quantitative estimation of antigens. In this method, the antigen-antibody precipitation is made more sensitive than in double immunodiffusion by the incorporation of the antiserum in the agar solution before the gel is made. Thus, the antiserum is uniformly distributed throughout the agar gel. Antigen is then allowed to diffuse from wells cut into the agar gel.

68. Resolution for ^{125}I-labeled specimens under electron microscope (EM) autoradiographic conditions was assessed experimentally. With this isotope the size of the silver halide crystal was the most important resolution-limiting factor. Heavy metal staining such as is routinely used in preparing animal tissues for EM autoradiography produced an improvement in resolution of approximately 15-20%.

69. Autoradiography is a technique using X- ray film to visualize molecules or fragments of molecules that have been radioactively labelled. Autoradiography has many applications in the laboratory. Autoradiography can, for example, be used to analyze the length and number of DNA fragments after they are separated from one another by a method called gel electrophoresis. Autoradiography is defined as "self" "radioactivity" "writing," and it has become a collection of techniques where radioactive particles/energy (beta/gamma)localized within a solid substance is imaged by close opposition to a detection media. The original detection media was X-ray film, or another type of media that contained a photographic emulsion, and the radioactivity was derived from a compound with an attached radioisotope label of relatively low energy, such as a beta particle.

70. Sucrose gradient centrifugation is a type of centrifugation often used to purify enveloped viruses (with densities 1.1-1.2 g/cm^3), ribosomes, and membranes and so on. This method is also used to purify exosomes. There are two methods - equilibrium centrifugation and non-equilibrium centrifugation.

71. The Feulgen technique selectively stains DNA, and under controlled conditions, can be used for the photometric determination of DNA content. Acid hydrolysis removes purine bases from the DNA, thereby unmasking free aldehyde groups. The aldehyde groups then react with Schiff's reagent, which results in the purple staining. RNA is not hydrolyzed by the HCl treatment and, thus, the reaction is DNA-specific. The technique involves treating sections with dilute hydrochloric acid. This treatment hydrolysis DNA and removes the bases. The sugar remains and reacts as an aldehyde, including condensation with Schiff's reagent when it is subsequently applied. The typical red colouration where DNA was present is thus formed.

72. Quinacrine mustard, an alkylating agent, was the first chemical to band chromosomes viewed under a fluorescence microscope. Quinacrine dihydrochloride has subsequently been substituted by quinacrine mustard. The alternating bands of bright and dull fluorescence are called Q bands. The bright bands are primary composed of DNA rich in adenine and thymine, while the dull bands are rich in guanine and cytosine. Q bands are especially useful for distinguishing the human Y chromosome and various chromosome polymorphisms involving satellites and centromeres of specific chromosomes.

73. Reverse banding (R-banding) involves the incubation of slides containing metaphase chromosomes in hot phosphate buffer and stained with Giemsa. The banding pattern that results is essentially the reverse of G bands. R bands are GC-rich. The AT-rich regions are selectively denatured by heat leaving the GC-rich regions intact. Fluorochromes that are GC specific also produce a reverse chromosome banding pattern. R-banding is helpful for analyzing the structure of chromosome ends, since these areas usually stain light with G-banding.

74. Ninhydrin is also used in amino acid analysis of proteins. Most of the amino acids, except proline, are hydrolyzed and react with ninhydrin. Also, certain amino acid chains are degraded. Therefore, separate analysis is required for identifying such

amino acids that either react differently or do not react at all with ninhydrin. The rest of the amino acids are then quantified colorimetrically after separation by chromatography.

75. As ribosomal particles were first isolated from cell lysates by ultracentrifugation, the ribosomes and their sub-particles were named according to their sedimentation characteristics during centrifugation. The sedimentation properties of a particle depend on its molecular size and geometrical shape. The sedimentation characteristics also depend on the physical properties of the solution through which the particle sediments. The two eukaryotic ribosomal subunits have sedimentation coefficients of 40×10^{-13} and 60×10^{-13}. As one Svedberg (S) unit is 10^{-13}, the two ribosomal subunits are referred to as the 40S and the 60S ribosomal subunits. The molecular mass of the 40S and 60S particles are 1.5 and 3.0 million g/mol, respectively. Thus, the complete ribosome has a mass of approximately 4.5 million g/mol. The complete ribosome is referred to as the 80S ribosome. Thus with increase in size S increases.

76. Restriction enzymes are DNA-cutting enzymes found in bacteria (and harvested from them for use). Because they cut within the molecule, they are often called restriction endonucleases. Since the restriction enzymes did not cut the genome into fragments, that means either the genome has no specific recognition sequence for the selected restriction endonuclease or the genome is not made up of DNA, i.e., the viral genome has RNA as the genetic material.

77. The procedure for creating a DNA fingerprint consists of first obtaining a sample of cells containing DNA (e.g., from skin, blood, or hair), extracting the DNA, and purifying it. The DNA is then cut at specific points along the strand with substances called restriction enzymes. This produces fragments of varying lengths that are sorted by placing them on a gel and then subjecting the gel to an electric current (electrophoresis): the shorter the fragment the more quickly it will move toward the positive pole (anode). The sorted, double-stranded DNA fragments are then subjected to a blotting technique in which they are split into single strands and transferred to a nylon sheet. The fragments undergo autoradiography in which they are exposed to DNA probes—pieces of synthetic DNA that have been made radioactive and that bind to the nonidentical bits. A piece of X-ray film is then exposed to the fragments, and a dark mark is produced at any point where a radioactive probe has become attached. The resultant pattern of these marks can then be analysed.

78. DNA microarrays are created by robotic machines that arrange minuscule amounts of hundreds or thousands of gene sequences on a single microscope slide. To determine which genes are turned on and which are turned off in a given cell, a researcher must first collect the messenger RNA molecules present in that cell. The researcher then labels each mRNA molecule by using a reverse transcriptase enzyme (RT) that generates a complementary cDNA to the mRNA. During that process fluorescent nucleotides are attached to the cDNA. If a particular gene is very active, it produces many molecules of messenger RNA, thus, more labeled cDNAs, which hybridize to the DNA on the microarray slide and generate a very bright fluorescent area. Genes that are somewhat less active produce fewer mRNAs, thus, less labeled cDNAs, which results in dimmer fluorescent spots. If there is no fluorescence, none of the messenger molecules have hybridized to the DNA, indicating that the gene is inactive.

79. The dye Cibacron Blue F3GA has a high affinity for many proteins and enzymes. It has therefore been attached to various solid supports such as Sephadex, Sepharose, polyacrylamide, and the like. In the immobilized form the dye has rapidly been exploited as an affinity chromatographic medium to separate and purify a variety of proteins including dehydrogenases, kinases, serum albumin, interferons, several plasma proteins, and a host of other proteins. Such a diversity shown by the blue dye in binding several unrelated classes of proteins has generated considerable work in terms of studies of the chromophore itself and also the immobilized ligand.

NUMERICAL TYPE QUESTIONS

1. Given, Molar absorption coefficient of NADH at 340 nm = 6220 L/M/CM

Molar absorption coefficient of NAD = 0

Therefore, Net molar absorption coefficient of the solution, ε = 6220 L/M/CM

Concentration of NADH and NAD = 10^{-4} M = 10×10^{-6}M

The total concentration of the mixture, $C = 20 \times 10^{-6}$ M

Path length of the cuvette, l = 5min = 0.5 cm

Hence, absorbance of the solution,

$A = \varepsilon Cl = 6220 \times 20 \times 10^{-6} \times 0.5 = 0.0622$

2. In native PAGE gels, both the size and the charge of the polypeptide will affect migration. Thus the differences in size between the alpha and beta globins cannot be resolved using the methods described in this paper, and the alpha and beta globin subunits appear as one band in SDS-PAGE gels. In native PAGE haemoglobins A and S will migrate differently due to differences in charge, despite the fact they are the same size. Haemoglobin in native form consists of four monomers each with molecular mass of about 15 kDa (mass varies among species), associated with a haeme group. Samples containing large amounts of haemoglobin may produce as many as four bands corresponding to the monomer, dimer, trimer, or tetramer of haemoglobin.

3. We know, from Lambert-Beer law,

Absorbance, $A = \varepsilon Cl$

Where ε = molar conductivity of the medium, C = concentration of the sample, l = path length

We know, Transmittance, T is given by, $A = -\log T$

So A is proportional to T which is proportional to the path length

Therefore, $A_1/A_2 = l_1/l_2 \Rightarrow A_2 = A_1 \times l_2/l_1 = 100\text{-}80 = 20$

$\Rightarrow A_2 = 40$

Therefore, % of light transmitted = (100 – 40) % = 60%

4. We know, from Lambert-Beer law,

Absorbance, $A = \varepsilon Cl$

Where ε = molar conductivity of the medium, C = concentration of the sample, l = path length

Given, A = 0.54, l = 0.5 cm, $\varepsilon = 5.4 \times 10^3 M^{-1} cm^{-1}$

Therefore, $C = A/\varepsilon l = 0.54/5.4 \times 10^3 \times 0.5 = 2 \times 10^{-4} M = 0.2 \times 10^{-3} M = 0.2$ mM

EXERCISE – II

MCQ TYPE QUESTIONS

1. The density of DNA (g/mol or g/cm^3) as a function of its G + C content is given by following equation.

$$\rho = 1.66 + 0.098\ X_{G+C}$$

$$X_{G+C} = \frac{\rho - 1.66}{0.098}$$

as G + C is the mole fraction of G + C in DNA. So it can be expressed in terms of mole % G + C as follows

$$\%\ X_{G+C} = \frac{\rho - 1.66}{0.098} \times 100$$

2. **Circular dichroism (CD)** is dichroism involving circularly polarized light, i.e., the differential absorption of left- and right-handed light. Left-handed circular (LHC) and right-handed circular (RHC) polarized light represent two possible spin angular momentum states for a photon, and so circular dichroism is also referred to as dichroism for spin angular momentum.

3. In dead-end filtration, rate of filtration is inversely proportional to the viscosity of the solution.

4. Chromatography involves a solute or solute extract being dissolved in a mobile phase, which may be a gas, a liquid or a supercritical fluid. The mobile phase is then forced through an immobile, immiscible stationary phase. The distribution of analytes between these can often be described quite simply as analyte is in equilibrium between the two phases.

$$A_{mobile} = A_{stationary}$$

The time between sample injection and an analyte peak reaching a detector at the end of the column is termed the retention time (t_R).

The dissociation constant for solute-adsorbent binding is K_c. So, K_c is inversely proportional to the retention time of the column. So, retention time (t_R) of a column decreases with increasing K_c value.

5. In ion exchange chromatography, be it cation based or anion based resin used, the elution of the substrate bound to the resins is done through increasing concentrations of salt in a stepwise gradient form of elution. For hydrophobic column chromatography, ligands bound to the column through various interactions can be eluted by decreasing the level of polarity in the solution. In Gel filtration chromatography, molecules are based on their different molecular weights through the process of moving from the column by the use of any kind of aqueous solution of particular pH. In Chromatofocusing or also known as isoelectric focusing, samples of proteins are precipitated at the isoelectric point using amphotytes of different charges.

6. In transmission electron microscope to increase the electron opacity samples are impregnated with heavy metals such as uranyl acetate. Uranyl acetate is a water-soluble uranium compound and is often used as stains in electron microscopy.

7. The helix content of a protein can be determined using – circular dichroism spectroscopy. CD spectrum of unknown protein = x % alpha-helical + y % beta-sheet + z % random coil

8. Membrane filtration traps contaminants larger than the pore size in the surface of the membrane. If contaminants are smaller than the desired particle, decrease the membrane pore size and trap the product while passing the contaminants through the membrane. Heat-labile fluids such as serum are sterilized by membrane filtration.

A widely used method for vapor sterilization is the autoclave. It can be used to sterilize various heat-stable liquids

Gamma irradiation is a physical means of sterilisation or decontamination where products are exposed to gamma rays. It can be used to sterilize polypropylene tubes.

In UV irradiation, Ultraviolet lamps are used to sterilize workspaces and tools used in biology laboratories and medical facilities. It can be used to sterile Biological safety cabinets.

9. Protein secondary structure can be determined by the Circular Dichromism where as XRD can be used for the Tertiary structure prediction of the protein. Freeze drying is an unit operation for the concentration increase in a solution of limited solute. Ultra centrifugation is the unit operation for sedimentation.

10. All the properties in column A are phenomena unit for downstream process in column B.

11. Svedberg Unit stands for sedimentation coefficient for centrifugation process. Partition coefficient for two layer aqueous extraction unit. Rejection coefficient is the unit for ultra filtration. Dialysis unit is activity coefficient.

12. • The **Diaphragm Pressure Gage** uses the elastic deformation of a diaphragm (i.e. membrane) instead of a liquid level to measure the difference between an unknown pressure and a reference pressure.

• In an antifoam system; if the foam contacts a **rubber sheathed electrode**, an electric unit actuates a solenoid valve to allow the passage of a sterile antifoam agent into the fermentor.

• A final widely used method for the determination of cell number is a turbidometric measurement or light scattering. This technique depends on the fact that as the number of cells in a solution increases, the solution becomes increasingly turbid (cloudy). The solution looks turbid because light passing through it is scattered by the microorganisms present and the turbidity is proportional to the number of microorganisms in the solution. The turbidity of a culture can be measured using a photometer or a spectrophotometer. Photometers, such as the Klett-Summerson device, use a red, green or blue filter providing a broad spectrum of light.

• A **rotameter** is a device that measures the flow rate of liquid or gas in a closed tube.

13. Electroporation, or electropermeabilization, is a significant increase in the electrical conductivity and permeability of the cell plasma membrane caused by an externally applied electrical field. It is usually used in molecular biology as a way of introducing some substance into a cell, such as loading it with a molecular probe, a drug that can change the cell's function, or a piece of coding DNA.

NUMERICAL TYPE QUESTIONS

1. Expected molar extinction sufficient of polypeptide chain

$$E_{280nm} = (5 \times 5690) + (14 \times 1280)$$
$$= 46370\ M^{-1}\ cm^{-1}$$

Now, conc.of dehydrogenase

$$C = \frac{0.32}{37} \times 10^{-3} = 8.648 \times 10^{-6}$$

$$\therefore \quad A/E = C \Rightarrow A = EC$$
$$= 46370 \times 8.648 \times 10^{-6}$$
$$A = 0.401$$

2. (*i*) $H = \frac{A}{u} + Bu + C$

The minimum HETP occurs when

$$u = \sqrt{A/B} = 10^{-4}\ m/sec.$$
$$A = 3 \times 10^{-8}\ m^2/sec$$
$$B = 3\ sec$$
$$C = 6 \times 10^{-5}\ m$$

(*ii*) $\therefore \quad H = 6.6 \times 10^{-4}$

Now, $= 6.6 \times 10^{-2}\ cm$

(*iii*) $H = \frac{L}{N}$, N = No. of theoretical plates

L = Length of the column

$$\therefore \quad H = \frac{L}{N} = \frac{66cm}{6.6 \times 10^{-2}\ cm} = 1000$$

■■

CHAPTER 4

Cell Biology

Introduction

Cells are the structural and functional units of all living organisms. The smallest organism consists of a single cell and are microscopic whereas larger organisms are multicellular. Multicellular organisms contain many different types of cells, which vary in size, shape, and specialized functions. Yet no matter how large and complex the organism each of its cells retains. Some individuality and independence, have essentially the same biochemical machinery and all share the same genetic code.

Cellular Dimensions

Most cells are microscopic, invisible to the unaided eye. Animal and plant cells are typically 5 to 100 μm in diameter that limits the dimensions of a cell. The lower limit is probably set by the minimum number of each type of biomolecules required by the cell. The upper limit of a cell size is probably set by the cell. The upper limit of a cell size is probably set by the rate of diffusion of solute molecules in aqueous system.

- According to second law of thermodynamics, every living organism (like other systems) has a tendency to increase disorderliness (entropy).
- To maintain organisation at the molecular, cellular and organism to collect energy and nutrients from its surroundings, and use them to reduce entropy.
- All supplies for chemical reactions have to pass through cell membrane (cell surface).
- As a cell increases in size, its metabolic needs and chemical activities increase proportionately. However, its surface area increases at a slower rate than its volume. Hence, the surface : volume ratio of a cell decreases with increasing size.

Therefore, a large cell volume to accomodate many organelies at the same time limits the exchange of materials through the surface.

Types of Cell

Cells are identified as being of one of the two types :

(*i*) Prokaryotic cell

(*ii*) Eukaryotic cell

The main difference between these types is that prokaryotic cell lacks a nuclear envelope, their DNA occupies a space in cell called a **nucleoid.** From an evolutionary point of view, porkaryotes are considered to be ancestors of eukaryotes.

Organization of Eukaryotic Cell

A Eukaryotic cell is surrounded by a plasma membrane, has a membrane bound nucleus and contains a number of other distinct subcellular organelles. These organelles are membrane-bounded structure, each having a unique role and each containing a specific complement of proteins and other molecules.

Three major changes must have occured as prokaryotes gave rise to eukaryotes.

(*i*) The cells acquired more DNA, the mechanism to fold it compactly descrete complexes with specific proteins, the complex body called chromosome and to divide it equally between daughter cells during cell division.

(*ii*) A system of intracellular membrane developed, including a double membrane surrounding the DNA (i.e. nucleus)

(*iii*) Photosynthetic cyanobacteria become endosymbions of same aerobic eukaryotes, these cells became the photosynthetic precursor of modern green algae and plants.

Comparison of Prokaryotic and Eukaryotic Cells

Characteristic	Prokaryotic cell	Eukaryotic cell
Size	Generally small (1-10 m)	Generally large (5-100 μm)
Genome	DNA with nonhistone proteins; genome in nucleoid not surrounded by membrane.	DNA complexed with histone proteins in chromosomes; chromosomes in nucleus with membrane envelope.
Cell division	Fission or budding or mitosis	Mitosis including mitotic spindle, centrioles in many species, Meiotic division also occur during gametogenesis.
Membrane-bounded	Absent	Mitochondria, chloroplasts (in plants, some algae, endoplasmic reticulum Golgi complexes, lysosomes (in animals), etc.
Nutrition	Absorption; some photosynthesis	Absorption, ingestion; photosynthesis in some species
Energy metabolism	No mitochondria oxidative enzymes bound in plasma membrane, great variation in metabolic pattern.	Oxidative enzymes packaged in, mitochondira, more unified pattern of oxidative metabolism.
Cytoskeleton	None	Complex with microtubules, intermediate filaments, actin filaments
Intracellular movement	None	Cytoplasmic streaming, endocytosis, phagocytosis, mitosis, vesicle transport.

Plasmalemma

Models of Membrane Structure

Three distinct layers are seen in electron micrographs of plasmalemma of tissues fixed with osmium tetraoxide. Two of these layers are relatively dense and *osmiophilic* in nature; each of them is about 20 Å thick. The two osmiophilic layers are separated by a relatively light *osmiophobic layer* of about 35 Å thickness (Fig.). The three layers (two osmiophilic and one osmiophohic) together are known as "***unit membrane***", this term was coined by **Robertson**. The average thickness of a unit membrane is about 75 Å, but there is a considerable variation in the thickness of unit membranes of different tissues.

For example, the thickness of plasma lemma in rabbit RBCs is about 215 Å. *The structures of other membranes found in a cell are similar to that of plasma lemma.*

In 1935, **Davson** and **Danieli** proposed the *lipid-protein* ***bilayer*** *model* of unit membrane structure. According to this model, two layers of lipid molecules are arranged in the centre of a unit membrane, each layer being only one molecule thick. The polar ends of all lipid molecules in a layer are oriented towards the outside, while their non-polar ends are situated towards the centre of membrane. On the outside of each of the two layers of lipid molecules, a layer of protein molecules of one molecule thickness is located (Fig. a). However, many properties of plasma lemma and other biological membranes are difficult to explain according to this simple model.

A *fluid-mosaic model* has been proposed to account for the various properties of biological membranes. According to this model, plasma lemma (and other biological membranes) are made up of two layers of lipid molecules (lipid bilayer), each layer being only one molecule thick. The lipid molecules are arranged in the same manner as in the lipid-protein bilayer model: the polar ends of lipid molecules are oriented towards the outside of the membrane, while the non-polar ends face inside. However, the protein molecules are embedded either fully or partially throughout the lipid bilayer.

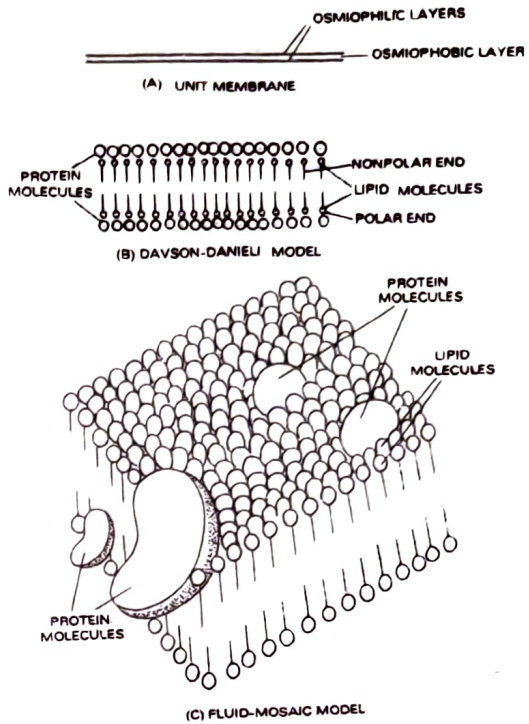


Fig. Plasma lemma. (a) Unit membrane seen under electron microscope after osmium staining. (b) The Davson-Danieli model of unit membrane organisation, and (c) the fluid-mosaic model of unit membrane (note that in this model, protein molecules are embedded within the lipid bilayer).

Biological membranes have a dynamic structure; both lipid and protein molecules can move laterally, membrane surfaces can fuse to yield a single membrane, and membranes can bud off as vesicles. The proteins move more slowly than do lipids. Specific flippases may be involved in the transport of different lipids from one lipid monolayer into the other. In response to stimuli, protein molecules can move into or be excreted out of a cell.

Protein molecules embedded in a membrane and having some part exposed on any one or both the sides of the membrane are called ***transmembrane proteins***. The exposed parts of these molecules are available for various reactions and are hydrophilic, while the part embedded in the membrane is hydrophobic. A class of transmembrane proteins called *receptors* have active sites, which bind to specific molecules present outside the cell; this binding induces conformational changes in the part of receptor located in the cytoplasm which, in turn, triggers other changes in the cell. Receptors provide a means for cells to respond to external stimuli.

Biological membranes are asymmetric in that they have two distinct surfaces called (1) cytoplasmic and (2) non-cytoplasmic faces. The ***cytoplasmic face*** of a membrane is in contact with the cytosol. The lipid monolayer comprising this face has more highly charged and unsaturated lipids. In addition, charged groups are concentrated on this surface. The ***non-cytoplasmic face*** of plasma lemma forms the outer cell surface, while in the case of membranes within cells it forms the surface bounding the lumen of the compartment formed by the membrane. This face has much less unsaturated lipids than does the cytoplasmic face, and saccharides are present only on this face.

Mechanism of Solute Transport Across Membrane

Living cells are surrounded by lipid bilayers, the plasma membranes in which various proteins are embedded. These membranes are relatively impermeable to hydrophilic substances, such as ions and a variety of metabolites, and thus provide a selective permeability barrier. The mechanisms by which such materials are moved across extracellular or intracellular membranes are generally termed ***transport processes.***

Table. Classification of Transport Processes

I. Nonmediated transport—i.e., simple diffusion

II. Mediated transport

A. Facilitated diffusion—i.e., carrier-mediated passive transport

B. Active transport

1. Primary (ATPase-linked pumps)
2. Secondary (gradient-coupled)

Solutes may traverse membranes in the different ways are given in the table . These processes are either **mediated or nonmediated**. Nonmediated transport of a solute is simply physical diffusion from higher solute concentration to low solute concentration.

In contrast most solutes transfer across biological membranes by specific pathways characterized

(*i*) by specificity for the substance transported and

(*ii*) by saturation kinetics, i.e., the transport system can become saturated with the solute transported. Such mediated transport is made possible by proteins capable of reversibly binding specific substrates; these transport molecules have been variously termed *transport systems, carriers, porters, translocases, or permeases.*

Several classes of mediated transport have been recognized. *Facilitated diffusion* refers to processes whereby a compound moves across the membrane by mechanisms that do not require the explicit provision of energy. Entry of glucose into hepatocytes, erythrocytes and muscle cells along its concentration gradient is such a process.

Active transport is the term employed for processes by which a molecule moves across the membrane *against* its concentration or, for ions, its electrochemical gradient. The required energy may be supplied the either of two ways.

(*i*) Energy is provided by simultaneous hydrolysis of ATP (ATPase activity) or of some other high-energy compound on the surface of the protein that is serving as the porter. Such processes are referred to as *primary active transport* and also as *pumps.*

(*ii*) Transport of a metabolite is coupled to the simultaneous movement of a second substance, which is moving down its concentration gradient, a process for which ΔG is large and negative. The second molecule may be moving in the same direction as the first (*symport*) or in the opposite direction (*antiport*). Such processes are referred to as *secondary* or *gradient-coupled active transport.*

Transport processes perform a number of vital functions. They maintain cellular volume, pH, and tonicity. They maintain gradients of K^+ and Na^+ between the intra- and extracellular fluid compartments, which are necessary for the normal function of excitable cells. They bring important metabolites into the cell while extruding products of cellular metabolism for excretion or for delivery to another type of cell.

Nonmediated Transport – Simple Diffusion

In simple diffusion, the transport rate or net flux is directly proportional to the concentration gradient of the substance across the membrane. Several types of forces may drive such nonmediated transport processes. If the membrane is permeable to the substance and its concentration is higher on one side of the membrane than the other, the solute will tend to be transferred across the membrane down its concentration gradient. Differing hydrostatic pressures of fluids on opposite sides of a membrane may provide the driving force for passive movement of fluid from higher to lower pressure. A third type of driving force for nonmediated diffusion is the electric potential developed by separation of charges across a membrane with different finite permeabilities for the ions involved. In each of the cases mentioned, transport against a gradient requires provision of energy (active transport), regardless of the specific driving force involved, passive diffusion across permeable membranes occurs spontaneously only in the direction for which the change in free energy (ΔG) for the transport process is negative.

That the rate of solute transfer by free diffusion is proportional to the driving force is an important principle summarized in Fick's first law of diffusion which states

$$J = -DA\frac{dc}{dx}$$

where J = rate of flux of solute, A is the area of membrane available for diffusion, and

dc/dx = concentration gradient which provides the driving force for diffusion.

D= diffusion coefficient which depends on the physical properties of solute and solvent, such as the shape and size of the solute particle and viscosity of the solvent.

When a solute bears a net charge, then its rate of diffusion across a membrane is influenced not only by its concentration gradient but also by the electric potential gradient across the membrane.

Another factor regulating diffusion is the solubility of the substance in the membrane, as reflected by its partition coefficient between its aqueous environment and the membrane. Some solutes are weak acids or bases. For such substances the rate of diffusion into cells depends on the concentration of the nonionized, i.e., lipid-soluble, form. This is in turn critically dependent on the pH of the extracellular fluid, as can be discerned from the Henderson-Hasselbach equation.

$$pH = pK + \log\frac{[A^-]}{[HA]}$$

If only the undissociated form, HA, can diffuse across the membrane, as is the case with most weak acids, then a 1 unit decrease in pH resulting in a tenfold increase in [HA] produces a tenfold increase in the rate of diffusion of the solute or drug into the cells.

Mediated Transport

A. Facilitated Diffusion

Transport by facilitated diffusion is analogous to receptor-mediated phenomena. It is mediated by specific interactions of solutes with transport proteins located in cellular membranes. Since the number of such units in the membrane is finite, facilitated diffusion processes are usually characterized by saturability. This is in contrast with the situation for nonmediated transport processes, in which the rate of transport shows a linear dependence on solute concentration. The maximal rate of flux which is achieved when all transport sites are occupied by solute is referred to as the *transport maximum,* or T_m.

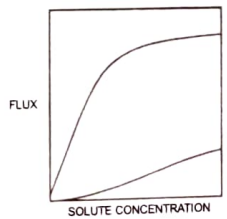


Fig. Dependence of trans-membrane solute flux on solute concentration.

The upper curve shows the dependence on the solute concentration of the flux of a solute that is mediated by a facilitated diffusion mechanism. The lower line shows the expected dependence of solute flux on concentration for a nonmediated transport process.

Another "receptor-like" characteristic of the transport proteins involved in facilitated diffusion is their specificity. This is apparent in the competition for the transport sites between compounds of similar structure. In several nonelectrolyte transport systems of mammalian cells there is also *strict stereospecificity* with D sugars transported but not L sugars or L amino acids transported but not D amino acids.

An example of such a facilitated diffusion system is the hexose transport system of human erythrocytes. The specific transport protein (MW = 55,000) is a glycoprotein which has been purified and reconstituted into phospholipid vesicles. It binds sugars with stereospecificity for the D form. Each carrier molecule is able to translocate about 500 glucose molecules per second.

Facilitated diffusion processes are driven only by the gradients of the transported solutes and do not require an external supply of energy. In most cases the molecular properties of the transport systems involved are unknown. In general, however, they are integral transmembrane proteins and are often glycoproteins. As with many receptor proteins, it is hypothesized that such transport proteins may exist in at least two distinct conformational states. In one state the ligand binding site is accessible to the aqueous phase on one side of the membrane. In the other conformational state, the binding site is accessible at the other side of the membrane. The affinity of binding the transported ligand may be different in the two states. The energy necessary for conversion between the two states is derived from the negative free energy of the ligand's binding to its transport site and/or from the concentration gradient of the ligand (or from another energy source in the case of active transport). The *term gating* is used to describe the phenomenon whereby conformational changes triggered by substrate binding accomplish the transport of a solute molecule from one side of a membrane to the other by opening a channel.

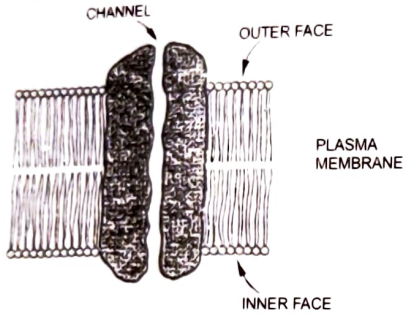


Fig. Model for an integral membrane protein involved in mediated transport.

The integral membrane proteins involved in gating probably have structures similar. Such proteins are oligomers with a small number (two to four) of similar or identical subunits spanning the membrane. Each polypeptide chain has the same orientation in the membrane. Such aggregates may generate a narrow water-filled *channel* down the center of the aggregate through which translocation of solute occurs. Hydrophilic groups on the protein presumably line this channel. If the aggregate is a dimer of two identical polypeptide chains, it would have a twofold axis of symmetry parallel to this channel and perpendicular to the plane of the membrane.

B. Active Transport

Active transport refers to net movement of a solute against its concerntration or (for ions) electrochemical gradient. The energy required for such "uphill" transport may be provided either by (direct coupling to an energy-producing reaction(primary active transport)) or by coupling to the net transport of another solute down its gradient (secondary active transport). In mammalian tissues the [Na^+*pump* and the Ca^{2+} *pump* are examples of primary active transport in which the "uphill" transport of ions is linked to simultaneous hydrolysis of ATP by specific enzymes (ATPases)]. Secondary active transport systems exist in certain tissues for glucose and other sugars and for neutral amino acids.

Secondary Active Transport

In some mammalian cells the transport of sugars, neutral amino acids, and some other solutes is often accomplished by *sodium symport systems*. Examples are the transfer of glucose against its concentration gradient from the intestine into the blood and the reabsorption of glucose from the renal tubule. Such mechanisms are closely related to facilitated diffusion but with the carrier representing a two-substrate rather than a single-substrate system. Thus the carrier has binding sites for both Na^+ and another solute. The free energy made available by the inward Na^+ movement *down* its electrochemical gradient is used for the symport (cotransport) of the other solute *against* its concentration gradient. (As Na^+ and the other solute (e.g., glucose) are discharged into the cytoplasm, the Na^+ pump (see below) returns Na^+ to the extracellular environment, thus maintaining the Na^+ electrochemical gradient. Since this latter process requires the hydrolysis of ATP (one step removed), such processes are termed *secondary active transport*. Some secondary active transport mechanisms such as Na^+-Ca^{2+} exchange in cardiac muscle cells are *antiport* systems, in which a solute is transported against its concentration gradient by the same carrier which is transporting another solute in the opposite direction down its concentration gradient). In mammalian cells, antiports, like symports, are often driven by Na^+ gradients maintained by the Na^+ pump.

Primary Active Transport

Na^+ Pump and Na^+, K^+-ATPase

In most animal cells the intracellular [K^+] is relatively high and constant, between 120 and 160 mM, whereas [Na^+] is much less, being 10 to 20 mM. In contrast, extracellular fluid contains a relatively high [Na^+], about 150 *mM*, and [K^+] is usually about 4mM. Therefore, concentration gradients of these two ions exist across all cell membranes. The constancy of the high intracellular [K^+] is maintained by an energy-requiring extrusion of Na^+ out of the cell with replacement by K^+.

The analogy to a mechanical pump is apt in that energy is utilized to move the ions against an opposing concentration gradient. Erythrocyte membranes have an ATPase that requires Mg^{2+} and both Na^+ and K^+ for activation. The enzyme is spatially asymmetric, i.e., it is stimulated by extracellular but not intracellular K^+ and by intracellular but not extracellular Na^+. Addition of either Na^+ or K^+ alone does not affect the rate of enzymic activity, but when both ions are present, ATP hydrolysis is markedly accelerated. Extensive evidence indicates that the Na^+, K^+-ATPase is the enzymic equivalent of the system responsible for active transmembrane Na^+ transport, the Na^+ pump.

Na^+, K^+-ATPase (MW ≈ 250,000) is involved in the transport of Na^+ and K^+ across the plasma membrane of nearly all eukaryotic cells. The purified enzyme consists of a catalytic α subunit (MW ≈ 95,000) and a glycoprotein β subunit (MW ≈ 40,000). As ATP is hydrolyzed, the larger subunit is phosphorylated; the phosphorylated intermediate has been characterized as an aspartyl-β-phosphate residue in the enzyme. This subunit has the binding sites for Na^+, K^+, and ouabain, a cardiac glycoside that inhibits active Na^+-K^+ transport in most cells.

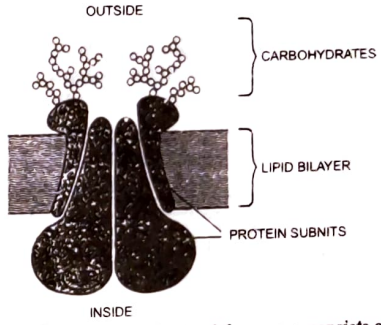


Fig. A model of the Na^+, K^+-ATPase. Each functional unit of the enzyme consists of two large α subunits that span the membrane and two smaller β glycoprotein subunits that are exposed on the outside of the cell. Each functional unit has one binding site for cardiac glycosides, one phosphorylation site, three binding sites for sodium, and two binding sites for potassium.

1. $ATP + Na^+_{inside} + ATPase_a \xrightleftharpoons{Mg^{2+}} Na\text{-}ATPase_a\sim P + ADP$
2. $Na\text{-}ATPase_a\sim P \rightleftharpoons Na\text{-}ATPase_b\sim P$
3. $Na\text{-}ATPase_b\sim P + K^+_{outside} + H_2O \rightleftharpoons ATPase_a + P_i + Na^+_{outside} + K^+_{inside}$

Fig. A schematic representation of the transport of Na^+ and K^+ by the Na^+,K^+-ATPase. The two forms of the enzyme, a and b, designate possible conformational alterations occurring during the exchange of ions. The action of ouabain inhibits the K^+– dependent reaction 3, the hydrolysis of phosphorylated Na^+– $ATPase_b$. Although depicted as an Na^+/K^+ of 1, experimental ratios of 3Na^+ transported in to 2K^+ transported out have been reported for the red cell membrane and the squid axon, as well as for experimentally prepared vesicles into which the purified enzyme has been incorporated.

A model for the structure of the Na^+, K^+ -ATPase as shown in the above figure. It has been suggested that the large subunit extends through the full thickness of the cell membrane with the K^+ and ouabain sites on its outer surface and Na^+ and ATP sites on its inner surface. The glycoprotein may orient the enzyme when it takes its position in the membrane bilayer since oligosaccharides generally project from the outer surface of the plasma membrane, as described previously. The isolated enzyme contains about one-third phosphoglycerides by weight; these are required for its activity.

The mechanism of action of the ATPase has been postulated as follows. Transfer of the terminal phosphate of ATP to the enzyme occurs in a reaction dependent upon Mg^{2+} and Na^+, in the absence of K^+. When K^+ is added to the intermediate, phosphate is released as P_i. The vectorial equation for the reaction is as follows, and the sequence suggested.

$$3Na^+_{inside} + 2K^+_{outside} + ATP^{4-} + H_2O \longrightarrow 3Na^+_{outside} + 2K^+_{inside} + ADP^{3-} + P_i^{2-} + H^+$$

Thus the energy made available by ATP hydrolysis drives the active transport process by leading to conformational changes in the enzyme. The mechanism is cyclical; binding of Na^+ causes phosphorylation of the enzyme leading to a conformational change which seems to move Na^+ to the other side of the membrane. Subsequent binding of K^+, associated with hydrolysis of the phosphorylated intermediate, leads to translocation of K^+.

The Na^+ K^+-pump of mammalian cells plays multiple roles. These include: maintenance of osmotic equilibrium a cross fragile cell membranes, thus preventing their rupture: maintenance of high intracellular [K^+] required for various metabolic processes; maintenance of Na^+ and K^+ gradients required for propagation of nerve impulses; maintenance of the Na^+ and K^+ gradients which drive various secondary active transport processes, such as those involved in solute reabsorption by the kidney and intestinal absorption of nutrients.

Calcium Transport and the Calcium Pump

The regulation of the free cytosolic [Ca^{2+}] in mammalian cells is complex. Since [Ca^{2+}] is much higher extracellularly (≈ 1mM) than intracellularly (<1 M), there tends to be a movement of Ca^{2+} down its electrochemical gradient into the cell. There are at least two distinct channels that move Ca^{2+} into cells, one being a relatively specific Ca^{2+} channel independent of membrane potential and the other being a potential-dependent calcium permeability channel. Calcium efflux against its electrochemical gradient occurs via at least two pathways. One is by a specific ATP-dependent Ca^{2+} pump which is linked to a Ca^{2+}, Mg^{2+}-activated ATPase. The other is a Na^+- Ca^{2+} exchange mechanism which utilizes the energy of the inwardly directed Na^+ gradient to accomplish "uphill" Ca^{2+} extrusion (gradient coupled active transport). This process depends on the maintenance of the Na^+ gradient by the Na^+, K^+-pump.

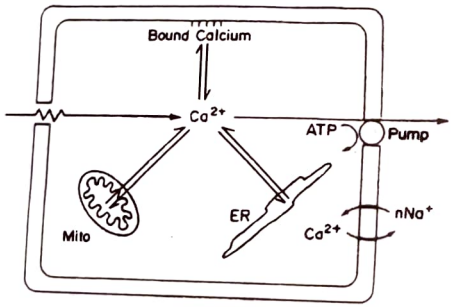


Fig. Factors regulating the cytosolic concentration of Ca^{2+} in cells. Pump refers to the energy-dependent Ca^{2+} extrusion mechanism. Mito and ER refer to mitochondria and endoplasmic reticulum, respectively; these contain high concentrations of calcium. "Bound calcium" refers to calcium bound to specific sites on plasma and other cell membranes.

Within the cell there are reservoirs of Ca^{2+} maintained in the mitochondria and smooth endoplasmic reticulum by energy-dependent uptake processes. In secretory cells an additional pool of Ca^{2+} is found in secretory granules. The mitochondrial pool appears to be the largest.

Certain hormones may alter cytosolic [Ca^{2+}]. They may promote an influx of extracellular Ca^{2+} or Ca^{2+} which is loosely bound to plasma membrane sites.

An important example of the functioning of a Ca^{2+}-ATPase in the active transport of Ca^{2+} is the system in the membranes of the sacroplasmic reticulum of striated muscle. The sarcoplasmic reticulum is an intracellular network of tubules and vesicles which surrounds the myofibrils. (Active transport of Ca^{2+} by the sarcoplasmic reticulum is driven by ATP hydrolysis in the presence of Mg^{2+} and serves to pump Ca^{2+} against its concentration gradient from the cytosol into the sarcoplasmic reticulum). The consequent lowering of [Ca^{2+}] in the region of the contractile fibers of the muscle terminates the contraction cycle. The major subunit of the Ca^{2+}-transporting ATPase (MW= 100,000) makes up ≈ 85 percent of all the proteins of the sarcoplasmic reticulum. As with other transport systems, the Ca^{2+}-ATPase of sarcoplasmic reticulum is an oligomer containing four such subunits. In addition to this subunit, which spans the sarcoplasmic reticulum membrane, the enzyme system also contains a lower molecular weight proteolipid subunit.

As with the Na^+, K^+-ATPase discussed, the enzyme is transiently phosphorylated during the transport process. Each cycle of phosphorylation-dephosphorylation is associated with the transport of 2 Ca^{2+} for each ATP hydrolyzed; 1 Mg^{2+} appears to move in the opposite direction. The phosphorylation site is the β-*carboxyl* group of an aspartic residue of the 100,000 dalton subunit. Conformational changes in the protien caused by the cycle of phosphorylation-dephosphorylation presumably trigger the translocation of Ca^{2+}. The system can be solubilized with detergents and its Ca^{2+}-transporting and ATPase activities reconstituted in phospholipid vesicles.

Calmodulin

A variety of cellular metabolic processes is modulated by alteration of the activity of Ca^{2+}-sensitive enzymes. In many cases these effects are mediated by specific "receptor" proteins which first bind Ca^{2+} and then interact with enzymes to modify their activity. The most important "calcium receptor" is *calmodulin*, or Ca^{2+}-dependent regulatory protein (CDR). This protein is ubiquitous in eukaryotic cells (MW = 16,700). It is an acidic heat-stable protein with a structure that has been highly conserved throughout evolution. Its properties are very similar to those of troponin C, and there is 45 precent direct homology in their amino acid sequinces (78 precent for conservative replacements). The close sequence homologies between calmodulin, troponin C, and the parvalbumins (which are Ca^{2+}-binding proteins found in myoplasm suggest that these molecules represent a family of Ca^{2+}-binding proteins which have evolved from a single ancestral calcium-binding protein. Each molecule of calmodulin contains four binding sites for Ca^{2+} with $K_D \approx 10^{-5}$M.

Calmodulin-Ca^{2+} complexes function as regulators of a wide variety of metabolic processes. Regulation of enzymic activities by Ca^{2+} and calmodulin occurs in a two-stage process as follows:

$$\text{Calmodulin}_{\text{inactive}} + Ca^{2+} \rightleftharpoons (\text{Calmodulin}^*.Ca^{2+})_{\text{active}}$$

$$E_{\text{less active}} + (\text{Calmodulin}^*.Ca^{2+})_{\text{active}} \rightleftharpoons (E^*.\text{Calmodulin}^*.Ca^{2+})_{\text{active}}$$

where E denotes a regulated enzyme and * denotes a new conformation. Thus, calmodulin by itself is inactive. It becomes active only after binding Ca^{2+}, which leads it to assume a more helical conformation, which is the active species capable of reversibly binding to and activating various enzymes.

Table. Some Enzymes Regulated by Calmodulin

Phosphodiesterase—"high K_M for cAMP"
Adenylate cyclase (brain)
Ca^{2+}-ATPase(erythrocyte, synaptosomal)
Myosin light chain kinase
NAD kinase
Phosphorylase kinase
Phospholipase A_2
• Other Ca^{2+}-dependent protein kinases
• Guanylate cyclase

In addition, calmodulin also appears to be involved in the prosesses of microtubule assembly and disassembly, and neurotransmitter release.

In the case of phosphorylase kinase, not only does exogenous calmodulin. Ca^{2+} activate the enzyme, but one of the four different subunits of the enzyme is identical to calmodulin. This enzyme is completely dependent on Ca^{2+} for activity.

Ionophores and Ion Channels

Ionophores, can interact with metal ions and mediate their transport across both artificial and natural membranes. There are two major groups of such agents.

(*i*) *ion carriers;* and (*ii*) *ion channels.*

(*i*) **Ion carriers** : An example of an ion carrier is the macrocyclic peptide valinomycin, which is an antibiotic synthesized by cartain fungi. It contains six amide and six ester bonds with side chains consisting of hydrophobic alkyl radicals. Valinomycin binds numerous cations of the alkali, alkaline-earth, and transition metal series. Its complex with K^+ is 10^3 to 10^4 times more stable than that with Na^+. Valinomycin increases K^+ conductance in virtually all membrane systems studied and at concentrations as low as 10^{-9} M. Valinomycin lowers the energy barrier for transport of K^+ across membranes and in a sense catalyzes its translocation in an enzyme-like fashion. Binding of the ion in the central cavity of the carrier is followed by a variety of conformational changes. Within the membrane the cation is screened from the hydrophobic lipid environment by the hydrocarbon skeleton and side chains of the carrier while held tightly by six ion-dipole bonds formed with the inwardly pointing ester carbonyls of valine. Thus, by forming a complex with valinomycin the cation is made lipid-soluble and its passage through the membrane is facilitated.

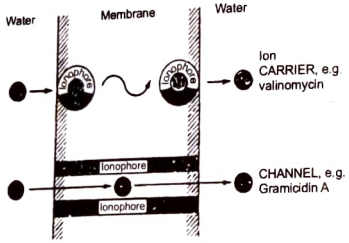


Fig. Schematic representation of the action of ionophores on membranes. M^+= metal ion.

(*ii*) **Ion channel:** Examples of ion channels are the gramicidin antibiotics. Ion channels such as the gramicidins do not complex cations. Rather they induce ion permeability of membranes at concentrations as low as 10^{-10} M by forming dimers which in effect provide "tubes" which span the membrane and through which the cations pass. The exact structure of these dimers is unknown.

Cilia and Flagella

Cilia and **flagella** are hairlike, motile organelles that project from the surface of a variety of eukaryotic cells. Bacteria also possess structures referred to *as flagella,* but prokaryotic flagella are simple filaments that bear no evolutionary relationship to their eukaryotic counterparts.

Cilia and flagella are essentially the same structure. A cilium can be likened to an oar as it moves the cell in a direction perpendicular to the cilium itself. In its power stroke, the cilium is maintained in a rigid state as it pushes against the surrounding medium. In its recovery stroke, the cilium becomes flexible, offering little resistance to the medium. Cilia tend to occur in large numbers on a cell's surface, and their beating activity is usually coordinated. In multicellular organisms, cilia move fluid and particulate meterial through various tracts.

Flagella exhibit a variety of different beating patterns (*waveforms*), depending on the cell type, e.g., the single-called alga pulls itself forward by waving its two flagella in an asymmetric manner that resembles the breaststroke of a human swimmer. This same algal cell can also push itself through the medium using a symmetric beat resembling that of a sperm. The degree of asymmetry in the pattern of the beat in the algal cell is regulated by the internal calcium ion concentration.

Structure of Cilia and Flagella

The entire ciliary or flagellar projection is covered by a membrane that is continuous with the plasma membrane of the cell. The core of the cilium, called the **axoneme,** contains an array of microtubules that runs longitudinally through the entire organelle. With rare exceptions, the axoneme of a motile cilium or flagellum consists of nine peripheral doublet microtubules surrounding a central pair of single microtubules. This same microtubular structure, which is known as the "9 + 2 array," is seen in axonemes from protists to mammals and serves as another of the many reminders that all living eukaryotes have evolved from a common ancestor. All microtubules of the axoneme have the same polarity; their plus ends are at the tip of the projection, and their minus ends at the base. Each peripheral doublet consists of one complete microtubule, the A *tubule,* and one incomplete microtubule, the *B tubule,* the latter containing 10 or 11 subunits rather than the usual 13.

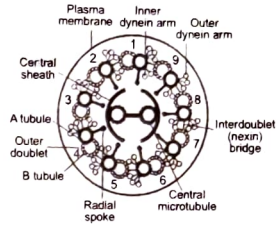


Fig. Structure of flagellar axoneme

The basic structure of the axoneme was first described in 1952; in plants by Irene Manton and in animals by Don Fawcett and Keith Porter. As the resolving power of electron microscopes improved, certain of the less obvious components became visible. The central tubules were seen to be enclosed by projections that form the *central sheath,* which is connected to the A tubules of the peripheral doublets by a set *of radial spokes.* The doublets are connected to one another by an *interdoublet bridge* composed of an elastic protein, nexin. Of particular importance was the observation that a pair of "arms"—an *inner arm* and an *outer arm*—project from the A tubule. A longitudinal section, that is, one cut through the axoneme parallel to its long axis, reveals the continuous nature of the microtubules and discontinuous nature of the other elements.

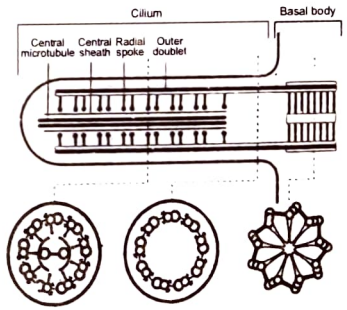


Fig. Basal bodies and axonemes.

A cilium or flagellum emerges from a *basal body.* The A and B tubules of the basal body elongate to form the doublets of the cilium or flagellum. If a cilium or flagellum is sheared from the surface of a living cell, a new organelle is regenerated as an outgrowth of the basal body. As with other microtubular structures, the growth of an axoneme occurs at the plus (outer) ends of its microtubules. How does a cell organize and maintain a construction site at the outer tip of an axoneme situated micrometers from the body of the cell where the synthesis of building materials takes place?

A process known as **intraflagellar transport,** which depends on the activity of both plus end-and minus end- directed microtubular movement.

A Sliding of Microtubular Doublets that Involves Dynein

Ciliary and flagellar motion have been compared to muscle contraction. The key process in both appears to be the sliding of the microtubular doublets over each other, together with the associated making and breaking of cross bridges between adjacent doublets. The central microtubules extend to the very tip; whereas subfibers B terminate at a distance of 1 to 2 m from the tip, and subfibers A terminate slightly beyond subfibers B. Doublets 1 and 6 are represented in the normal position, as well as during the effective and return strokes.

Observe that in the cross section of the normal position the 9 doublets are visible; whereas in the others, subfiber B is missing in certain doublets situated in the convex side of the cilium (an indication that there is sliding between the two subfibers.)

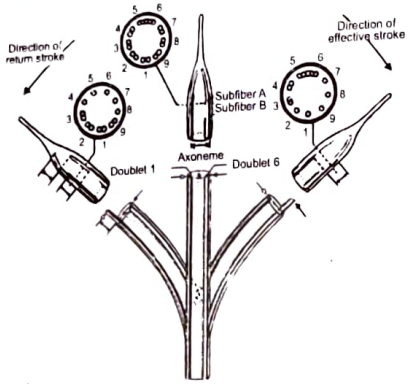


Fig. Diagram representing the sliding hypothesis in ciliary motion. The longitudinal section shows the tip of a cilium with doublets 1 and 6. The cross-section shows the 9 doublets of microtubules in the normal position, and cilium bent in the direction of the effective stroke (right) and of the return stroke (left). Note that between subfiber A and B there is a displacement and that the cilium as a whole undergoes a certain rotation. These changes can be recognized by observing the number of microtubules in the cross-section and the position of doublets 1 and 6.

Recent experimental work on ciliary motion has shown notable similarities with the sliding mechanism involved in the interaction of actin and myosin in muscle. The dynein arms attached to subfiber A have been compared with the crossbridges of myosin and it has been postulated that they form intermittent attachments, by which one doublet is able to push the adjacent one towards the tip of the axoneme. Two adjacent doublets (N and N + 1) are represented, together with the attachment of subfiber A of N to subfiber B of N +1, by way of the dynein arm.

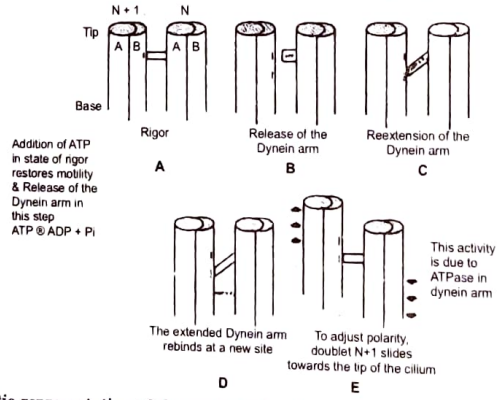


Fig. Schematic representation of the mechanochemical cycle involved in ciliary movement.

Under normal conditions, this attachment is not observed in an intact cilium. However, if the ciliary membrane is extracted with a detergent, the axoneme enters in a state of *rigor* in which the attachment is then produced (A). (Addition of ATP to axonemes) in the state of rigor restores motility and causes release of the dynein arm (B). This release is independent of the hydrolysis of ATP.

In this cycle the next step would be the re-extension of the dynein arm (C) and its rebinding at an angle, with a new, more proximal, site on subfiber B (D). This step apparently needs the hydrolysis of ATP to ADP + Pi. In the last step, the arm returns to the rigor position and displacement of the doublets results (E). Thus the (sliding of the microtubular doublets, which is at the base of ciliary motion, is apparently due to a mechanochemical cycle involving ATP and the ATPase present in the dynein arm.

Experimentally, in glycerin-extracted cilia, flagella, or spermatozoon tails (or in cilia demembranated by detergents), the addition of ATP restores rhythmical activity. On the other hand, extraction of dynein or the use of antidynein anti-serum leads to inhibition of movements.

In sperm flagella it has been observed that the beating is regulated by Ca^{++}, cAMP, and other factors that appear to act on the state of phosphorylation of certain proteins in the axoneme.

Gas Vesicles

Types of Vesicle Transport

Materials are carried between compartments by membrane-bound vesicles, typically 60-100 nm in diameter, that bud from donor membranes and fuse with acceptor membranes. If one scrutinizes electron micrographs for vesicles caught in the act of budding, one finds that most of these membranous buds are covered on their cytosolic surface by a "fuzzy," electron-dense layer. Further analysis reveals that the darkstaining layer consists of a **protein coat** formed from soluble proteins that assemble on the cytosolic surface of the donor membrane at sites where budding takes place. Each coated bud pinches off to form a **coated vesicle.**

Function of Protein coats :

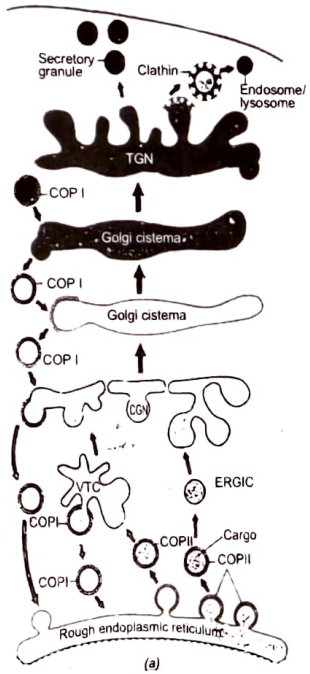
These have at least two distinct functions :

(1) they act as a mechanical device that causes the membrane to curve and form a budding vesicle, and

(2) they provide a mechanism for selecting the components to be carried by the vesicle. Selected components include

(*a*) cargo consisting of secretory, lysosomal, and membrane proteins to be transported, and

(*b*) the machinery required to target and dock the vesicle to the correct acceptor membrane protein coats are able to make these selections by virtue of their specific affinity for the cytosolic "tails" of integral proteins that reside in the donor membrane.



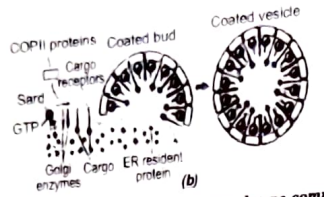


Fig. Movement of materials by vesicular transport between membrane compartment of biosynthetic pathway.

Several distinct classes of coated vesicles have been identified; they are distinguished by the proteins that make up their coat, their appearance in the electron microscope and their role in cell trafficking.

Three Best Studied Coated Vesicles Are :

1. **COPII-coated vesicles** move materials from the ER "forward" to the ERGIC and Golgi complex. (The ERGIC is the intermediate compartment situated between the ER and Golgi complex.) (COP is an acronym for coat proteins.)

   COPII-Coated Vesicles : Transporting cargo from the ER to the Golgi complex

   The COPII coat contains 5 proteins that were first identified in mutant yeast cells that were unable to carry out transport from the ER to the Golgi complex. Proteins selected by COPII-coated vesicles include

   (*i*) enzymes that act at later stages in the biosynthetic pathway such as the glycosyl transferases of the Golgi complex

   (*ii*) membrane proteins involved in the docking and fusion of the vesicle with the target compartment

   (*iii*) membrane protein that are able to bind soluble cargo mutation in one of these cargo receptors has been linked to an inherited bleeding disorder. Persons with this disorder fail to secrete certain coagulation factors that promote blood clotting among the COPII coat proteins is a small GTP-binding protein called Sar1, which plays a regulatory role, in this case regulating the assembly and dissembly of the vesicle coat. Upon binding a molecule of GTP, Sar1 undergoes a conformational change that causes its N-terminal and helix to attach itself to the cytosolic leaflet of the ER bilayer disassembly is triggered by hydrolysis of the bound GTP to produce a Sar1-GDP subunit, which has decreased affinity for the vesicle membrane. Dissociation of Sar1-GDP from the membrane is followed by the release of the other COPII subunits.

2. **COPI-coated vesicles** move materials in a retrograde direction

   (*i*) from the ERGIC and Golgi stack "backward" toward the ER and

   (*ii*) from *trans* Golgi cisternae "backward" to *cis* Golgi cisternae.

   COPI-Coated Vesicles: Transporting Escaped proteins back to the ER

   The coat contains a GTP binding protein, called ARF1, whose bound GTP must be hydrolyzed before the coat can disassemble. COPI-coated vesicles have been implicated in the movement of

   (*i*) Golgi resident enzymes in a trans to *cis* direction

   (*ii*) ER resident enzymes from the ERGIC and the Golgi complex back to the ER.

3. **Clathrin-coated vesicles** move materials from the TGN to endosomes, lysosomes, and plant vacuoles. They also move materials from the plasma membrane to cytoplasmic compartments along the endocytic pathway. They have also been implicated in trafficking from endosomes and lysosomes.

Retaining and Retrieving Resident ER Proteins

If vesicles continually bud from membrane compartments, how does each compartment retain its unique composition? What determines, for example, whether a particular protein in the membrane of the ER remains in the ER or proceeds on to the Golgi complex? Studies suggest that proteins are maintained in an organelle by a combination of two mechanisms:

(*i*) *retention* of resident molecules that are excluded from transport vesicles, and

(*ii*) *retrieval* of "escaped" molecules back to the compartment in which the normally reside.

Proteins that normally reside in the ER, those both in the lumen and in the membrane, contain short amino acid sequences at their C-terminus that serve as retrieval signals, ensuring their return to the ER if they should be accidentally carried forward to the ERGIC or Golgi complex. The retrieval of "escaped" ER proteins from these compartments is accomplished by specific receptors that capture the molecules and return them to the ER in COPI-coated vesicles. Soluble resident proteins of the ER lumen (such as protein disulfide isomerase and the molecular chaperones that facilitate folding) typically possess the retrieval signal "lys-asp-glu-leu" (or KDEL in single-letter nomenclature). These proteins are recognized and returned to the ER by the *KDEL receptor*, an integral membrane protein that shuttles between the *cis* Golgi and the ER compartments. If the KDEL sequence is deleted from an ER protein, the escaped proteins are not returned to the ER but instead are carried forward through the Golgi complex. Conversely, when a cell is genetically engineered to express a lysosomal or secretory protein that contains an added KDEL C-terminus, that protein is returned to the ER rather than being sent on to its proper destination. Resident membrane proteins of the ER, such as the SRP receptor, also have a retrieval signal at their C-terminus (usually KKXX, where K is lysine and X is any amino acid), which binds to the COPI coat, facilitating their return to the ER. Each membrane compartment in the biosynthetic pathway may have its own unique retrieval signals, which helps explain how each compartment can maintain its unique complement of proteins despite the constant movement of vesicles in and out of that compartment.

Beyond the Golgi Complex : Sorting Proteins at the TGN

Sorting and Transport of Lysosomal Enzymes : Lysosomal proteins are synthesized on membrane-bound ribosomes of the ER and carried to the Golgi complex along with other types of proteins. Once in the Golgi cisternae, soluble lysosomal enzymes are specifically recognized by enzymes that catalyze the two-step addition of a phosphate group to certain mannose sugars of the N-linked carbohydrate chains.

Lysosomal enzymes are transported from the TGN in clathrin-coated vesicles coats of these vesicles contain

(1) an outer honeycomb-like lattice composed of the protein clathrin, which forms a structural scaffold, and

(2) an inner shell composed of protein adaptors, which cover the surface of the vesicle membrane that faces the cytosol. The term "adaptor" describes a molecule that physically links two different types of materials. Lysosomal enzymes are escorted from the TGN by a recently discovered family of adaptor proteins called **GGAs.**

As indicated in the inset of GGA molecule has several domains, each capable of grasping a different protein involved in vesicle formation. The outer ends of the GGA adaptors bind to clathrin molecules, holding the clathrin scaffolding onto the surface of the vesicle. On their inner surface, the GGA adaptors bind to a sorting signal in the cytosolic tails of the mannose 6-phosphate receptors. The MPRs, in turn, bind to soluble lysosomal enzymes within the vesicle lumen. As a result of these interactions with GGA adaptors, the MPRs in the TGN membrane and lysosomal enzymes within the TGN lumen become concentrated into clathrin-coated vesicles. As in the formation of COPI and COPII vesicles, production of clathrin-coated vesicles begins with the recruitment to the membrane of a small GTP-binding protein, in this case ARF1, which sets the stage for the binding of the other coat proteins. Once the vesicle has budded from the TGN, the clathrin coat is lost and the uncoated vesicle proceeds to its destination, which may be an early endosome, late endosome, or plant vacuole. Before they reach one of these organelles, the MPRs dissociate from the lysosomal enzymes and return to the TGN for another round of lysosomal enzyme transport.

Targeting Vesicles to a Particular Compartment

Vesicle fusion requires specific interactions between different membranes. Vesicles from the ER, for example, fuse with the ERGIC or *cis* Golgi network and not with a *trans* cisterna. Selective fusion is one of the factors that ensures a directed flow through the membranous compartments of the cell. Despite a major research effort, we still do not fully understand the mechanisms by which cells target vesicles to particular compartments. It is thought that a vesicle contains specific proteins associated with its membrane that govern the movements and fusion potential of that vesicle. To understand the nature of these proteins, we will consider the steps the occur between vesicle budding and vesicle fusion.

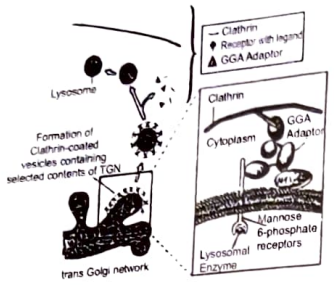
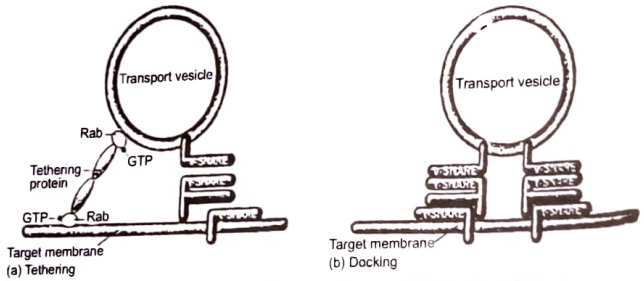


Fig. The formation of clathrin

(i) Movement of the vesicle toward the specific target compartment. In many cases, membranous vesicles must move considerable distances through the cytoplasm before reaching their eventual target. These types of movement are mediated largely by micro tubules, which act like railroad tracks carrying cargo containers along a defined pathway to a predetermined destination. The membranous carriers seen in, for example, were observed to move from the ERGIC to the Golgi complex over microtubules.



Fig, Steps in the trageting of transport vesicle to target membranes

(ii) Tethering vesicles to the target compartment. Microscopic studies indicate that vesicles are often "terthered" to a presumed target compartment, such as a Golgi cisterna, by extended, fibrous proteins. It is hypothesized that tethering is an early stage in the process of vesicle fusion that requires specificity between the vesicle and target compartment. Much of this specificity may be conferred by a family of small GTP-binding proteins called **Rabs,** which associate with membranes by a lipid anchor. With over 60 different Rab genes identified in humans, these proteins constitute the most diverse group of proteins involved in membrane trafficking. More importantly, different Rabs become associated with different membrane compartments, which suggests a role in targeting specificity. In their GTP-bound state, Rabs are thought to recruit cytosolic tethering proteins to mambrane surfaces.

(iii) Docking vesicles to the target compartment. At some point during the process leading to vesicle fusion, the membranes of the vesicle and target compartment come into close contact with one another as the result of an interaction between the cytosolic regions of integral proteins of the two membranes. The key proteins that engage in these interactions are called **SNAREs** and they constitute a family of more than 35 membrane proteins whose members are localized to specific subcellular compartments. Although SNAREs vary considerably in structure and size, all contain a segment in their cytosolic domain called a **SNARE motif** that consists of 60-70 amino acids capable of forming a coiled coil with another SNARE motif. SNAREs can be divided into two categories, **v-SNAREs,** which become incorporated into the membranes of transport vesicles during budding, and **t-SNAREs,** which are located in the membranes of target compartments.

(iv) Fusion between vesicle and target membranes. When artificial lipid vesicles (liposomes) containing purified t-SNAREs are mixed with liposomes containing a purified v-SNARE, the two types of vesicles fuse with one another, but not with themselves. This finding indicates that interactions between t-and v-SNAREs are capable of pulling two lipid bilayers together with sufficient force to cause them to fuse. However, a large body of evidence suggests that, although interaction between v-and t-SNAREs is required for membrane fusion, it is not sufficient by itself to bring about fusion *within a cell.* According to one view regarding the regulated secretion of neurotransmitter molecules, the four stranded SNARE bundle remains locked in an inactive conformation by the presence of another bound protein called complexin. Vesicles at this stage remain docked at the membrane and ready to discharge their contents almost instantaneously once they receive an activating signal, in the form of a rise in Ca^{2+} concentration (as discussed below). Regardless of how it is regulated, once the lipid bilayers of the two membranes merge, the SNAREs that previously projected from separate membranes now reside in the same membrane. Dissociation of the four-stranded SNARE complex is achieved by a doughnut-shaped, cytosolic protein called NSF that attaches to the SNARE bundle and twists it apart.

Exocytosis

The fusion of a secretory vesicle or secretory granule with the plasma membrane and subsequent discharge of its contents is called **exocytosis.** Exocytosis probably occurs on a rather continual basis in most cells, as proteins and other materials are delivered to both the plasma membrane and extracellular space. However, the best studied examples of exocytosis are those that occur during regulated secretion, most notably the release of neurotransmitters into the synaptic cleft. In these cases, membrane fusion produces an opening through which the contents of the vesicle or granule are released into the extracellular space. The arrival of a nerve impulse at the terminal knob of a neuron leads to an increase in the influx of Ca^{2+} and the subsequent discharge of neurotransmitter molecules by exocytosis. In this case, fusion is regualted by a calcium binding protein (synaptotagmin) present in the membrane of the synaptic vesicle. In other types of cells, exocytosis is generally triggered by release of Ca^{2+} from cytoplasmic stores. Contact between the vesicle and plasma membranes is thought to lead to the formation of a small, protein-lined "fusion pore". Some fusion pores may simply reclose, but in most cases, the pore rapidly dilates to form an opening for discharge of the vesicle contents. Regardless of the mechanism, when a cytoplasmic vesicle fuses with the plasma membrane, the luminal surface of the vesicle membrane becomes part of the outer surface of the plasma membrane, whereas the cytosolic surface of the vesicle membrane becomes part of the inner (cytosolic) surface of the plasma membrane.

Endoplasmic Reticulum

The **endoplasmic reticulum (ER)** is divided into two subcompartments, the **rough endoplasmic reticulum (RER)** and the **smooth endoplasmic reticulum (SER).** Both types of ER comprise a system of membranes that enclose a space, or lumen, that is separated from the **surrounding cytosol.** As will be evident in the following discussion, the composition of the **luminal** or **cisternal space** inside the ER membranes is quite different from that of the surrounding **cytosolic space.**

Fluorescently labeled proteins and lipids are capable of diffusing from one type of ER into the other, indicating that their membranes are interconnected. In fact, the two types of ER compartments share many of the same proteins and engage in certain common activities, such as the synthesis of certain lipids and cholesterol. At the same time, however, numerous proteins are found only in one or the other type of ER. As a result, the two types of ER have important structural and functional differences.

The rough ER has ribosomes bound to its cytosolic surface, whereas the smooth ER lacks associated ribosomes. The RER is typically composed of a network of flattened sacs (**cisternae**). The RER is continuous with the outer membrane of the nuclear envelope, which also bears ribosomes on its cytosolic surface. In contrast, the membranous elements of the SER are typically tubular and form an interconnecting system of pipelines curving through the cytoplasm. When cells are homogenized, the SER fragments into smooth-surfaced vesicles, whereas the RER fragments into rough-surfaced vesicles.

Different types of cells contain markedly different rations of the two types of ER, depending on the activities of the cell. For example, cells that secrete large amounts of proteins, such as the cells of the pancreas or salivary glands, have extensive regions of RER. We will return to the function of the RER shortly, but first we will describe the activities of the SER.

Function of Smooth Endoplasmic Reticulum (SER)

(*i*) Synthesis of steroid hormones in the endocrine cells of the gonad and adrenal cortex.

(*ii*) Detoxification in the liver of a wide variety of organic compounds, including barbiturates and ethanol, whose chronic use can lead to proliferation of the SER in liver cells. Detoxification is carried out by a system of oxygen-transferring enzymes (oxygenases), including the *cytochrome P450* family. These enzymes are noteworthy for their lack of substrate specificity, being able to oxidize thousands of different hydrophobic compounds and convert them into more hydrophilic, more readily excreted derivatives. The effects are not always positive. For example, the relatively harmless compound benzo[a]pyrene formed when meat is charred on a grill is converted into a potent carcinogen by the "detoxifying" enzymes of the SER. Cytochrome P450s metabolize many prescribed medications and genetic variation in these enzymes among humans may explain differences from one person to the next in the effectiveness and side effects of many drugs.

(*iii*) Release of glucose from glucose 6-phosphate in liver cells by the enzyme glucose 6-phosphatase. Large reserves of glycogen are stored in the liver as granules on the cytosolic side of SER membranes. When the need for chemical energy arises, glycogen is converted enzymatically to glucose 6-phosphate. Glucose 6-phosphatase in the SER membranes then removes the phosphate group, generating glucose molecules that move into the bloodstream where they are carried to the body's tissues.

(*iv*) Sequestering calcium ions within the cytoplasm of skeletal and cardiac muscle cells. The regulated release of Ca^{2+} from the SER (or sarcoplasmicreticulum in the case of muscle cells) triggers contraction.

Functions of the Rough Endoplasmic Reticulum

The apical surface of the cell faces a duct that carries the secreted proteins out of the organ. The cytoplasm at the apical end of the cell is filled with secretory granules whose

contents are ready to be released into the duct upon arrival of the appropriate signal. The polarity of these glandular epithelial cells reflects the movement of secretory proteins through the cell from their site of synthesis to their site of discharge. The rough ER is the starting point of the biosynthetic pathway: it is the site of synthesis of the proteins, carbohydrate chains, and phospholipids that journey through the membranous compartments of the cell.

Synthesis of Proteins on Membrane-Bound versus Free Ribosomes. The role of the rough endoplasmic reticulum as the site of synthesis of secretory proteins was demonstrated by Jamieson and Palade, as described earlier.

Further experiments reveled that polypeptides are synthesized at two distinct locales within the cell.

1. Certain polypeptides are synthesized on ribosomes attached to the cytosolic surface of the RER membranes. These include-
   - (*a*) proteins secreted from the cell,
   - (*b*) integral membrane proteins, and
   - (*c*) soluble proteins that reside within compartments of the endomembrane system, including the ER, Golgi complex, lysosomes, endosomes, vesicles and plant vacuoles.
2. Other polypeptides are synthesized on "free" ribosomes, that is, ribosomes that are not attached to the RER, and subsequently released into the cytosol. This class includes
   - (*a*) proteins destined to remain in the cytosol (such as the enzymes of glycolysis and the proteins of the cytoskeleton),
   - (*b*) peripheral proteins of the inner surface of the plasma membrane (such as spectrins and ankyrins that are only weakly associated with the membrane's cytosolic surface)
   - (*c*) proteins that are transported to the nucleus, and
   - (*d*) proteins to be incorporated into peroxisomes, chloroplasts and mitochondria. Proteins in the latter two groups are synthesized to completion in the cytosol and then imported *posttranslationally* into the appropriate organelle across its boundary membrane(s).

The site of synthesis of a protein was determined by the sequence of amino acids in the N-terminal portion of the polypeptide, which is the first part to emerge from the ribosome during protein synthesis. They suggested the following :

1. Secretory proteins contain a **signal sequence** at their N-terminus that directs the emerging polypeptide and ribosome to the ER membrane.
2. The polypeptide moves into the cisternal space of the ER through a protein-lined, aqueous channel in the ER membrane. It was proposed that the polypeptide moves through the membrane as it is being synthesized, that is, *cotranslationally*.

Synthesis of Secretory, Lysosomal or Plant Vacuolar Proteins on Membrane-Bound Ribosomes

The entire process of membrane-bound protein synthesis can be studied in vitro using purified ER membranes or artificial vesicles containing purified ER proteins. The steps during the synthesis of a secretory, lysosomal, or plant vacuolar protein. The synthesis of the polypeptide begins after a messenger RNA binds to a free ribosome, that is, one that is *not* attached to a cytoplasmic membrane. In fact, all ribosomes are thought to be identical; those employed in the synthesis of secretory, lysosomal, or plant vacuolar proteins are taken from the same population (*pool*) as those used for production of proteins that remain in the cytosol. Polypeptides synthesized on membrane-bound ribosomes contain a signal sequence—which includes a stretch of 6-15 hydrophobic amino acid residues—that targets the *nascent* polypeptide to the ER membrane and leads to the compartmentalization of the polypeptide within the ER lumen. (A *nascent* polypeptide is one in the process of being synthesized). Although the signal sequence is usually located at or near the N-terminus, it occupies an internal position in some polypeptides.

As it emerges from the ribosome, the hydrophobic signal sequence is recognized by a **signal recognition particle (SRP),** which consists in mammalian cells of six distinct polypeptides and a small RNA molecule, called the 7S RNA. The SRP binds to both the signal sequence on the nascent polypeptide and the ribosome, arresting further synthesis of the polypeptide until the complex collides with and attaches specifically to the rough ER membrane. Interaction between an SRP and the signal sequence is a key step in this process because it ultimately determines whether a protein will end up in the cytosol or the endomembrane system.

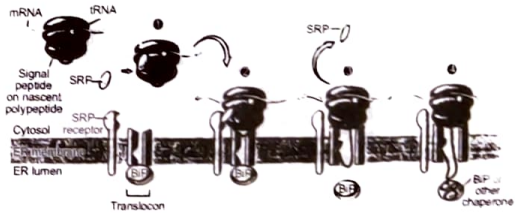


Fig. A schematic model for the synthesis of a secretory protein (or a lysosomal enzyme) on a membrane-bound ribosome of the RER.

The SRP serves as a tag that enables the entire complex (SRP-ribosome-nascent polypeptide) to bind specifically to the cytosolic surface of the ER membrane. Binding occurs through at least two distinct interactions: one between the SRP and the **SRP receptor,** and the other between the ribosome and the **translocon,** the protein-lined channel through the membrane. As the ribosome becomes tightly bound to the ER membrane, the SRP is released from its ER receptor, and the signal sequence on the nascent polypeptide is inserted into the narrow aqueous channel of the translocon (step 3). In subsequent steps, translation resumes, and the polypeptide is translocated through the channel into the ER lumen (step 4). Upon termination of translation and movement of the completed polypeptide through the translocon, the membrane-bound ribosome is released.

Several of the steps involved in synthesis and trafficking of secretory proteins are regulated by the binding or hydrolysis of GTP. **GTP-binding proteins (or G proteins)** play key regulatory roles in many different cellular processes. G proteins can be present in at least two alternate conformations, one containing a bound GTP molecule and the other a bound GDP molecule. GTP-bound and GDP-bound versions of a G protein have different conformations and thus have different abilities to bind other proteins. Because or this difference in binding properties, G proteins act like "molecular switches," the GTP-bound protein typically turns the process on and hydrolysis of the bound GTP turns it off. Both the SRP and the SRP receptor are G proteins. The hydrolysis of GTP bound to these two proteins occurs between steps 2 and 3 and triggers the release of the signal sequence by the SRP and its insertion into the translocon.

Processing of Newly Synthesized Proteins in the Endoplasmic Reticulum

As a nascent polypeptide enters the RER cisterna, it is acted on by a variety of enzymes located within either the membrane or the lumen of the RER. The N-terminal portion containing the signal peptide is removed from most nascent polypeptides by a proteolytic enzyme, the **signal peptidase.** Carbohydrates are added to the nascent protein by the enzyme *oligosaccharyltransferase*. Both the signal peptidase and oligosaccharyltransferase are integral membrane proteins that reside in close proximity to the translocon and act on nascent proteins as they enter the ER lumen.

The RER lumen contains molecular chaperones, such as BiP and calnexin. **Molecular chaperones** recognize and bind to unfolded or misfolded proteins and give them the opportunity to attain their correct (native) three dimensional structure. The ER lumen also contains a number of protein-processing enzymes, such as *protein disulfide isomerase* (PDI), which catalyzes the formation and reshuffling of disulfide bonds between cysteine residues of polypeptide chains. Disulfide bonds play an important role in maintaining the stability of proteins that are present at the extracellular surface of the plasma membrane or secreted into the extracellular space.

The endoplasmic reticulum is ideally constructed for its role as a port of entry for the biosynthetic pathway of the cell. Its membrane provides a large surface area to which many ribosomes can attach (an estimated 13 million per liver cell). The lumen of the ER cistenae provides a local environment that favors the folding and assembly of proteins and a compartment in which secretory, lysosomal, and plant-cell vacuolar proteins can be segregated from other newly synthesized proteins. The segregation of newly synthesized proteins in the ER cisternae removes them from the cytosol and allows them to be modified and dispatched toward their ultimate destination, whether outside the cell or within one of the cytoplasm's membranous organelles.

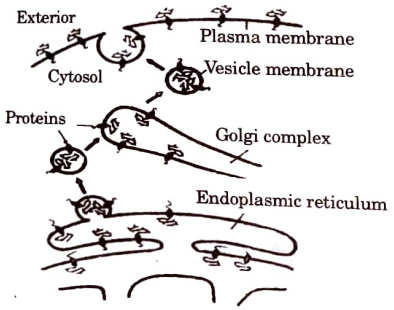


Fig. Maintenance of membrane as

Synthesis of Membrane Lipids

Most membrane lipids are synthesized entirely within the endoplasmic reticulum. The major exceptions are (1) sphingomyelin and glycolipids, whose synthesis begins in the ER and is completed in the Golgi complex, and (2) some of the unique lipids of the mitochondrial and chloroplast membranes, which are synthesized by enzymes that reside in those membranes. The enzymes involved in the synthesis of phospholpids are themselves integral proteins of the ER membrane with their active sites facing the cytosol. Newly synthesized phospholipids are inserted into the half of the bilayer facing the cytosol. Some of these lipid molecules are later flipped into the opposite leaflet through the action of enzymes that actively translocate lipid molecules across the bilayer. Lipids are carried from the ER to the Golgi complex and plasma membrane as part of the bilayer that makes up the walls of transport vesicles.

The membranes of different organelles have markedly different lipid composition , which indicates that changes take place as membrane flows through the cell. There are several factors that may contribute to such changes :

1. Most membranous organelles contain enzymes that modify lipids already present within a membrane, converting one type of phospholipid (e.g., phosphatidylserine) to another (e.g., phosphatidylcholine) (step 1).

2. When vesicles bud from a compartment, some types of phospholipids may be preferentially included within the membrane of the forming vesicle, while other types may be left behind (step 2).
3. Cells contain **phospholipid-transfer proteins** that can bind and transport phospholipids *through the aqueous cytosol* from one membrane compartment to another. These enzymes facilitate the movement of specific phospholipids from the ER to other organelles. This is especially important in delivering lipid to peroxisomes, mitochondria and chloroplasts, which are not part of the normal flow of membrane along the biosynthetic pathway.

Glycosylation in the Rough Endoplasmic Reticulum. Nearly all of the proteins produced on membrane-bound ribosomes—whether integral components of a membrane, soluble lysosomal or vacuolar enzymes, or parts of the extracellular matrix—become glycoproteins. Carbohydrate groups have key roles in the function of many glycoproteins, particularly as binding sites in their interactions with other macromolecules. They also aid in the proper folding of the protein to which they are attached. The sequences of sugars that comprise the oligosaccharides'of glycoproteins are highly specific, if the oligosaccharides are isolated from a purified protein, their sequence is consistent and predictable.

The addition of sugars to an oligosaccharide chain is catalyzed by a group of membrane-bound enzymes called **glycosyltransferases** that transfer a specific monosaccharide from an appropriate sugar donor to an appropriate sugar acceptor. The donor molecule is always a nucleotide sugar, such as CMP-sialic acid, GDP-mannose, or UDP-N-acetylglucosamine.

Golgi Apparatus

The *Golgi apparatus* or *Golgi bodies* of eukaryotic cells are organelles that play a variety of functions, including,

(1) the packaging of secretory materials that are to be discharged from the cell,

(2) the **processing** of proteins (e.g., **glycosylation, phosphorylation, sulfation,** and **selective proteolysis**) synthesized by ribosomes of the rough endoplasmic reticulum,

(3) the synthesis of certain polysaccharides and glycolipids,

(4) the sorting of proteins destined for various locations in the cell,

(5) the proliferation of new membranous elements for the plasma membrane, and

(6) the reprocessing of plasma membrane components that have re-entered the cytosol during endocytosis.

The Golgi apparatus was first described in 1898 by C. Golgi, a pioneer of cytology and cytochemistry. Although Golgi referred to the structure as the "internal reticular apparatus" of the cell, the organelle was later renamed in his honor. Golgi received the Noble Prize in 1906 for his many cytological findings and his cytochemical and histochemical innovations. To visualize these organelles, Golgi employed stains containing silver, osmium, and other heavy metals, and it was believed by many other cytologists that the organelle was an *artifact* produced by precipitation of the metal within the cell.

Structure of the Golgi Apparatus

Golgi bodies differ somewhat in organization from one type of tissue or cell to another, they characteristically take the form of a stack of flattened, oval **cisternae** surrounded at their circumferencial edges and above and below by vesicles and tubular structures. Golgi bodies are also referred to as **dictyosomes,** which means "stacklike bodies." The dictyosome does not include the vesicles that fuse with or are discharged from the margins of the cisternae. In most animal cells, the number of Golgi cisternae is five or six, the lumen of each cisternae varying in width from about 500 to 1000 nm. The Golgi bodies of plant cells often have 20 or more cisternae. The margins of each cisternae are gently curved so that the entire Golgi body takes on a bow-like appearance. The cisternae at the *convex* end of the dictyosome comprises what is called the **cis face** or **forming face,** and the cisternae at the *concave* end comprises the **trans face** or **maturing face.** The small vesicles that are adjacent to the *cis face* are believed to fuse with and contribute additional structure to the Golgi body. The vesicles near the *trans face* are larger and are believed to be formed from the uppermost cisternae. Small vesicles may also be discharged from the margins of the cisternae that are *between* the *cis* and *trans faces*.

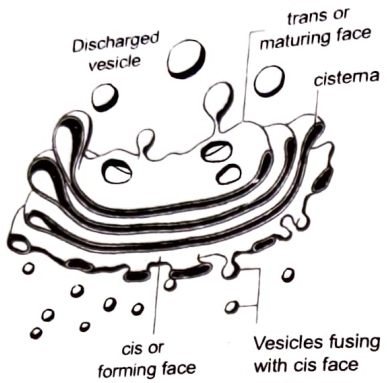


Fig. Diagrammatic reconstruction of the Golgi apparatus based on electron-microscopic study.

In many cells, particularly those in which the Golgi bodies, main function is related to secretion, the faces of the Golgi bodies are arranged in a specific manner. The forming face is located next to either the nucleus or a specialized portion of the endoplasmic reticulum that lacks bound ribosomes and is called "transitional" ER. The maturing face is usually directed toward the plasma membrane. It is believed that the nuclear membrane and endoplasmic reticulum are the source of the small vesicles that fuse with the *cis face* whereas some of the larger vesicles arising from the maturing face are **secretory vesicles** and subsequently fuse with the plasma membrane. When vesicles released from the maturing face of the dictyosome form an internal cell structure, such as the developing **acrosome** of sperm cells. Golgi bodies are sometimes referred to as elements of the **endomembrane** or **vacuolar system** of cells.

The cisternae membranes of Golgi bodies are *smooth*, that is, they are not lined by ribosomes or other particles. At their margins, the cisternae may be perforated, creating **fenestrae,** or they may give rise to tubular extensions that branch and anastomose with each other.

The size and number of Golgi bodies vary from one type of cell to another and according to the cell's metabolic activity. The organelle is therefore to be thought of as being almost continuously in a state of flux. The size and number of Golgi bodies increase during periods of active cellular secretion. In plant cells, the number of Golgi bodies increases during cell division when these organelles secrete materials that form the *cell plate*, which then develops into the cell wall that separates the two new cells. The *goblet cells* found in the intestinal epithelium contain only a single, large Golgi body located in the region of the cell where *mucigen granules* are stored prior to their secretion. In these cells, the size of the Golgi body increases dramatically during periods of digestive activity.

Origin and Development of Golgi Bodies

The proposed sources of new Golgi bodies are

(*i*) vesicles dispatched from the endoplasmic reticulum,

(*ii*) vesicles dispatched from the outer membrane of the nuclear envelope,

(*iii*) vesicles formed by imaginations of the plasma membrane, and

(*iv*) division of Golgi bodies at ready present in the cell. The most widely accepted view is that Golgi bodies are formed from vesicles dispatched from the ER. These vesicles are called **transition vesicles.** Transition vesicles migrate to the forming face of the Golgi body fuse there with existing cisterna membranes, and in so doing contribute to the organelle's growth. Presumably, a complete Golgi body can be formed in this manner.

Aggregations of transition vesicles occur in areas of the cytoplasm referred to as **zones of exclusion** which are free of ribosomes. These zones are often surrounded by membranes of the endoplasmic reticulum or the nuclear envelope. Small Golgi bodies, which are presumed to represent early stages in the development of the organelle, are found in the zones of exclusion. The cells of dormant seeds of higher plants generally lack Golgi bodies but they do display zones of exclusion containing aggregations of small vesicles.

Mitochondria

Mitochondria are organelles present in the cytoplasm of all eukaryotic cells. They provide an energy-transducing system by which the chemical energy contained in foodstufs is converted, by oxidative phosphorylation, into high-energy phosphate bonds (ATP). Mitochondria may be observed in the living cell; visibility is increased by the vital stain, ***Janus green.*** They display passive and active motion and show changes in volume and shape that are related to their function. Swelling of mitochondria can be induced by Ca^{++}, various hormones, and certain drugs.

They are rod-shaped, with a diameter of about 0.5 μm and a variable length that may range up to 7 μm. There are 1000 to 1600 mitochondria in a liver cell and 300,000 in some oocytes. Green plants contain fewer mitochondria than animal cells. The distribution of mitochondria may be related to their function as suppliers of energy. Their orientation in the cell may be influenced by the organization of the cytoplasmic matrix and vacuolar system.

Mitochondrial Structure

Mitochondrion consists of two membranes and two compartments. An outer limiting membrane, about 6 nm thick, surrounds the mitochondrion. Within this membrane, and separated from it by a space of about 6 to 8 nm, is an inner membrane that projects into the mitochondrial cavity complex infoldings called *mitochondrial crests.* This inner membrane divides the mitochondrion into two chambers or compartments :

(*i*) the outer chamber contained between the two membranes and in the core of the crests, and

(*ii*) the inner chamber filled with a relatively dense proteinaceous material usually called the *mitochondrial matrix.* This is generally homogeneous, but in some cases it may contain a finely filamentous material, or small, highly dense granules.

These granules contain phospholipids, which give them an affinity for calcium when this is added to the fixative. There is no evidence, however, for the normal presence of calcium phosphate in the granules.

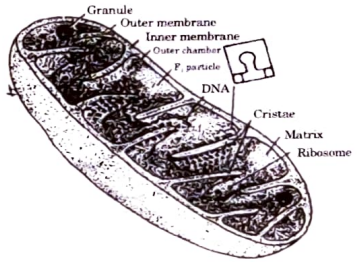


Fig. Three-dimensional diagram of a mitochondrion cut longitudinally. The main features are shown. Observe that the cristae are folds of the inner membrane and that on their matrix side they have the F_1 particles. The inset shows an F_1 particle with the head piece and stalk.

The mitochondrial crests are, in general, incomplete septa or ridges that do not interrupt the continuity of the inner chamber; thus, the matrix is continuous within the mitochondrion.

Within the mitochondrial matrix are small ribosomes and a circular DNA. The outer and inner membranes and the crests can be considered to be fluid molecular films with a compact molecular structure; the matrix is gellike and contains a high concentration of soluble proteins and smaller molecules.

Particles—On the M Side of the Inner Mitochondrial Membrane

The use of negative staining has enabled recogition of other details of mitochondrial structure. If a mitochondrion is allowed to swell and break in a hypotonic solution and is then immersed in phosphotungstate, the inner membrane in the crest appears convered by particles of 8.5 nm that have a stem linking each with the membrane. These so called "elementary" or "F_1" particles, are regularly spaced at intervals of 10 nm on the inner surface of these membranes. According to some estimates, there are between 10^4 and 10^5, elementary particles per mitochondrion. These particles correspond to a special ATPase involved in the coupling of oxidation and phosphorylation.

The presence of such F_1 particles on the matrix side (M side) confers to the inner mitochondrial membrane a characteristic asymmetry that is of fundamental importance to its function. It is interesting to recall here that bacterial plasma membranes contain F_1 particles with a similar asymmetric distribution (i.e., toward the bacterial matrix). The asymmetric location of the F_1 particles in the membranes is related to the direction of the proton pump (i.e., the flow of H^+).

Isolation of Mitochondrial Membranes

In recent years the two mitochondrial membranes and the compartments they limit have been separated. A so-called *mitoplast*, which includes the inner membrane and matrix, has been produced by separating these two elements from the outer membrane with digitonin. The mitoplast has pseudopodic processes and is able to carry out oxidative phosphorylation.

1. Outer and Inner Membranes—Structural and Chemical Differences

The outer membrane fraction has a 40% lipid content (compared to 20% in the inner membrane), contains more cholesterol, and is higher in phosphatidyl inositol; on the other hand, it is lower in cardiolipin. (The very prominent band of 29,000 daltons, porin is clearly observed in SDS gel electrophorests).

2. Mitochondrial Enzymes Are Highly Compartmentalized

The *outer membrane* contains, as in the endoplasmic reticulum (ER), an NADH-cytochrome-c-reductase system that consists of a flavoprotein and cytochrome b_5. The most specific enzyme system of the outer membrane is *monoamine oxidase*, which may serve as an enzyme marker. This membrane also contains kynurenine hydroxylase, fatty acid coenzyme A ligase, a phospholipase, and various enzymes of the phospholipid metabolism. The supernatant, obtained after treatment with digitonin is considered to originate from the outer compartment. In this compartment are found adenylate kinase, nucleoside diphosphokinase; DNAse I, and 5′-endonuclease.

The *mitoplast fraction* contains the components of the respiratory chain and oxidative phosphorylation and the soluble enzymes present in the matrix.

The *inner membrane* carries all the components of the respiratory chain and the oxidative phosphorylation system. It also has several *carriers or translocators* for the permeation of phosphate, glutamate, aspartate, ADP, and ATP. It is calculated that the respiratory chain represents about 20% and the phosphorylating system about 15% of the total protein content of the membrane.

The inner membrane contains a remarkably high concentration of cardiolipin (polyglycero phosphatides), which appear to be important in all systems involving electron transport.

Table. Enzyme distribution in mitochondria

Outer membrane	Inner membrane
Monoamine oxidase	Respiratory chain enzymes
Rotenone-insensitive NADH-cytochrome-c reductase	ATP synthetase
	Succinate dehydrogenase
Kynurenine hydroxylase	(β-Hydroxybutytate dehydrogenase
Fatty acid CoA ligase	Camitine fatty acid acyl transferase
Space between outer and inner membranes	Matrix
adenylate kinase	Malate and isocitrate dehydrogenases
Nucleoside diphosphokinase	Fumarase and aconitase
	Citrate synthestase
	α-Kelo acid dehydrogenases
	β-Oxidation enzymes

The *inner compartment* or *mitochondrial matrix* contains all the soluble enzymes of the citric acid or Krebs cycle and those involved in the oxidation of fatty acids. The matrix also contains DNA, ribosomes, and other RNA species and enzymes involved in the synthesis of protein.

Table. Enzyme activity of various mitochondrial components

Component	Inner Membrane	Outer Membrane	Matrix	Outer Chamber
Enzyme marker	Cytochrome oxidase	Monoamine oxidase	Malate dehydrogenase	Adenylate kinase
Specific activity of marker in component	9315	551	3895	6690
Specific activity in whole mitochondria	1980	22	2608	421
% Protein in component	21.3	4.0	66.9	6.3

Function of Mitochondria

In this organelle there are more than 70 enzymes and coenzymes, as well as numerous cofactors and metals that operate in an orderly and integrated fashion. The whole metabolic machinery of the mitochondria needs only the entrance of phosphate, ADP, and acetylcoenzyme A (acetyl-CoA) to produce ATP, CO_2 and H_2O.

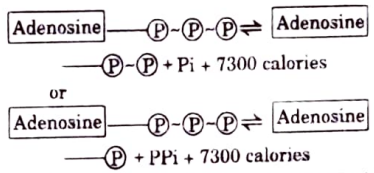
The energy that the cell has at its disposal exists as chemical energy primarily locked in high energy bonds. The cell uses only part of the *total energy* (H), also called *enthalpy*. contained in a chemical compound. This portion of the total energy, the *free energy* (G), does not dissipate as heat. Expressed as an energy change :

$$\Delta H = \Delta G + T\Delta S$$

The equation shows that the change in total energy (ΔH) is equal to the change in free or available energy (ΔG) plus the unavailable energy ($T\Delta S$), which dissipates as heat. (In this equation, T is the temperature in Kelvin, and S is the *entropy* of the system).

Chemical Energy and ATP

ATP is composed of the purine base adenine, a ribose moiety, and three molecules of phosphoric acid. (Adenine plus ribose forms the nucleoside *adenosine*). ATP and the closely related molecule ADP are the most important compounds in energy transformation. If phosphate is represented by P, the simplified formula of ATP and its transformation into ADP or AMP is as follows:



Note that the release of any one of the two terminal phosphates of ATP yields about 7300 calories per mole, instead of the 3000 calories from common chemical bonds. The high energy ~P bond enables the cell to accumulate a great quantity of energy in a very small space and keep it ready for use whenever it is needed.

Other nucleotides having high energy bonds, such as cytosine triphosphate (CTP), uridine triphosphate (UTP), and guanosine triphosphate (GTP), are involved in biosynthetic reactions. The energy source for these nucleotide triphosphates derives ultimately from ATP, however.

Chloroplast

Chloroplasts are organelles found in the green tissues of plants and are responsible for the absorption of light energy, the synthesis of carbohydrates, and the evolution of molecular oxygen. The sum of these three processes is called **photosynthesis.** Light energy captured by the chloroplast is converted into potential chemical energy in the form of carbohydrates and starts the "energy chain" in nature. The oxygen evolved during the capture of light energy becomes the ultimate oxidizing agent for cellular metabolic reactions. Mitochondrial oxidations, as well as other oxidations in plant, animal, and many microbial cells, depend on this primary source of oxygen.

A chloroplast is any membrane-encased organelle containing chlorophyll that belongs to a group of related organelles in plants called **plastids.** The plastids have a variety of morphological forms, carry out diverse functions, and store many different compounds. For example, the **amyloplast** is the starch-storing plastid of potato tubers, and the chromoplast is the lycopene-containing plastid that gives the fruit of tomatoes its red colour. Each of the diverse plastids is believed to arise from a *common proplastid* precursor. All the major groups of plants, with the exception of the fungi, contain chloroplasts. There are two fundamental types of chloroplasts each associated with 1 of two types of photosynthesis : C_3 photosynthesis & C_4 photosynthesis. C_3 photosynthesis converts (or "fixes") carbon-dioxide into three-carbon acids whereas C_4 photosynthesis fixes CO_2 into four carbon acids. Plants that carry out C_4 photosynthesis (called C_4 plants) are generally grasses or plants that are endemic to environments with little water. C_3 plant (those that carry out C_3 photosynthesis) are commonly broad-leafed flowering plants, cone-bearing plants, or those living in areas of adequate moisture. The simpler plants, such as the algae, contain only one or a few chloroplasts per cell, whereas vascular plants, such as the cone-bearing and flowering plant have many chloroplasts in each cell.

Photosynthesis occurs in some prokaryotic organisms such as the cyanobacteria (or blue-green algae and the anaerobic photosynthetic bacteria. These prokaryotes do not have true chloroplasts; instead, they have lammellated structures called **chromatophores** that carry out only the light-absorbing reactions of photosynthesis but not the carbohydrate-synthesizing reactions. For many laboratory studies of photosynthesis, the single-called plant *chlorella* is employed. This alga has one cup-shaped chloroplast that practically fills the cell. The organism can be easily and conveniently cultured with artificial lighting in the laboratory in solutions of inorganic salts.

For studies with Chloroplasts of higher plants leaves are generally used, and spinach and parsle leaves are probably the most popular source. In these leaves, the chloroplasts are found in greatest numbers in two internal tissues, the **palisade mesophyll** and the **spongy mesophyll.** Both tissues lie between an upper and lower epidermis and have thin cell walls that are easily broken. In spinach, there are between 20 and 40 chloroplasts in each palisade parenchyma cell. In the palisade parenchyma, the chloroplasts lie along the edges of the cell, the center of the cell being filled with large vacuoles. In the spongy parenchyma, the chloroplasts are more randomly distributed throughout the cytoplasm of the cell. In many genera, cytoplasmic streaming (i.e., *cyclosis*) moves the chloroplasts about the cell, and in a few instances an active amoeboid-type of movement by chloroplasts has been observed. Their positions in the cell and movement about the cell maximize the exposure of the chloroplasts to light.

The shape of most chloroplasts in higher plants is spheriod, avoid, or discoaid. Other irregular shapes sometimes occur but are more common in lower plants. For example, in algae, cup-shaped, spiral, star-shaped, and digitate forms of chloroplast are observed. The shape and structure of chloroplast can also be altered by the presence of starch granules. During periods of active photosynthesis, the sugar formed in the chloroplasts are polymerized into starches that precipitate as small granules. The starch granules are usually ellipsoidal and may be up to 1.5μm long.

Fine Structure of the Chloroplast

The chloroplasts of C_3 plants are composed of 2 membrane layers similar to those of mitochondria. Each membrane is about 50 Å thick and the two membranes are separated by a space of about 70-100 Å. The **outer membrane,** which lacks folds or projections, serves to delimit the organelle and regulate transport of materials between the cytoplasm and the interior of the organelle. The **inner membrane** parallels the outer membrane, but inward folds of this membrane are extensive.

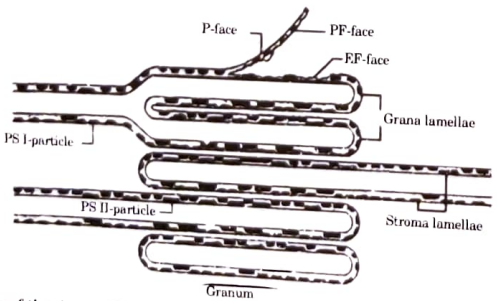


Fig. Diagram of the thylakoid lamellae ang grana showing the arrangement of p the membranes

The inner membrane gives rise to a series of internal parallel membrane sheets called **lamellae.** The lamellae are suspended in a grunular fluid or matrix that appears somewhat electron-dense in electron photomicrographs. This matrix is referred to as the **stroma.** The lamellae form a complex series of membranes through the stroma.

Most of the lamellae in the chloroplasts of higher plants are organized to form disk-shaped sacs called **small thylakoids.** The small thylakoids are often arranged in stacks called **grana** (one stack is a *granum*) having diameters of 300 to 600 nm. Because the thylakoids are disk-shaped, the grana appear much like a stack of coins. A typical chloroplast has between 40 and 60 grana, and each granum may be composed of 2 to 100 small, flattened thylakoids. Frequently, a small portion of the thylakoid extends redially into the stroma forming a branching tube, or **large thylakoid,** that communicates with other small thylakoids and grana. Collectively, the branching and anastomosing network is called the **stroma lamellae.**

Structure of the Thylakoid

The adjacent membranes of neighbouring thylakoids within each of the grana form thick layers called **grana lamellae.** Electron photomicrographs of grana lamellae fixed with glutaraldehyde and stained with osmium reveal a five-layered arrangement consisting of three dark 40-Å-thick osmiophilic layers enclosing two 1Å-thick osmiophobic spaces. Freeze-fracture techniques indicate that the grana membranes contain numerous particles. The particles, which are primarily protein in composition appear to be of two basic sizes, 105 and 104 Å in diameter. The stroma lamellae contain mainly the smaller-size particles. The larger particle are abundant on the *EF* face of the grana membrane and are associated with photosystem II of photosynthesis. The smaller particles are associated with photosystem I and are present in about equal numbers on the *PF* face of both the grana and stroma membranes.

Stroma Structures

The granular stroma contains a variety of particles. The presence of starch granules was noted earlier. Electron micrographs also reveal a number of osmiophilic granules and groups of ellipsoidal structure called **stromacenters.** Strands of DNA and ribosomes are also scattered through the stroma.

The chloroplasts of C_4 plants have the same general structures as those of C_3 plants (an outer enclosing membrane, grana and stroma thylakoids, astroma with DNA, ribosomes, and stromacenters), but there are some notable differences. The C_4 plant chloroplasts all have a **peripheral reticulum,** which is a group of anastomosing tubules about the periphery. In many C_4 plants such as corn there are two types of chloroplasts. Those found in the mesophyll cells are similar to the chloroplasts of C_3 plant but lack starch grains. The chloroplasts found in the cells surrounding the leaf veins (called *bundle sheath cells*) contain starch granules but have elongated grana.

Chemical Composition of Chloroplasts

The organic constituent present in greatest quantity in the chloroplast is protein, which may represent up to 70% of the dry weight. In leaf cells, 75% of the total cell nitrogen is found within the chloroplasts. Both structural and soluble proteins have been identified, but only a few of these have been extracted and purified. A peptide analysis by SDS gel electrophoresis shows compositional differences between stroma and grana lamellae, but the differences are primarily quantitative rather than qualitative.

Table. Chemical Composition of Spinach Chloroplasts

Component	Percentage of Chloroplast Dry Weight	
	Chlorplasts Isolated in Water	Values Corrected for Loss of Soluble Protein
Total protein	50	69
Water-insoluble protein	50	31
Water-soluble protein	0	38
Total lipid	34	21
Chlorophyll	8	5
Carotenoids	1.1	0.7
Ribonucleic acids	—	1.0-7.5
Deoxyribonucleic acids	—	0.02-0.1
Carbohydrate (starch, etc.)	Variable	

Essentially all the pigments and cytochromes are located in the lamellae. The stroma lacks these compounds but contains DNA and RNA, which are not present in the lamellae. Most of the RNA is associated with the ribosomes of the stroma. The amount of DNA is low; estimates are 10^{-15} to 10^{-14} grams per chloroplast or about 0.03% of its dry weight. However, this is enough information to account for the synthesis of some chloroplast proteins, including some of the enzymes of photosynthesis. The disposition of chloroplast DNA during chloroplast division is nuclear.

Lipid and lipid-soluble pigments account for 34% of the dry weight of the spinach chloroplast, exceedingly large number of different lipid compounds have been identified. The more common lipid are the galactosyl diglycerides, phospholipids, quinones (including vitamin K), and sterols.

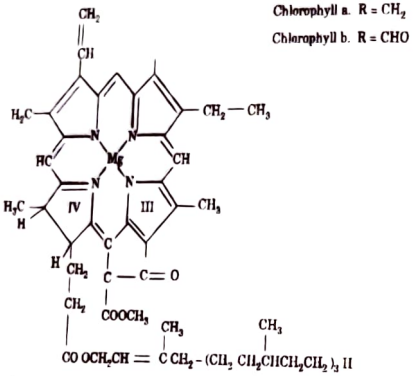


Fig. Chemical structure of chlorophyll a and b

1. Chlorophylls

The green pigments of chloroplasts and the sources of the colour of green plants are the **chlorophylls.** Although a large number of chemically distinct chlorophylls have been identified in a variety of different plants, the structures of these chlorophylls are basically the same. All photosynthetic plants have been found to contain chlorophyll *a*, but the presence of the secondary chlorophylls *b*, *c*, *d*, and *e* depends on the type of plant. Higher plants usually have chlorophyll *b*. In the photosynthetic bacteria, a chlorophyll called **bacteriochlorophyll** occurs in place of chlorophyll a. Together, chlorophylls *a* and *b* represent about 8% of the dry weight of spinach chloroplasts, with an *a*:*b* weight ratio of 2.1 to 3.5. In most plants, the *a*:*b* ratio varies according to the light intensity to which the plants are expossed. For example, alpine plants, which receive light of high intensity, have an average ratio of 5.5. The ratio is 2.3 in shade plants.

Table. Absorption Maxima of Plant and Bacterial Pigments

Pigment	Wavelength (nm)	Occurrence
Chlorophyll *a*	430,670	All green plants
Chlorophyll *b*	455,640	Higher plants; green algae
Chlorophyll *c*	445,625	Diotoms; brown algae
Bacteriochlorophyll	365,605,770	Purple and green bacteria
α-Carotene	420,440,470	Leaves; some algae
β-Carotene	425,450,480	Some plants
γ–Carotene	440,460,495	Some plants
Luteol	425,445,475	Green leaves; red and brown algae
Violaxanthol	425,450,475	Some leaves
Fucoxanthol	425,450,475	Diotoms; brown algae
Phycoerythrins	490,546,576	Red and blue-green algae
Phycocyanins	618	Red and blue-green algae
Allophycoxanthin	654	Red and blue-green algae

Each chlorophyll has a characteristic light absorption spectrum. Extracted chlorophyll *a* has absorption maxima at 430 and 670 nm, whereas the absorption maxima of chlorophyll *b* occur at 455 and 640 nm. The absorption spectrum and maxima of plant pigments vary according to the solvent used for extraction. *Invivo* studies of chlorophyll *a* indicate that its native absorption maximum occurs at 677 nm. One very important form of chlorophyll *a* that is readily bleached by light has an absorption maximum at 700 nm. This form, which represents only about 0.1% of the total chlorophyll *a* molecules present in a sample, is called P_{700} or chlorophyll a_1. The roles of the various chlorophylls in photosynthesis are discussed later in the chapter.

2. The Carotenoids

The carotenoids are all long-chain isoprenoid compounds having an alternating series of double bonds. Although these compounds are synthesized only in plant tissue and participate in photosynthesis, they also serve as precursors of vitamin A in animal tissues. Most carotenoids are yellow, orange, or red. Most of these pigments are located in the chloroplast lamellae and are believed to function as accessory pigments for light absorption during photosynthesis.

Development of Chloroplasts

New chloroplasts are produced by the division of mature chloroplasts or by development from proplastids in the cell. The cells of young shoots of higher plants may contain 20 to 40 of these small avoid bodies. Apparently, proplastids can develop into a number of different plastid types in addition to chloroplasts. As a proplastid develops into a chloroplast, its inner membrane gives rise to infoldings that form the lamellae and thylakoids. Proplastids, as well as mature chloroplasts, increase in numbers by a form of division.

The control of chloroplast development is not fully understood. The differentiation of new organelles appears to rely on an interaction between genetic information present in the cell nucleus and information present in the chloroplast itself. Chloroplasts contain DNA that is transcribed and translated within the orgenelle to form some of the chloroplast proteins; however, other chloroplast proteins are derived from transcription and translation of nuclear DNA.

Ribosomes

Ribosomes are particles of about 200 Å diameter; they are composed of RNA and protein. Generally, ribosomes are attached to the outer surfaces of ER membranes; this converts smooth ER elements into rough ER. But some ribosomes are present in the hyaloplasm unattached to any membrane.

The weight of ribosomes and other macro molecules (large molecules) is determined by ultracentrifugation (centrifugation at very high speeds of 50,000 rpm or more). The sedimentation rate of macromolecules is expressed as ***s (sedimentation coefficient)***, which is the average velocity per unit of acceleration and is expressed as cm/sec^2. The size or weight of a macromolecule is expressed in ***S units*** or *Svedberg* (the inventor of **ultracentrifuge)** ***units***, which has a reasonably reliable relationship with *s*. On the basis of their sedimentation rate, ribosomes are classified into three groups: (1) 80 *S* size animal ribosomes, (2) 80 *S* plant ribosomes, and (3) 70 *S* eukaryotic organelle (mitochondria and plastids) and prokaryotic ribosomes. In the conditions of low Mg^{++} concentration, each plant ribosome (80 *S*) dissociates into a larger subunit of 60 S and a smaller subunit of 36 *S;* the 80 *S* animal ribosomes dissociate into 60 *S* and 45 *S* subunits; while 70 *S* prokaryotic and organelle ribosomes dissociate into 50 *S* and 30 *S* subunits.

The smaller subunit of a ribosome is placed onto the larger subunit much like a cap-on-the-head. Animal and plant ribosomes are composed of 55-60 percent protein and 40-45 percent RNA, while prokaryotic and organelle ribosomes have only 37 percent protein but 63 per cent RNA. Although organelle ribosomes have been grouped with the prokaryotic ones, they differ from the latter both in the size and the type of RNA molecules they contain. Organelle ribosomes themselves seem to constitute a separate, but highly variable, category of ribosomes.

In E.coli, each 50S larger subunit has one molecule each of a 23S and a 5S (about 120 nucleotides) RNA, while the 30s smaller subunit has one molecule of a 16S RNA (about 1600 nucleotides). The larger subunit has 34 different proteins (L_1 to L_{34}), while the smaller subunit has 21 different proteins. One protein of the smaller subunit (S_{20}) is found in the larger subunit also (L_{26}), while one protein of the larger subunit exists in two different forms (L_7 and L_{12}), which differ only by acetylation.

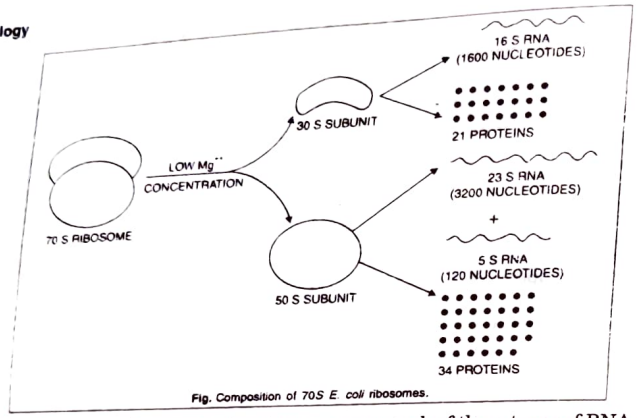


Fig. Composition of 70S *E. coli* ribosomes.

In mammalian ribosomes, the larger subunit has one copy each of three types of RNA molecules (28S, 5.8S and 5S), while the smaller subunit has one RNA molecule of *18S size*. Plant ribosome larger subunits have a 25S and a 5S RNA molecule, while the smaller subunits have one copy each of a 16S and a 5S RNA. Detailed three dimensional models of ribosomes have been proposed; a description of these models is beyond the scope of this book.

Ribosomes are synthesised and assembled in nucleolus, within the nucleus, from where they migrate into the cytoplasm. *Ribosomes are essential for protein synthesis* as mRNAs can support protein synthesis only when they are attached to ribosomes. Several (5-8) ribosomes may bind to a single mRNA molecule during protein synthesis; such ribosomes together with the mRNA constitute a ***polysome***.

Lysosomes

Lysosomes are vesicles of 400-800 μm formed by budding off of Golgi bodies; they contain hydrolytic enzymes. The main enzyme present in lysosomes is acid phosphatase; other enzymes are, acid DNAase, acid RNAase, β-galactosidase etc. The membrane surrounding a lysosome has a unit membrane organisation. Different vesicles contain different enzymes. Lysosomes are of two types: (1) primary and (2) secondary. ***Primary lysosomes*** are produced by Golgi bodies and contain hydrolytic enzymes only. They fuse with food vacuoles produced through phagocytosis and pinocytosis to generate ***secondary lysosomes***. Secondary lysosomes contain both hydrolytic enzymes as well as food-materials. The food particles are ultimately digested by the hydrolytic enzymes and absorbed into the hyaloplasm; the undigested portion of food materials remains in the secondary lysosomes, which are now known as ***residual bodies***.

The enzymes present in lysosomes are capable of digesting any living organism. When a white blood cell (WBC) ingests (by phagocytosis) a bacterium or some other organism, all the lysosomes of WBC fuse with the food vacuole thus produced. As a result, the bacterium as well as the WBC itself are lysed. In some situations, the enzymes present in lysosomes are released into the cytoplasm, which leads to a lysis of the concerned cell *(autolysis)*. Thus the function of lysosomes is to digest (lyse) the food particles and microorganisms ingested by a cell and also to cause autolysis of the cell, if required.

Peroxisomes are small vesicles surrounded by a unit membrane and contain the enzyme catalase. They are involved in a series of oxidizing reactions in the cells.

Spherosome

Spherosomes are vesicles of 0.5-1 μ diameter and contain, upto 98 percent, lipid; they also contain some acid phosphatase. Spherosomes are found in plant cells only; they are absent in animal and prokaryotic cells, Spherosomes are surrounded by a single unit membrane. According to some scientists, spherosomes function in lipid storage, but some others do not agree with this view.

Centriole

Centrioles are hollow cylindrical structures of about 1,200-1,500 Å diameter and of about 3,000-5,000 Å in length. Centrioles are always present in pairs,one centriole being oriented vertical to the other (Fig. 2.9 *a*). They are confined to animal cells, and are not found in plant cells. However, basal bodies found at the base of flagella of plant cells are identical with centrioles in their ultrastructure. Generally, new centrioles arise during interphase in association with pre-existing centrioles; they organise at an interval from and right angles to the two pre-existing centrioles in a cell (Fig. *a*).

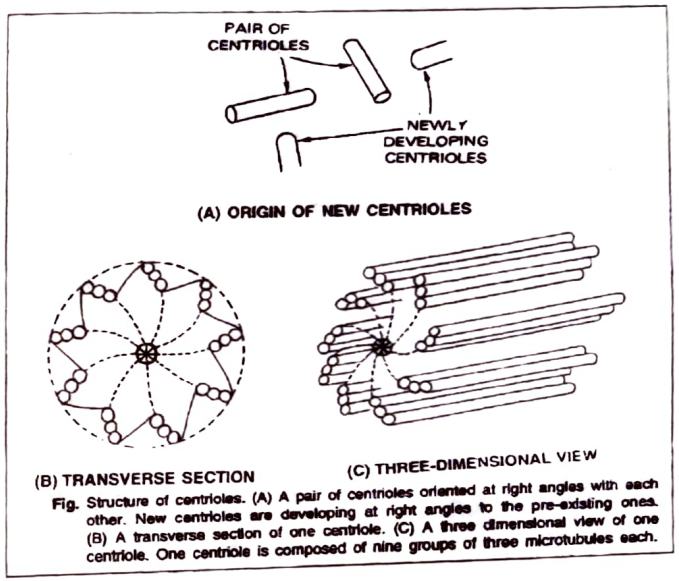


Fig. Structure of centrioles. (A) A pair of centrioles oriented at right angles with each other. New centrioles are developing at right angles to the pre-existing ones. (B) A transverse section of one centriole. (C) A three dimensional view of one centriole. One centriole is composed of nine groups of three microtubules each.

A newly organised centriole, called ***procentriole,*** remains only 1/2 of the parent centriole length, and plays no role in the following mitotic division. However, it becomes a parental centriole when the pair is passed into one of the daughter cells following mitosis; it then functions as parent centriole and provides a template function for the organization of a new procentriole. But at least in some cases, preexisting centrioles are not essential for the organisation of new centrioles, which arise *de novo*.

Electron micrographs of transverse sections of centrioles reveal nine fibrils arranged at the periphery of a centriole, Each fibril is composed of three microfibrils arranged in a single plane so that each fibril has a flat topography. Each microfibril consists of a single microtubule. The fibrils of a centriole are oriented at an angle on the periphery. The inner microfibril of each of the nine fibrils is connected with the outer microfibril of a neighbouring fibril; at the same time, it is also connected with a central microfibril located in the centre of the centriole. Thus the transverse ultrastructure of a centriole is somewhat similar to the wheel of a bullock-cart (or any other wheel with spokes). The role of centriole during cell division is unclear. Once it was believed that the microtubules of the spindle apparatus originate from centrioles (in animals). But it is now clear that microtubules originate from the dense amorphous region surrounding the centrioles present in *centrosomes* (the body at the poles from which spindle apparatus develops). It is likely that centrioles play a role in the orientation of the spindle apparatus. In both plant and animal cells, centrioles serve as the basal bodies of flagella.

Nucleus

Nuclei take up relatively deep stain with basic dyes, and are generally spherical in shape. But in differentiated tissues, they may exhibit considerable variation in shape. The size of nuclei varies with the physiological state and the degree of differentiation of cells. In physiologically active and undifferentiated cells they are relatively larger in size, while in physiologically inactive and differentiated cells they are relatively smaller. Nuclear size is also affected by the chromosome number (or the amount of DNA) of a species, as well as by the ploidy level of a cell or individual. In general, a cell contains a single nucleus, but in some tissues, *e.g.*, early stages of endosperm development, and in some organisms, *e.g.*, some protozoa, some fungi etc., two or more nuclei are present within a single cell. In contrast, some specialized cells, *e.g.*, RBCs and companion cells of phloem, do not contain any nucleus.

Nucleus is the store-house of almost all genetic information needed for the functioning of a cell/ organism. It produces ribosomes, tRNA (transfer RNA) and mRNA, which (together with the protein synthesis machinery of cells) produce the various structural and enzymatic proteins essential for cellular organization and biochemical reactions. In essense, nucleus governs the development of almost all the traits of an organism by providing the information necessary for the synthesis of various structural and functional proteins. The significance of nuclei in the development was first demonstrated by a German scientist, Hammerling in 1934, through transplantation studies in the unicellular alga ***Acetabularia***. He clearly demonstrated that the development of characters in this alga *was* governed by the nucleus and not by the cytoplasm.

Pea nuclei contain 14 percent DNA, 12 percent RNA and 74 percent protein on dry weight basis. In electron micrographs, the following three distinct types of structures are discernible in interphase nuclei: (1) nuclear envelope, (2) nucleolus, and (3) chromatin fibers.

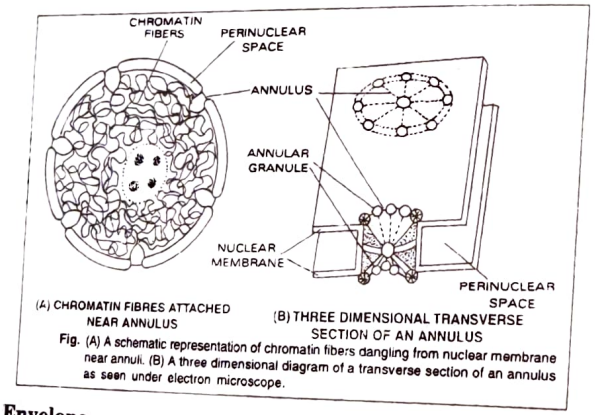


Fig. (A) A schematic representation of chromatin fibers dangling from nuclear membrane near annuli. (B) A three dimensional diagram of a transverse section of an annulus as seen under electron microscope.

Nuclear Envelope

Nucleus is enclosed by two concentric membranes each being 70-80 Å in thickness. There is a space of about 200-300 Å between the two membranes; this space is known as the *perinuclear space*. The two membranes together with the perinuclear space constitute ***nuclear envelope***. The outer membrane of envelope is continuous with ER present in the cytosol. The inner surface of the inner membrane of nuclear envelope is covered with a dense network of proteinaceous fibers connected with the proteins present in the inner membrane; this network is called **nuclear lamina**. Lamina is composed of 3 types of proteins called **lamins** A, B and C. The outer layer of lamina holds the nuclear pores in place, while the inner layer provides attachment for chromosomes. The lamina is disrupted at the nuclear pores and the nucleoloplasmic annulus is connected with the lamina. At the end of G_2 phase, M phase kinase phosphorylates lamins leading to dissociation of the lamina. At the end of mitosis, lamins are dephosphorylated by phosphatases; this allows their reassociation into a nuclear lamina.

The two membranes of nuclear envelope come into contact at structures called *nuclear pore complex* or simply *nuclear pores*, or even *annuli* (singular, *annulus*: due to their ring shaped organization). The nuclear pore complexes are *pores* of 200-400 Å in diameter are arranged in a hexagonal pattern at an interval of 800 Å from each other and from the centre of the hexagons thus formed. The ultrastructural details of annuli are rather complex (Fig.). At the peripheri of an annulus, eight annular granules are embedded in each of the two nuclear membranes; the granules are located at equal distance from each other. In the centre of the annulus, another annular granule is located. The eight annular granules within a membrane (in an annulus) are connected with each other and with the central granule (Fig. *a*).

Nuclear pore complexes are released intact from the nuclear envelope following a detergent treatment; these appear as annular (ring-shaped) rosettes made of 8 spokes. The various findings about nuclear pores have been used to propose a model of their organization (Fig). According to this model, the pore complex is composed of 8 identical subunits. Each subunit has a large number of individual proteins, which are organized to form an elongated structure with the following features: (1) coaxial arms at both the ends, (2) a radial arm which extends outside the pore complex and anchors in the membrane, and (3) a central spoke which extends into the pore (Fig). The coaxial armsof the 8 subunits join to form the two rings or annuli of the pore complex, the outer ring projects into **cytosol and** is called ***cytoplasmic ring,*** while the inner ring projects into the nucleus and is known as ***nucleoloplasmic ring.*** The central spokes of the 8 subunits join to form the central ring. The pore diameter at the openings is ~ 20 nm; this narrows down to ~ 48 nm due to the central ring. A structure called ***transporter*** is located within the region of the central ring; the transporter contains a pore of ~10nm in diameter. Nuclear pore complex provides passage for the transport of all the materials into and out of the nucleus.

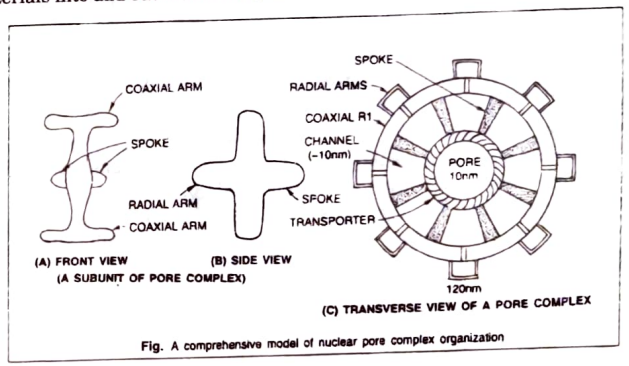


Fig. A comprehensive model of nuclear pore complex organization

Nucleolus

Interphase nuclei contain relatively dense spherical bodies called *nucleolus*. Generally, nucleoli disappear during cell division in the late prophase of mitosis and meiosis, and the nucleolar material is distributed onto the chromosomes. At the end of cell division, the nucleolar materials dissociate from chromosomes, and fuse together to organise a new nucleolus. Nucleolus is ordinarily attached to a specific region (***nucleolar organiser region;*** **NOR)** of specific chromosomes **(nucleolar organiser chromosomes; NOC)** of the genome. Ordinarily, two nucleoli are present in an interphase nucleus, but during meiotic prophase a single nucleolus can be seen (since the two homologous nucleolus organiser chromosomes pair to form a single bivalent). But in certain special situations, one interphase nucleus may contain more than two nucleoli.

The average chemical composition of nucleoli is as follows : protein, 70 percent, **RNA** 30 percent, and a small amount of DNA. The ultra structure of nucleoli reveals four regions: (1) amorphous region, (2) granular region, (3) fibrillar region, and (4) chromatin fibers. The central region of a necleolus is amorphous and is composed mainly of protein. The amorphous region is surrounded by nucleolonema, which consists of a *granular portion* mostly containing ribosomes of 150-200 Å diameter, and *a fibrillar region* primarily containing RNA fibers of 50-150 Å diameter. In addition, *chromatin fibers* from the nucleolar organiser region also pass through nucleoli.

The main function of nucleoli is production and organisation of ribosomes. Ribosomal RNA (rRNA) is produced by genes located in the nucleolar organiser region of NOCs.

Chromatin Fibers

In transverse sections, transversely and obliquely-cut ends of chromatin fibers are seen throughout an interphase nucleus, except for the major part of nucleolus. The average diameter of chromatin fibers is about 230Å. In interphase nuclei, chromatin fibers are relatively much more condensed in heterochromatin than are those in euchromatin. As a result, heterochromatic regions appear much more dense than do euchromatic regions. During interphase, one end (telomore) of each chromosome is attached at the periphery of an annulus. The rest of chromosome dangles free in the nucleus in a highly, but loosely, folded state. Thus interphase nuclei are filled with a mass of folded chromatin fibers (and nucleolus). Chromatin fibers are the basic units of chromosome structure: a more detailed discussion on them follows later.

Cell Growth and Cell Division

Growth an increase in size or mass of a developing/living system—is an irreversible process that occurs at all organizational levels.

Cell Cycle

All cells are produced by divisions of Pre-existing cell. Continuity of life depends on cell division. Cell cycle can be defined as the entire sequence of events happening from the end of one nuclear division to the beginning of the next. The cell cycle involves the following three cycles.

1. **Chromosome cycle.** In it DNA synthesis alternates with mitosis (or karyokinesis or nuclear division). During DNA synthesis, each double-helical DNA molecule is replicated into two identical daughter DNA molecules and during mitosis the duplicated copies of the genome are ultimately separated.
2. **Cytoplasmic cycle.** In it cell growth alternates with cytokinesis (or cytoplasmic division). During cell growth many other components of the cell (RNA, proteins and membranes) become double in quantity and during cytokinesis cell as a whole divides into two. Usually the karyokinesis is followed by the cytokinesis but sometimes the cytokinesis does not follow the karyokinesis is followed by the cytokinesis but sometimes the cytokinesis does not follow the karyokinesis and results into the multinucleate cell, e.g., cleavage of egg in *Drosophila*.
3. **Centrosome cycle.** Both of the above cycles require that the centrosome be inherited reliably and duplicated precisely in order to form the two poles of the mitotic spindle; thus, centrosome cycle forms the third component of cell cycle.

Stages of Cell Cycle

Howard and Pele (1953) have divided cell cycle into four phases or stages : G_1, S, G_2 and M phase. The G_1 phase, S phase and G_2 phase are combined to from the classical interphase.

1. G_1 Phase

After the M phase of previous cell cycle, the daughter cells begin G_1 of interphase of new cell cycle. G_1 is a resting phase. It is called first gap phase, since no DNA synthesis takes place during this stage; currently, G_1 is also called first growth phase, since it involves synthesis of RNA, proteins and membranes which leads to the growth of nucleus and cytoplasm of each daughter cell towards their mature size.

During G_1 phase, chromatin is fully extended and not distinguishable as discrete chromosomes with the light microscope. This is a time of resumption of normal cell metabolism which has slowed down during the previous cell division. Thus, G_1 involves transcription of three types of RNAs, namely rRNA, tRNA and mRNA; rRNA synthesis is indicated by the appearance of nucleolus in the interphase (G_1 phasee) nucleus. Proteins synthesized during G_1 phase (1) regulatory proteins which control various events of mitosis; (2) enzymes (e.g., DNA polymerase) necessary for DNA synthesis of the next stage; and (3) tubulin and other mitotic apparatus proteins.

G_1 phase is most variable as to duration (Table; it either occupies 30 to 50 percent of the total time of the cell cycle or lacks entirely in rapidly dividing cells (e.g., blastomeres of early embryo of frog and mammals). Terminally differentiated somatic cells (i.e., end cells such as neurons and striated muscle cells) that no longer divide, are arrested usually in the G_1 state; such a type of G_1 phase is called G_0 phase.

Different stages of a mitotic cell cycle and their duration in hours.

Parts of cell cycle	Phases	Description of Phases	*Vicia cells*	Mouse L cells	Human Hela cells
Interphase	G_1	Pre-DNA-synthesis phase	12	12	12
	S	DNA-synthesis phase	6	6-8	10
	G_2	Post-DNA synthesis phase	12	3-4	3
Mitosis	M	Mitotic phase	1	1	1

2. S Phase

During the S phase or synthetic phase of interphase, replication of DNA and synthesis of histone proteins occur. New histones are required in massive amounts immediately at the beginning of the S period of DNA synthesis to provide the new DNA with nucleosomes. Thus, at the end of S phase, each chromosome has two DNA molecules and a duplicate set of genes. S phase occupies roughly 35 to 45 per cent of cell cycle.

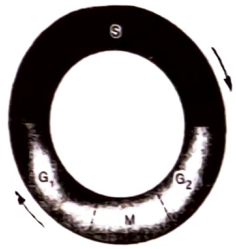


Fig. The cell cycle or mitotic cycle, showing relative duration of phases (e.g., interphase and mitotic phase) in a growing cell. S, synthesis of DNA; G_1 the first gap or growth phase; G_2 the second gap or growth phase; and M, mitotic phase.

3. G_2 phase

This is a second gap or growth phase or resting phase of interphase. During G_2 phase, synthesis of RNA and proteins continues which is required for cell growth. It may occupy 10 to 20 percent time of cell cycle. As the G_2 phase draws to a close, the cell enters the M phase.

General Events of Interphase

The interphase is characterized by the following features :

The nuclear envelope remains intact. The chromosomes occur in the form of diffused, long, coiled and indistinctly visible chromatin fibre. The DNA amount becomes double. Due to accumulation of ribosomal RNA (rRNA) and ribosomal proteins in the nucleolus, the size of the latter is greatly increased. In animal cells, a daughter pair of centrioles originates near the already existing centriole and, thus, an interphase cell has two pairs of centrioles.

In animal cells, net membrane biosynthesis increases just before cell division (mitosis). This extra membrane seems to be stored as blebs on the surface of the cells about to divide.

4. M phase or Mitotic phase

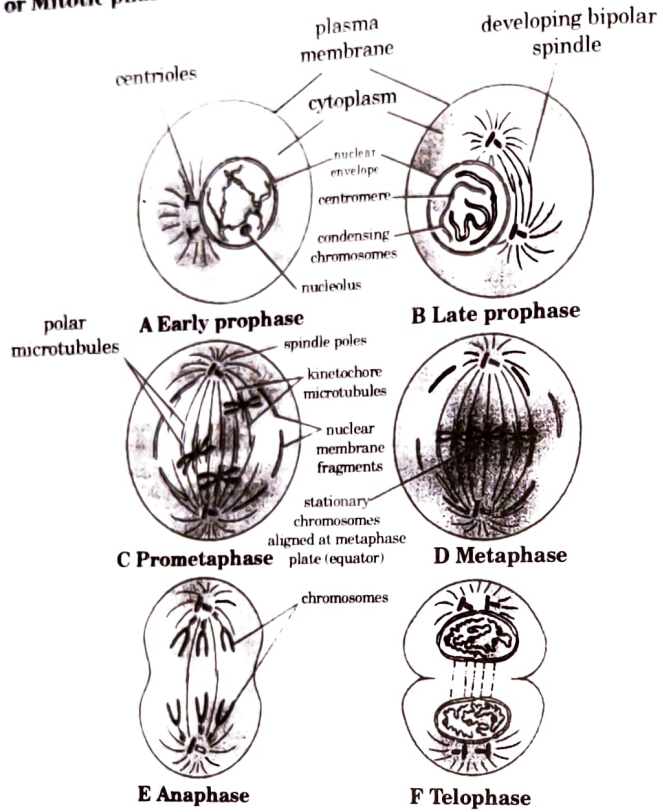


Fig. Diagrammatic summary of mitosis in the animal cell.

The mitosis (Gr., *mitos* = thread) occurs in the somatic cells and it is meant for the multiplication of cell number during embryogenesis and blastogenesis of plants and animals.

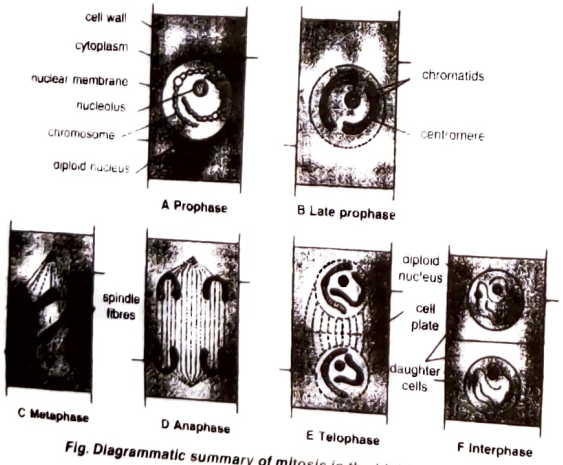


Fig. Diagrammatic summary of mitosis in the hig

1. **Prophase:** The appearance of thin-thread like condensing chromosomes marks the first phase of mitosis, called **prophase** (Gr., *pro*-before; *phasis* = appearance). The cell becomes spheroid, more refractile and viscous.

   Each prophase chromosome is composed of two coiled filaments, the **chromatids**, which are the result of the replication of DNA during the S phase. As prophase progresses, the chromatids become shorter and thicker and two sister chromatids of each chromosome are held together by a special DNA-containing region, called the **centromere or primary constriction**. During prophase proteins of the trilaminar **kinetochores** (one for each chromatid) start depositing or organizing on the centromere of each chromosome (see Darnell et al., 1986). Further, during early prophase, the chromosomes are evenly distributed in the nuclear cavity; as prophase progresses, the chromosomes approach the nuclear envelope, causing the central space of the nucleus to become empty.

   In the cytoplasm, the most conspicuous change is the formation of the spindle or **mitotic apparatus**. In the early prophase, there are two pairs of centrioles, each one surrounded by the so-called aster which is composed of microtubules radiating in all directions. The two pairs of centrioles migrate to opposite poles of the cell along with the asters and become situated in antipodal positions. Between the separating centrioles forms a spindle. The microtubules of the spindle are arranged like two cones base to base, broad at the centre or equator of the cell and narrowing to a point at either end or pole. Mitotic spindle contains three main types of fibres : (1) polar fibres, which extend from the two poles of the spindle toward the equator; (2) kinetochore fibres, which attach to the kinetochores of centromeres of each mitotic chromosomes and extend toward the poles; and (3) astral fibres, which radiate outward from the poles toward the periphery or cortex of cell. In cells of most higher plants, however, spindle forms without the aid of centrioles and lacks asters.

   Lastly, during prophase, the nucleolus gradually disintegrates. Degeneration and disappearance of the nuclear envelope marks the end of prophase. This process is incompletely understood. However, following two factors may be involved in this process:

   1. Enzymatic action either by some mitochondrial enzymes, cytosolic MPF kinase or nuclear RNA (or ribozyme)
   2. Physical action, i.e., physical stress exerted by microtubules which become attached to the nuclear envelope.

   There are variations available with respect to the dissolution of nuclear envelope and the nucleolus. In a number of primitive classes of plants and animals the nuclear envelope does not dissolve during mitosis.

   Prometaphase: The breaksdown of nuclear envelope signals the commencement of prometaphase and enables the mitotic spindle to interact with the chromosomes. This stage is characterized by a period of frantic activity during which the spindle appears to be tryping to contain and align the chromosomes at the metaphase plate. In fact, at this stage the chromosomes are violently rotated and oscillated back and forth between the spindle poles because their kinetochores are capturing the plus ends of microtubules growing from one or the other spindle pole and are being pulled by the captured microtubules. The kinetochores thereby act as a "cap" that tends to protect the plus end from depolymerizing, just as the centrosome at the spindle pole tends to protect the minus end from depolymerizing. Thus, sister chromatids become attached by their kinetochores to opposite poles; balanced bipolar forces hold chromosomes on the metaphase plate.

2. **Metaphase:** During metaphase (Gr., *meta* = after; *phasis* = appearance) the chromosomes are shortest and thickest. Their centromeres occupy the plane of the equator of the mitotic apparatus (a region known as the **equatorial or metaphase plate**).

3. **Anaphase:** The anaphase (Gr., *ana* = up; *phasis* = *appearance*) begins abruptly with the synchronous splitting of each chromosome into its sister chromatids, called *daughter chromosomes*,each with one kinetochore. Synchronous splitting of each centromere during prophase is evidently caused by an increase in cytosolic Ca^{2+}. In fact, Ca^{2+} -containing membrane vesicles accumulate at spindle poles and release calcium ions to initiate anaphase. Anaphase involves the following two steps :

(*i*) **Anaphase A:** Durinng it, there is poleward movement of chromatids due to shortening of the kinetochore microtubules. During their poleward migration, the centromeres (and kinetochores) remain foremost so that the chromosomes characteristically appear U, V or J-shaped.

(*ii*) **Anaphase B:** It involves separation of poles themselves accompanied by the elongation of the polar microtubules. The astral microtubules also help in anaphase B by their attractive interaction with cell cortex.

4. **Telophase:** The end of the polar migration of the daughter chromosomes marks the beginning of the telophase; which in turn is terminated by the reorganization of two new nuclei and their entry into the G_1 phase of interphase. In general terms, the events of prophase occur in reverse sequence during this phase. A nuclear envelope reassembles around each group of chromosomes to form two daughter nuclei. The mitotic apparatus except the centrioles disappears; high viscosity of the cytoplasm decreases; the chromosomes resume their long, slender, extended form as their coils relax; and RNA-synthesis restarts causing the nucleolus to reappear.

Cytokinesis

Both DNA synthesis and mitosis are coupled to cytoplasmic divison, or cytokinesis—the constriction of cytoplasm into two separate cells. During cytokinesis, the cytoplasm divides by a process, called **cleavage**. The mitotic spindle plays an important role in determining where and when cleavage occurs. Cytokinesis usually begins in anaphase and continues through telophase and into interphase. The first sign of cleavage in animal cells is **puckering** and **furrowing** of the plasma membrane during anaphase. The furrowing invariably occurs in the plane of the metaphase plate, at right angles to the long axis of the mitotic spindle. A cleavage furrow tends to form midway between asters originating from two centrosomes in fertilized sand dollar eggs.

Cleavage is accomplished by the contraction of a ring composed mainly of actin filaments. This bundle of filaments, called **contractile ring**, is bound to the cytoplasmic face of the plasma membrane by unidentified attachment proteins. The contractile ring assembles in early anaphase, once assembled, it develops a force large enough to bend a fine glass needle inserted into the cell. Evidently this force is generated due to muscle-like sliding of actin and myosin filaments in the contractile ring. The actin-myosin interaction pulls the plasma membrane down into a furrow. During a normal cytokinesis, the contractile ring does not get thicker as the furrow invaginates, suggesting that it continuouly reduces its volume by losing filaments. When cleavage ends, the contractile ring is finally dispensed with altogether and the plasma membrane of the cleavage furrow narrows to form the midbody, which remains as a tether (Tether means a rope for confining a beast within certain limits) between two daughter cells. The midbody contains the remains of the two sets of polar microtubules, packed tightly together with dense matrix material.

Cytokinesis greatly increases the total cell-surface area as two cells form from one. Therefore, the two daughter cells resulting from cytokinesis require more plasma membrane than in the plant cell. Lastly, prior to cytokinesis, in M phase large membrance-bounded organelles such as Golgi apparatus and the endoplasmic reticulum break up into smaller fragments and vesicles; this may ensure their even distribution into daughter cells during cytokinesis.

Physiology of Cell Cycle and Mitosis

Following aspects of cell cycle and mitosis need somewhat more detailed explanation:

1. **Regulation of mitotic chromosome cycle:** Mitotic chromosome cycles is found to be regulated by the following three control factors (i.e., diffusible proteins) :

(*i*) The S-phase activator that normally appears in the cytoplasm only during S-phase and 'switches on' DNA synthesis

(*ii*) The M-phase promoting factor (**MPF**) that is present only in M-phase cytoplasm and causes 'omosome condensation.

(*iii*) The DNA-dependent M-phase delaying factor that is present in S-phase cytoplasmand inhibits the process leading to onset of MPF production.

MPF is a large-sized protein comprising two subunits-an inert subunit and a kinase subunit which can phosphorylate (and activate) the inert subunit (called self-activation) and other molecules. Thus MPF kinase directly phosphorylates several substances, including histone H1, thereby promoting chromosome condensation; and it may be through a cascade of phosphorylation that MPF triggers all the complex events of mitosis such as nuclear envelope breakdown and cytoskeletal change (*e.g.*, formation of mitotic spindle).

2. **Dissolution and formation of nuclear envelop during mitosis:** At least three parts of the nuclear envelope complex must be considered during its breakdown (at prophase) and reassembly (at telophase) :

(*i*) outer and inner nuclear membranes

(*ii*) the underlying nuclear lamina of lamin proteins and

(*iii*) the nuclear pores.

A prophase many proteins become phosphorylated by MPF and phosphorylation of nuclear lamins help regulate the disintegration and reconstruction of the nuclear envelope. The phosphorylation of the lamins occurs at many different sites in each polypeptide chain and causes them to disassemble, thereby disrupting the nuclear lamina. Subsequently, perhaps in response to a different signal (see Telophase), the nuclear envelope proper breaks up into small membrane vesicles.

It is reported that in less than an hour (prophase to prometaphase) almost the entire 4000 pores disappear from the nuclear membranes of cultured mammalian cells. These pore complexes have been found on chromosomes during mitosis.

The sudden transition from metaphase to anaphase is thought to initiate dephosphorylation of many proteins, including histone H1 and the lamins, that were phosphorylated at prophase. Shortly thereafter, at telophase, nuclear membrane vesicles associated with the surface of individual chromosomes are fused to re-form the nuclear membranes which partially enclose clusters of chromosomes before coalescing to re-form the complete nuclear envelope (Fig). During this process the presynthesized nuclear pores reassemble and the dephosphorylated lamins reassociated to form the nuclear lamina; one of the lamina protein (lamin B) remains with the nuclear membrane fragments throughout mitosis and may help in nuclear assembly.

3. **Role of cytoskeleton in mitosis:** It has been said that the chromosomes in mitosis are like the corpse at a funeral : they provide the reason for the proceedings but play no active part in them. The active role is played by their distinct cytoskeletal structures that appear short-termly in M phase. The first to form is a **bipolar mitotic spinde**, composed of microtubules and their associated protein; it is meant for poleward migration of daughter chromosomes during mitosis. The second cytoskeletal structure required in M phase in animal cells is a contractile ring of microfilaments and myosin that forms slightly later just beneath the plasma membrane; it is meant for cytokinesis of the cell. The third cytoskeletal component is meshwork of intermediate filaments that surrounds the interphase nucleus; it elongates during mitosis to enclose the two daughter nuclei and finally divides in half by the cleavage furrow.

Working of mitotic spindle during anaphase: During anaphase A, a surprisingly large force acts on a chromosome as it moves from the metaphase plate to the spindle pole. By hydrodynamic analysis it has been calculated that to move a chromosome, a force about 10^{-11} dynes is needed, and that the entire displacement–from equator to the pole of a chromosome-may require the use of about 30 ATP molecules. As each chromosome moves polewards, its kinetochore microtubules disassemble, so that they have nearly disappeared at telophase. The site of subunit loss can be determined by injecting labelled tubulin into cells during metaphase. The labelled subunits are found to be added to the kinetochore end of kinetochore microtubules and then lost as anaphase A proceeds indicating that kinetochore "eats" its way poleward along its microtubules at anaphase (Fig.) However, typically microtubule disassemble at kinetochores, poles or at both sites is probably necessery for equator-to-pole movement.

Further, the mechanism by which kinetochore of the chromosome moves up the spindle during anaphase A is still unknown. However, the following three models throw some light on it

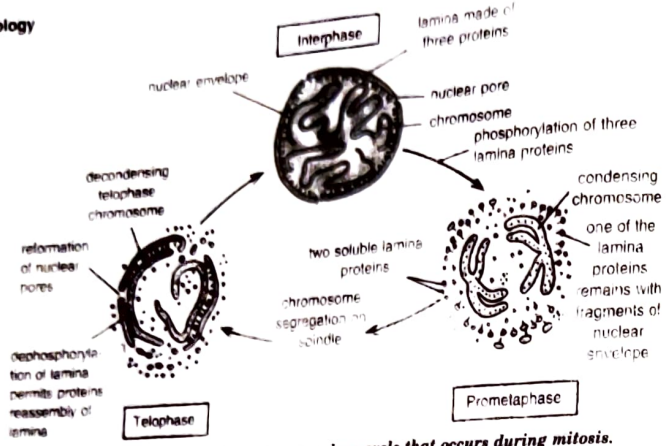
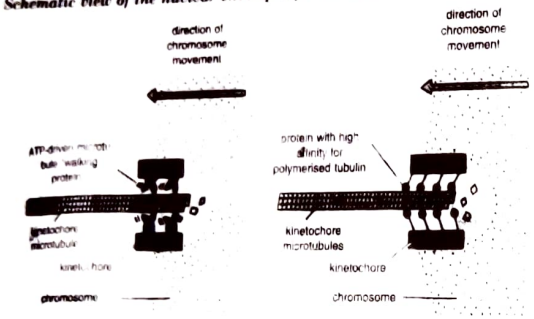


Fig. Schematic view of the nuclear-envelope cycle that occurs during mitosis.



A. ATP-driven chromosome movement drives microtubule disassembly

B. Microtubule disassembly drives chromosome movement.

Fig. ***Two alternative models to explain the generation of poleward force by the kinetochore for its chromosome during anaphase. A—Microtubule-walking proteins that resemble dynein or kinesin are part of the kinetochore, and they use the energy of ATP hydrolysis to pull the chromosome along its bound microtubules: B—Chromosome movement is driven by microtubule disassembly : as tubulin subunits dissociate, the kinetochore tends to slide poleward in order to restore its binding to the walls of the microtubule (After Alberts et al, 1989).***

1. The kinetochore hydrolyzes ATP to move along its attached microtubule, with the plus end of the microtubule depolymerizing as it becomes exposed (Fig.A).
2. The depolymerization of the microtubule itself causes the kinetochore to move passively to optimize its binding energy on the microtubule (Fig.B).
3. A system of elastic protein filaments might connect the kinetochore to the pole and pull the kinetochore steadily poleward. Thus, in this case, microtubules merely regulate movements of the chromosomes.

In mammalian cells, anaphase B begins shortly after the chromatids have begun their voyage to the poles and stops when the spindle is about 1.5 to 2.0 times its metaphase length (15 times increase-in certain protozoa). Thus, anaphase B increases the distance between two spindle poles and in contrast to anaphase A, is accompanied by the polymerization of polar microtubules at their plus ends. Further the polar microtubules from each half-spindle overlap in a central region near the spindle equator (*e.g.*, diatoms). During anaphase B these two sets of antiparallel polar microtubules appear to slide away from each other in the region of overlap. Dynein-like force generating protein may be involved in such a directed movement of chromosomes.

Significance of Mitosis

The mitosis has the following significance for living organisms :

1. The mitosis helps the cell in maintaining proper size.
2. It helps in the maintenance of an equilibrium in the amount of DNA and RNA in the cell.
3. The mitosis provides the opportunity for the growth and development to organs and the body of the organisms.
4. The old decaying and dead cells of body are replaced by the help of mitosis.
5. In certain organisms, the mitosis is involved in asexual reproduction.
6. The gonads and the sex cells depend on the mitosis for the increase in their number.
7. The cleavage of egg during embryogenesis and division of blastema during blastogenesis, both involve mitosis.

Meiosis and Reproductive Cycle

The term meiosis (Gr., *meioum* = to reduce or to diminish) was coined by J.B. Farmer in 1905. Meiosis produces a total of four haploid cells from each original diploid cell. These haploid cells either become or give rise to gametes, which through union (fertilization) support sexual reproduction and a new generation of diploid organisms. Thus, meiosis is required to run the reproductive cycle of eukaryotes such as microorganisms Chloamydomonas, Neurospora; bryophytes; plants and animals.

Process of Meiosis

Meiosis superficially resembles two mitotic divisions without an intervening period of DNA replication. The first meiotic division includes a long prophase in which the homologous chromosomes become closely associated to each other and interchange of hereditary material takes place between them. Further, in the first meiotic division the reduction of chromosome number takes place and, thus,two haploid cells are resulted by this division. The first meiotic division is also known as the **heterotypic division**. in the second meiotic division the haploid cell devides mitotically and results into four haploid cells. The second meiotic division is also known as the **homotypic division.** In the homotypic division pairing of chromosomes, exchange of the genetic material and reduction of the chromosome number do not occur.

- Meiosis
  - Division I or Heterotypic division
    - Prophase I-
      - Proleptotene
      - Leptotene
      - Zygotene
      - Pachytene
      - Diplotene
      - Diakinesis
    - Prometaphase I
    - Metaphase I
    - Anaphase I
    - Telophase I
  - Interphase
  - Division II or Homotypic division
    - Prophase II
    - Metaphase II
    - Anaphase II
    - Telophase II

Heterotypic Division or First Meiotic Division

Meiosis starts after an interphase which is not very different from that of an intermitotic interphase. In the beginning of the first meiotic division the nucleus of the meiocyte starts to swell up by absorbing the water from the cytoplasm and the nuclear volume increases about three folds. After these changes the cell passes to the first stage of first meiotic division which is known as prophase.

Prophase I

The first prophase is the longest stage of the meiotic division. It includes following substages:

1. **Proleptotene or Prolepto-nema:** (Gr. pro = before; leptas = thin; nema = thread). The proleptotene stage closely resembles with the early mitotic prophase. In this stage the chromosomes are extremely thin, long,uncoiled, longitudinally single and slender thread-like structures.

2. **Leptotene or Leptonema:** In the leptotene stage the chromosomes become more uncoiled and assume a long thread-like shape. The chromosomes at this stage take up a specific orientation inside the nucleus; the ends of the chromosomes converge toward one side of the nucleus, that side where the centrosome lies (the bouquet stage). The centriole duplicates and each daughter centriole migrates towards the opposite poles of the cell. On reaching at the poles each centriole duplicates and thus, each pole of cell possesses two centrioles of a single diplosome.

3. **Zygotene or Zygonema:** (Gr., *zygon* = adjoining). In the zygotene stage, the pairing of homologous chromosomes takes place. The homologous chromosomes which come from the mother (by ova) and father (by sperm) are attracted towards each other and their pairing takes place. The pairing of the homologous chromosomes is known as synapsis (Gr., *synapsis* = union). The synapsis begins at one or more points along the length of the homologous chromosomes.

   The pairing of the homologous chromosome is very exact and specific (i.e., alignment of chromosomes is exactly gene-for-gene). The paired homologous chromosomes are joined by a roughly 0.2-μm thick, protein-containing framework called a synaptonemal complex (SC). This complex extends along the whole length of the paired chromosomes and is usually anchored at either end to the nuclear envelope. SC helps to stabilize the pairing of homologous chromosomes and to facilitate the cytogenetical activity, called recombination or crossing over (occurring during pachynema). SC is not found in those organisms in which crossing over does not occur (e.g., the male fruitfly, *Drosophila melanogaster*).

4. **Pachytene or Pachynema:** (Gr., *pachus* = thick). In the pachynema stage the pair of chromosomes become twisted spirally around each other and cannot be distinguished separately. In the middle of the pachynema stage each homologous chromosome spilts lengthwise to form two chromatids. Actually, the doubling of the DNA molecule strands which is necessary for the subsequent duplication of chromosomes occurs earlier, before the beginning of meiotic prophase. Through the earlier part of the meiotic prophase, however, the DNA molecule in each chromosome behaves as a single body. In the pachynema stage, this is now changed, the two chromatids of each chromosome containing half of the DNA present in the chromosome at start, become partially independent of one another, although they still continue to be linked together by their common centromere. Each synaptonemal pair at this point is commonly referred to as bivalent or dyads because it consists of two visible chromosomes, or as a quadrivalent or tetrad because of the four visible chromatids.

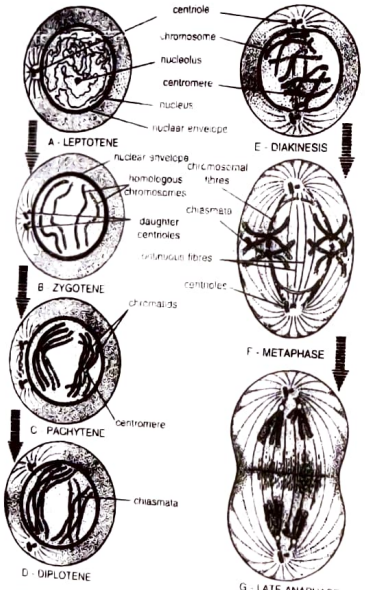


Fig. Different stages of first meiotic division

During pachynema stage an important genetic phenomenon called "crossing over" takes place. The crossing over involves reshuffling, redistribution and mutual exchange of hereditary material of two parents between two homologous chromosomes. After the division of chromatids, the interchange of chromatid segments takes place between the nonsister chromatids of the homologous chromosomes. The broken chromatid segments are united with the chromatids due to the presence of an enzyme, ligase. The process of interchange of chromatin material between one non-sister chromatid of each homologous chromosome is known as the crossing over which is accompanied by the chiasmata formation.

Stern and Hotta (1969) have reported that during the pachytene and zygotene stage, synthesis of small amount of DNA takes place. This DNA amount is utilized in the repairing of broken DNA molecule of the chromatids during the chiasmata formation and crossing over.

The nucleolus remains prominent up to this stage and it is found to be associated with the nucleolar organizer region of the chromosome.

5. **Diplotene or Diplonema:** In diplonema, unpairing or desynapsis of homologous chromosomes is started and chiasmata are first seen. At this phase the chromatids of each tetrad are usually clearly visible, but the synaptonemal complex appears to be dissolved, leaving participating chromatids of the paired homologous chromosome physically joined at one or more discrete points called **chiasmata** (singular, chiasma; Gr., *chiasma* = cross piece). These points are where crossing over took place. Often there is some unfolding of the chromatids at this stage, allowing for RNA synthesis and cellular growth.

6. **Diakinesis:** In the diakinesis stage the bivalent chromosomes become more condensed and evenly distributed in the nucleus. The nucleolus detaches from the nucleolar organizer and ultimately disappers. The nuclear envelope breaks down. During diakinesis the chiasma moves from the centromere towards the end of the chromosomes and the intermediate chiasmata diminish. This type of movement of the chiasmata is known as terminalization. The chromatids still remain connected by the terminal chiasmata and these exist up to the metaphase.

Prometaphase

In the prometaphase the nuclear envelope disintegrates and the microtubules get arranged in the form of spindle in between the two centrioles which occupy the position of two opposite poles of the cell. The chromosomes become greatly coiled in the spiral manner and get arranged on the equator of the spindle.

Metaphase I

Metaphase I consists of spindle fibre attachment to chromosomes and chromosomal alignment at the equator. During metaphase I, the microtubules of the spindle are attached with the centromeres of the homologous chromosomes of each tetrad. The centromere of each chromosome is directed towards the opposite poles. The repulsive forces between the homologous chromosomes increase greatly and the chromosomes become ready to separate.

Anaphase I

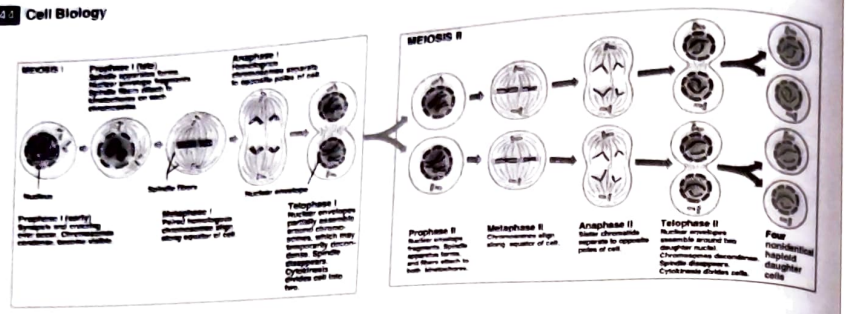
At anaphase I homologous are free from each other and due to the shortening of chromosomal fibres or microtubules each homologous chromosome with its two chromatids and undivided centromere move towards the opposite poles of the cell. The chromosomes with single or few terminal chiasma usually separate more frequently than the longer chromosomes containing many chiasmata.

The actual reduction and disjunction occurs at this stage. Here it should be carefully noted that the homologous chromosomes which move towards the opposite poles are the chromosomes of either paternal or maternal origin. Moreover, because during the chiasma formation out of two chromatids of a chromosome, one has changed its counterpart, therefore, the two chromatids of a chromosome do not resemble with each other in the genetical terms.

Telophase I

The arrival of a haploid set of chromosomes at each pole defines the onset of telophase I, during which nuclei are reassembled. The endoplasmic reticulum forms the nuclear envelope around the chromosomes and the chromosomes become uncoil. The nucleolus reappears and, thus, two daughter chromosomes are formed. After the karyokinesis, cytokinesis occurs and two haploid cells are formed.

Both cells pass through a short resting phase of interphase. During interphase, no DNA replication occurs, so that chromosomes at the second prophase are the same double-strandedstructures that disappeared at the first telophase. In case of *Trillium* telophase I and interphase I do not occur and the a naphase I is followed by prophase II directly.



Homotypic or Second Meiotic Division

The homotypic or second meiotic division is actually the mitotic division which divides each haploid meiotic cell into two haploid cells. The second meiotic division includes following four stages.

Prophase II

In the prophase second, each centriole divides into two and, thus, two pairs of centrioles are formed. Each pair of centrioles migrates to the opposite pole. The microtubules get arranged in the form of spindle at the right angle of the spindle of first meiosis. The nuclear membrane and the nucleolus disappear. The chromosomes with two chromatids become short and thick.

Metaphase II

During metaphse II, the chromosomes get arranged on the equator of the spindle. The centromere divides into two and, thus, each chromosome produces two monads or daughter chromosomes. The microtubules of the spindle are attached with the centromere of the chromosomes.

Anaphase II

The daughter chromosomes move towards the opposite poles due to the shortening of chromosomal microtubules and stetching of interzonal microtubules of the spindle.

Telophase II

The chromatids migrate to the opposite poles and now known as chromosomes. The endoplasmic reticulum forms the nuclear envelope around the chromosomes and the nucleolus reappears due to synthesis of ribosomal RNA (rRNA) by rDNA and also due to accumulation of ribosomal proteins.

After the karyokinesis, in each haploid meiotic cell, the cytokinesis occurs and, thus, four haploid cells are resulted. These cells have different types of chromosomes due to the crossing over in the prophase I.

Significance of Meiosis

The meiosis has the greatest significance for the biological world because of its following uses:

1. The meiosis maintains a definite and constant number of the chromosomes in the organisms
2. By crossing over, the meiosis provides an opportunity for the exchange of the genes and, thus causes the genetical variations among the species. The variations are the raw materials of the evolutionary process.

Thus the meiosis is a peculiar taxonomic, genetical and evolutionary process.

Comparison Between Mitosis And Meiosis

Mitosis	Meiosis
1. Mitosis occurs continuously in the body or somatic cells.	1. Meiosis occurs in the germ cells (the cells of the testes or ovaries) during the process of gametogenesis.
2. The whole process completes in one sequence or phase.	2. The whole process completes in two successive divisions which occur one after the other.
3. The prophase is of short duration and includes no substage.	3. The prophase is of longer duration and it completes in six successive stages, viz., proleptotene, leptotene, zygotene, pachytene, diplotene and dikinesis.
Prophase	
4. The homologous chromosomes (paternal and maternal) duplicate into two chromatids. The two chromatids separate and form new chromosomes. Each daughter cell receives the daughter chromosome or chromatids of each homologous chromosome and, thus, having the chromosome number like the parental cells.	4. Out of two homologous chromosomes only one type of chromosome either maternal or paternal moves to the daughter cells. A daughter cell, thus, receives only a maternal or paternal chromosome of the homologous pair and the number of chromosome remain half than the paternal cells.
5. No pairing or synapsis takes place between the homologous chromosomes.	5. Pairing or synapsis occurs between the homologous chromosomes.
6. Duplication of chromosomes takes place in the early prophase.	6. Duplication or splitting of chromosomes takes place in the late prophase (pachytene stage).
7. No chiasma formation or crossing over takes place	7. Chiasma formation or crossing over takes place.
8. The exchange of the genetic material between the homologous chromosomes does not occur.	8. The exchange of the genetic material takes place between the non-sister chromatids of homologous chromosomes.
Metaphase	
9. The chromatids occur in the form of dyads.	9. The chromatids of two homologous chromosomes occur as the tetrads.
10. The centromeres of the chromosomes remain directed toward the equator and the arms of the chromosomes remain directed towards the poles.	10. The centromeres of the chromosomes remain directed towards the poles and the chromosomal arms remain directed towards the equator.
Anaphase	
11. The chromosomes are the monads, i.e., having single chromatid.	11. The chromosomes are the diads, i.e., having two chromatids and single centromere.
12. The chromosomes are long and thin.	12. The chromosomes are short and thick.
Telophase	
13. The telophase always occur.	13. The first telophase is sometimes omitted.
Significance	14. In meiotic division the chromosome number is reduced to half in the daughter cells than the parental cells.
14. The chromosome number in each daughter cell remains the same like the parent cell.	
15. A diploid cell produces four diploid cells by a mitotic division.	15. A diploid cell produces four haploid cells by a meiotic division

Cell Signaling

Signal transduction is a process of converting extracellular signal into cellular responses. All cells receive and respond to signals from their surroundings. This is accomplished by a variety of signal molecules that are secreted or expressed on the surface of one cell and bind to receptors expressed by other cells, thereby integrating and coordinating the functions of the many individual cells that make up organisms. Each cell is programmed to respond to specific extracellular signal molecules.

Signal Molecules

Signal molecules are chemically heterogenous compounds. These molecules are divided into two categories - membrane bound and secretory signal molecules. Some signal molecules remain bound to the surface of the cells and mediate contact dependent signaling. In most cases, signal molecules are secreted by signaling cells. Secreted signal molecules are divided into three general categories based on the distance over which signals are transmitted : *endocrine*, *paracrine* and *autocrine* signal molecules.

Endocrine signaling molecules (hormones) are secreted by specialized endocrine cells and carried through the circulation to act on target cells at distant body sites.

Paracrine signaling molecules released by one cell acts on neighbouring target cells. An example is provided by the action of neurotransmitters in carrying signals between nerve cells at a synapse.

Autocrine signaling molecules produce effect on same cell that produce it. One important example of such is the response of cells of the vertebrate immune system to foreign antigens. Certain types of T lymphocytes respond to antigenic stimulation by synthesizing a growth factor that drives their own proliferation, thereby increasing the number of responsive T lymphocytes and amplifying the immune response.

Signal molecules can be further classified into *extracellular* and *intracellular* molecules. Extracellular signal molecules are synthesized and released by signaling cells and bind to either cell surface receptors or intracellular receptors. Extracellular signaling molecules are synthesized and released by signaling cells and produce a specific response only in target cells that have receptors for the signalling molecules. Most of the extracellular signaling molecules bind to receptors expressed on the target cell surface called cell surface receptors. Cell-surface receptors belong to three major classes :

1. G protein coupled receptors
2. Ion-channel receptors and
3. Enzyme linked receptors

Binding of extracellular signaling molecules to cell surface receptor leads to increase (or decrease) of low molecular weight intracellular signaling molecules termed secondary messengers. These low-molecular weight signaling molecules include cyclic AMP (cAMP), cyclic GMP (cGMP), diacylglycerol (DAG); inositol 1, 4, 5-trisphosphate (IP_3), phosphoinositides and calcium.

Some extracellular signaling molecules like steroids, retinoids spontaneously diffuse through the plasma membrane and bind to *intracellular receptors* present in cytoplasm or nucleus. By virtue of their hydrophobic nature, they traverse the plasma membrane and exert their effects within the cell, where they control the activities of specific genes. These regulatory effect occur at the level of transcription of responsive genes Steroid and related hormones bind specific receptor proteins within the cell which interact with hormone response element (HRE). Binding of the complex to HRE affects transcription rates of nearby genes.

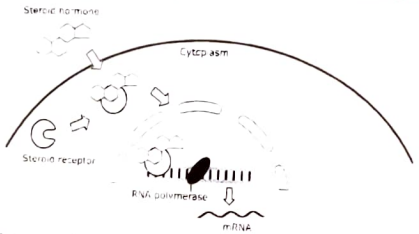


Fig : Steps in the activation of a gene by a steriod hormone, such as the glucocorticoid.

The intracellular nuclear receptors belong to two major classes. The type I receptors include the steroid hormone receptors. Glucocorticoid receptor is representative of this class. Glucocorticoid receptor exists in the cytoplasm complexed with a hsp90. When the receptor binds to its glucocorticoid ligand it changes conformation, dissociates from hsp90, and move into the nucleus to activate genes after binding with glucocorticoid response elements (GREs).

The *type* II receptors present in the nucleus. One representative example is thyroid hormone recepter Type II receptors bind to their response elements in both the presence and absence of their ligands. In the absence of ligand receptor can repress transcription, whereas binding of the receptors along with their ligands can stimulate transcription. Thus, the same protein can act as either an activator or a repressor.

Example of signal molecules

Signal molecules that interact with cell surface receptor

Epinephrine, Non-epinephrine, Glucagons, Insulin, Gastrin, Secretin, Cholecystokinin, ACTH

Signal molecules (hormones) that interact with cytosolic or nuclear receptor

Steroid hormone (Progesterone, Estradiol, Testosterone, Cortisol, Corticosterone, Aldosterone),

Steroid like hormone (α-ecdysone) and Non-steroid hormone (Thyroxine, Retenoid, derived from retinol)

Agonists and Antagonists

Structural analogs of signaling molecules fall into two classes: *agonists* and *antagonists*. Structural analogs which mimic the function of a signaling molecule by binding to its receptor and causing the normal response are called *agonists*. Whereas *antagonists* are structural analogs which bind to the receptor but induces no response. Hence an antagonist acts as an inhibitor of the natural signaling molecule (or agonist) by competing for binding sites on the receptor, thereby blocking the physiological activity of the signaling molecule. For example alprenolol and isoproterenol act as antagonists and agonist to the epinephrine receptor, respectively.

G Proteins Coupled Receptors (GPCR) and G-Protein

GPCRs are the largest family of cell surface receptors that transmits signals to intracellular targets via the intermediary action of guanine nucleotide-binding proteins called **G-protein.** GPCRs are present in all eukaryotes. It includes the receptors for many neurotransmitters, neuropeptides, and peptide hormones. In addition, the G protein-coupled receptor family includes a large number of receptors that are responsible for smell, light, and taste. Despite the chemical and functional diversity of the signal molecules that bind to them, all GPCRs have a similar structure. GPCRs consists of single polypeptide of multipass nature. They crosses the lipid bilayer seven times and are therefore sometimes called *serpentine* receptors. An example of GPCR that responds to chemical signals is the *β-adrenergic receptor.* This protein receptor binds a hormone, epinephrine (also called adrenaline).

GPCRs are linked to trimeric G proteins (simply G-protein). G proteins are **GTPase switch proteins.** There are two classes of GTPase switch proteins: **trimeric G proteins** and **monomeric G protein.** These proteins are turned *on* when bound to GTP and turned *off* when bound to GDP. In the absence of a signal, G protein is bound to GDP, signals activate the release of GDP, and the subsequent binding to GTP over GDP is favoured by the higher concentrations of GTP in the cell. The intrinsic GTPase activity of these GTP-binding proteins hydrolyzes the bound GTP to GDP and P_i, thus converting the active form back to the inactive form.

G-proteins are attached to the cytoplasmic face of plasma membrane. Ligand binding to GPCR leads to activation of an associated signal-transducing G protein. All G proteins contain three subunits: α, β and γ. α and γ subunits of G-proteins are anchored to the membrane lipids covalently. The β subunit is always complexed with the γ subunit, forming the heterodimeric structure $G_{\beta\gamma}$. The α subunit is a GTPase switch protein that alternates between an active state with bound GTP and an inactive state with bound GDP. Binding of a trimeric G protein to an activated receptor leads to dissociation of GDP, binding of GTP to G_α, and dissociation of G_α-GTP from$G_{\beta\gamma}$. G_α-GTP and $G_{\beta\gamma}$ can specifically interact with effector proteins leading to changes in their activity. The intrinsic GTP ase activity of G_α inactivates G_α-GTP by catalyzing GTP hydrolysis. G_α-GDP then dissociates from its effector and reassociates with $G_{\beta\gamma}$. The GTPase activity of the G_α determines the length of time that the signal remains on. In many cases, a protein termed RGS (Regulator of G protein Signaling) accelerates GTP hydrolysis by the G_α subunit.

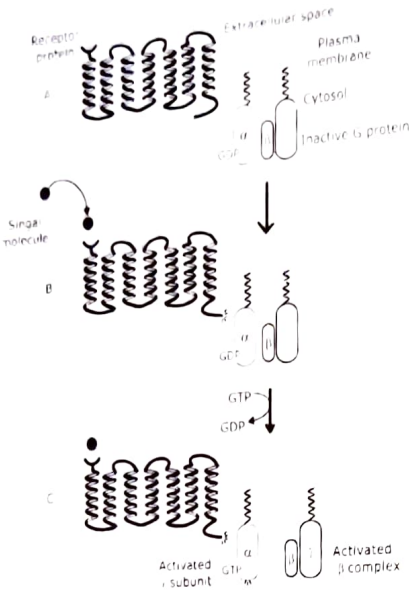


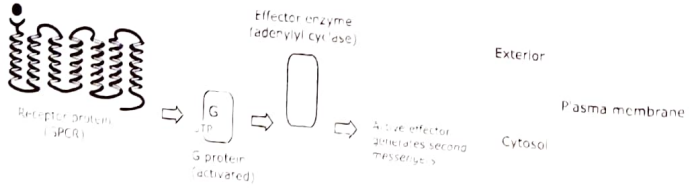
Figure : GPCR typically have seven membrane-spanning domains. In the absence of signal molecule, the heterotrimeric G-protein complex is in an inactive GDP-bound form. On binding of hormone to the receptor, there is a conformational change of the receptor and activation of the G-protein complex. This results from the exchange of GDP with GTP on the α subunit, after which α and βγ dissociate. The α or βγ subunit binds to and activates the effector.

There is a large variety of G proteins which are activated by a wide variety of receptors. The activation of an individual G protein may cause it to stimulate or to inhibit a particular effector. Two of the classic G proteins are G_s, which stimulates adenylate cyclase, and G_q, which stimulates phospholipase-C-β.

Table : Classes of trimeric G proteins and their functions

Family members	*Action mediated by*	*Functions*
G_s	α	activates adenylyl cyclase; activates Ca^{2+} channels
G_{olf}	α	activates adenylyl cyclase in olfactory sensory neurons
G_i	α	inhibits adenylyl cyclase
	βγ	activates K^+ channels
G_0	βγ	activates K^+ channels; inactivates Ca^{2+} channels
	α and βγ	activates phospholipase C-β
G_t	α	activates cyclic GMP phosphodiesterase
G_q	α	activates phospholipase C-β

In the case of Gs, the Gsα subunit activates effector, adenylate cyclase. Because this G protein is involved in enzyme *activation*, it is called stimulatory G protein (Gs). The Gsα subunit binds and hydrolyzes GTP and activates plasma-membrane-bound enzyme adenylate cyclase. Adenylate cyclase is a large multipass transmembrane protein with its catalytic domain on the cytosolic side of the plasma membrane. Activated adenylate cyclase synthesizes cAMP from ATP.



A best-studied example of a GPCR that responds to epinephrine is the β-*adrenergic receptor.* β-adrenergic receptor is coupled to G proteins Gs, which activate *adenylate cyclase.* Once activated, adenylate cyclase catalyzes synthesis of cAMP. The epinephrine-stimulated increase in cAMP and subsequent activation of protein kinase A enhance the conversion of glycogen to glucose 1-phosphate by stimulating glycogen degradation **protein kinase A** promotes glycogen degradation indirectly by phosphorylating and thus activating a kinase, glycogen phosphorylase kinase, that in turn, phosphorylates and activates glycogen phosphorylase, the enzyme that degrades glycogen. The entire process is reversed when epinephrine is removed and the level of cAMP drops.

The effector protein for some of G proteins is a Na^+ or K^+ channel. For example, the *muscarinic acetylcholine receptors* in cardiac muscle are GPCRs. These receptors are linked via a trimeric G protein (Gi) to K^+ channels. Binding of acetylcholine triggers activation of the Giα subunit that leads to opening of associated K^+ channels. The subsequent efflux of K ions causes hyperpolarization of the plasma membrane.

cAMP and Protein Kinase A

cAMP directly activates *cAMP-dependent protein kinase* termed protein kinase A, PKA. cAMP also induces opening of certain types of ion channels in the plasma membrane of some highly specialized cells. PKA catalyzes the transfer of the terminal phosphate group from ATP to specific serines or threonines of selected target proteins, thereby regulating their activity. In the inactive state, PKA consists of a complex of two catalytic subunits and two regulatory subunits. The binding of cyclic AMP to the regulatory subunits alters their conformation, causing them to dissociate from the complex. The released catalytic subunits are thereby activated to phosphorylate specific substrate protein molecules. The active catalytic subunits phosphorylate a number of target proteins (cytosolic as well as nuclear) on serine and threonine residues.

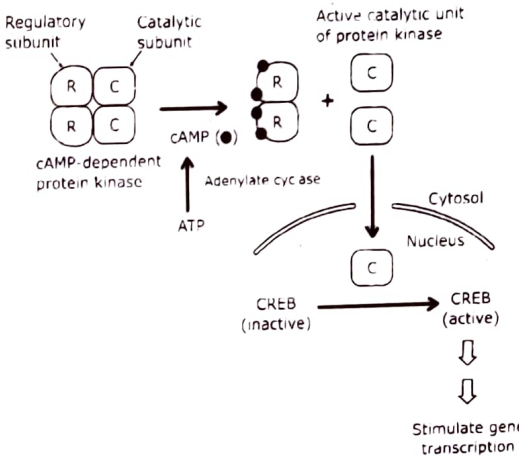


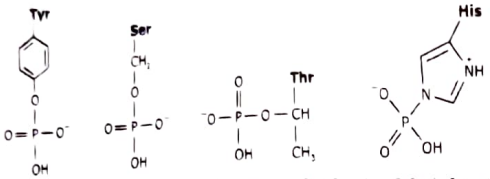
Figure : The activation of cyclic-AMP-dependent protein kinase (PKA). The binding of cyclic AMP to the regulatory subunits induces a conformational change, causing these subunits to dissociate from the complex, thereby activating the catalytic subunits. Each regulatory subunit has two cyclic-AMP-binding sites. Once the catalytic subunits are freed and active, they can migrate into the nucleus (where they can phosphorylate gene regulatory proteins), while the regulatory subunits remain in the cytoplasm.

One of the major nuclear targets for PKA is the transcription factor **CREB** (cAMP Response Element Binding Protein), which is activated by phosphorylation. Phosphorylation at a single *Ser* residue greatly increases the activity of CREB bound to the **CRE** (cAMP Response Element), which is found in genes whose transcription is induced by cAMP.

Protein Kinases

Activation of all cell-surface receptors leads to changes in protein phosphorylation through the activation of protein kinase. There are many types of protein kinases involved in signal transduction. They all have the same basic catalytic activity: they add a phosphate group to an amino acid in a target protein. The phosphate is provided by ATP. Protein kinases can be classified by the types of amino acids that they phosphorylate in the protein target. Four groups of protein kinases are distinguished by the types of amino acid targets :

- Protein serine/threonine kinases (phosphorylate either serine or threonine in the target protein)
- Protein tyrosine kinases (phosphorylate tyrosine in the target protein)
- Dual specificity kinases (phosphorylate target proteins on either tyrosine or serine/threonine)
- Histidine kinases (phosphorylate target proteins on histidine)



Protein kinases may be membrane bound or cytosolic on the basis of their location in the cell. *Cytosolic protein kinases* are most often Ser/Thr protein kinases. The activity of protein kinases is opposed by the activity of protein *phosphatases*, which remove phosphate groups from specific substrate proteins.

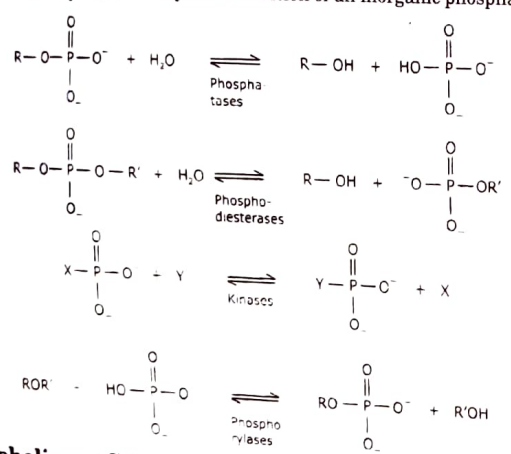
Differences betweeen *phosphatase, phosphodiesterase, kinase* and *Phosphorylases*

A **phosphatase** is an enzyme that removes a phosphate group from its substrate by hydrolysing phosphoric acid monoesters into a phosphate ion and a molecule with a free hydroxyl group.

A **phosphodiesterase** is an enzyme that breaks a phosphodiester bond.

A **kinase,** alternatively known as a *phosphotransferase,* is an enzyme that transfers phosphate groups from high-energy donor molecules, such as ATP, to substrates.

A **phosphorylase** is an enzyme that catalyze the addition of an inorganic phosphate to an acceptor.



G_q and Phospholipase C-β

Many G-protein-coupled receptors exert their effects via G proteins like G_q that activate the plasma-membrane-bound enzyme **phospholipase** C-β (PLC-β). The activated phospholipase C cleaves phosphatidylinositol 4,5-bisphosphate (PIP_2) to generate two secondary messenger : *inositol 1,4,5-trisphosphate* (IP_3) and *diacylglycerol* (DAG). The membrane phospholipid PIP_2 is a minor component of the plasma membrane, localized to the inner leaflet of the phospholipid bilayer. One secondary messenger

diacylglycerol remains associated with the plasma membrane, the other second messenger produced by PIP_2 cleavage, IP_3, is a small polar molecule that is released into the cytosol, where it acts to signal the release of calcium ion from endoplasmic reticulum. IP_3 acts to release calcium ion from the endoplasmic reticulum by binding to receptors that are ligand-gated calcium ion channels. As a result, cytosolic calcium ion levels increase, which affects the activities of a variety of target proteins, including protein kinases and phosphatases. Diacylglycerol together with phosphatidylserine and calcium ion, helps to activate the enzyme **protein kinase C** (PKC), which is recruited from the cytosol to the cytosolic face of the plasma membrane. When activated, PKC phosphorylates specific serine or threonine residues on target proteins.

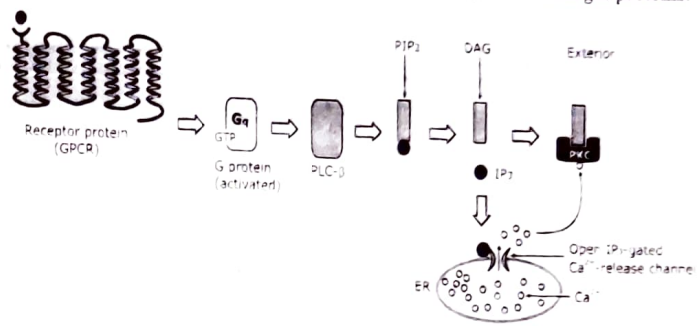


Figure : The activated receptor binds to a specific trimeric G protein (G_q), causing the α subunit to dissociate and activate phospholipase C-β. Two intracellular mediators are produced when PIP_2 is hydrolyzed by phospholipase C-β : IP_3, which diffuses through the cytosol and releases Ca^{2+} ' from the ER, and DAG, which remains in the membrane and helps activate the enzyme protein kinase C.

Mode of Action of Cholera and Pertussis Toxin

Cholera toxin is a protein complex secreted by the bacterium *Vibrio cholerae.* The cholera toxin is an oligomeric complex made up of six protein subunits: a single copy of the A subunit, and five copies of the B subunit.

The A subunit has catalytic property that ADP-ribosylates G proteins. The pentameric protein binds to the surface of the intestinal epithelium. The A subunit ribosylates the *Arg* residue of alpha subunit of Gs protein. ADP ribose is provided by the intracellular NAD^+ This ADP ribosylation alters the A subunit so that it can no longer hydrolyze its bound GTP, causing it to remain in an active state that stimulates adenylyl cyclase indefinitely. The resulting prolonged elevation in cyclic AMP levels within intestinal epithelial cells causes a large efflux of chloride ion and water into the gut, thereby causing the severe diarrhea that characterizes cholera.

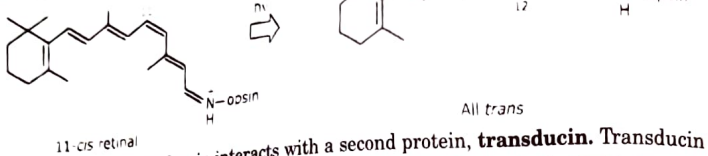
Pertussis toxin, an exotoxin that causes pertussis (whooping cough), catalyzes the ADP ribosylation of the *Cys* residue of α subunit of G_i.G_i is an inhibitory G protein. In activated state $G\alpha_i$ binds GTP and dissociate from the βγ complex. The released α_i inhibits the adenylate cyclase.The ADP ribosylation of $G\alpha_i$ prevents the Gi complex from interacting with receptors. So the complex remains bound to GDP and is unable to inhibit adenylate cyclase.

Calcium ion / Calmodulin Complex

Many of the effects of calcium ion are mediated by the protein calmodulin, which is activated by calcium ion binding when the concentration of cytosolic calcium ion increases, calcium ion/calmodulin then binds to a variety of target proteins, including protein kinases. Calcium activates the regulatory protein calmodulin, which stimulates many enzymes and transporters. Calmodulin (17 KDa protein) is a small, highly conserved calcium binding cytosolic acidic protein found in all eukaryotic cells. It binds up to four calcium ions and acts as an important intracellular receptor for regulatory calcium signals, calcium ion binds to calmodulin in a cooperative fashion. Thus a small change in the level of cytosolic calcium ion leads to a large change in the level of active calmodulin. As it binds calcium, calmodulin undergoes conformational changes which can increase its affinity for target proteins. It acts both directly, through interaction with key target enzymes, and indirectly, via specific kinases. Calmodulin participates in the regulation of several biological processes including energy and biosynthetic metabolism, cell motility, exocytosis, cytoskeletal assembly, and intracellular modulation of both cAMP and calcium concentrations.

Rhodopsin and G_t

Vision depend on G-protein-coupled receptors that regulate cyclic-nucleotide-gated ion channels. The first step in vision is absorption of light energy by rhodopsin. **Rhodopsin** is an integral protein with seven membrane-spanning alpha helices, present in the rod cells. The light-absorbing pigment (chromophore) 11-cis-retinal is covalently attached to opsin, the protein component of rhodopsin. When a photon is absorbed by the retinal component of rhodopsin, the energy causes a photochemical change; **11-cis-retinal** is converted to **all-trans-retinal**. This change in the structure of the chromophore causes conformational changes in the rhodopsin molecule.



In its excited conformation, rhodopsin interacts with a second protein, **transducin.** Transducin (G_t) is a heterotrimeric GTP-binding protein. It can bind either GDP or GTP. In the dark, GDP is bound, all three subunits of the protein. (α, β and γ) remain together, and no signal is sent. When rhodopsin is excited by light, it interacts with transducin, catalyzing the replacement of bound GDP by GTP from the cytosol. Transducin then dissociates into $G_{t\alpha}$ and $G_{t\beta\gamma}$ and the $G_{t\alpha}$-GTP carries the signal from the excited receptor to the effector molecule *cGMP phosphodiesterase* (PDE). PDE enzyme converts cyclic GMP to 5′-GMP. The cyclic GMP (cGMP) levels in the cytosol fall. This drop in cyclic GMP concentration leads to a decrease in the amount of cyclic GMP bound to the plasma membrane Na^+ channels, allowing more of these highly cyclic-GMP-sensitive channels to close. Closure of these channels inhibits the influx of Na^+. These channels are kept open in the dark by cyclic GMP that has bound to them. In this way light causes a hyperpolarization rather than a depolarization of the plasma membrane. So, the activation of rhodopsin molecules by light results in hyperpolarization.

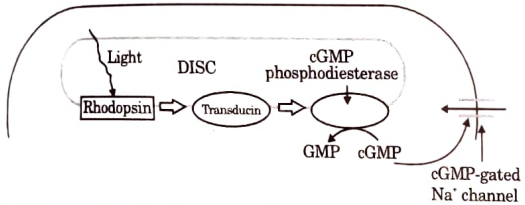


Figure : Light activates rhodopsin, which activates transducin to bind GTP. This activates phosphodiesterase, which catalyzes the conversion of cGMP to GMP. The resulting decrease in the cytoplasmic cGMP concentration causes cGMP-gated ion channels to close.

A number of mechanisms operate in rods to allow the cells to revert quickly to a resting, dark state. First, the Na^+ channels that close in response to light are also permeable to calcium ion, so that when they close, the normal influx of calcium ion is inhibited, causing the calcium ion concentration in the cytosol to fall. The decrease in calcium ion concentration stimulates guanylyl cyclase to replenish the cyclic GMP, rapidly returning its level to where it was before the light was switched on. A specific calcium ion-sensitive protein mediates the activation of guanylyl cyclase in response to a fall in calcium ion levels. In contrast to calmodulin, this protein is inactive when calcium ion is bound to it and active when it is calcium ion-free.

Second, a *rhodopsin-specifie kinase,* a serine kinase, phosphorylates the cytosolic tail of activated rhodopsin and partially inhibits the ability of the rhodopsin to activate transducin. An inhibitory protein called **arrestin** then binds to the phosphorylated rhodopsin, further inhibiting rhodopsin's activity. At the same time as rhodopsin is being shut off, a *regulator of G protein signaling* (RGS) protein, which acts like GTPase activating protein, binds to activated transducin, stimulating the transducin to hydrolyze its bound GTP to GDP, which returns transducin to its inactive state.

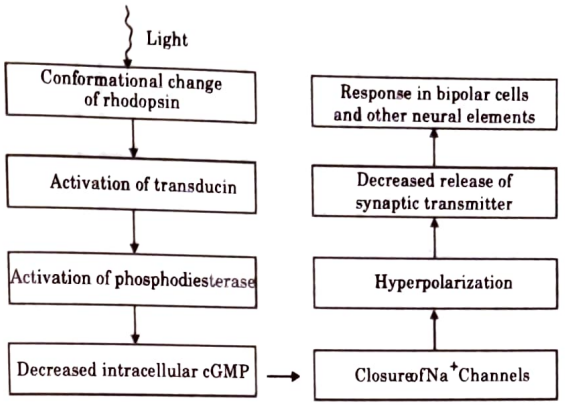
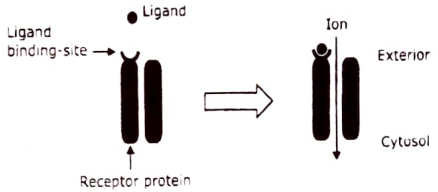


Figure : Sequence of events involved in phototransduction in rods.

Ion-Channel-Linked Receptors

Ion-channel linked receptors, also known as *transmitter-gated ion channels* or *ionotropic receptors,* are involved in rapid synaptic signaling between electrically excitable cells. This type of signaling is mediated by a small number of neurotransmitters that transiently open or close an ion channel formed by the protein to which they bind, briefly changing the ion permeability of the plasma membrane and thereby the excitability of the postsynaptic cell. The ion-channel-linked receptors belong to a large family of homologous, multipass transmembrane proteins.



Enzyme-Linked Receptors

Enzyme-linked receptors are a second major type of cell-surface receptor. Like G-protein-linked receptors, enzyme-linked receptors are transmembrane proteins with their ligand-binding domain on the outer surface of the plasma membrane. Instead of having a cytosolic domain that associates with a monomeric G protein, however, their cytosolic domain either has an intrinsic enzyme activity or associates directly with an enzyme.

Several classes of enzyme-linked receptors have thus far been identified like Receptor tyrosine kinases (the most numerous of the enzyme-linked receptors), Tyrosine-kinase-associated receptors, Receptor like tyrosine phosphatases, Receptor serine/threonine kinases, Receptor guanylyl cyclases and others.

A. Receptor tyrosine kinases

The receptor tyrosine kinases (RTKs) are the second major type of cell-surface receptors. The ligands for RTKs may be insulin, Epidermal growth factor (EGF), Transforming growth factor-α (TGFα), Platelet-derived growth factor (PDGF), Fibroblast growth factor (FGF) and Colony stimulating factor-1 (CSF-1).

All RTKs comprise an extracellular domain containing a ligand-binding site, a single hydrophobic transmembrane α helix, and a cytosolic domain that includes a region with protein-tyrosine kinase activity. Most RTKs are monomeric, and ligand binding to the extracellular domain induces formation of receptor dimers.

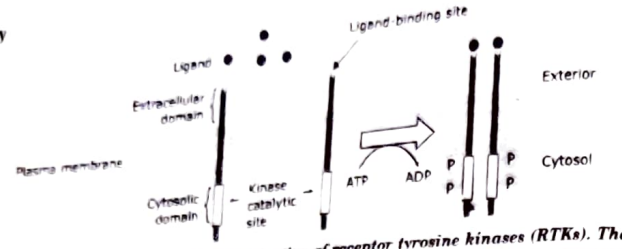


***Figure :** General structure and ligand-induced activation of receptor tyrosine kinases (RTKs). The cytosolic domain of RTKs contains a protein tyrosine kinase catalytic site. Ligand binding causes a conformational change that promotes formation of a functional dimeric receptor, bringing together two intrinsic or associated kinases, which then phosphorylate each other on a tyrosine residue in the cytosolic domain. The activated kinase then phosphorylates other tyrosine residues in the receptor's cytosolic domain. The resulting phosphotyrosines function as docking sites for various signal-transduction proteins.*

Ligand binding leads to activation of the intrinsic protein-tyrosine kinase activity of the receptor and *autophosphorylation* of tyrosine residues in its cytosolic domain. The activated receptor also can phosphorylate other protein substrates. Several types of proteins with which the activated receptor may interact. Some of these proteins are a GTPase activating protein (GAP), phospholipase C-γ (PLC-γ), and the *Src-like* nonreceptor protein tyrosine kinases. Three major pathways can transduce a signal from the activated RTK :

1. Ras-MAP kinase pathway;
2. IP_3/DAG pathway;
3. PI-3 kinase pathway.

Ras-MAP Kinase Pathway

Ligand-induced activation of an RTK stimulates its tyrosine kinase activity, which subsequently stimulates several signal-transduction pathways including the *Ras-MAP kinase pathway. Ras* is an intracellular monomeric GTPase switch protein and functions in transducing signals from many different RTKs. *Ras* belongs to Ras superfamily which also includes Rho, Rab and Ran.

Ras superfamily

Subfamily	Function
Ras	Regulates cell growth through serine-threonine protein kinases
Rho	Reorganizes cytoskeleton through serine-threonine protein kinases
Rab	Plays a key role in endocytotic pathways
Ran	Functions in the transport of RNA and protein into and out of the nucleus

Ras cycles between an inactive GDP-bound form and active GTP-bound form. Ras cycling requires the assistance of two proteins, GEF (guanine nucleotide exchange factor) and GAP (GTPase-activating protein). Ras activation is accelerated by a protein called GEF, which binds to the Ras-GDP complex, causing dissociation of the bound GDP. Because GTP is present in cells at a higher concentration than GDP, GTP binds spontaneously to *empty* Ras molecules, with release of GEF. The average lifetime of a GTP bound to Ras is about 1 minute. GAP, which binds to Ras-GTP and accelerates its intrinsic GTPase activity. Binding of a ligand to an RTK lead to activation of Ras. Two cytosolic proteins - adaptor protein and GEF provide the key links between RTK and Ras.

Adaptor proteins bind to an activated receptor, and then other proteins(s) bind to them, and may therefore become substrates for the receptor. Adaptor has no catalytic activity of its own, but serves merely to bring other proteins to the receptor. The cytosolic adaptor protein GRB2 binds to a specific phosphotyrosine on an activated, ligand-bound receptor and to the cytosolic GEF, Sos protein. Sos then promotes formation of active Ras GTP. An Src homology domain, SH2 *domain*, in GRB2 binds to a specific phosphotyrosine residue in the activated receptor. The SH2 domain was originally identified as a conserved amino acid sequence in the non-catalytic region of tyrosine kinases encoded by viral oncogenes such as v-src. GRB2 also contains two SH3 *domains*, which bind to and activate Sos. SH3 binds to proline-rich motifs.

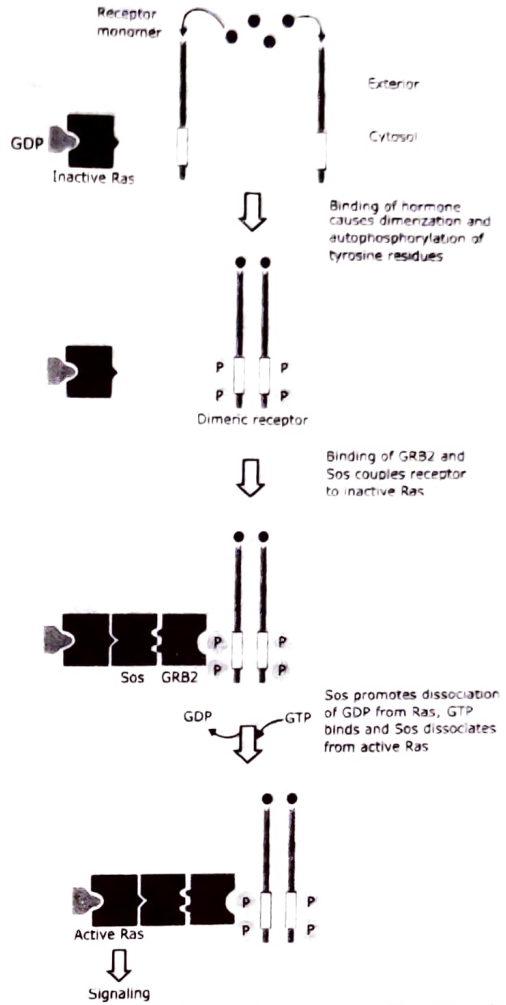


***Figure :** Activation of Ras. An adapter protein, GRB2, and guanine nucleotide-exchange factor, Sos link most activated receptor tyrosine kinases to Ras. The SH2 domain in GRB2 binds to a phosphotyrosine in activated RTKs, while its two SH3 domains bind Sos, thereby bringing Sos close to membrane-bound RasGDP and activating its nucleotide exchange activity. Binding of Sos to inactive Ras causes a large confor-mational change that permits release of GDP and binding of GTP, forming active Ras. GAP, which accelerates GTP hydrolysis, is localized near RasGTP by binding to activated RTKs.*

MAP Kinase Pathways

Activation of a receptor tyrosine kinase may itself trigger multiple pathways: for example, activation of EGF receptor activates the Ras dependent pathway as well as Ras-independent pathways involving second messengers. The activation of Ras triggered by activated receptor tyrosine kinases are short-lived. Ras activates a downstream serine/threonine phosphorylation cascade that includes a **MAP-kinase** (mitogen-activated protein kinase). Activated Ras induces a *kinase cascade* that culminates in activation of MAP kinase. This serine/threonine kinase, which can translocate into the nucleus, phosphorylates many different proteins -including transcription factors that regulate expression of important cell-cycle and differentiation-specific proteins. Activation of MAP kinase in two different cells can lead to similar or different cellular responses, as can activation in the same cell by stimulation of different RTKs.

Many serine/threonine kinases participate in this phosphorylation cascade, but three of them constitute the core module of the cascade. The last of the three is called MAP-kinase. Downstream serine/threonine phosphorylation cascade follow the following pattern

1. Activated Ras binds to the N-terminal domain of **Raf** (MAP-kinase-kinase-kinase), a Ser/Thr kinase.
2. Raf activity leads to the activation of **Mek** (MAP-kinase-kinase). Raf directly phosphorylates Mek, which is activated by phosphorylation on two serine residues. Mek is an unusual enzyme with dual specificity, which can phosphorylate both threonine and tyrosine. Its target is the **Erk** (MAP kinase).
3. MEK phosphorylates and activates MAP kinase. The MAP kinases are serine/threonine kinases. After this point in the pathway, all the activating events take the form of serine/threonine phosphorylations. MAP kinase phosphorylates many different proteins, including nuclear transcription factors, that mediate cellular responses.

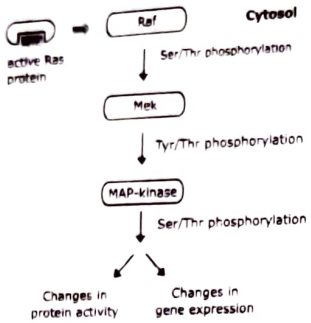


Figure : MAP kinase pathway. Activated Ras triggers the downstream kinase cascade, culminating in activation of MAP kinase. Interaction of the Raf N-terminal regulatory domain with RasGTP leads to activation of Raf kinase activity. Note that in contrast to many other protein kinases, activation of Raf does not depend on phosphorylation. Hydrolysis of RasGTP to RasGDP releases active Raf, which phosphorylates and thereby activates Mek. Active Mek then phosphorylates and activates MAP kinase, another serine/threonine kinase. MAP kinase phosphorylates many different proteins, including nuclear transcription factors, that mediate cellular responses.

The ultimate effect of the MAP kinase pathway is a change in the pattern of transcription. In the classic MAP kinase pathway, MAP kinase moves into the nucleus, where it phosphorylates target transcription factors. In alternative pathway, it phosphorylates a cytoplasmic factor; this may be a transcription factor that then moves to the nucleus or a protein that regulates a transcription factor (for example, by releasing it to go to the nucleus).

IP3/DAG Pathway

Many RTKs can initiate the P_3/DAG pathway by activating another isoform of phospholipase C, the phospholipase C-γ. The SH2 domains of PLC-γ bind to specific phosphotyrosines of the activated receptors, thus positioning the enzyme close to its membranebound substrate PIP2. Membrane-bound PLC-γ after activation cleaves PIP2 to generate two second messengers, DAG and IP3. Signaling via the IP3/DAG pathway leads to an increase in cytosolic calcium and to activation of protein kinase C.

PI-3 Kinase Pathway

In addition to initiating the IP_3/DAG pathway, some activated RTKs also initiate another phosphoinositide pathway, the *PI-3 kinase pathway*. PI-3 kinase binds to phosphotyrosine residues in the cytosolic domain of activated RTKs. PI-3 kinase catalyzes the formation of PI 3,4-bisphosphate or PI 3,4,5-trisphosphate which act docking sites for various signal-transducing proteins.

A primary binding target of PI 3,4,5-trisphosphate is **protein kinase B** (PKB), a serine/threonine kinase. Besides its kinase domain, PKB contains a PH domain that tightly binds the 3-phosphate in PI 3,4,5-trisphosphate. In an inactive form protein kinase B present in the cytosol. In case of rise in PI 3,4,5-trisphosphate, protein kinase B binds to them and is localized at the cell surface. PI 3,4,5-trisphosphates also recruits kinase, PDK1, to the plasma membrane via binding of its PH domain. Active PDK1 causes phosphorylation of protein kinase B. Fully activated protein kinase B then dissociates from the plasma membrane and phosphorylate its many target proteins.

Insulin Signaling Pathways

The insulin receptor is a dimeric (linked together by disulphide bonds) receptor tyrosine kinase. Binding of the hormone activates the receptor, which then phosphorylates itself and other proteins at various tyrosine residues. The phosphorylated insulin receptor next phosphorylates *insulin receptor substrates* (IRS) on tyrosine residues. Phosphorylated IRS binds to the SH2 domains of a variety of proteins that are directly involved in mediating different effects of insulin.

Phosphorylated IRS molecule acts as docking site for the lipid kinase, PI-3 kinase that leads to the activation of protein kinase B (PKB). Other signal transduction proteins interact with IRS including Grb2, an adaptor protein that contains SH3 domains, which in turn associates with the Sos and elicits activation of the G protein, **Ras.** Ras activates the MAP kinase pathway leading to mitogenic responses. These pathways act in a coordinated manner to regulate glucose, lipid and protein metabolism.

B. Tyrosine kinases activated receptors (cytokine receptors)

Rather than possessing intrinsic enzymatic activity, many enzyme linked receptors act by stimulating intracellular protein-tyrosine kinases with which they are noncovalently associated. This family of receptors includes the receptors for most cytokines (e.g., interleukin-2 and erythropoietin) and for some polypeptide hormones (e.g., growth hormone) whose cytosolic domains are closely associated with a member of a family of cytosolic protein tyrosine kinases, the *JAK kinases*. Like RTK, the cytokine receptors contain N-terminal extracellular ligand-binding domains, single transmembrane *alpha* helices, and C-terminal cytosolic domains. However, the cytosolic domains of the cytokine receptors are devoid of any known catalytic activity. Instead, the cytokine receptors function in association with nonreceptor protein-tyrosine kinases, which are activated as a result of ligand binding.

The first step in signaling from cytokine receptors is thought to be ligand-induced receptor dimerization and cross-phosphorylation of the associated non-receptor protein-tyrosine kinases. These activated kinases then phosphorylate the receptor, providing phosphotyrosine-binding sites for the recruitment of downstream signaling molecules. The nonreceptor protein-tyrosine kinases associated with the cytokine receptors fall into two major families. Many of these kinases are members of the *Src* family, which consists of Src and closely related proteins. Src possess protein-tyrosine kinase activity. In addition to Src family members, the cytokine receptors are associated with nonreceptor protein-tyrosine kinases belonging to the Janus kinase, or **JAK,** family. Members of the JAK family appear to be universally required for signaling from cytokine receptors. In contrast, members of the Src family play key roles in signaling from antigen receptors on B and T lymphocytes but do not appear to be required for signaling from most cytokine receptors.

JAK-STAT Pathway

The JAK kinases take their name (originally Janus kinases) from the characteristic presence of two kinase domains in each molecule. Several members of the family are known (JAK1,2,3, etc.); each associates with a specific set of cytokine receptors. The JAK kinases are tyrosine kinases whose major substrates are transcription factors called STATs (Signal transducers and activators of transcription). There are several types of STATs; each STAT is phosphorylated by a particular set of JAK kinases. All STAT proteins contain an N-terminal SH2 domain that binds to a phosphotyrosine in the receptor's cytosolic domain, a central DNA-binding domain, and a C-terminal domain with a critical tyrosine residue.

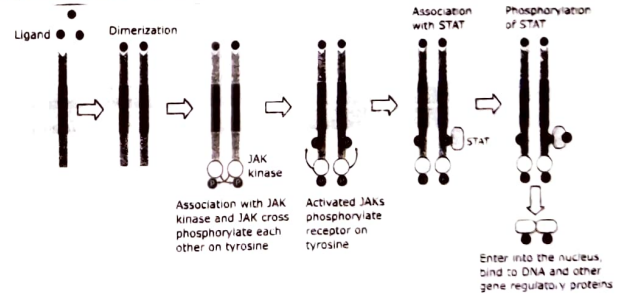


Figure : Cytokine receptors associate with and activate JAK kinases. STATs bind to the complex and are phosphorylated. They dimerize and translocate to the nucleus. The complex binds to DNA and activates transcription.

Once a STAT is bound to the receptor, the C-terminal tyrosine is phosphorylated by an associated JAK kinase. STAT phosphorylation leads to the formation of both homodimers and heterodimers. The basis for dimerization is a reciprocal interaction between an SH2 domain in one subunit and a phosphorylated Tyr in the other subunit. The STAT dimers move into the nucleus, where they bind to specific *enhancer* sequences controlling target genes.

Nitric oxide

Nitric oxide (NO) is a paracrine signaling molecule. Like the steroid hormones, NO is able to diffuse directly across the plasma membrane of its target cells. Nitric oxide is synthesized from the amino acid *arginine* by the enzyme *nitric oxide synthase*. Once synthesized, NO diffuses out of the cell and can act locally to affect nearby cells. One well-characterized example of NO action is the dilation of blood vessels. The acetylcholine, a neurotransmitter acts on endothelial cells to stimulate NO synthesis. NO then diffuses to neighbouring smooth muscle cells where it interacts with iron bound to the active site of the enzyme guanylyl cyclase. This increases enzymatic activity, resulting in synthesis of the second messenger cyclic GMP, which induces muscle cell relaxation and blood vessel dilation.

NO ⇨ Guanylyl cyclases ⇨ cGMP ⇨ PKG

Effect of Viagra

Concentration of cGMP decreases because a specific phosphodiesterase convert cGMP to the inactive 5'-GMP. Sildenafil (Viagra) causes cGMP levels to remain high by inhibiting the activity of phosphodiesterase.

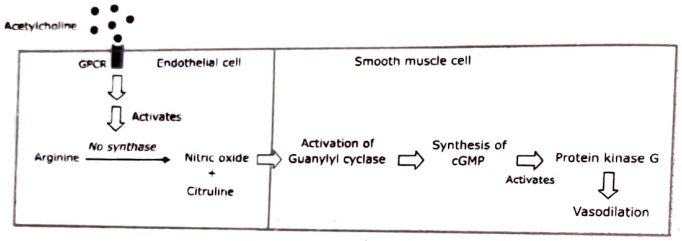


Fig. Action of nitric oxide.

Two-component signaling systems

Two-component signaling system is the most common form of signaling pathway that responds to extracellular events in bacteria and plants. A two component system consists of a sensor that is an auto-phosphorylating histidine kinase and an *effector* that can catalyze transfer of phosphate from the sensor to itself.

The sensor protein is a *histidine kinase* that is located in the membrane. It can be activated by binding a ligand that is in the extracellular medium. Activation causes the kinase to autophosphorylate. The reaction transfers the phosphate from ATP on to a histidine residue in the kinase.

The sensor interacts with an effector protein (also called a *response regulator*). The effector protein has two domains - regulatory domain and effector domain. The regulatory domain catalyzes transfer of the phosphate group from the histidine on the sensor to an aspartic acid residue in its own domain. This activates the effector domain. The usual end target of a two-component pathway is the regulation of gene transcription.

Chemotaxis in bacteria

Movement of organisms in specific directions in response to chemical stimulus is called *chemotaxis*. Chemotaxis in bacteria depends on a signaling pathway that terminates at the flagellar motor.

Bacterial flagellar motion is rotatory in nature. Membrane proteins MotA and MotB along with FliG create a **proton** channel that drives rotation of the flagellum. The rotation of flagella can be *clockwise* or *counterclockwise*. When the flagella rotate clockwise, forward motion ceases and the cells *tumble*. Whereas counterclockwise rotations of flagella impart forward motion *(run)*.

In the absence of a chemical gradient, bacteria *(E.coli)* move in a random fashion that includes runs, where the cell is swimming forward and tumbles when the cell stops. Following a tumble, the direction of the next run is random. However, if a gradient of a chemical attractant is present, these random movements become biased. As the bacteria senses that it is moving toward higher concentrations of the attractant runs become longer and tumbles less frequent. If the bacteria is sensing a repellent, the same general mechanism applies, although in this case it is the decrease in concentration of the repellent that promotes runs.

Bacterial chemotaxis is mediated by a transmembrane receptors and a phosphorylation relay system. It depends on a two component signaling pathway activated by *histidine kinase* associated receptors. The chemotaxis receptors are methylated during adaptation and so are also called *methyl-accepting chemotaxis proteins* (MCPs).

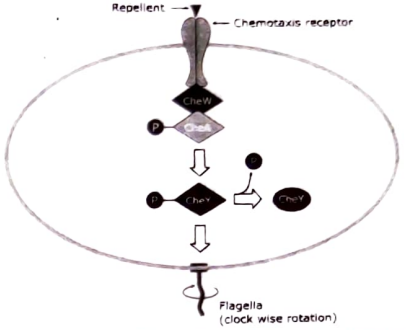


Fig. The two-component signalling pathway that enables chemotaxis receptors to control the flagellar motor during bacterial chemotaxis.

The phosphorylation relay system enables the chemotaxis receptors to control the flagellar motor. The binding of a repellent increases the activity of the receptor, which binds CheW (adaptor protein) and CheA (histidine autokinase), thereby stimulating CheA to phosphorylate itself on histidine. CheA quickly transfers its covalently bound, high-energy phosphate directly to aspartate of CheY to generate CheY-phosphate. The phosphorylated CheY dissociates from the receptor, diffuses through the cytosol and binds to the flagellar motor and causes it to rotate clockwise, resulting in *tumbling*. The binding of an attractant has the opposite effect. It decreases the activity of the receptor and therefore decreases the phosphorylation of CheA and CheY, which results in counterclockwise flagellar rotation and run. CheZ accelerates the dephosphorylation of CheY-phosphate, thereby inactivating it. A methyltransferase, CheR, catalyzes methylation of the MCP.

Chemotaxis proteins

1. CheA, a cytoplasmic sensor kinase
2. CheW, an adaptor protein linking the sensor protein with CheA
3. CheY, the response regulator controlling the flageller motor
4. CheZ, an Asp-specific protein phosphatase for signal termination
5. CheR, a methyltransferase catalyzing methylation of the MCP.

Cancer

Malignant Growth

Cells which have escaped normal control regulating its growth and division, producing a clone of dividing daughter cells which invade adjacent tissues and may interfere with their activities are called cancer cells. The cells which undergo rapid, abnormal and uncontrolled growth at the cost of remaining cells are called neoplastic cells. These proliferating cells form a mass of undifferentiated cells. This cells mass is called tumour or neoplasm or cancerous growth.

Cancer cells are characterized by three properties

(*i*) diminished or unrestricted control of growth

(*ii*) invasion of local tissues and

(*iii*) spread of metastasis, to other parts of the body.

Tumours

The tumours are classified as *benign* or *malignant*. Their study is known as *oncology*.

A. Benign Tumours

Tumours that remains localized in a specific area at the site of origin, forming a single mass enclosed in a capsule, are called *benign tumours*. Usually these are not fatal.

Benign tumours can be removed effectively with surgery. They usually do not reappear.

B. Malignant Tumours

The malignant tumours are *cancerous*. Their cells are poorly differentiated with high rate of mitosis, nuclear polymorphism and abnormal mitosis. Malignant tumours contain rapidly growing, actively moving cells that tend to invade other tissues. The movements of malignant cells from their original site to other body areas is called 'metastasis'. When metastasizing cells gain access to blood vessel, the circulating blood carries those cells throughout the body. When they lodge at distant body site, the cells continue to grow, forming new tumours. The rate of cell production in malignant cancerous tissue exceeds the rate of cell death and leads to tumour formation. These outgrow their blood supply and the rapidly increasing number of cells compress the small blood vessels. This is called *ischaemic necrosis*. Many malignant tumours grow progressively and relentlessly. The growth rate fluctuates greatly in different tumours. Cancers or malignant tumours are uncapsulated and invasive. These show excessive growth and infiltration into adjacent tissues. Most cancer cells show tiny pseudopods indicative of their invasive nature. The malignant tumours are usually poorly differentiated. Their cells may exhibit different degree of resemblance to the tissue cells from which these have originated. The partial loss of differentiation is termed as *dysplasia* and complete loss, so that the tumour no longer resembles the tissue of its origin is called *anaplasia*. Cancer cells can enter the cell cycle repeatedly and in this way they are immortal.

Cancer can be caused by physical, chemical or biogenic agents. There agents damage or alter DNA, so that cancer is truly a disease of the genome. Initiation and promotion are well-established phenomenon in chemical careinogenesis. Initiation probably damage DNA, but the mechanism involved in promotion remain to be clarified. Phorbolesters are much studied promoting agents, they activate protein kinase C, which exerts a veriety of effects.

Characteristics of Cancer Cells

(*i*) Cancer cells have abnormally enlarged nuclei containing abnormal number of chromosomes.

(*ii*) Nucleolus displays hypertrophy.

(*iii*) Endoplasmic reticulum and ribosomes are more abundant as compared to normal cells.

(*iv*) Cytoskeleton undergoes deploymerization and disintegration, and the cells become independent and their movements uncoordinated.

(*v*) Cytoskeleton undergoes deploymerization and disintegration, and the cells become independent and their movements uncoordinated.

(*vi*) Adhesiveness decreases.

(*vii*) Secrete large amounts of proteolytic enzyms which act on inert serum protein, plasminogen and form plasmin which removes many extrinsic proteins from cell surface and signals the cell to divide.

(*viii*) Negative surface charge increase.

(*ix*) Warburg (1920) suggested that aerobic respiration is depressed in tumour cells. Despite a normal oxygen up take, cancer cells tend to use several times the normal glucose requirement and produce lactic acid under aerobic conditions. This places a burden on the liver, which must use ATP to get rid of lactate.

(*x*) They have ability to induce local blood vessel formation (*angiogenesis*). It is required to bring nutrients and oxygen to a cancerous tumour whose growth is not contained in a capsule.

Cancer cells release a growth factor, that causes neighbouring blood vessels to branch into the cancerous tissue.

(*xi*) They have ability to spread (metastasis)

(*xii*) They loss cell cycle control.

(*xiii*) Heritability, transplantability, genetic mutability and loss of density dependent inhibition are other characteristics of cancer cells.

Types of Cancers

About 200 different types of cancers have been identified. These are broadly classified into following groups :

(*i*) **Carcinoma :** It includes tumours of brain, breast, skin, cervical region. These are derived from epithelial tissue, originating either from ectoderm or endoderm. These occur as solid tumours, located in the nervous tissue on body surface or in the associated glands. Carcinoma are the commonest cancer and may be

—Squamous carcinoma—cancer of epithelial tissue

—Adeno Carcinoma—cancer of glands.

(*ii*) **Sarcoma :** Sarcoma are cancers (tumours) of connective tissue, cartilage, bone and muscles which are mesodermal origin. Sarcoma are named according to the tissue in which these are formed :

Fibrosarcoma—Sarcoma of fibrous connective tissue

Liposarcoma—Cancer of Adipose tissue

Chondrosarcoma—Cancer of cartilage cells (chondroblast cells)

Osteosarcoma—Cancer of Bone

Leiomyo Sarcoma—Cancer of smooth muscles.

(*iii*) **Lymphoma :** Lymph nodes, bone marrow, liver & spleen produce excessive lymphocytes. Hodgkin's disease and 'mycosis fungoides' are examples of lymphoma.

(*iv*) **Leukaemias :** These include the neoplastic proliferation of leukocytes (white blood corpuscles-WBC) precursors in the bone marrow and are described as *mycloproliferative disorders* : These lead to increase in the number of leukocytes that infiltrate into various other tissues and organs, producing enlargement of organs or tumours and can be

⇒ Monocytic leukaemia

⇒ Megakaryocytic leukaemia

⇒ Erythraemic myclosis

⇒ Promyelocytic leukaemia

⇒ Lymphoblastic leukaemia

⇒ Myeloblastic leukaemia in white blood cell cancer

⇒ Melanomas—Cancer of pigment cells

⇒ Gilomas—Cancer of glial cells of brain

⇒ Glomangioma—A benign tumour that develops from an arteriovenous glomus (cluster of blood cells) of the skin.

Causes of Cancer

The process of conversion of a normal cell to malignancy is called *carcinogenesis* and the agents which cause this are called carcinogens. Two major classes of genes contribute to causing cancer. **Oncogenes** must be activiated to cause cancer. **Tumour suppressors**, which normally hold mitosis in check, must be inactivated or removed to eleminate control of the cell cycle and initiate cancer.

Oncogenes

Oncogenes are genes that normally activate cell division in specific situations. When an oncogene activates cell division in the wrong time or place, cancer may result. Oncogenes are not alien to the cell, they are normal, essential genes that have undergone a mutation. In its normal non-mutated state, it is a proto-oncogene, a gene that can be transformed into on oncogene.

The **src** gene is now known to be an *oncogene*. It is possible that the **src** gene is under the control of a viral promotor that has turned it on.

The oncogenes most frequently involved in human cancers belong to the **ras** gene family. An alteration of only a single nucleotide pair is sufficient to convert a normally functioning **ras** proto-oncogene to an oncogene. The **ras K** oncogene is found in about 25% of lung cancers, 50% of colon cancer and 90% of pancreatic cancer. The **ras N** oncogene is associated with leukaemias and lymphomas and both **ras** oncogenes are frequently found in thyroid cancers.

Activation of proto-oncogenes to oncogenes is achieved by at least five mechanisms (promoter and enhancer insertion, chromosomal translocation, gene amplification, point mutation).

Tumour Suppressor Genes

Tumour suppressor gene is so called because it prevents cancer from occurring. Researchers have identified about a half a dozen tumour suppressor genes. When these gene malfunction, a tumour results. Tumour suppressor genes are now recognized as key players in the genesis of cancer. The effect of oncogenes on cell growth has been compared to putting one's foot on the accelerator of an automobile, whereas the action of tumour suppressor genes resembles taking one's foot off the brakes. Important tumour suppressor genes include RBI and P 53, both of which are nuclear phosphoproteins and probably affect the transcription of genes involved in regulating events in the cell cycle. A genetic model for colorectal cancer invokes the interplay of tumour suppressor genes and the K-ras oncogene. Mutations in mismatch repair genes have been found to be associated with hereditary nonpolyposis colon cancer, and loss of responsiveness to the growth inhibitory effect of TGF β appears important at least in development of this type of tumour. A number of cancer susceptibility genes have been isolated. They include RB 1, P 53, BRCA 1, BRCA 2. Tumour progression reflects an instability of the tumour genome, probably due to at least in part to defects in DNA repair systems, activation to additional oncogenes, and inactivation to additional tumour suppressor genes.

The **RB tumour suppressor gene** was discovered by Alfred Knudson. RB tumour-suppressor gene has now been found to malfunction in cancers of breast, prostate, bladder. Loss of RB tumour suppressor gene through chromosomal deletion is particularly frequent in a type of lung cancer called small cell lung carcinoma.

It is now known that signaling proteins often regulate the transcription of genes whose products are needed for cell division, and this appears to be true of the RB protein. When a particular factor attaches to a receptor, the RB protein is activated. An active RB protein turns off the expression of a proto-oncogene, whose product initiates cell division. Translocations of genes may occur as result of mistakes during mitosis, or they may be caused by carcinogens or viral infection.

Another major tumour-suppressor gene is called P 53, a gene that is more frequently mutated in human cancers than any other known gene. It has been found that the P 53 protein acts an a transcription factor and as such is involved in turning on the expression of a gene called WAF 1 or Clp 1. The product of this gene is a cell cycle inhibitor. A number of cell proteins combine with the P 53 protein. When a protein comines with P 53, it is hard to tell whether it is activating P 53 or is being activated by P 53. This intricate situation is now being unraveled.

Discovering the normal function of the P 53 gene is a top priority in biomedical research so far, the normal form of gene is implicated in cell cycle control, is cell's expression of other genes, and in several metabolic processes. Understanding P 53 will open the door to understanding many types of cancers.

Carcinogens

While oncogenes and tumour-supperessors play a part in causing cancer, other possible causes include certain chemicals, nutrient deficiencies, and radiations.

Direct carcinogens cause cancer when they are applied to fibroblast cells growing in culture. Benzene is an example for this.

Procarcinogens are safe outside the body, but inside they are metabolized to produce intermediate compounds that cause cancer. They include certain organic dyes, cigarettes, nitrites & nitrates used to preserve processed meats.

Promoters are chemicals that make other carcinogens more powerful. They include alcohol, certain hormones and various chemicals in cigrette smoke.

Polycyclic hydrocarbons produce cancer at the site of local application. Aromatic amines like 2-naphthylamine and benzidine cause cancer of urinary bladder. A20-dyes cause liver cancer and amino-fluorenes also cause cancer of liver and urinary bladder. Alkylating agents like mustard gas, methylnitrosourea and nitrosamine and various inorganic compounds like asbestos fibres, arsenic, beryllium, nickel, cadmium, and chromates are all carcinogenic.

Radiations

X-ray, α, β and γ rays and ultraviolet light all induce tumours in man and animals. While passing through the tissue, these release energy which alters various macromolecules of the cell including nucleic acids. This causes increased rate of mutations in the irradiated cells.

Diagnosis And Treatment of Cancer

The traditional medical methods for treating cancers are quite direct and aggressive. Malignant neoplasm are removed surgically, burned with radiation, or poisoned chemically.

Chemotherapy uses toxic chemicals to kill rapidly dividing cells. Many of these drugs are altered forms of the nucleotides found in DNA. Magnetic Resonance Imaging (MRI) uses an electromagnet to detect tumours. Position Emission Tomography (PET) can reveal the altered metabolic activity of cancers and differentiate them from normal tissue. Therefore, the early diagnosis is the most important factor. Depending on the site, diagnosis is made by various means of which *biopsy is the most important one.* Other are use of devices for visualization of hollow organs, roentgenography including computerized axial tomography (CAT), mammography, ultrasound, cytology such as 'Papanicolaou test', and palpation for lumps. Some of these techniques and devices will demonstrate an increase in the size or change in the shape of an organ, but such alterations may be due to either a benign or malignant growth. Colon cancer can be spotted early with an at home hemoccult (hidden blood) test. Most women have a yearly Papsmear to detect Cervical cancer.

List of Some Important Tumour Viruses

Class	Members
(A) DNA viruses	
1. Papoviruses	Polyoma virus, SV 40 virus, human papilloma virus (e.g. HPV-16)
2. Adenoviruses	Adenoviruses 12, 18 & 31
3. Herpes virus	Epstein-Barr-virus, Herpes Simplex virus type-2 (HSV-2), Luck's disease virus, Marck's disease virus
4. hepadnovirus	hepatitis B virus
(B) RNA viruses	
1. Retroviruses type C	Murine sarcoma and Leukaemia viruses, Avian sarcoma & Leukaemia viruses
	Human T-cell leukaemia viruses I & II
2. Retrovirus type B	Mouse mammary tumour virus

Some Differences Between Oncogenes and Tumour Suppressor Genes

Oncogenes	Tumour Suppressor Genes
1. Mutation in one of the two alleles sufficient for activity, act dominant to wild type	Mutations in both alleles or a mutation in one followed by a loss of or reduction to homozygosity in the second
2. "Gain of function" of a protein that signals cells division	Loss of function of a protein
3. Mutation arises in somatic tissues, not inherited	Mutation present in germ cell (can be inherited) or somatic cells
4. Some tissue preference	Strong tissue preference (e.g. effect of RB II gene in retina)

Some New and Developing Cancer Therapies

Therapy	Example or Underlying Principle
1. Anti-angiogenesis	Angiostatin, endostatin & other molecules
2. Antisense therapy (measurement of protease specific antigen (PSA) is useful to monitor prostatic cancer recurrence or persistance after surgency). Increase of PSA shows metastases.	Directed against key oncogenes
3. Apotosis promoters	Develop agents that stimulate apoptosis selectively in tumour cells.
4. Cell cycle inhibitors	Develop agents capable of selectively inhibiting key steps of the cell cycle of cancer cells.
5. Differentiation stimulators	All trans-retinoic acid in the treatment of acute promyelocytic leukaemia.
6. Gene Therapy	Replace key tumour suppressor proteins (e.g. p 53, p RB) or mutant repair enzyme.
7. Farnesyl/transferase inhibitors	RAS proteins are attached to surface membranes via farnesyl, inhibiting its attachment inhibits function of **ras** proteins
8. Glycosylation inhibitors	Alter patterns of sugar on cells surface glycoproteins in order to inhibit metastasis.
9. Metastasis Inhibitors	Inhibitors of various proteases (e.g. metalloproteases) may inhibit metastasis
10. Monoclonal Antibodies	Directed against oncogene encoded receptors or adhesion molecules on the surface of tumour cells against the receptor
11. Prevention	Aspirin to help prevent colorectal cancer. Anti smoking measures.
12. Protein kinase Inhibitors	Many tumour cells have elevated amounts of various protein kinases involved in signaling etc.
13. Ribozymes	Can degrade RNA molecules (e.g. specific m RNAs) that are relatively unique to tumour cells.
14. Telomerase inhibitors	May act on tumour cells in which telomerase is active
15. Vaccines	Inject peptides from relatively "tumour- specific" proteins to boost the T-cell Immune response

EXERCISE – I

MCQ TYPE QUESTIONS

1. Besides giving out secretory vesicles, the Golgi apparatus is also concerned with the formation of
   (a) lysosomes
   (b) plastids
   (c) grana of chloroplasts
   (d) cell plates after cell division

2. The inner membrane of the mitochondria is usually highly convoluted forming a series of infolding known as
   (a) thylakoids (b) lamellae
   (c) cristae (d) grana

3. Centromere is required for
   (a) replication of DNA
   (b) chromosome segregation
   (c) poleward movement of chromosomes
   (d) cytoplasmic cleavage

4. The desmosomes are concerned with
   (a) cell division
   (b) cell adherence
   (c) cytolysis
   (d) cellular excretion

5. The prokaryotic flagella consists of
   (a) helical arranged protein molecule
   (b) "9 + 2" membrane enclosed structure
   (c) unit membrane enclosed fibre
   (d) protein membrane enclosed fibre

6. Organelle associated with glycosylation of protein is
   (a) ribosome
   (b) peroxisome
   (c) endoplasmic reticulum
   (d) mitochondria

7. Element necessary for middle lamella
   (a) Ca (b) Zn
   (c) K (d) Cu

8. In fluid mosaic model of plasma membrane
   (a) upper layer is non polar and hydrophilic
   (b) upper layer is polar amd hydrophobic
   (c) phospholipids form a bimolecular layer in middle part
   (d) proteins form a middle layer.

9. According to widely accepted "fluid mosaic model" cell membranes are semi-fluid, where lipids and integral proteins can diffuse randomly. In recent years, this model has been modified in several respects. In this regard which of the following statement is not correct?
   (a) proteins in cell membrane can travel within the lipid bilayer.
   (b) proteins can also undergo flip-flop movements in the lipid bilayer.
   (c) proteins can remain confined within certain domains of the membrane.
   (d) many proteins remain completely embedded within the lipid bilayer.

10. Which of the following statement regarding cilia is not correct?
   (a) cilia contain an outer ring of nine doublet microtubules surrounding two singlet microtubules.
   (b) the organised beating of cilia is controlled by fluxes of Ca2+ across the membrane
   (c) cilia are hairlike cellular appendages
   (d) microtubules of cilia are composed of tubulin

11. Which of the following statements regarding mitochondrial membrane is not correct ?
   (a) the outer membrane resembles a sieve
   (b) the outer membrane is permeable to all kinds of molecules
   (c) enzymes of the electron transfer chain are embedded in the outer membrane
   (d) the inner membrane is highly convoluted forming a series of infoldings

12. Select the wrong statement from the following
   (a) both chloroplasts and mitochondria have an internal compartment ,the thylakoid space bounded by the thylakoid membrane
   (b) both chloroplast and mitochondria contain DNA
   (c) the chloroplasts are generally much larger than mitochondria
   (d) both chloroplast and mitochondria contain an inner and outer membrane

13. Which is not a constituent of cell membrane?
   (a) glycolipids
   (b) proline
   (c) phospholipids
   (d) cholesterol

14. The two sub units of ribosome remain united at a critical ion level of
(a) magnesium (b) calcium
(c) copper (d) manganese

15. In germinating seeds fatty acids are degraded exclusively in the
(a) peroxisomes (b) mitochondria
(c) proplastids (d) glyoxysomes

16. Plasmodesmata are
(a) locomotory structures
(b) membranes connecting the nucleus with the plasmalemma
(c) connections between adjacent cells
(d) lignified cemented layers between cells

17. Cytoskeleton is made up of
(a) callose deposits
(b) cellulosic fibrils
(c) proteinaceous filaments
(d) calcium carbonate granules

18. Zonulaadherens is a kind of
(a) filament (b) desmosome
(c) membrane (d) mesosome

19. In nucleoplasm, a conspicuous body of spherical shape attached to a particular chromosome on a definite position is called -
(a) plasmid (b) karyolymph
(c) nucleolus (d) nuclear reticulum

20. Which of the following is responsible for the mechanical support, protein synthesis and enzyme transport?
(a) cell membrane
(b) mitochondria
(c) dictyosome
(d) Endoplasmic reticulum

21. Genes present in the cytoplasm of eukaryotic cells are found in
(a) mitochondria and inherited via egg cytoplasm
(b) lysosomes and peroxisomes
(c) golgi bodies and smooth endoplasmic reticulum
(d) plastids and inherited via male gamete

22. The nucleus is separated from the surrounding cytoplasm by a nuclear membrane, which is
(a) single layerd with pores
(b) Double layered with pores
(c) single layered without pores
(d) double layered without pores

23. Karyolymph is –
(a) nuclear pore (b) nuclear sap
(c) SPM membrane (d) none of these.

24. Oxysomes are present on
(a) cell wall of algae
(b) outer membrane of mitochondria
(c) Inner membrane of mitochondria
(d) none of the above

25. Protein packaging is done in
(a) golgi apparatus
(b) ribosomes
(c) endoplasmic reticulum
(d) nucleolus

26. Phagosomes and pinosomes are collectively referred to as
(a) lysosomes (b) glyoxysomes
(c) spherosomes (d) Endosomes

27. In the flagellar membrane which enzyme catalyses ATP ?
(a) Cytoplasmic dyenin (b) Asconicdyenin
(c) Kinesis (d) Myosin

28. Extension of plasma membrane in prokaryotes is
(a) ER (b) Mesosome
(c) Ribosome (d) None

29. Action of lysozyme is
(a) Physiological (b) Anatomical
(c) Morphological (d) None

30. In higher plants the shape of the chloroplast is
(a) Discoid (b) Cup-shaped
(c) Girdle shaped (d) Reticulate

31. The strength and rigidity of a cell wall is due to–
(a) suberin (b) lignin
(c) cellulose (d) pectin

32. Term basal body is associated with the development of
(a) Cilia and flagella (b) Cell plate
(c) Phragmoplast (d) Kinetochore

33. Lipid molecule in plasma membrane are arranged in
(a) scattered (b) series
(c) alternate (d) head parallel

34. Largest number of chloroplast is found in
(a) palisade tissue
(b) spongy tissue
(c) transfusion tissue
(d) bundle sheath cells

35. Cell membrane is ----------to ions
(a) permeable
(b) differentially permeable
(c) semipermeable
(d) impermeable

36. The prokaryotic cells are characterised by
(a) absence of nuclear membrane
(b) presence of distinct chromosomes
(c) absence of chromatin material
(d) presence of distinct nuclear membrane

37. A major breakthrough in the studies of cells came with the development of electron microscope. This is because
(a) the electron microscope is more powerful than the light microscope as it uses a beam of electrons which has wavelength much longer than that of photons
(b) the resolving power of the electron microscope is much higher than that of the light microscope
(c) the resolving power of the electron microscope is 200-350 nm as compared to 0.1 -0.2 nm for the light microscope
(d) electron beam can pass through thick materials, whereas light microscopy requires thin sections

38. Biological organisation starts with
(a) cellular level
(b) organismic level
(c) atomic level
(d) sub microscopic molecular level

39. Keeping in view the 'fluid mosaic model' for the structure of cell membrane, which one of the following statement is correct with respect to the movements of lipids and proteins from one lipid monolayer to the other(described as flip flop movement)?
(a) while proteins can flip-flop,lipids can't
(b) neither lipids, nor proteins can flip-flop
(c) both lipids and proteins can flip-flop
(d) while lipids can rarely flip-flop, proteins cannot.

40. Which of the following position is the base to define the formation place of cell plate at the end of nuclear division?
(a) Microtubulus
(b) Vesicles
(c) Endoplasmic reticulum
(d) Golgi body

41. What is not seen during mitosis in somatic cells?
(a) Spindle fibres
(b) Chromosome movement
(c) Disappearance of nucleolus
(d) Synapsis

42. Which of the following structure will not be common to mitotic cell of a higher plant?
(a) Cell plate (b) Centromere
(c) Centriole (d) Spindle fibre

43. The number of chromosomes after I phase of meiotic division in reduction division
(a) Remain unchanged (b) Become doubled
(c) Become halved (d) None of the above.

44. Meiosis can be observed in
(a) Root tips (b) Cambium
(c) Anther (PMC) (d) Pollen grains

45. During cell division chromosome attaches with spindles
(a) Kinetochore
(b) Centrosome
(c) Centriole
(d) Secondary constriction

46. Cell division cannot be stopped in which phase of the cell cycle
(a) G1- phase (b) G2 -phase
(c) S-phase (d) Prophase

47. The replication of centrioles occurs during
(a) Early prophase (b) Late prophase
(c) Late telophase (d) Interphase

48. Chromonemata start associating into bivalent chromosomes during
(a) Zygotene (b) Leptotene
(c) Pachytene (d) Diplotene

49. In the somatic cell cycle
(a) A short interphase is followed by a long mitotic phase
(b) G2 phase follows mitotic phase
(c) In G1 phase DNA content is double the amount of DNA present in the original cell
(d) DNA replication takes place in S-phase

50. The study of chromosomes at meiotic diplotene shows that
(a) The intimately paired chromosomes repel each other and begin to separate
(b) The pairing of homologous chromosomes which had initiated in the earlier stage is completed
(c) The homologous chromosomes remain united by chiasmata
(d) None of the above

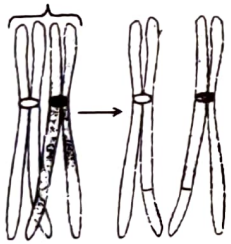
51. In which of the following stage, the chromosome is thin and like long thread
(a) Leptotene (b) Zygotene
(c) Pachytene (d) Diakinesis

52. At which phase of meiosis, the two cells, each with separated sister chromatids move towards opposite poles
(a) Metaphase I (b) Metaphase
(c) Anaphase I (d) Anaphase II

53. Which phase comes in between the G1 and G2 phase of cell cycle?
(a) M-phase (b) G_o - phase
(c) S-phase (d) Interphase

54. Calcium dependent kinases can control
(a) Cell cycle activities
(b) DNA replication
(c) Cell surface receptors
(d) Membrane structure

55. Given below is the representation of a certain event at a particular stage of a type of cell division. Which is this stage?



(a) Prophase I during meiosis
(b) Prophase II during meiosis
(c) Prophase of Mitosis
(d) Both prophase and metaphase of mitosis

56. One of the following serves as a temporary storage place for proteins and other compounds synthesized by endoplasmic reticulum
(a) Lysosomes
(b) Sphaerosome
(c) Microsomes
(d) Dictyosomes

57. Peroxisomes are involved in the synthesis of
(a) Glucose
(b) Fatty ardds
(c) Glycine and serine
(d) Nucleotides

58. One of the cell organelle is said to function as "trigger of cell division"
(a) Sphaerosome
(b) lysosome
(c) Lomasome
(d) Glyoxysome

59. Bacterial two component system includes
(a) Sensory kinase and response regulator
(b) Stimulus and response
(c) Signal and receptor
(d) Sensory kinase and phosphotransferase

60. Cholera toxin produced by Vibrio cholerae acts through
(a) Activating GTPase activity
(b) decreasing cAMP production
(c) Ribosylation of G-α substrate
(d) none

61. Opsins use a photo-isomerization reaction to translate electromagnetic radiation into cellular signals using
(a) GPCR
(b) RTK
(c) Two component pathway
(d) Serine/Threonine pathway

62. Incorrect statement about NO is
(a) Intracellular receptor present in smooth muscles
(b) Gyanylyl Cyclase is attached with NO receptor
(c) activated macrophage produce it
(d) cAMP is produced for its function

63. Glucagon produces
(a) cGMP
(b) cAMP
(c) DAG
(d) NO

64. Cytokine receptor follows
(a) GPCR
(b) RTK
(c) JAK-STAT
(d) Serine-threonine pathway

65. VEGF follows the pathway
(a) JAK-STAT
(b) RTK
(c) GPCR
(d) two component

66. β-γ complex of G-protein
(a) is not a signalling molecule
(b) activates second messenger of LGIC
(c) binds with GDP
(d) binds with Adenylate Cyclase to produce AMP

67. Toll-like receptors follow which signalling
(a) cAMP production
(b) activation of NF-AT
(c) Activation of NF-KB
(d) Ca^{++}/Calmodulin

68. Location of genetic system in prokaryotic cells is
(a) Nucleoid (b) Chromatin body
(c) Mitochondria (d) Both (a) & (b)

69. Functionally related genes are not clustered in
(a) Prokaryotic cells (b) Eukaryotic cells
(c) Viruses (d) All of these

70. Histones are not present in
(a) Bacteria (b) Algae
(c) Fungi (d) (a) and (b)

71. Sexuality in prokaryotic cells
(a) Haploid
(b) Diploid
(c) Partial diploid
(d) None of these

72. Mesosomes are present in
(a) Bacteria
(b) Fungi
(c) Protozoa
(d) Virus

73. Cytoplasmic membrane does not carry out respiration and photo synthesis in
(a) Eukaryotic cells
(b) Prokaryotic cells
(c) Both
(d) None of these

74. Eukaryotic cytoplasmic membrane contains
(a) Photosynthetic machinery
(b) Sterols
(c) Peptidoglycan
(d) Respiratory machinery

75. Locomotor organelles in Eukaryotic cells are
(a) Simple fibril
(b) Unifibrilled with '9 + 1' microtubules
(c) Multifibrilled with '9 + 2' microtubules
(d) (a) and (b)

76. Energy yielding metabolic pathway in eukaryotic cells is
(a) Glycolysis (Anaerobic)
(b) Glycolysis (aerobic)
(c) Nitrogen fixing
(d) All of these

77. Cell membranes are structurally identical in all animal cells
(a) True
(b) False
(c) Animals of particular environment
(d) Animals of particular species

78. Membranes are components of all of the following except
(a) Microtubule (b) Nucleus
(c) Golgi apparatus (d) Mitochondria

79. Which of the following is present in prokaryotic cell ?
(a) Mitochondria (b) Ribosomes
(c) Chloroplasts (d) Nuclear membrane

80. The foreign particle enters or passes through the
(a) Microtubule
(b) Ribosome
(c) Nucleus
(d) Plasma membrane

81. Bacterial cells are prokaryotic; in comparison, to typical eukaryotic cell, they would
(a) Be smaller
(b) Have a smaller nucleus
(c) Have a plasma membrane
(d) Have nuclear membrane

82. Which of the following correctly matches an organelle with its function ?
(a) Mitochondria – photosynthesis
(b) Nucleus – movement
(c) Lysosome – movement
(d) Central vacuole – storage

83. Mitochondria and chloroplasts share several common features, for e.g.,
(a) Both are capable of semi-autonomous growth and respiration
(b) Neither are compounds of the endo-membrane system
(c) Each contains a small amount of DNA
(d) All of these

84. Match the following :

1. Smooth ER	A. Consists of microtubules
2. Centrioles	B. Digests damaged cells
3. Peroxisomes	C. Synthesizes ribosome's
4. Nucleolus	D. Detoxifies alcohol in liver cells
5. Lysosome	E. Breaks down the by-product of cell respiration

(*a*) 1-D; 2-A; 3-E; 4-C; 5-B
(*b*) 1-D; 2-B; 3-A; 4-E; 5-C
(*c*) 1-B; 2-A; 3-E; 4-D; 5-C
(*d*) 1-C; 2-B; 3-D; 4-E; 5-A

NUMERICAL TYPE QUESTIONS

1. Ribosome that occur exclusively in mitochondria is–

2. Root cells of wheat has 2n = 42 chromosomes. Which one of the following is the basic chromosome number of wheat

3. How many mitotic divisions are needed for a single cell to make 128 cells?

4. How many reduction divisions are necessary for the formation of 200 grains of wheat

5. The number of mitotic cell division required to produce 256 cells from single cell would be

6. In an organism, if the normal diploid number of chromosomes is 8. How many chromatids are present in each daughter cell at the end of meiosis I?

7. Which type of ribosomes are present in the bacteria and where they are distributed ?

EXERCISE - II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2016

1. In a typical mitotic cell division cycle in eukaryotes, **M** phase occurs immediately after the
(*a*) G_0 phase
(*b*) **S** phase
(*c*) G_1 phase
(*d*) G_2 phase

2. ATP biosynthesis takes place utilizing the H^+ gradient in mitochondria and chloroplasts. Identify the correct sites of H^+ gradient formation.
(*a*) Across the outer membrane of mitochondria and across the inner membrane of chloroplast
(*b*) Across the inner membrane of mitochondria and across the thylakoid membrane of chloroplast
(*c*) Within the matrix of mitochondria and across the inner membrane of chloroplast
(*d*) Within the matrix of mitochondria and within the stroma of chloroplast

3. Which one of the following statements regarding G proteins is **INCORRECT**?
(*a*) GDP is bound to G protein in the resting stage
(*b*) GTP bound a subunit cannot reassemble with βγ dimer
(*c*) All G proteins are trimeric
(*d*) Activation of G protein may result in activation or inhibition of the target enzymes

2015

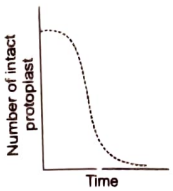
4. Which one of the following is INCORRECT about a typical apoptotic cell ?
(*a*) Phosphatidylserine is presented on the outer cell surface
(*b*) Cytochrome *c* is released from mitochondria
(*c*) Mitochondrial membrane potential does not change
(*d*) Annexin-V binds to the cell surface

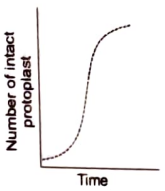
5. The cytokinetic organelle in plant cells is
(*a*) centriole (*b*) phragmoplast
(*c*) proplastid (*d*) chromoplastid

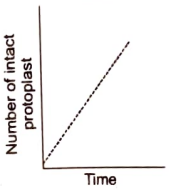
6. Choose the **CORRECT** sequence of steps involved in cytoplast production.
(*a*) Digestion of cell wall → protoplast viability → cybrid formation → osmotic stabilizer
(*b*) Osmotic stabilizer → digestion of cell wall → protoplast viability → cybrid formation
(*c*) Protoplast viability → osmotic stabilizer → digestion of cell wall → cybrid formation
(*d*) Osmotic stabilizer → digestion of cell wall → cybrid formation → protoplast viability

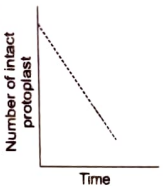
2014

7. If protoplasts are placed in distilled water, they swell and burst as a result of endosmosis. The plot representing the kinetics of burst is

(*a*)


(*b*)


(*c*)


(*d*)


2012

8. Choose the correct signal transduction pathway.
(*a*) Hormone → 7 TM receptor → G protein → cAMP → PKA
(*b*) Hormone → G protein → 7 TM receptor → cAMP → PKA
(*c*) Hormone → 7 TM receptor → G protein → PKA → cAMP
(*d*) Hormone → 7 TM receptor → cAMP → G protein → PKA

9. Match the entries in Group I with those in Group II.

Group I	Group II
P. MTT	1. Dihydrofolate reductase
Q. Annexin V	2. Succinate ehydrogenase
R. Methotrexate	3. Microtubules
S. Taxo	4. Phosphatidylserine

Codes :

	P	Q	R	S
(*a*)	3	1	4	2
(*b*)	2	4	1	3
(*c*)	2	3	4	1
(*d*)	4	2	1	3

10. Determine the correctness or otherwise of the following **Assertion** (A) and **Reason** (R).
Assertion: Plants convert fatty acids into glucose.
Reason: Plants have peroxisomes.
(*a*) Both (A) and (R) are true but (R) is not the correct reason for (A)
(*b*) Both (A) and (R) are true and (R) is the correct reason for (A)
(*c*) (A) is true but (R) is false
(*d*) (A) is false but (R) is true

2011

11. Apoptosis is characterized by
(*a*) necrosis
(*b*) programmed cell death
(*c*) membrane leaky syndrome
(*d*) cell cycle arrest process

12. A cell in G_1 of interphase has 12 chromosomes. How many chromatids will be found per cell during metaphase II of meiosis?
(*a*) 6 (*b*) 12
(*c*) 18 (*d*) 24

2010

13. Ames test is used to determine
(*a*) the mutagenicity of a chemical
(*b*) carcinogenicity of a chemical
(*c*) both mutagenicity and carcinogenicity of a chemical
(*d*) toxicity of a chemical

14. The formation of peptide cross-links between adjacent glycan chains in cell synthesis is called
(*a*) Transglycosylation (*b*) Autoglycosylation
(*c*) Autopeptidation (*d*) Transpeptidation

15. Determine the correctness or otherwise of the following Assertion (*a*) and the Reason (*r*)
Assertion : Cytoplasmic male sterility (cms) is invariably due to defect(*s*) in mitochondrial function.
Reason : cms can be overcome by pollinating a fertility restoring (Rf) plant with pollen from a non cms plant.
(*a*) both (*a*) and (*r*) are true and (*r*) is the correct reason for (*a*)
(*b*) both (*a*) and (*r*) are true and (*r*) is not the correct reason for (*a*)
(*c*) (*a*) is false but (*r*) is true
(*d*) (*a*) is true but (*r*) is false

2009

16. Caspases are involved in the process of
(*a*) DNA replication
(*b*) Mutation and recombination
(*c*) Antibody synthesis
(*d*) Aposptosis

2008

17. Which one of the following is the growth factor used for growth of tissues and organs in plant tissue culture?
(*a*) Cysteine (*b*) Cytokinin
(*c*) Cytidylate (*d*) Cyclic AMP

18. Some living cells (e.g. plant cell) have the capacity to give rise to whole organism. The term used to describe this property is
(a) morphogenesis (b) androgenesis
(c) totipotency (d) organogenesis

2007

19. A bioremedial solution to reduce oxides of nitrogen and carbon in flue gases is to integrate flue gas emission to
(a) micro-algal culture (b) fish culture
(c) mushroom culture (d) sericulture

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (a)	2. (c)	3. (c)	4. (b)	5. (a)	6. (c)	7. (a)	8. (c)	9. (b)	10. (b)
11. (c)	12. (a)	13. (b)	14. (a)	15. (d)	16. (c)	17. (c)	18. (b)	19. (c)	20. (d)
21. (a)	22. (b)	23. (b)	24. (c)	25. (a)	26. (d)	27. (a)	28. (b)	29. (a)	30. (a)
31. (b)	32. (a)	33. (d)	34. (a)	35. (b)	36. (a)	37. (b)	38. (d)	39. (d)	40. (b)
41. (d)	42. (c)	43. (c)	44. (c)	45. (a)	46. (c)	47. (d)	48. (a)	49. (d)	50. (c)
51. (a)	52. (d)	53. (c)	54. (a)	55. (a)	56. (d)	57. (c)	58. (b)	59. (a)	60. (d)
61. (a)	62. (d)	63. (b)	64. (c)	65. (b)	66. (b)	67. (c)	68. (d)	69. (b)	70. (a)
71. (c)	72. (a)	73. (a)	74. (b)	75. (c)	76. (a)	77. (b)	78. (a)	79. (b)	80. (d)
81. (a)	82. (d)	83. (d)	84. (a)						

Numerical Type Questions

1. 55S 2. 7 3. 7 4. 250 5. 8 6. 8
7. 70s, cytoplasm

EXERCISE – II

MCQ Type questions

1. (d)	2. (b)	3. (c)	4. (c)	5. (b)	6. (b)	7. (a)	8. (a)	9. (b)	10. (a)
11. (b)	12. (b)	13. (c)	14. (d)	15. (a)	16. (d)	17. (b)	18. (c)	19. (a)	

EXPLANATIONS

EXERCISE – I

MCQ TYPE QUESTIONS

1. The Golgi Apparatus is present in all eukaryotic cells. It is a central stack, pile of parallel, flattened intercommunicating sacs or cisternae and many peripheral tubules and vescicles. It is located near the nucleus. The two poles of a Golgi apparatus are called cis and trans face. The Golgi complex gives rise to primary lysosomes by budding from the trans face of cisternae.

2. The cristae extend inward to varying degrees and may fuse with those from the opposite side, dividing the mitochondria into compartments. They increase the surface area of the mitochondria to hold a variety of enzymes for oxidation-reduction reaction during respiration.

3. The centromere is the part of a chromosome that links sister chromatids. During mitosis, spindle fibres attach to the centromere via the kinetochore. It appears as a constriction when chromosomes contract during cell division.

4. A desmosome is a cell structure specialized for cell-to-cell adhesion. Desmosomes help to resist shearing forces and are found in simple and stratified squamous epithelium. The intercellular space is very wide (about 30 nm). Desmosomes are also found in muscle tissue where they bind muscle cells to one another. They act as spot-welds and keep cells firmly together.

5. Prokaryotic flagella consist of a single thread. It is composed of flagellin arranged in helical chains so as to form a hollow core.

6. The bonding of sugar residue to another organic compound is glycosylation. Glycoproteins are formed in the lumen of the granular or rough ER.

7. The middle lamella is a pectin layer which cements the cell walls of two adjoining cells together. Plants need this to give them stability and so that they can form plasmodesmata between the cells. It is the first formed layer which is deposited at the time of cytokinesis. The cell plate that is formed during cell division itself develops into middle lamella. The middle lamella is made up of calcium and magnesium pectate.

8. According to the fluid mosaic model of S.J. Singer and G.L. Nicolson (1972), the plasma membrane that surrounds these cells has two layers (a bilayer) of phospholipids (fats with phosphorous attached, which at body temperature are like vegetable oil fluid. Each phospholipid molecule has a head that is attracted to water and a tail that repels water. Both layers of the plasma membrane have the hydrophilic heads pointing toward the outside; the hydrophobic tails form the inside of the bilayer.

9. There are many different proteins embedded in the membrane. Except for flip-flop movements rest occurs.

10. Cilia are fine hair like vibratile. Their movement either propel the microbe or move the medium past a fixed cell. The cilia are enclosed by a unit membrane. Within the membrane is a fluid matrix having a supporting axial shaft. The shaft is composed of eleven microtubules. Two microtubules are single and lie at centre with a gap in between. They are called central singlets. The remaining nine microtubules are double and lie in a ring. The two microtubule forming a doublet are named as A and B subtubules. The microtubules single and double both are composed of globular protein tubulin. The arms of A microtubule contain a protein dynein. The latter is ATPase enzyme.

11. The outer membrane of mitochondria is smooth, freely permeable to most small molecules, contains fewer enzymes and is poor in proteins. It has porin proteins which form channels for the passage of molecules through it. It allows uptake of substrates and release of ATP. The inner membrane is semipermeable and regulates the passage of materials into and out of mitochondria. It is rich in enzymes and carrier proteins. It is usually produced into numerous infolds called cristae. It bears minute regularly spaced lollipop shaped particles known as oxysomes.

12. Both chloroplasts and mitochondria contain DNA and double membrane bound organelles having an inner membrane and outer membrane. Mitochondria occur in cytoplasm of both plants and animals cells. Mitochondria contain two chambers. The inner membrane forms mitochondrial cristae. Chloroplast is a also a double membranous organelle but found only in plants. The membrane bound matrix of chloroplast is stroma and inside the stroma thylakoids are present which form grana.

13. The cell membrane is a biological membrane that separates the interior of all cells from the outside environment. The cell membrane is selectively permeable to ions and organic molecules and controls the movement of substances in and out of cells. The basic function of the cell membrane is to protect the cell from its surroundings. It consists of the lipid bilayer with embedded proteins. Cell membranes are involved in a variety of cellular processes such as cell adhesion, ion conductivity and cell signalling and serve as the attachment surface for several extracellular structures, including the cell wall, glycocalyx, and intracellular cytoskeleton.

14. The ribosome is a large and complex molecular machine, found within all living cells, that serves as the primary site of biological protein synthesis (translation). Ribosomes link amino acids together in the order specified by messenger RNA (mRNA) molecules. Ribosomes consist of two major subunits-the small ribosomal subunit reads the mRNA, while the large subunit joins amino acids to form a polypeptide chain. These two subunits are associated by Mg^{2+} ions.

15. Glyoxysomes are specialized peroxisomes found in plants (particularly in the fat storage tissues of germinating seeds) and also in filamentous fungi. As in all peroxisomes, in glyoxysomes the fatty acids are hydrolyzed to acetyl-CoA by peroxisomal? oxidation enzymes. Besides peroxisomal functions, glyoxysomes possess additionally the key enzymes of glyoxylate cycle (isocitratelyase and malate synthase) which accomplish the glyoxylate cycle bypass.

16. Plasmodesmata are microscopic channels which traverse the cell walls of plant cells and some algal cells, enabling transport and communication between them.

17. The cytoskeleton (also CSK) is a cellular scaffolding or skeleton contained within a cell's cytoplasm. The cytoskeleton is present in all cells. Eukaryotic cells contain three main kinds of cytoskeletal filaments, which are microfilaments, intermediate filaments, and microtubules. The cytoskeleton provides the cell with structure and shape.

18. Zonula adherens, intermediate junction, or "belt desmosome" are protein complexes that occur at cell-cell junctions in epithelial tissues, usually more basal than tight junctions. An adherens junction is defined as a desmosome whose cytoplasmic face is linked to the actin cytoskeleton. They can appear as bands encircling the cell (zonulaadherens) or as spots of attachment to the extracellular matrix.

19. Nucleolus is a naked, roughly rounded darkly stained structure that is attached to chromatin at specific spot called nucleolar organizer region or NOR. It is area of synthesis of rRNA.

20. The endoplasmic reticulum (ER) is an organelle of cells in eukaryotic organisms that forms an interconnected network of membrane vesicles. Endopalsmic reticulum functions as cytoskeleton or intracellular and ultrastructural skeletal framework by providing mechanical support to colloidal cytoplasmic matrix. Proteins and enzymes synthesised by ribosomes enter the channels of rough endoplasmic reticulum for both intracellular use as well as extracellular transport.

21. In eukaryotic cells genes present in the cytoplasm are found in the mitochondria and inherited via egg cytoplasm. Material inheritances are the genetic effects that are transmitted via the maternal line. There are many such affect some resulting from the purely maternal origin of mitochondria or chloroplasts; others form effects on the phenotype of the embryo by easily developing within the female.

22. A nuclear membrane is the double lipid bilayer membrane which surrounds the genetic material and nucleolus in cells. The nuclear membrane consists of two lipid bilayers-the inner nuclear membrane, and the outer nuclear membrane. The space between the membranes is called the perinuclear space, a region contiguous with the lumen (inside) of the endoplasmic reticulum. The outer nuclear membrane is also contiguous with the endoplasmic reticulum. The inner nuclear membrane encloses the nucleoplasm, and is covered by the nuclear lamina, a mesh of intermediate filaments which stabilizes the nuclear membrane as well as being involved in chromatin function and gene expression. The nuclear membrane is punctured by thousands of nuclear pore complexes-large hollow proteins about 100 nm across, with an inner channel about 40 nm wide. They link the inner and outer nuclear membranes.

23. NUCLEAR SAP - or nucleoplasm. It is transparent, semi-fluid and colloidal substance which fills the nucleus. It contains nucleosides and a number of enzymes which are required for the synthesis and functioning of DNA, RNA, nucleoproteins etc

24. Oxysomes are the structures which are present on the surface of the folded Inner Membrane of the Mitochondria. They are also called f0-f1 particles or ATP synthase. They play the most important role in the formation of energy for the functioning of the cell. It contains enzymes ATP synthase responsible for synthesis of ATP.

25. Proteins synthesised by rough ER reach the cisternae of the golgicomlex. Here, they combine with carbohydrates to form glycoproteins. In animals golgi apparatus is involved in the packaging and the exocytosis of the following materials–

    (*i*) zymogen of exocrine pancreatic cells

    (*ii*) mucus secretions by goblet cells

    (*iii*) secretion of collagen.

    (*iv*) formation of yolk and vitelline membrane.

26. Phagosomes and pinosomes are collectively referred to as endosomes. An endosome is a membrane-bound compartment inside eukaryotic cells. It is a compartment of the endocytic membrane transport pathway from the plasma membrane to the lysosome. Molecules internalized from the plasma membrane can follow this pathway all the way to lysosomes for degradation, or they can be recycled back to the plasma membrane. Molecules are also transported to endosomes from the Golgi and either continue to lysosomes or recycle back to the Golgi. Endosomes provide an environment for material to be sorted before it reaches the degrading lysosome.

27. A eukryotic flagellum is a bundle of nine fused pairs of microtubuledoublets surrounding two central single microtubules. The so-called "9 + 2" structure is characteristic of the core of the eukaryotic flagellum called an axoneme. Each of the outer 9 doublet microtubules extends a pair of dynein arms (an "inner" and an "outer" arm) to the adjacent microtubule; these dynein arms are responsible for flagellar beating, as the force produced by the arms causes the microtubule doublets to slide against each other and the flagellum as a whole to bend. These dynein arms produce force through ATP hydrolysis.

28. Mesosomes are convoluted multidimensional membranous structures that formed by the invagination of the plasma membrane of prokaryotes. They are of two types-septal and lateral. Septalmesosome is in contact with nucleoid and takes part in separation of replicated nucleods and septum formation. Lateral mesosome is equivalent to mitochondria. It is the site of respiration.

29. Lysozymes, also known as muramidase or N-acetylmuramideglycanhydrolase, are glycoside hydrolases. These are enzymes (EC3.2.1.17) that damage bacterial cell walls by catalyzing hydrolysis of 1,4-beta-linkages between N-acetylmuramic acid and N-acetyl-D-glucosamine residues in a peptidoglycan and between N-acetyl-D-glucosamine residues in chitodextrins. It does this by binding to the peptidoglycan molecule in the binding site within the prominent cleft between its two domains. This causes the substrate molecule to adopt a strained conformation similar to that of the transition state. In this stressed state, the glycosidic bond is easily broken. Lysozyme is abundant in a number of secretions, such as tears, saliva, human milk, and mucus.

30. Chloroplast is an organelle found within the cells of green plants and eukaryotic algae which contains the membranes, photosynthetic pigments (notably chlorophyll), and enzymes necessary for photosynthesis. In higher plants, chloroplasts are usually disk-shaped, arranged in a single layer in the cytoplasm. It has an inner and outer membrane, as well as a third membrane-bound sac called the thylakoid. Stacks of thylakoids, known as grana, are where the Light Reactions of photosynthesis take place.

31. Lignin is the complex polymorphic molecule composed of phenyl propanoid units associated with cellulose in cell walls of sclerenchyma, xylem vessels, and tracheids making them strong and rigid.

32. Basal body (also called a basal granule or kinetosome) is an organelle formed from a centriole, and a short cylindrical array of microtubules. It is found at the base of a eukaryotic undulipodium (cilium or flagellum) and serves as a nucleation site for the growth of the axoneme microtubules. Centrioles, from which basal bodies are derived, act as anchoring sites for proteins that in turn anchor microtubules within centrosomes, one type of microtubule organizing centre (MTOC). These microtubules provide structure and facilitate movement of vesicles and organelles within many eukaryotic cells. Basal bodies, however, are specifically the bases for cilia and flagella that extend out of the cell.

33. The plasma membrane is a lipoproteinaceous layer where there is a lipid bilayer with proteins sandwiched in between. The lipid molecules have polar heads arranged parallel with hydrophobic tails.

34. The mesophyll tissue of the leaf blade is differentiated into an upper palisade and lower spongy mesophyll. The former has larger chloroplast and the cells are elongated and closely packed whereas cells of spongy mesophyll have fewer chloroplasts and separated by large air spaces.

35. Selectively permeable and does not allow the passage of large solute molecules but is semi-permeable for water.

36. Prokaryotes (before nucleus) are single celled organisms without a nucleus (bacteria and archaea)

37. Microscopes are used for studying cellular structures. They are used to magnify small objects. In electron microscopes, a high energy beam of electrons is focused through electromagnetic lenses. It can magnify very small details with high resolving power. The increased resolution results from the shorter wavelength of the electron beam.

38. Molecular assemblies are large organised sets of molecular units that make up parts of organelles. For example one common macromolecular assembly is the microtubule which is important in forming structure in the cell related to maintaining the cell structure or related to cell movement.

39. According to fluid mosaic fluid there is rapid internal moyion involving flexing within each lipid molecule a rapid lateral diffusion of the lipids is possible and a slow 'flip-flop' motion, i.e., a transfer of lipid molecule from one side of bilayer to the other, is also possible. The lipid molecules might also rotate about their axes. The proteins of the membrane are concerned with the enzymatic activity of the membrane, with transport of molecules and with a receptor function whereas; the lipid bilayer provides the permeability barrier.

40. Vesicles provided by the golgi apparatus unite to form phragmoplast, which join to form cell plate. Cell plate is, first laid down in center and then proceeds towards periphery.

41. Synapsis is the pairing of homologous chromosome during zygotene stage of prophase I of meiotic division-I. The homologous chromosomes come from the mother and father.

42. Centrioles are organdies important in spindle formation during nuclear division. These are found commonly in most animal cells and some lower plant cells.

43. Because first phase of meiosis includes reduction division while second phase exhibits mitotic divisions.

44. As a result of it pollen grains formed.

45. Due to attachment of spindle fibres at the centromeres (kinetochore) of chromosomes, the chromosomes are arranged in the centre or at equator, or equatorial plate or metaphase due to their active movements.

46. The check points are basically present in the interphase.

47. Centrioles replicate in G2 of interphase of cell cycle but some authors reported this duplication in early prophase.

48. Pairing between the homologous chromosomes takes place in zygotene stage of prophase-I.

49. In the somatic cell cycle S-phase is a stage of replication of each chromosome by synthesis of a new DN molecule on the template of existing DNA.

50. This occurs for the exchange of genetic material during crossing over.

51. Leptotene is a stage in the prophase I of meiosis. In this stage chromosome appears thin and long.

52. Separation of sister chromatids or movement of them to the opposite pole is carried out at anaphase II. In anaphase-I chromosome separation and movement take place.

53. The sequence of interphase (I-Phase) is (G1 S G2)

54. Calcium Dependent Kinases (CDKs) have the ability to control cell cycle activities during cell division.

55. Diagram 1st represents crossing over that takes place in pachytene stage of prophase I during meiosis.

56. Dictysomes contain phospholipids, proteins, enzymes and vitamin C, which is synthesized by ER.

57. Peroxisomes are involved in synthesis of glycine and serine in the process of photo respiration.

58. Lysosome function or trigger of cell division or initiate cell division by digesting repressor molecule.

59. Two-component signal transduction systems enable bacteria to sense, respond, and adapt to changes in their environment or in their intracellular state. Each two-component system consists of a sensor protein-histidine kinase (HK) and a response regulator (RR). In the prototypical two-component pathway, the sensor HK phosphorylates its own conserved His residue in response to a signal(s) in the environment. Subsequently, the phosphoryl group of HK is transferred onto a specific Aspartate residue on the RR. The activated RR can then effect changes in cellular physiology, often by regulating gene expression. Two-component pathways thus often enable cells to sense and respond to stimuli by inducing changes in transcription. In bacteria, two-component systems (TCS) are widely used signal transduction devices which are engaged in a multitude of gene regulatory systems that respond to changing growth conditions.

60. When cholera toxin (CT) is released from the bacteria in the infected intestine, it binds to the intestinal cells known as enterocytes through the interaction of the pentameric B subunit of the toxin with the GM1 ganglioside receptor on the intestinal cell, triggering endocytosis of the toxin. Next, the A/B cholera toxin must undergo cleavage of the A1 domain from the A2 domain in order for A1 to become an active enzyme. Once inside the enterocyte, the enzymatic A1 fragment of the toxin A subunit enters the cytosol, where it activates the G protein $G_s\alpha$ through an ADP-ribosylation reaction that acts to lock the G protein in its GTP-bound form, thereby continually stimulating adenylate cyclase to produce cAMP. The high cAMP levels activate the cystic fibrosis transmembrane conductance regulator (CFTR), causing a dramatic efflux of ions and water from infected enterocytes, leading to watery diarrhoea. The effects of cholera involve the actions of other *Vibrio cholerae* toxins that aid the pathogen in its colonisation, coordinated expression of virulence factors, and toxin action. The toxic action of CT is initiated by binding of its B subunits to the high-affinity monosialo-ganglioside GM1 receptors. Each B subunit monomer has a binding site for GM1. Moreover, a single amino acid from a neighbouring B subunit also plays an important role in binding, explaining the much higher binding affinity of the CTB pentamer, as compared to that of the CTB monomer. Endocytosis of CT may follow one of three pathways: (i) lipid raft/caveolae mediated endocytic pathway, (ii) clathrin mediated endocytic pathway, or (iii) ADP-ribosylation factor 6 (Arf6)-associated endocytic pathway. CT then travels to the endoplasmic reticulum (ER) in a retrograde fashion.

61. Vision itself depends on a biochrome that consists of a protein, opsin, attached to a chromophore. The chromophore may be either retinal (vitamin A1), in which case the molecule is called rhodopsin; or 3-dehydroretinal (vitamin A2), in which case the molecule is called porphyropsin. When light enters the eye and strikes the visual biochrome, the molecule undergoes a chemical change that stimulates the receptor nerve and thereby produces a visual stimulus. They consist of an apoprotein, opsin covalently linked to the chromophore cis-retinal. Vision is affected through the absorption of a photon by cis-retinal which is isomerised to trans-retinal. This isomerisation leads to a change of conformation of the protein. Opsins are integral membrane proteins that belong to the rhodopsin-like superfamily of G-protein-coupled receptors (GPCRs). The activating ligands of the different superfamily members vary widely in structure and character, yet the proteins appear faithfully to have conserved a basic structural framework, believed to consist of 7 transmembrane (TM) helices.

62. When NO forms, it has a half-life of only a few seconds, in large part because superoxide anion has a high affinity for NO (both molecules have an unpaired electron making them highly reactive). Therefore, superoxide anion reduces NO bioavailability. NO also avidly binds to the haeme moiety of haemoglobin (in red blood cells) and the haeme moiety of the enzyme guanylyl cyclase, which is found in vascular smooth muscle cells and most other cells of the body. Therefore, when NO is formed by vascular endothelium, it

rapidly diffuses into the blood where it binds to haemoglobin and subsequently broken down. It also diffuses into the vascular smooth muscle cells adjacent to the endothelium where it binds to and activates guanylyl cyclase. This enzyme catalyzes the dephosphorylation of GTP to cGMP, which serves as a second messenger for many important cellular functions, particularly for signalling smooth muscle relaxation.

63. Glucagon is a pancreatic hormone which is produced by cells in the islets of Langerhans. Glucagon is a 29-amino-acid peptide that is produced specifically by the alpha cells of the islets. It has a high degree of similarity with several glucagon-like peptides that are secreted by cells scattered throughout the gastrointestinal tract. Glucagon signals through its receptor on the cell surface. The binding of glucagon to the extracellular loops of the glucagon receptor results in conformational changes of the latter, leading to subsequent activation of the coupled G proteins. At least two classes of G proteins are known to be associated with and involved in the signal transduction of the glucagon receptor, namely $G_s\alpha$ and G_q. The activation of $G_s\alpha$ leads to activation of adenylate cyclase, increase in intracellular cAMP levels, and subsequent activation of protein kinase A (PKA). The activation of G_q leads to the activation of phospholipase C, production of inositol 1, 4, 5-triphosphate, and subsequent release of intracellular calcium.

64. Cytokines are soluble extracellular proteins or glycoproteins that are crucial intercellular regulators and mobilizers of cells engaged in innate as well as adaptive inflammatory host defences, cell growth, differentiation, cell death, angiogenesis, and development and repair processes aimed at the restoration of homeostasis. Cytokines are released by various cells in the body, usually in response to an activating stimulus, and they induce responses through binding to specific receptors on the cell surface of target cells. Cytokines can be grouped by structure into different families and their receptors can likewise be grouped. The receptors and their corresponding cytokines have been divided into several families based on their structure and activities, including type I cytokine receptors, type II cytokine receptors, TNF receptor family, chemokine receptors, TGF-beta receptors, and members of immunoglobulin superfamily. Signalling by cytokine receptors depends upon their association with the Janus kinases (JAKs), which couple ligand binding to tyrosine phosphorylation of signalling proteins recruited to cytokine receptor complex. Among these signalling proteins are unique family of transcription factors named the signal transducers and activators of transcription (STATs).

65. Vascular endothelial growth factor (VEGF) is a chemical signal produced by cells that stimulates the growth of new blood vessels. It is part of the system that restores the oxygen supply to tissues when blood circulation is inadequate. VEGF is a sub-family of growth factors, specifically the platelet-derived growth factor family of cystine-knot growth factors. They are important signalling proteins involved in both vasculogenesis and angiogenesis. The binding of VEGF to VEGFR-2 leads to a cascade of different signalling pathways, resulting in the up-regulation of genes involved in mediating the proliferation and migration of endothelial cells and promoting their survival and vascular permeability. For example, the binding of VEGF to VEGFR-2 leads to dimerization of the receptor, followed by intracellular activation of the PLC gamma; PKC-Raf kinase-MEK-mitogen-activated protein kinase (MAPK) pathway and subsequent initiation of DNA synthesis and cell growth, whereas activation of the phosphatidylinositol 3' -kinase (PI3K)-Akt pathway leads to increased endothelial-cell survival. One of the most common intracellular signaling pathways triggered by RTKs is known as the mitogen-activated protein (MAP) kinase cascade, because it involves three serine-threonine kinases. The pathway starts with the activation of Ras, a small G protein anchored to the plasma membrane.

66. The hetero-trimeric guanine nucleotide-binding proteins (G-proteins) function to transduce signals from this vast panoply of receptors to effector systems including ion channels and enzymes that alter the rate of production, release or degradation of intracellular second messengers. G Protein-linked Receptors have an extracellular N-terminus and a cytosolic C-terminus separated by seven transmembrane alpha helices connected by peptide loops. The two classes of G Proteins are large hetero-trimeric G Proteins and small monomeric G Proteins. In heterotrimeric G proteins (G alpha, beta, gamma), when a messenger binds the G Protein-linked receptor, the receptor changes conformation to allow association of the trimeric G Protein with the receptor. G-alpha subunit binds the guanine nucleotide (GDP or GTP). This interaction causes the G alpha subunit to release the GDP, pick up a GTP and detach from the complex. Depending upon the G protein in question, the GTP-G alpha complex, the G beta- G gamma complex or both bind target protein(s). The G alpha will remain an activating messenger until the GTP is hydrolyzed by the G alpha subunit (GTP -> GDP +Pi). The "inactive" GDP-G alpha will then re-associate with the G-beta-G alpha complex to rapidly turn down this pathway when the original stimulatory signal is removed.

67. Toll-like receptor (TLR) signalling plays an essential role in the innate immune response. Activation of TLR signalling through recognition of pathogen-associated molecular patterns leads to the transcriptional activation of genes encoding for pro-inflammatory cytokines, chemokines and co-stimulatory molecules, which subsequently control the activation of antigen-specific adaptive immune response. TLRs are members of a larger superfamily of interleukin-1 receptors (IL-1Rs) that share significant homology in their cytoplasmic regions. In particular, TLRs and members of the IL-1R family share a conserved stretch of ~200 amino acids in their cytoplasmic region known as the Toll/IL-1R (TIR) domain. Triggering of the IL-1R or TLR causes the adaptor protein MyD88 to be recruited to the receptor complex, which in turn promotes association with the IL-1R-associated kinases IRAK4 and IRAK1. During the formation of this complex, IRAK4 is activated, leading to the hyper-phosphorylation of IRAK-1, which then induces the interaction of TRAF6 with the complex. The IRAK-4-IRAK-1-TRAF6 complex then interacts at the membrane with another preformed complex consisting of TAK1, TAB1, and TAB2. This interaction induces phosphorylation of TAB2 and TAK1, which then translocate together with TRAF6 and TAB1 to the cytosol. TAK1 is subsequently activated in the cytoplasm, leading to the activation of IKK. Inactive IKK sequesters NF-κB in the cytoplasm, but activation leads to phosphorylation and degradation of IκB and consequent release of NF-κB. Activation of TAK1 also results in the activation of MAP kinases and c-Jun NH2-terminal kinase (JNK).

68.

Genetic system	Prokaryotic cells	Eukaryotic cells
Locomotion	Nucleoid, chromatin body, or nuclear material	Nucleus, mitochondria chloroplasts

69. For eukaryotic cells nucleolus is present; functionally related genes are not clustered.

70. Histones are proteins (core proteins) which are not present in prokaryotic cells (bacteria).

71. In prokaryotic cells (sexuality) is partial diploid. That means zygote nature is merozygotic.

72. Mesosomes are present in bacteria (prokaryotes)

73. Cytoplasmic membranes of eukaryotic cells contain sterols, and they do not contain respiratory and photosynthetic machinery. Based on this reason Eukaryotic cells do not carry out respiration and photosynthesis.

74. Eukaryotic cytoplasmic membrane contain sterols.

75. Locomotor organelles in prokaryotes is simple fibri and in eukaryotes having multifibrilled with 9 + 2 microtubules.

76. Glycolysis is the pathway for anaerobic energy yielding mechanism for eukaryotic cells.

77. The structure of cell membrane varies with the cell type. Membranes that contain an electron transport chain are involved with active transport contain more proteins membranes with more cholesterol are less fluid. Since form determines the function, the structure of the membrane varies with its function.

78. Microtubules are part of the cytoskeletal structure and are made up of protein tubulin. The others all consist of selectively permeable plasma membranes.

79. Prokaryotes have no internal membranes. Therefore, they lack mitochondria, endoplasmic reticulum and nuclear membrane. They do have small ribosome's.

80. The enter/leave a cell. Substances must pass through plasma membrane. Plasma membrane defines the boundary of the cell.
81. The smaller prokaryotic cells range 0.5 to 2.0m. Eukaryotic cells are larger than prokaryotic cells.
82. But remember, plant cells have a central vacuole, while most animal cells have several vacuole located throughout the cytoplasm.
83. All of the features listed above appear to be related to the origin of these organelles.
84. The smooth ER detoxifies the cell, produces steroids like cholesterol and contains the rough ER, to the golgi.
    - Centriols consist of microtubules in a 9 triplet configuration.
    - Peroxines break down peroxide, which is poisionous by product of cell respiration.
    - The nucleolus is responsible for synthesizing rRNA, one of the main components of ribosomes.
    - Lysosomes contain hydrolytic enzymes that break down nutrients, waste products and all debris.

NUMERICAL TYPE QUESTIONS

1. 55 S mitochondrial ribosomes occur in variety of forms, the predominant form in multicellular organism being the 55 S ribosomes.
2. For wheat. $2n = 6x = 42$. Therefore, $x = 7$

   'x' represents basic or genomic number.
3. A single mitotic division results in the production of two cells from a single one.
4. For the formation of 100 grains of wheat 100 grains and 100 megaspores are required microspore mother cell produces 4 microspores (pollen grains) as a result of reduction division. Hence for the formation of 100 pollen grains, (100/4) = 25 PMC are needed and each will undergo one reduction division. Each megaspore mother cell will produce 4 megaspores in which one become functional and other three degenerate. Thus each MMC produces one megaspore. So 100 MMC are required to produce 100 megaspores. Total number of reduction divisions to produce 100 grains is 100 + 25 = 125.
5. Mitotic cells division is equational division where mother cell divides to form 2 daughter cells. So to produce 256 cells from a single cell, 8 mitotic divisions will occur which can be represented as 1 2 4 8 16 32 64 128 256.
6. Because each chromosome bears two chromatids and at the end of meiosis-I chromosome number becomes halved i.e., 4 and number of chromatids are $4 \times 2 = 8$.
7. Bacteria are prokaryotic cell which contain 70 s ribosomes in their structure. 70 s ribosomes are subdivided into 50s and 30s subunits. S-Sedimentation coefficient / Svedberg units.

   70s ribosome's are distributed in the cytoplasm

EXERCISE - II

MCQ TYPE QUESTIONS

1. G_2 phase, or Gap 2 phase, is the third and final subphase of Interphase in the cell cycle directly preceding mitosis. It follows the successful completion of S phase, during which the cell's DNA is replicated. G2 phase ends with the onset of prophase, the first phase of mitosis in which the cell's chromatin condenses into chromosomes.
2. During chemiosmosis, the free energy from the series of reactions that make up the electron transport chain is used to pump hydrogen ions across the membrane, establishing an electrochemical gradient. Hydrogen ions in the matrix space can only pass through the inner mitochondrial membrane through a membrane protein called ATP synthase.

   ATP synthesis in chloroplasts is very similar to that in mitochondria: Electron transport is coupled to the formation of a proton (H+) gradient across a membrane. The energy in this proton gradient is then used to power ATP synthesis. Two types of processes that contribute to the formation of the proton gradient are:

   (a) processes that release H+ from compounds that contain hydrogen, and

   (b) processes that transport H+ across the thylakoid membrane.
3. All G proteins are not trimeric. GPCR dimerization also was proposed to participate in arrestin-mediated desensitization based on the size of the cytoplasmic tip of rhodopsin and the presence of two cup-like domains in arresting both of which were implicated in receptor binding which one arrestin binds to two receptors in a dimer.
4. During apoptotic the mitochondrial membrane potential (mmp) decreases.
5. The cytokinetic organelle in plant cells is Phragmoplast.
6. Digestion of cell wall → protoplast viability → cybrid formation → osmotic stabilizer
7. The plot of swelling of protoplasts and bursting during endosmosis when protoplasts are kept or placed in distilled water is a sigmoidal curve where number of intact protoplasts decrease with time. So, the plot is represented by graph in (a).
8. In the G protein associated signal cascade system, binding of hormone to a cell surface receptor, like the 7TM (a 7 transmembrane alpha helices) which is a part of G protein. This will activate production of second messengers like cAMP by the enzyme adenylate cyclase. The cAMP in turn binds and activates down stream signal proteins line cAMP dependent protein kinase A (PKA).
9. MTT- binds to succinate dehydrogenase Annexin V-specifically binds to phosphatidyl serine

   Methotrexate- allosterically inhibits dihydrofolate reductase

   Taxol-suppresses microtubule dynamics.
10. Plants can make glucose from fatty acids, but this is only because they are able to use the glyoxylate cycle instead of the Krebs cycle.
11. Programmed cell death (PCD) is a death of a cell in any form, mediated by an intracellular program. PCD is carried out in a regulated process which usually confers advantage during an organisms life cycle. PCD also serves an important function during both plant and metazoan (multicellular animals) tissue development. Apoptosis is a form of programmed cell death. However, necrosis is a non physiological process that occurs as a result of infection or injury.
12. Meiosis is a two part cell division process in organism that sexually reproduces which results in gametes with one half the number of chromosomes of the parent cell. Chromosomal replication does not occur between meiosis I and meiosis II; meiosis I proceeds directly to meiosis II without going through interphase. The second part of the meiosis i.e. meiosis II resembles mitosis more than meiosis I. Chromosomal number which have already been reduced to haploid (n) by the end of meiosis I, remains unchanged after this division. In meiosis II, the phases are again analogous to mitosis: prophase II, metaphase II, anaphase II and telophase II. Meiosis begins with two haploid (n=2) cell and end with 4 haploid (n=2) cells.
13. Mutagenicity and carcinogenicity are not mutually exclusive biological characteristics. The Ames test is an assay of chemicals to determine their mutagenic potential. A positive result implies the carcinogenic property of chemicals as well. The variable tested is the chemical's potential to cause a reversion to growth and estimate frame-shift & point mutations. This allows detection of cancer-causing property based on mutagenic action.
14. Transpeptidation is a reaction involving the transfer of one or more amino acids from one peptide chain to another, as by transpeptidase action, or of a peptide chain itself, as in bacterial cell wall synthesis. The enzyme transpeptidase is also known as DD-transpeptidase, a bacterial enzyme that cross-links the peptidoglycan chains to form rigid cell walls.
15. Cytoplasmic male sterility (CMS) is a condition under which a plant is unable to produce functional pollen. CMS is a maternally inherited trait that is often associated with unusual open reading frames (ORFs) found in mitochondrial genomes. Male fertility can be restored by nuclear-encoded fertility restorer (Rf) gene(s). These fertility-restorer genes are thought to block or compensate for cytoplasmic dysfunctions that are phenotypically expressed during pollen development. On the one hand, sterility results from mitochondrial genes causing cytoplasmic dysfunction, and on the other, fertility restoration relies on nuclear genes that suppress cytoplasmic dysfunction.
16. Caspases, or cysteine-aspartic proteases or cysteine-dependent aspartate-directed proteases are a family of cysteine proteases that play essential roles in apoptosis (programmed cell death), necrosis, and inflammation.

17. Cytokinins (CK) are a class of plant growth substances (phytohormones) that promote cell division, or cytokinesis, in plant roots and shoots. They are involved primarily in cell growth and differentiation, but also affect apical dominance, axillary bud growth, and leaf senescence. Folke Skoog discovered their effects using coconut milk in the 1940s at the University of Wisconsin Madison.

18. Totipotency is the ability of a single cell to divide and produce all the differentiated cells in an organism, including extraembryonic tissues. Totipotent cells include spores and zygotes.

19. The advantages of using a microalgal-based system in integration with flue gas emission are that:

- High purity CO_2 gas is not required for algal culture. Flue gas containing varying amounts of CO_2 can be fed directly to the microalgal culture. This will simplify CO_2 separation from flue gas significantly.
- Some combustion products such as NOx or SOx can be effectively used as nutrients for microalgae. This could simplify flue gas scrubbing for the combustion system.
- Microalgae culturing may yield high value commercial products. Sale of these high value products can offset the capital and the operation costs of the process.
- The envisioned process is a renewable cycle with minimal negative impacts on environment.